PCOS ULTRASOUND CLASSIFICATION — UNIQUE MODEL + CLASSIFIER COMPARISON

Only the 3-seed experiments (green rows) were run under a fixed, documented seeding protocol and were re-obtained identically in independent runs, so they are the reproducible results; every experiment marked “Not reproducible” was run as a single exploratory experiment (no fixed seed controlling network training, or one seed without a repeat run) and is reported for comparison only.

1. Overall Comparison — Best to Worst by Test Accuracy

Rank	Model	Classifier	Test Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Repro-ducible?	Source notebook (.ipynb)
1	MobileNetV2	AdaBoost	91.71 ± 2.07 Seed / Value **2622715460 / 94.01** 512 / 89.99 2 / 91.14	91.67 ± 2.05 Seed / Value **2622715460 / 93.96** 512 / 90.00 2 / 91.04	91.71 ± 2.07 Seed / Value **2622715460 / 94.01** 512 / 89.99 2 / 91.14	91.52 ± 2.21 Seed / Value **2622715460 / 93.95** 512 / 89.63 2 / 90.97	**Reproducible**	3_Seed_exp.ipynb (cell 4) (seeded) MobileNet and EfficientNet B0-Adaboost.ipynb
2	MobileNetV2 + EfficientNet-B0	AdaBoost ensemble (soft vote)	92.44 ± 1.03 Seed / Value **2622715460 / 93.53** 512 / 92.30 2 / 91.49	92.38 ± 1.06 Seed / Value **2622715460 / 93.49** 512 / 92.26 2 / 91.38	92.44 ± 1.03 Seed / Value **2622715460 / 93.53** 512 / 92.30 2 / 91.49	92.33 ± 1.03 Seed / Value **2622715460 / 93.43** 512 / 92.15 2 / 91.40	**Reproducible**	(seeded) Ensemble.ipynb (cell 11)
3	MobileNetV2 + EfficientNet-B0 + EfficientNet-B3	AdaBoost ensemble (soft vote)	92.37 ± 1.16 Seed / Value **2622715460 / 93.46** 512 / 92.51 2 / 91.14	92.31 ± 1.19 Seed / Value **2622715460 / 93.41** 512 / 92.47 2 / 91.05	92.37 ± 1.16 Seed / Value **2622715460 / 93.46** 512 / 92.51 2 / 91.14	92.23 ± 1.21 Seed / Value **2622715460 / 93.37** 512 / 92.35 2 / 90.96	**Reproducible**	(seeded) Ensemble.ipynb (cell 11)
4	MiT-B2 + MobileNetV2	Gaussian Naive Bayes	92.85	92.93	92.85	92.88	Not reproducible‡	MIT more epochs.ipynb (cell 30)
5	EfficientNet-B3	AdaBoost	91.71 ± 0.88 Seed / Value 2622715460 / 91.35 **512 / 92.71** 2 / 91.08	91.62 ± 0.89 Seed / Value 2622715460 / 91.24 **512 / 92.64** 2 / 90.99	91.71 ± 0.88 Seed / Value 2622715460 / 91.35 **512 / 92.71** 2 / 91.08	91.59 ± 0.93 Seed / Value 2622715460 / 91.25 **512 / 92.64** 2 / 90.88	**Reproducible**	3_Seed_exp.ipynb (cell 8) (seeded) EfficientNet B3 with Adaboost.ipynb
6	EfficientNet-B0	AdaBoost	92.08 ± 0.49 Seed / Value **2622715460 / 92.64** 512 / 91.83 2 / 91.76	92.05 ± 0.50 Seed / Value **2622715460 / 92.62** 512 / 91.73 2 / 91.79	92.08 ± 0.49 Seed / Value **2622715460 / 92.64** 512 / 91.83 2 / 91.76	91.99 ± 0.43 Seed / Value **2622715460 / 92.49** 512 / 91.72 2 / 91.77	**Reproducible**	3_Seed_exp.ipynb (cell 12) (seeded) MobileNet and EfficientNet B0-Adaboost.ipynb
7	MobileNetV2 + EfficientNet-B3	AdaBoost ensemble (soft vote)	92.12 ± 0.73 Seed / Value **2622715460 / 92.57** 512 / 92.51 2 / 91.28	92.07 ± 0.70 Seed / Value **2622715460 / 92.50** 512 / 92.45 2 / 91.26	92.12 ± 0.73 Seed / Value **2622715460 / 92.57** 512 / 92.51 2 / 91.28	91.97 ± 0.80 Seed / Value **2622715460 / 92.50** 512 / 92.37 2 / 91.05	**Reproducible**	(seeded) Ensemble.ipynb (cell 11)
8	MobileNetV2	Direct (fine-tuned CNN)	88.96 ± 3.32 Seed / Value **2622715460 / 92.44** 512 / 85.83 2 / 88.62	89.13 ± 3.41 Seed / Value **2622715460 / 92.57** 512 / 85.76 2 / 89.07	88.96 ± 3.32 Seed / Value **2622715460 / 92.44** 512 / 85.83 2 / 88.62	88.39 ± 3.61 Seed / Value **2622715460 / 92.20** 512 / 85.02 2 / 87.94	**Reproducible**	3_Seed_exp.ipynb (cell 4) (seeded) MobileNet and EfficientNet B0-Adaboost.ipynb
9	EfficientNet-B3	SVM	92.37	92.53	92.37	92.43	Not reproducible	efficinetnet b3 baseline and ml.ipynb (cell 15)
10	EfficientNet-B3	Random Forest	92.37	92.40	92.37	92.17	Not reproducible	efficinetnet b3 baseline and ml.ipynb (cell 5)

Rank	Model	Classifier	Test Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Repro-ducible?	Source notebook (.ipynb)
11	EfficientNet-B0 + EfficientNet-B3	AdaBoost ensemble (soft vote)	91.76 ± 0.62 Seed / Value 2622715460 / 91.89 **512 / 92.30** 2 / 91.08	91.67 ± 0.66 Seed / Value 2622715460 / 91.80 **512 / 92.27** 2 / 90.96	91.76 ± 0.62 Seed / Value 2622715460 / 91.89 **512 / 92.30** 2 / 91.08	91.62 ± 0.62 Seed / Value 2622715460 / 91.78 **512 / 92.14** 2 / 90.94	**Reproducible**	(seeded) Ensemble.ipynb (cell 11)
12	EfficientNet-B3	XGBoost	92.17	92.19	92.17	91.96	Not reproducible	efficinetnet b3 baseline and ml.ipynb (cell 13)
13	InceptionV3	Direct (fine-tuned CNN)	91.76	91.80	91.76	91.52	Not reproducible	Inception v3 baseline and ml.ipynb (cell 3)
14	MobileNetV2	Gaussian Naive Bayes	91.42	91.85	91.42	91.54	Not reproducible	MobileNet with ML.ipynb (cell 8)
15	ResNet18	Random Forest	90.60	90.54	90.60	90.56	Not reproducible	ResNet with ML.ipynb (cell 4)
16	MobileNetV2	Random Forest	90.40	90.26	90.40	90.29	Not reproducible	MobileNet with ML.ipynb (cell 4)
17	VGG16	Direct (fine-tuned CNN)	90.33	90.56	90.33	89.90	Not reproducible	Vgg with Dice Loss (Baseline).ipynb (cell 5)
18	VGG16	SVM	90.05	90.55	90.05	90.20	Not reproducible	Xgboost and SVM and all 4 models.ipynb (cell 27)
19	EfficientNet-B3	KNN	90.05	89.97	90.05	89.78	Not reproducible	efficinetnet b3 baseline and ml.ipynb (cell 7)
20	EfficientNet-B3	Direct (fine-tuned CNN)	89.12 ± 1.33 Seed / Value **2622715460 / 90.05** 512 / 87.60 2 / 89.71	89.28 ± 1.21 Seed / Value **2622715460 / 90.10** 512 / 87.90 2 / 89.84	89.12 ± 1.33 Seed / Value **2622715460 / 90.05** 512 / 87.60 2 / 89.71	88.60 ± 1.53 Seed / Value **2622715460 / 89.69** 512 / 86.85 2 / 89.27	**Reproducible**	3_Seed_exp.ipynb (cell 8) (seeded) EfficientNet B3 with Adaboost.ipynb
21	EfficientNet-B0	Direct (fine-tuned CNN)	89.01 ± 0.58 Seed / Value 2622715460 / 89.03 **512 / 89.58** 2 / 88.42	89.04 ± 0.44 Seed / Value 2622715460 / 88.91 **512 / 89.52** 2 / 88.68	89.01 ± 0.58 Seed / Value 2622715460 / 89.03 **512 / 89.58** 2 / 88.42	88.57 ± 0.73 Seed / Value 2622715460 / 88.69 **512 / 89.23** 2 / 87.78	**Reproducible**	3_Seed_exp.ipynb (cell 12) (seeded) MobileNet and EfficientNet B0-Adaboost.ipynb
22	MobileNetV2	KNN	89.51	89.40	89.51	89.21	Not reproducible	MobileNet with ML.ipynb (cell 6)
23	InceptionV3	SVM	89.24	89.40	89.24	89.30	Not reproducible	Inception v3 baseline and ml.ipynb (cell 16)
24	MobileNetV2	SVM	88.96	89.40	88.96	89.11	Not reproducible	Xgboost and SVM and all 4 models.ipynb (cell 15)
25	MiT-B2 + MobileNetV2	KNN	88.96	89.24	88.96	88.39	Not reproducible‡	MIT more epochs.ipynb (cell 30)
26	ResNet18	KNN	88.90	88.70	88.90	88.71	Not reproducible	ResNet with ML.ipynb (cell 6)
27	ResNet18	XGBoost	88.69	88.60	88.69	88.64	Not reproducible	Xgboost and SVM and all 4 models.ipynb (cell 7)

Rank	Model	Classifier	Test Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Repro-ducible?	Source notebook (.ipynb)
28	ResNet18	Direct (fine-tuned CNN)	88.42	88.74	88.42	87.76	Not reproducible	ResNet with ML.ipynb (cell 4)
29	InceptionV3	Random Forest	87.74	88.02	87.74	87.85	Not reproducible	Inception v3 baseline and ml.ipynb (cell 6)
30	MiT-B2 + FPN	Random Forest	87.53	88.34	87.53	86.57	Not reproducible‡	MIT + RF and MIT only.ipynb (cell 33)
31	VGG16	KNN	87.47	87.36	87.47	86.92	Not reproducible	Vgg with ML.ipynb (cell 6)
32	ResNet18	AdaBoost	87.19	87.43	87.19	87.29	Not reproducible	ResNet with ML.ipynb (cell 10)
33	MobileNetV2	Direct — 5-fold CV	87.19	87.76	87.19	86.26	Not reproducible	MobileNet with Kfold.ipynb (cell 3)
34	InceptionV3	AdaBoost	87.06	87.37	87.06	87.18	Not reproducible	Inception v3 baseline and ml.ipynb (cell 12)
35	MiT-B2 + MobileNetV2	Random Forest	87.06	87.94	87.06	85.99	Not reproducible‡	MIT+ MobileNet+ ML.ipynb (cell 30)
36	VGG16	XGBoost	86.92	86.64	86.92	86.67	Not reproducible	Xgboost and SVM and all 4 models.ipynb (cell 25)
37	InceptionV3	XGBoost	86.78	86.82	86.78	86.80	Not reproducible	Inception v3 baseline and ml.ipynb (cell 14)
38	ResNet18	SVM	86.72	88.67	86.72	87.14	Not reproducible	Xgboost and SVM and all 4 models.ipynb (cell 9)
39	MobileNetV2	XGBoost	86.72	86.65	86.72	86.68	Not reproducible	Xgboost and SVM and all 4 models.ipynb (cell 13)
40	AlexNet	Direct — 5-fold CV	86.58	86.44	86.58	85.95	Not reproducible	AlexNet with kfold.ipynb (cell 3)
41	MiT-B2 + MobileNetV2	AdaBoost	86.51	87.48	86.51	85.32	Not reproducible‡	MIT+ MobileNet+ ML.ipynb (cell 30)
42	EfficientNet-B3	Gaussian Naive Bayes	86.31	87.69	86.31	84.95	Not reproducible	efficinetnet b3 baseline and ml.ipynb (cell 9)
43	MiT-B2 + FPN	Direct (fine-tuned CNN)	86.24	86.87	86.24	85.13	Not reproducible‡	MIT + RF and MIT only.ipynb (cell 32)
44	AlexNet	XGBoost	85.56	86.08	85.56	85.75	Not reproducible	Xgboost and SVM and all 4 models.ipynb (cell 19)
45	AlexNet	Direct (fine-tuned CNN)	85.42	85.49	85.42	85.45	Not reproducible	AlexNet with Dice Loss (Baseline).ipynb (cell 3)

Rank	Model	Classifier	Test Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Repro-ducible?	Source notebook (.ipynb)
46	InceptionV3	KNN	84.95	85.37	84.95	85.11	Not reproducible	Inception v3 baseline and ml.ipynb (cell 8)
47	AlexNet	SVM	84.74	87.11	84.74	85.26	Not reproducible	Xgboost and SVM and all 4 models.ipynb (cell 21)
48	VGG16	Direct — 5-fold CV	84.20	85.94	84.20	82.25	Not reproducible	Vgg kfold.ipynb (cell 3)
49	InceptionV3	Gaussian Naive Bayes	83.38	86.40	83.38	84.01	Not reproducible	Inception v3 baseline and ml.ipynb (cell 10)
50	VGG16	Gaussian Naive Bayes	83.04	82.70	83.04	82.82	Not reproducible	Vgg with ML.ipynb (cell 8)
51	ResNet18	Direct — 5-fold CV	82.63	82.35	82.63	81.29	Not reproducible	ResNet wthi Kfold.ipynb (cell 3)
52	AlexNet	KNN	82.36	82.24	82.36	80.79	Not reproducible	AlexNet with ML.ipynb (cell 10)
53	AlexNet	Random Forest	81.61	81.05	81.61	80.25	Not reproducible	AlexNet with ML.ipynb (cell 4)
54	VGG16	AdaBoost	76.63	76.34	76.63	72.32	Not reproducible	Vgg with ML.ipynb (cell 10)
55	AlexNet	Gaussian Naive Bayes	75.07	83.51	75.07	76.30	Not reproducible	AlexNet with ML.ipynb (cell 8)
56	VGG16	Random Forest	74.73	80.21	74.73	66.63	Not reproducible	Vgg with ML.ipynb (cell 4)
57	ResNet18	Gaussian Naive Bayes	67.57	74.25	67.57	69.12	Not reproducible	ResNet with ML.ipynb (cell 8)

- 57 unique rows from 68 results in the 21 ZIP notebooks plus the 6 seeded (3-seed) results and the 4 ensemble results from the attached seeded notebooks. Test set: 1,468 images (1,052 Appears abnormal, 416 Appears normal).
- Each model + classifier pair appears once; where a pair was run several times, the run with the highest test accuracy is kept.
- Green rows are the 3-seed experiments (seeds 2622715460, 512, 2): the main value is mean ± SD (sample SD, n = 3) over the three seeds, and below it the individual result of each seed is listed in a small Seed | Value table (seeds 2622715460, 512 and 2); the bold value is the seed with the highest test accuracy (it is also the highest seed on precision, recall and F1-score). Rows are still ranked by that best seed; the grey rows are single runs.
- Precision, Recall and F1-score are class-weighted averages.
- ‡ The MiT-B2 notebooks fix seed 42 but contain no repeat run or saved artefacts, so they are not verified as reproducible.
- Section 3 lists all further notes.

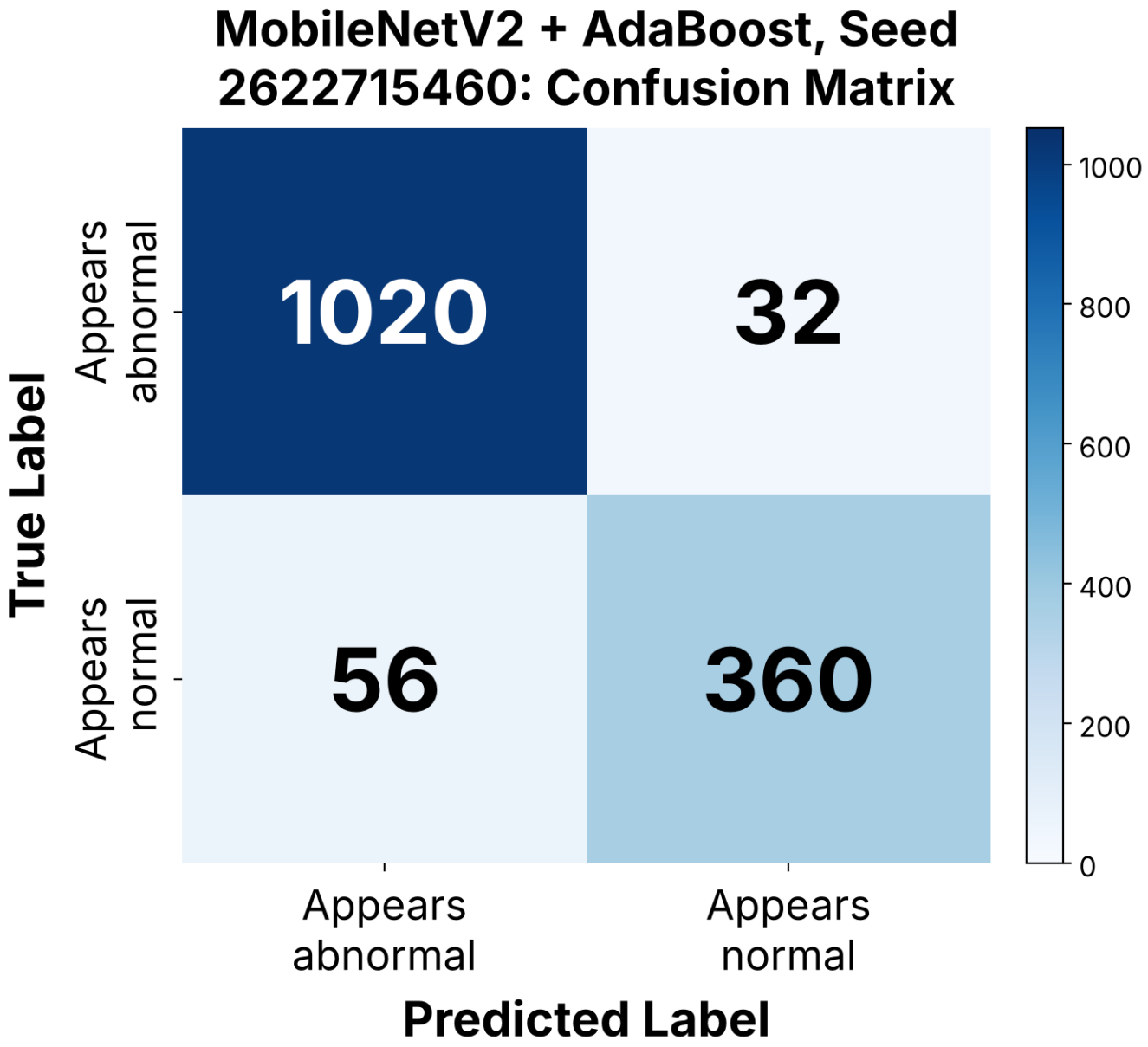
2. Results, Confusion Matrices and Graphs — Table Order

1. MobileNetV2 + AdaBoost — Seed 2622715460 (seed 1 of 3)

Source: 3_Seed_exp.ipynb, cell 4. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 4, seed 2622715460* (same numbers for that seed).

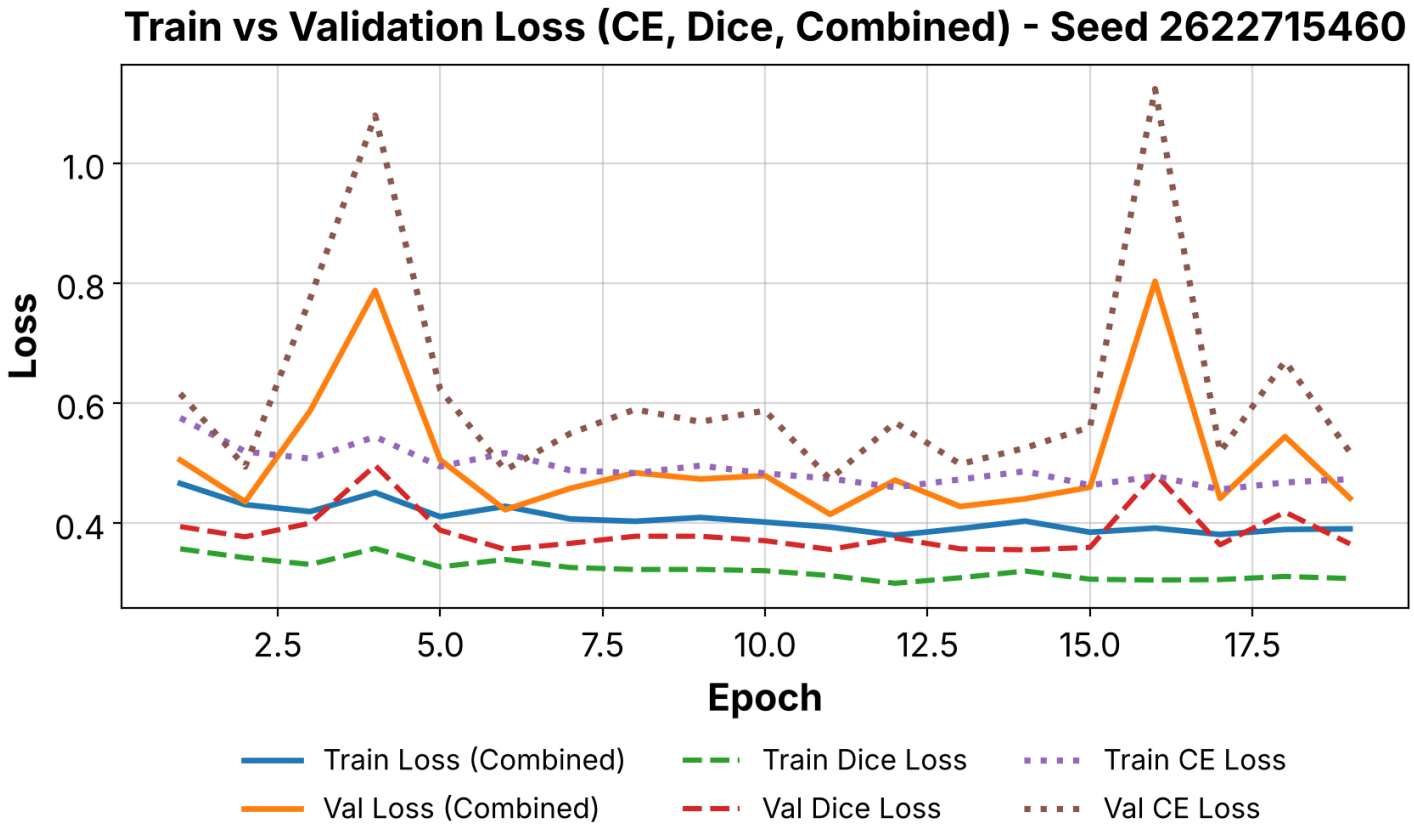
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**94.01%**	89.99%	91.14%	91.71 ± 2.07 %
Precision (weighted)	**93.96%**	90.00%	91.04%	91.67 ± 2.05 %
Recall (weighted)	**94.01%**	89.99%	91.14%	91.71 ± 2.07 %
F1-score (weighted)	**93.95%**	89.63%	90.97%	91.52 ± 2.21 %
AdaBoost validation acc.	67.29%	65.62%	63.54%	65.49 ± 1.88 %
Best validation acc. (CNN)	72.08%	71.25%	77.50%	73.61 ± 3.39 %

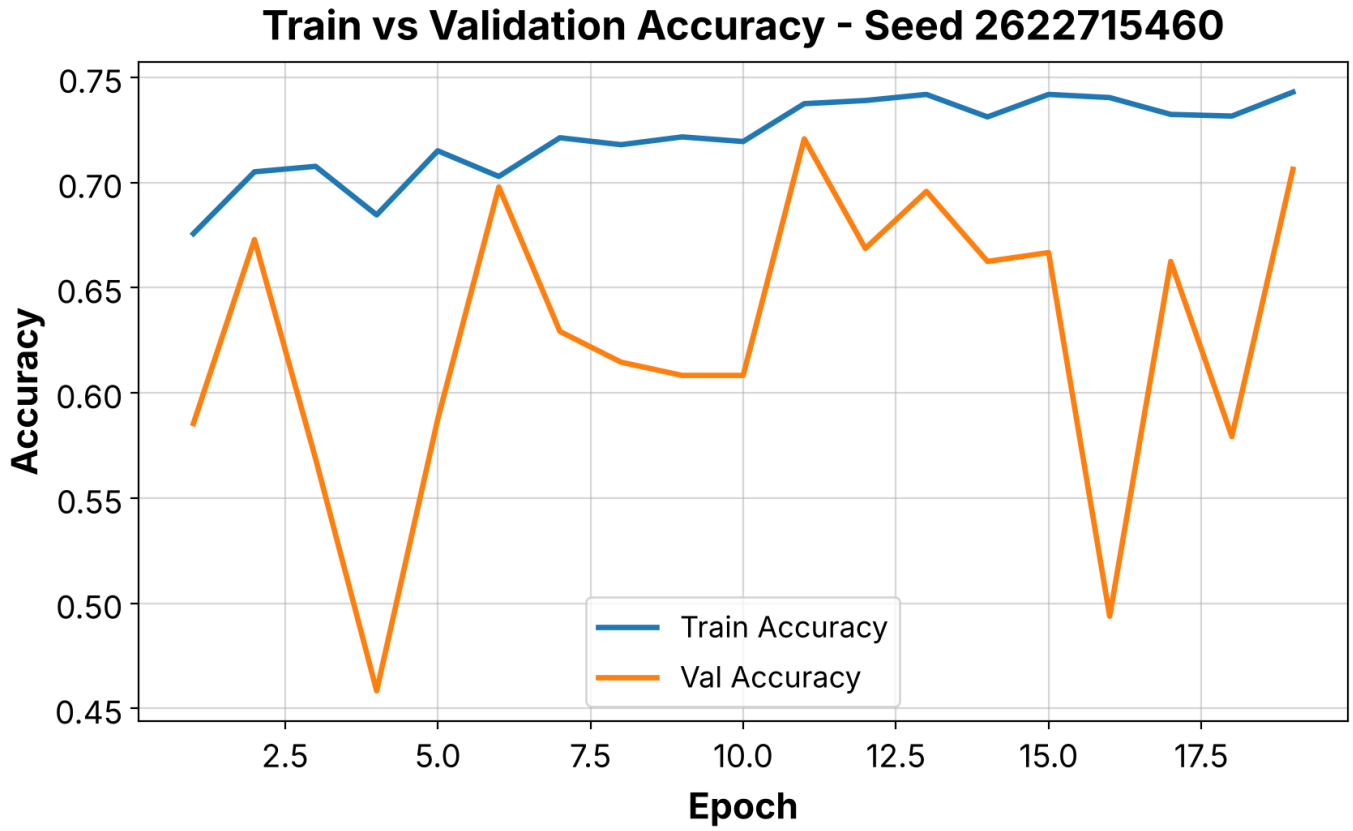
Confusion Matrix



1. MobileNetV2 + AdaBoost, Seed 2622715460 — graphs

Training / Validation Graphs



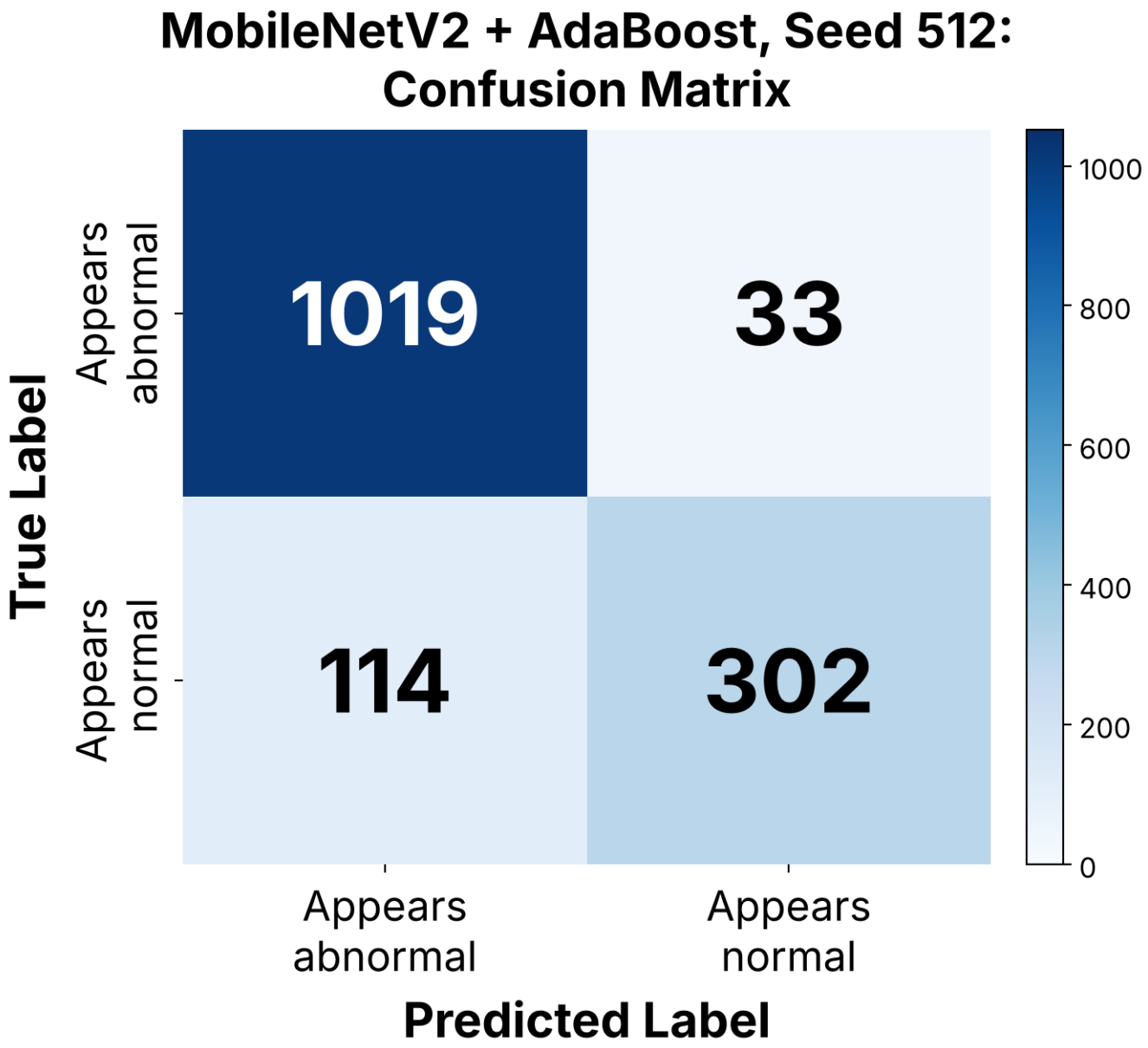


1. MobileNetV2 + AdaBoost — Seed 512 (seed 2 of 3)

Source: 3_Seed_exp.ipynb, cell 4. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 4, seed 2622715460* (same numbers for that seed).

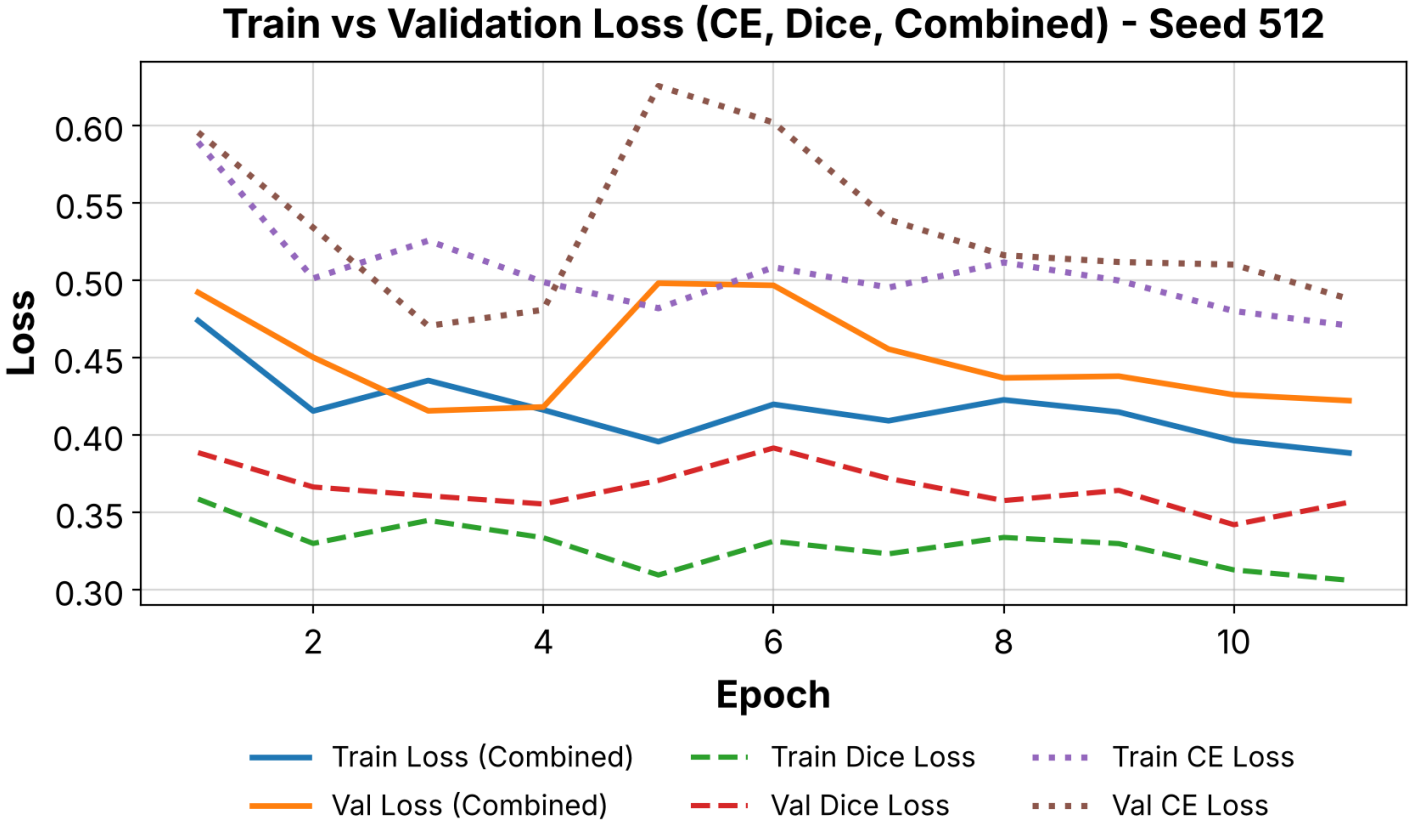
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	94.01%	**89.99%**	91.14%	91.71 ± 2.07 %
Precision (weighted)	93.96%	**90.00%**	91.04%	91.67 ± 2.05 %
Recall (weighted)	94.01%	**89.99%**	91.14%	91.71 ± 2.07 %
F1-score (weighted)	93.95%	**89.63%**	90.97%	91.52 ± 2.21 %
AdaBoost validation acc.	67.29%	65.62%	63.54%	65.49 ± 1.88 %
Best validation acc. (CNN)	72.08%	71.25%	77.50%	73.61 ± 3.39 %

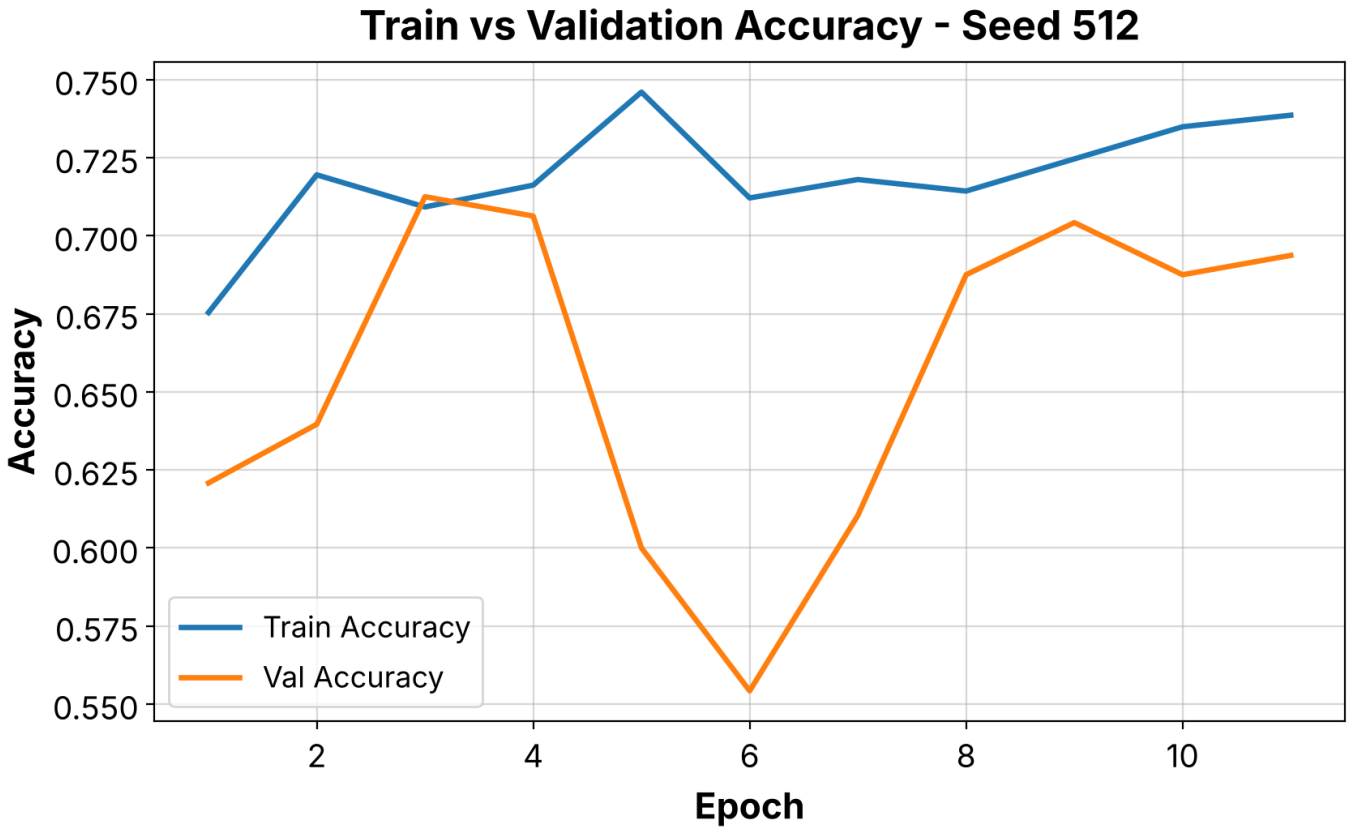
Confusion Matrix



1. MobileNetV2 + AdaBoost, Seed 512 — graphs

Training / Validation Graphs



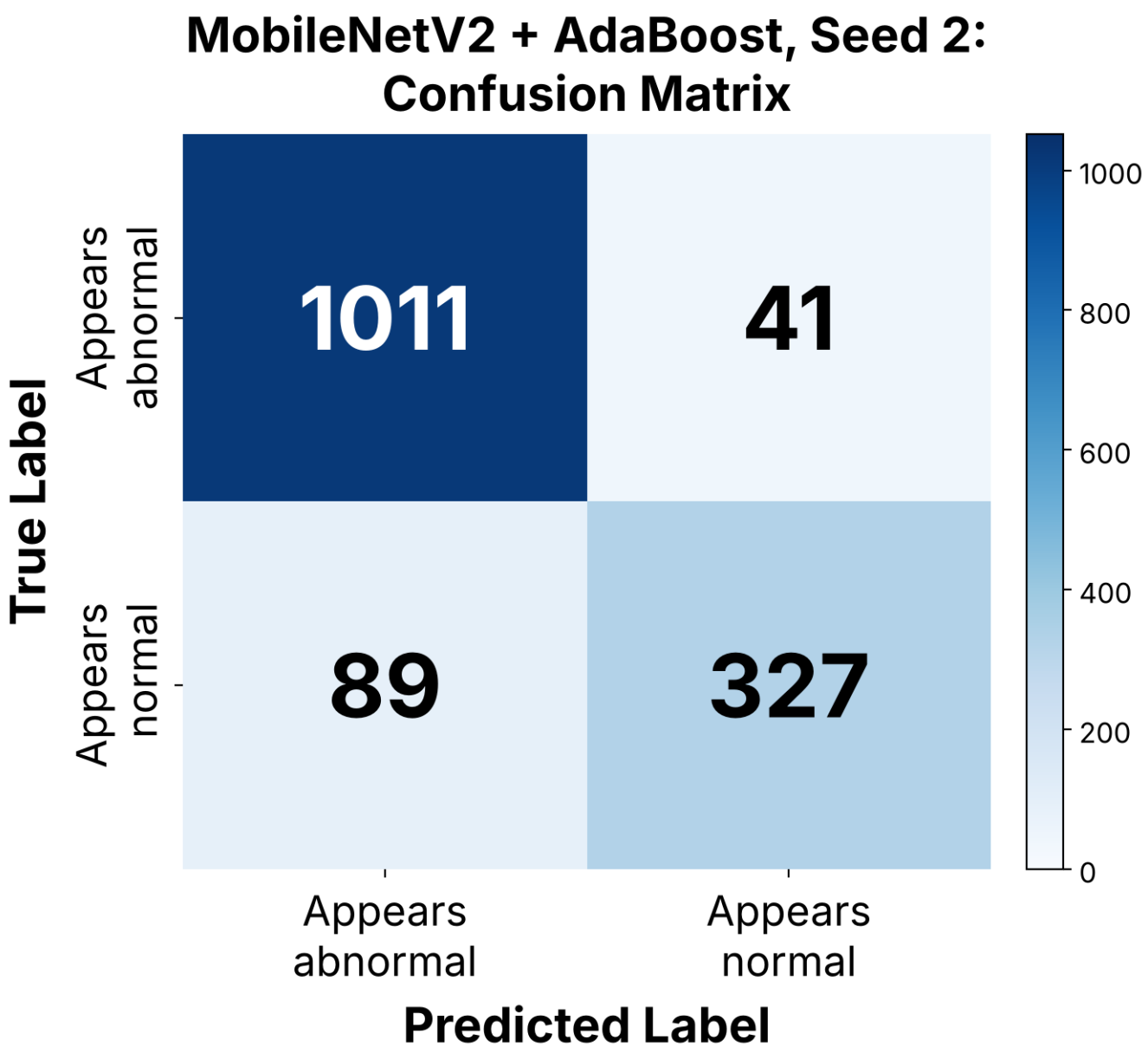


1. MobileNetV2 + AdaBoost — Seed 2 (seed 3 of 3)

Source: 3_Seed_exp.ipynb, cell 4. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 4, seed 2622715460* (same numbers for that seed).

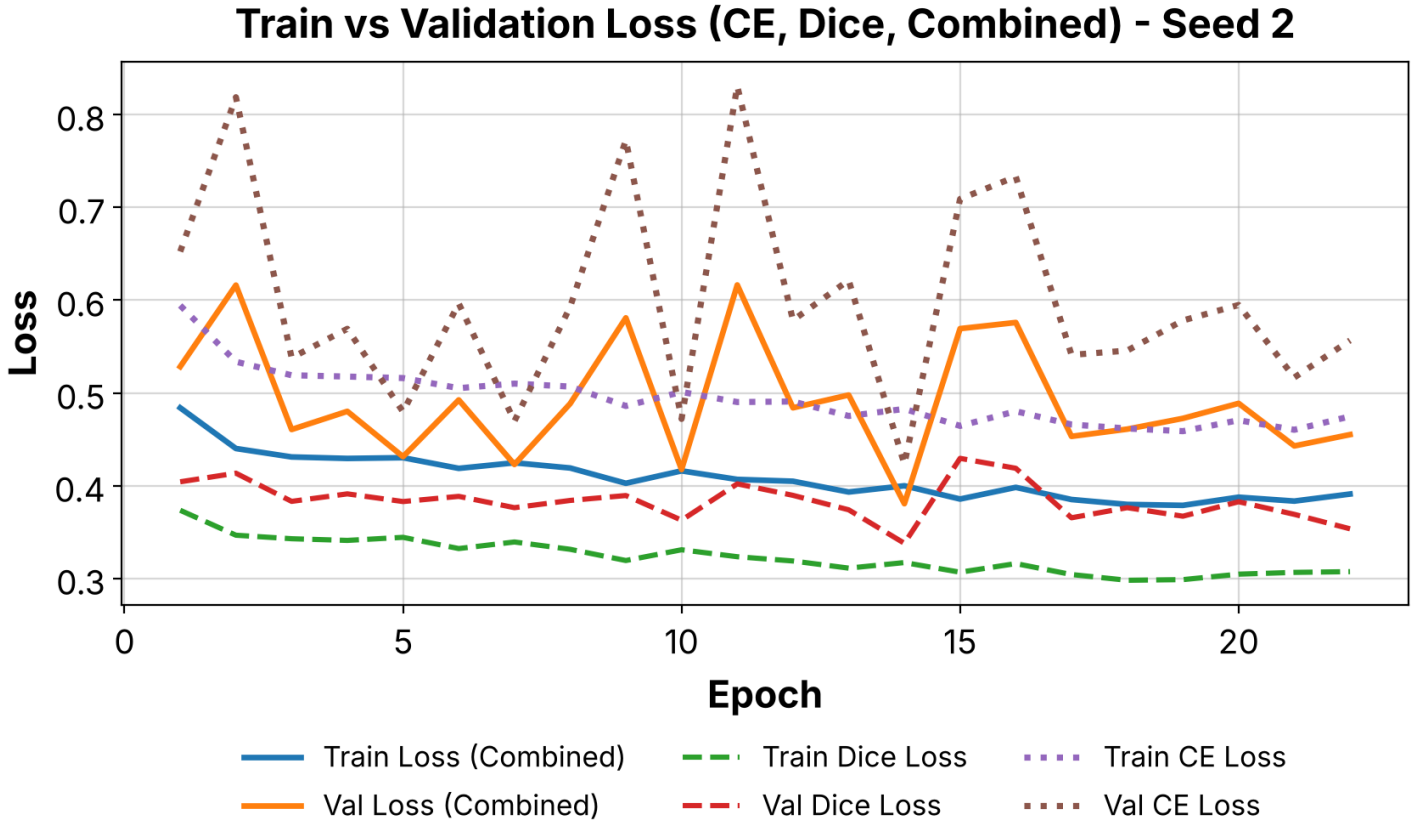
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	94.01%	89.99%	**91.14%**	91.71 ± 2.07 %
Precision (weighted)	93.96%	90.00%	**91.04%**	91.67 ± 2.05 %
Recall (weighted)	94.01%	89.99%	**91.14%**	91.71 ± 2.07 %
F1-score (weighted)	93.95%	89.63%	**90.97%**	91.52 ± 2.21 %
AdaBoost validation acc.	67.29%	65.62%	63.54%	65.49 ± 1.88 %
Best validation acc. (CNN)	72.08%	71.25%	77.50%	73.61 ± 3.39 %

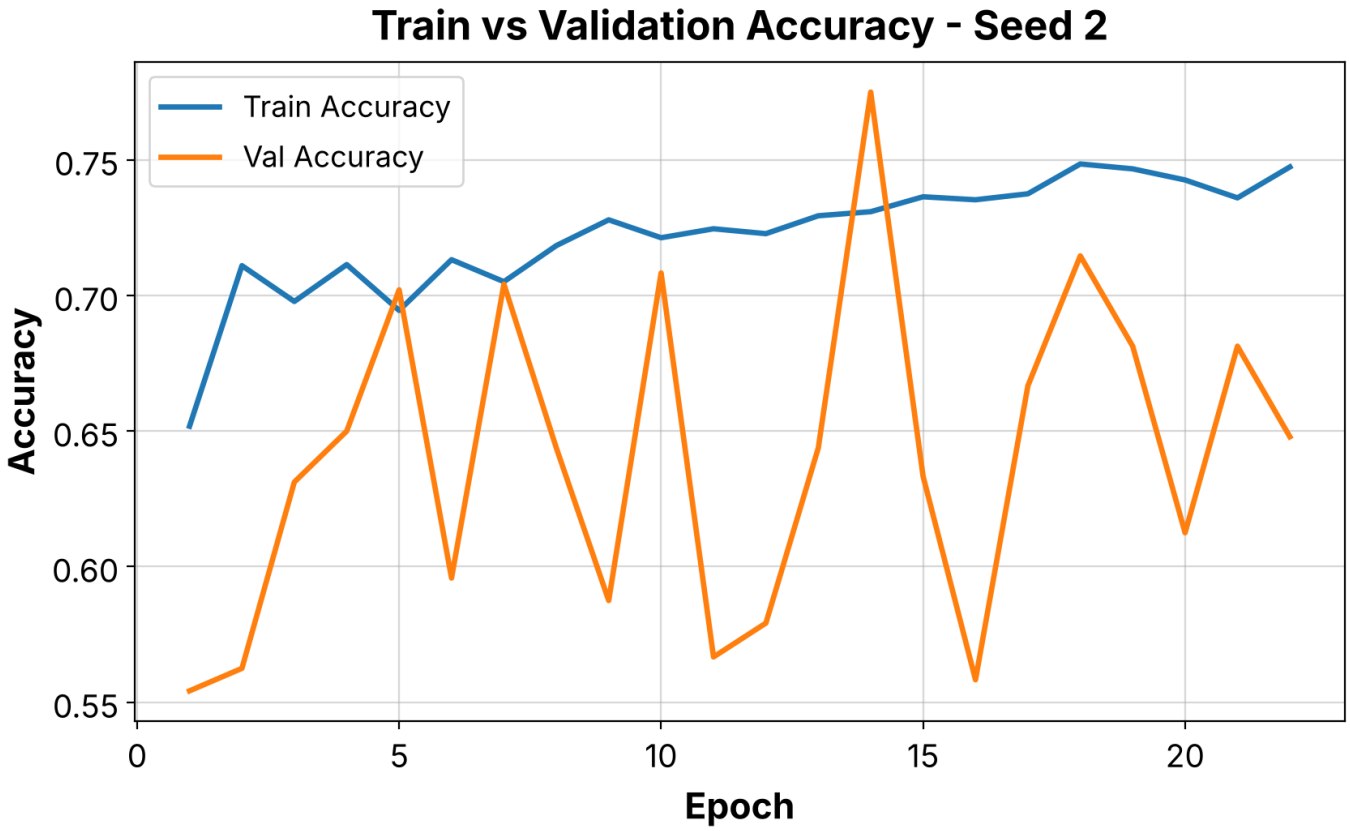
Confusion Matrix



1. MobileNetV2 + AdaBoost, Seed 2 — graphs

Training / Validation Graphs





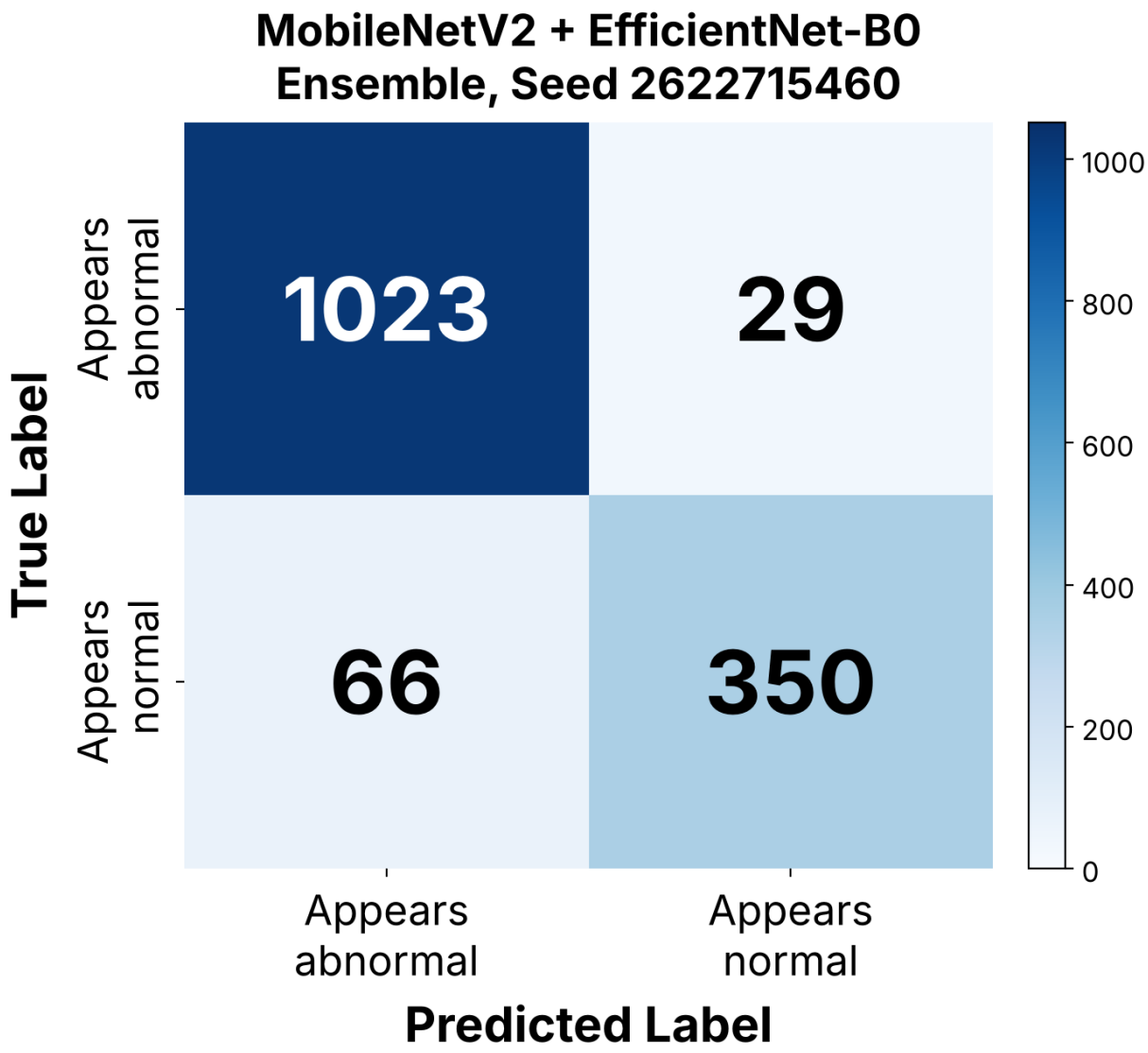
2. Ensemble — MobileNetV2 + EfficientNet-B0 (AdaBoost, soft voting) — Seed 2622715460 (seed 1 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**93.53%**	92.30%	91.49%	92.44 ± 1.03 %
Precision (weighted)	**93.49%**	92.26%	91.38%	92.38 ± 1.06 %
Recall (weighted)	**93.53%**	92.30%	91.49%	92.44 ± 1.03 %
F1-score (weighted)	**93.43%**	92.15%	91.40%	92.33 ± 1.03 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 1, EfficientNet-B0 = row 6.

Confusion Matrix



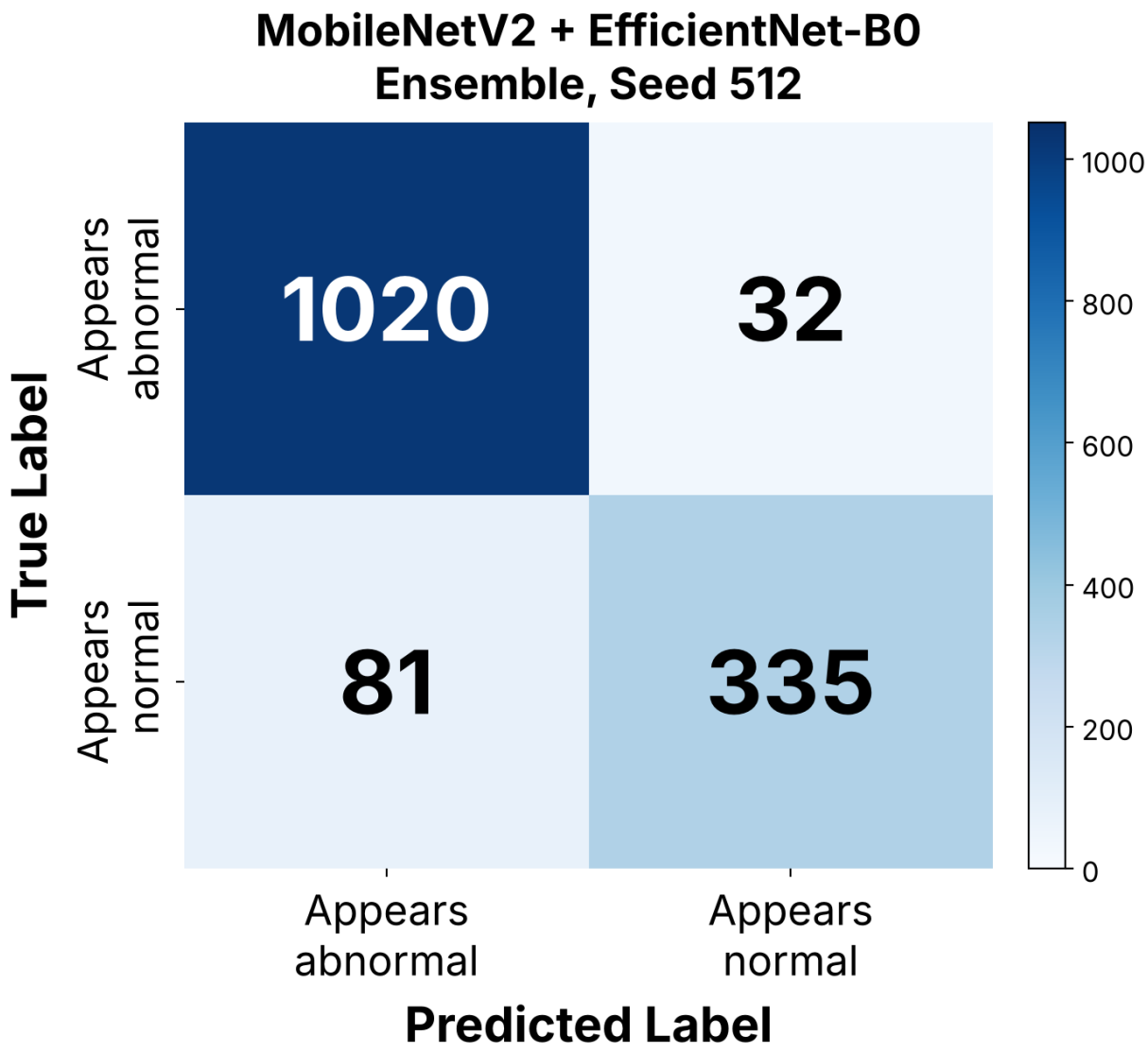
2. Ensemble — MobileNetV2 + EfficientNet-B0 (AdaBoost, soft voting) — Seed 512 (seed 2 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	93.53%	**92.30%**	91.49%	92.44 ± 1.03 %
Precision (weighted)	93.49%	**92.26%**	91.38%	92.38 ± 1.06 %
Recall (weighted)	93.53%	**92.30%**	91.49%	92.44 ± 1.03 %
F1-score (weighted)	93.43%	**92.15%**	91.40%	92.33 ± 1.03 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 1, EfficientNet-B0 = row 6.

Confusion Matrix



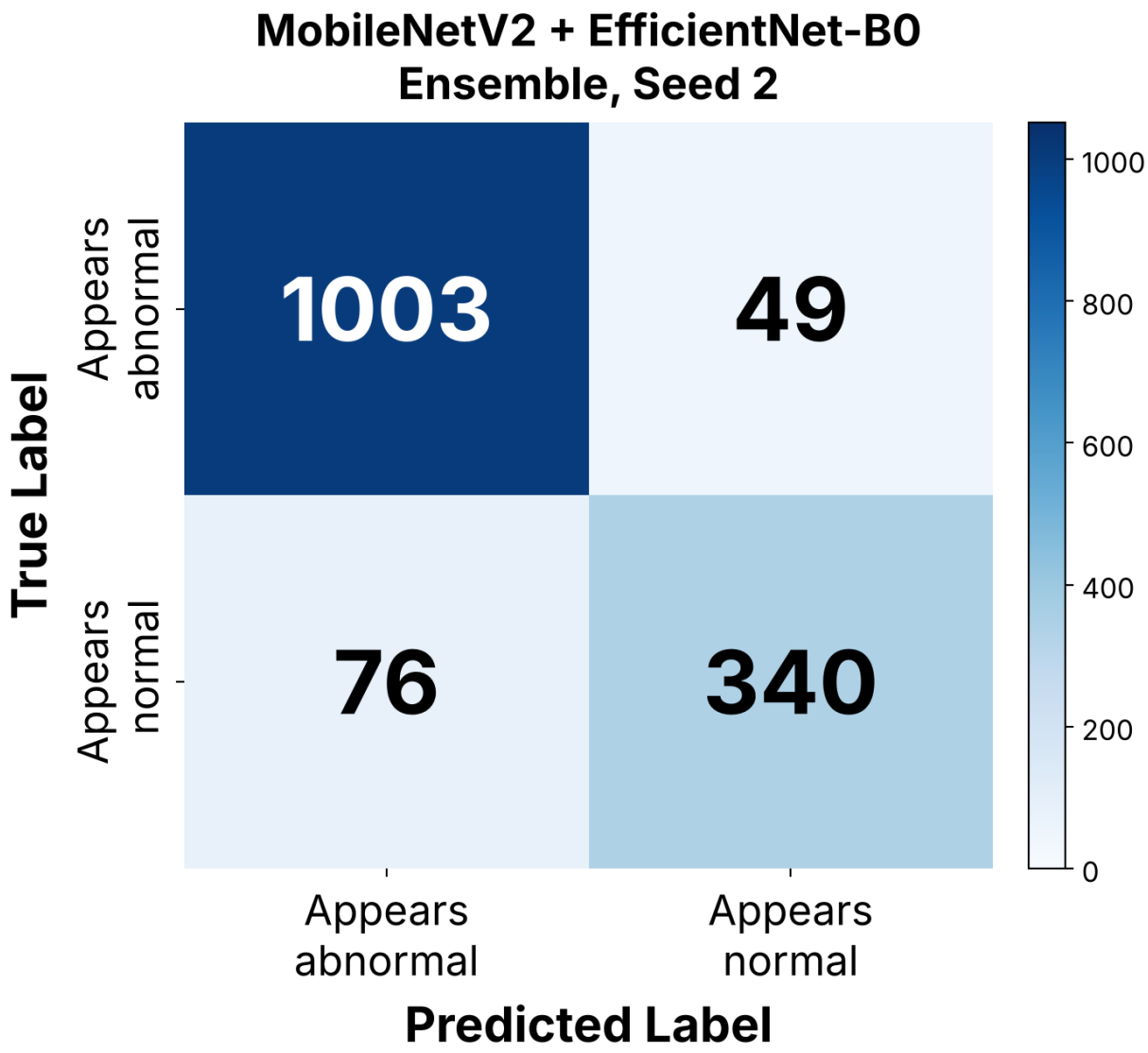
2. Ensemble — MobileNetV2 + EfficientNet-B0 (AdaBoost, soft voting) — Seed 2 (seed 3 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	93.53%	92.30%	**91.49%**	92.44 ± 1.03 %
Precision (weighted)	93.49%	92.26%	**91.38%**	92.38 ± 1.06 %
Recall (weighted)	93.53%	92.30%	**91.49%**	92.44 ± 1.03 %
F1-score (weighted)	93.43%	92.15%	**91.40%**	92.33 ± 1.03 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 1, EfficientNet-B0 = row 6.

Confusion Matrix



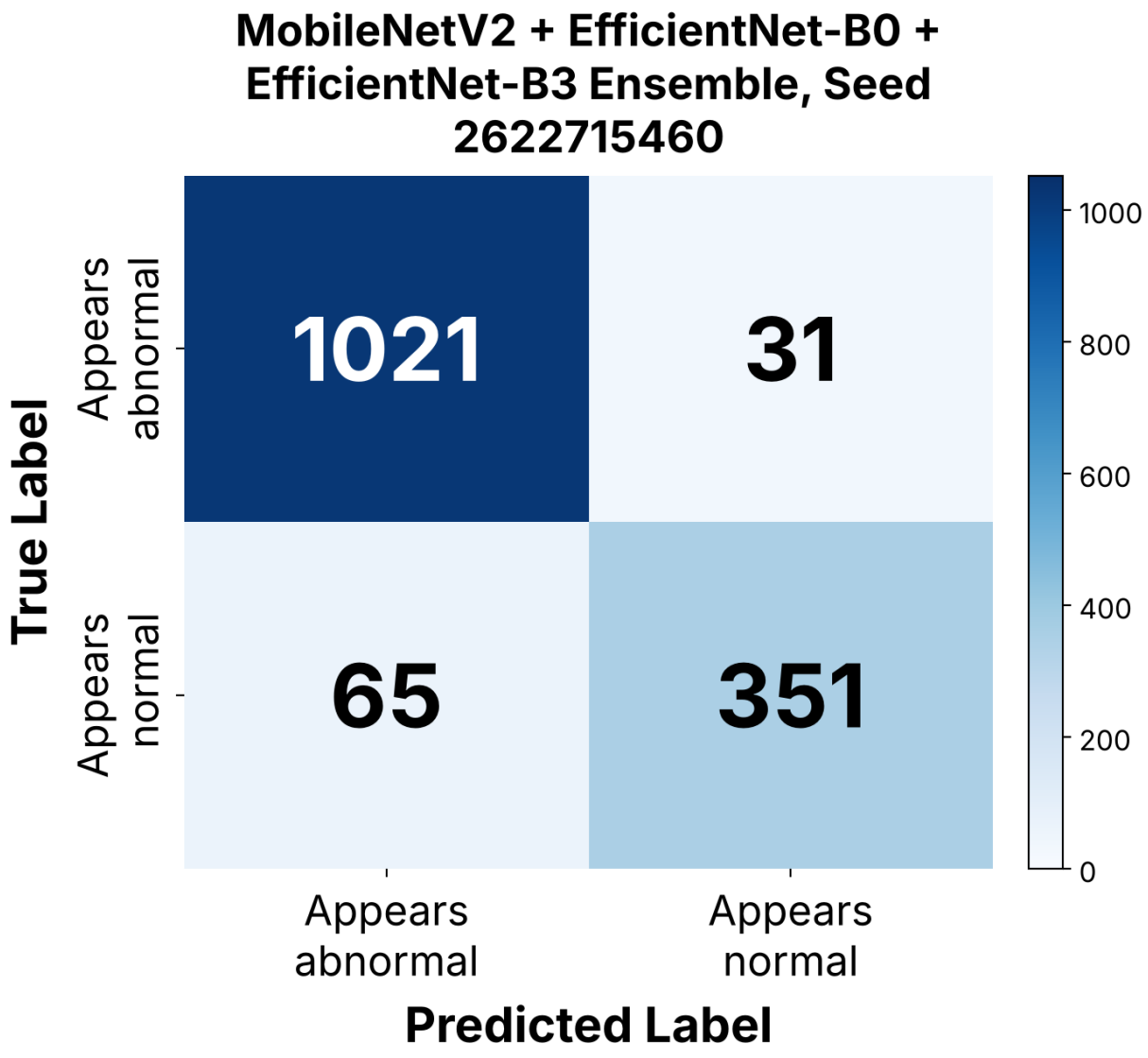
3. Ensemble — MobileNetV2 + EfficientNet-B0 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 2622715460 (seed 1 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**93.46%**	92.51%	91.14%	92.37 ± 1.16 %
Precision (weighted)	**93.41%**	92.47%	91.05%	92.31 ± 1.19 %
Recall (weighted)	**93.46%**	92.51%	91.14%	92.37 ± 1.16 %
F1-score (weighted)	**93.37%**	92.35%	90.96%	92.23 ± 1.21 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 1, EfficientNet-B0 = row 6, EfficientNet-B3 = row 5.

Confusion Matrix



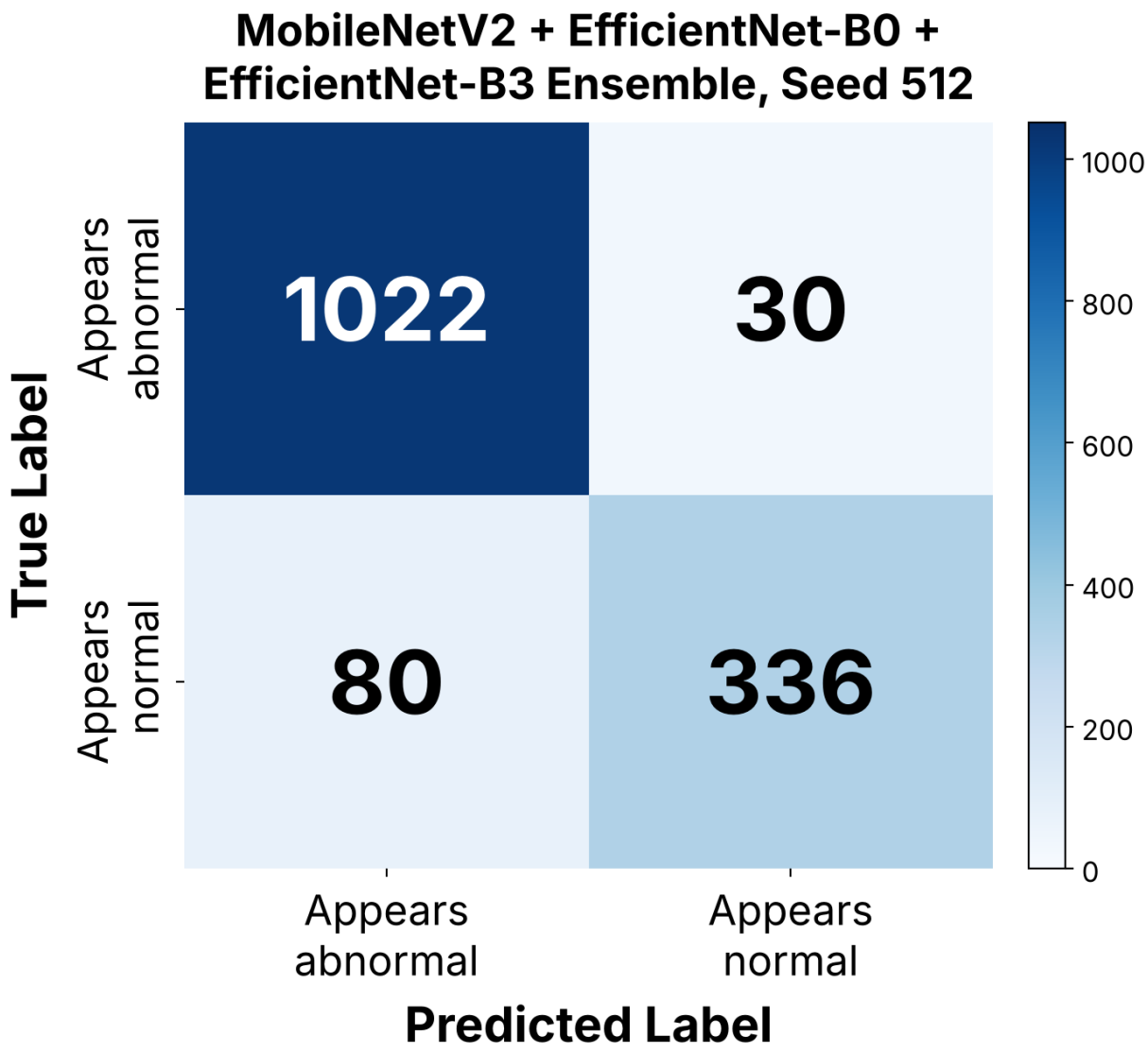
3. Ensemble — MobileNetV2 + EfficientNet-B0 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 512 (seed 2 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	93.46%	**92.51%**	91.14%	92.37 ± 1.16 %
Precision (weighted)	93.41%	**92.47%**	91.05%	92.31 ± 1.19 %
Recall (weighted)	93.46%	**92.51%**	91.14%	92.37 ± 1.16 %
F1-score (weighted)	93.37%	**92.35%**	90.96%	92.23 ± 1.21 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 1, EfficientNet-B0 = row 6, EfficientNet-B3 = row 5.

Confusion Matrix



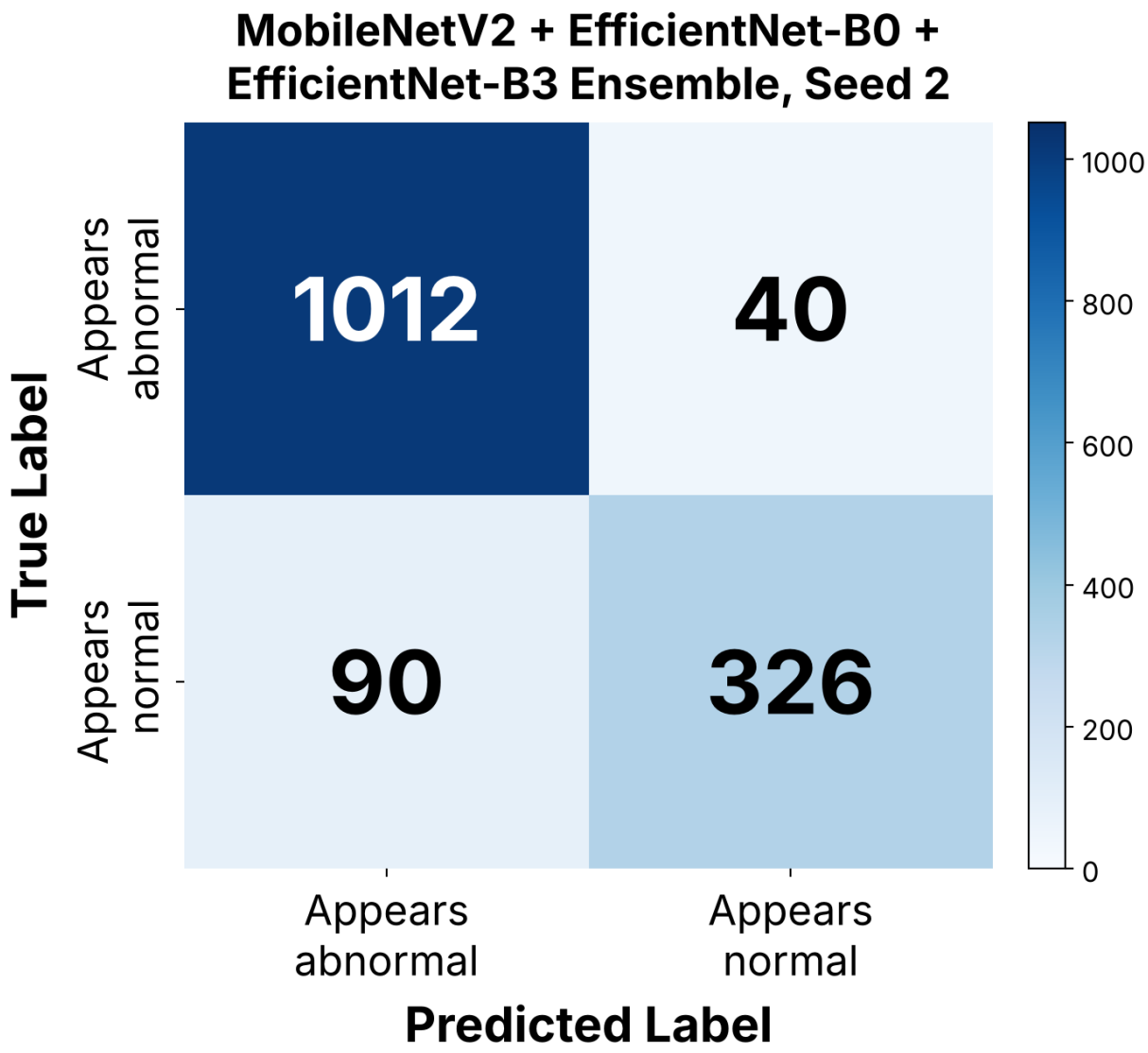
3. Ensemble — MobileNetV2 + EfficientNet-B0 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 2 (seed 3 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	93.46%	92.51%	**91.14%**	92.37 ± 1.16 %
Precision (weighted)	93.41%	92.47%	**91.05%**	92.31 ± 1.19 %
Recall (weighted)	93.46%	92.51%	**91.14%**	92.37 ± 1.16 %
F1-score (weighted)	93.37%	92.35%	**90.96%**	92.23 ± 1.21 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 1, EfficientNet-B0 = row 6, EfficientNet-B3 = row 5.

Confusion Matrix



4. MiT-B2 + MobileNetV2 + Gaussian Naive Bayes

Source notebook: MIT more epochs.ipynb (inside MIT + MobileNet.zip)

Accuracy: 92.85%

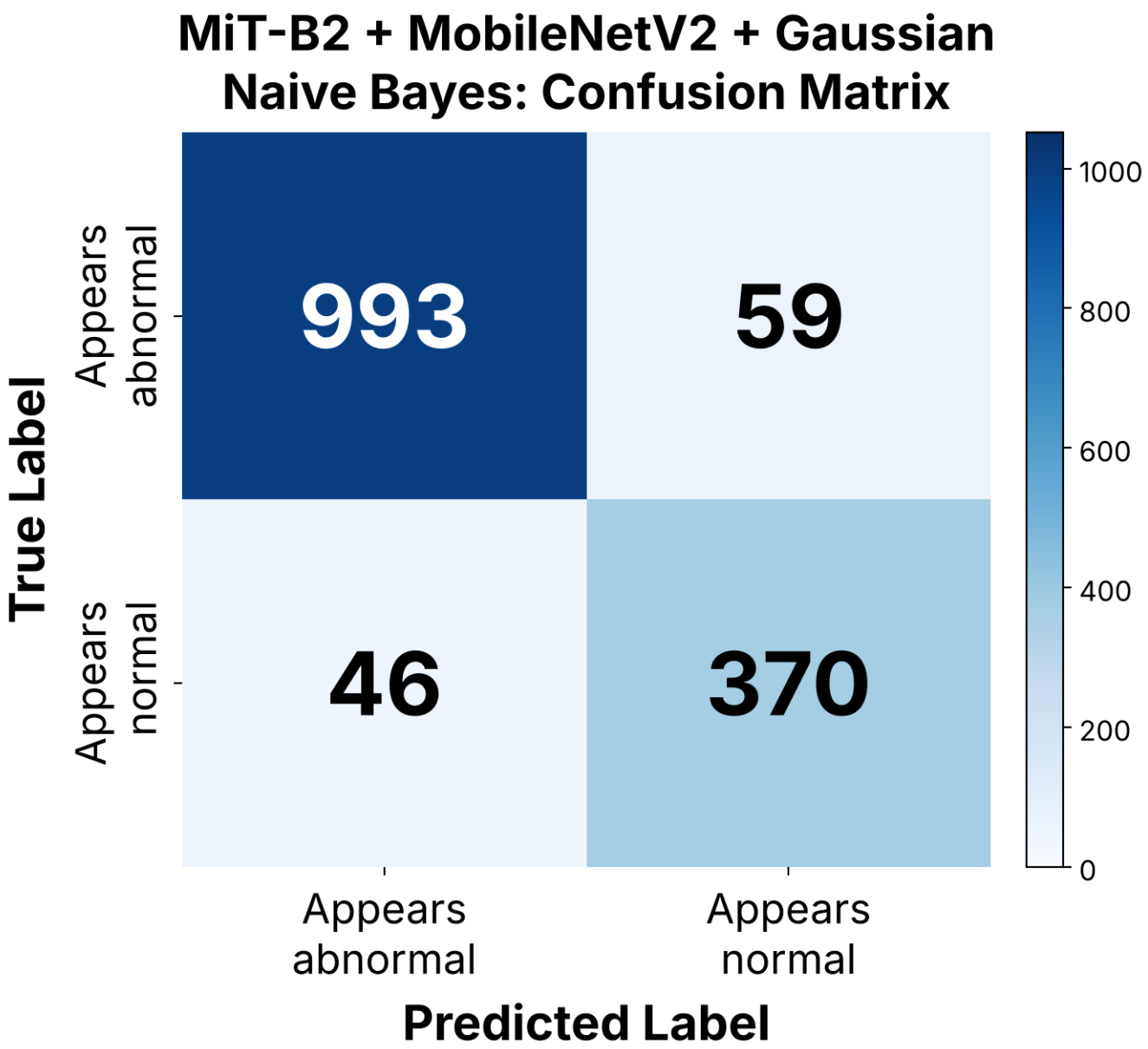
Precision: 92.93%

Recall: 92.85%

F1-score: 92.88%

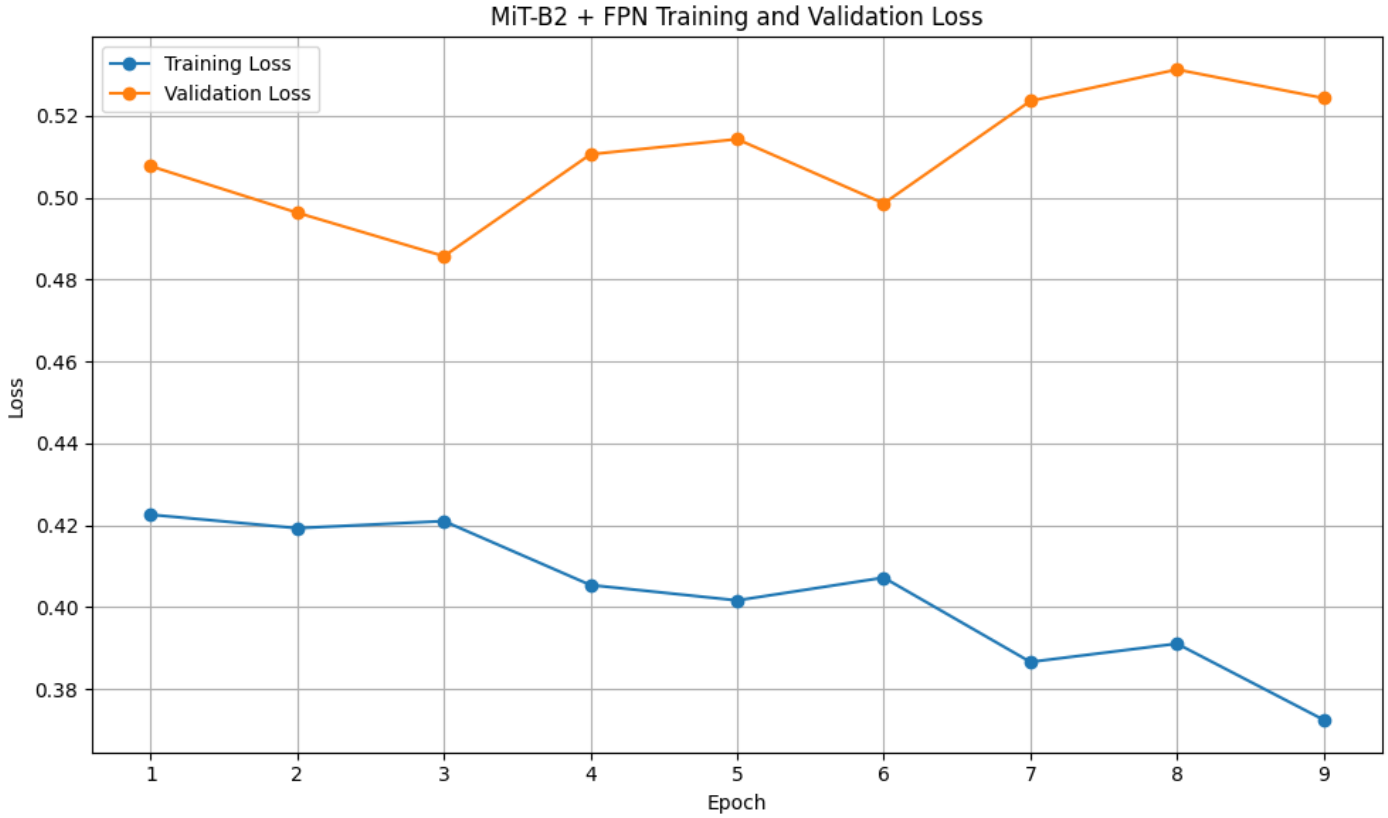
Same pair also run elsewhere with lower accuracy: 92.71%, 92.64% (not shown).

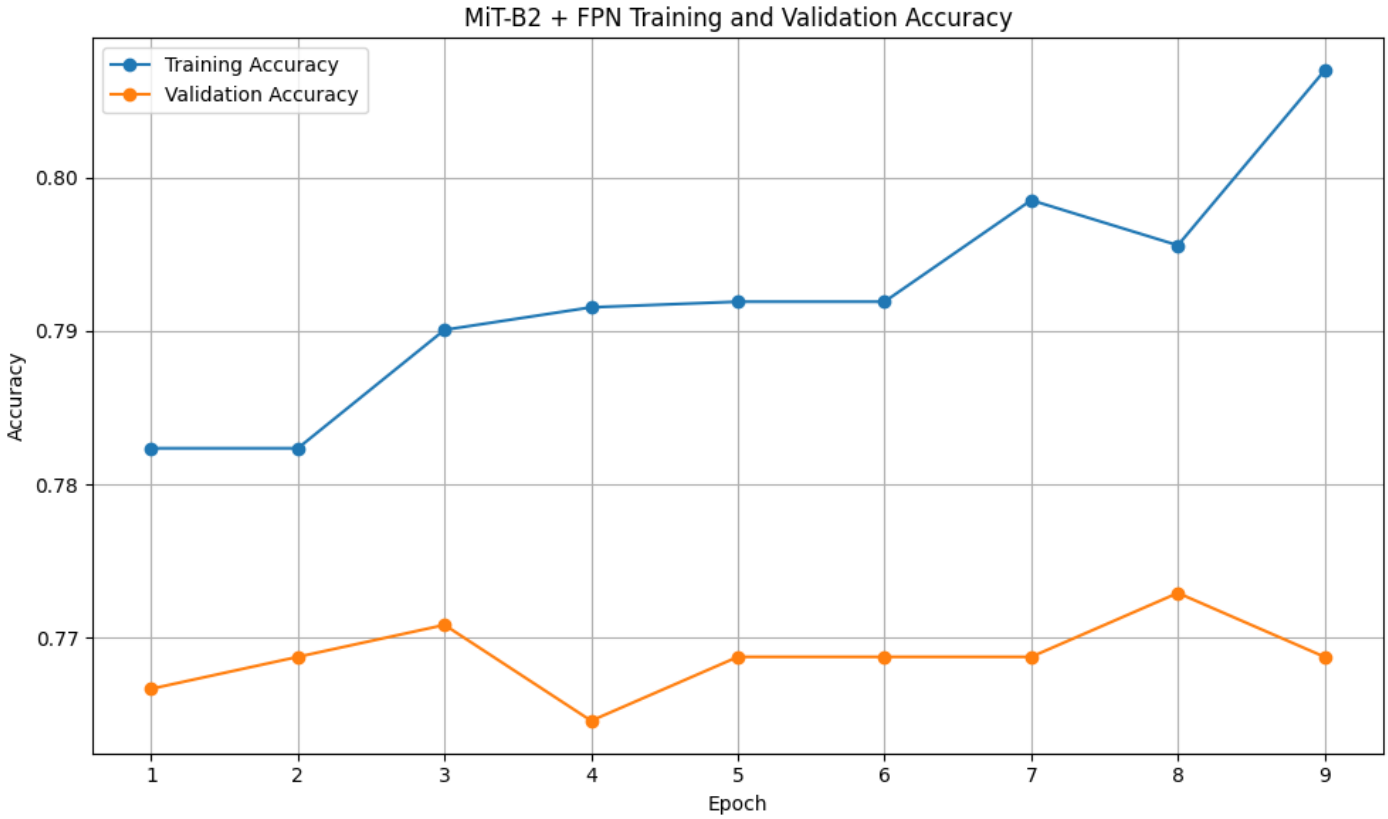
Confusion Matrix



4. MiT-B2 + MobileNetV2 + Gaussian Naive Bayes — graphs (1/2)

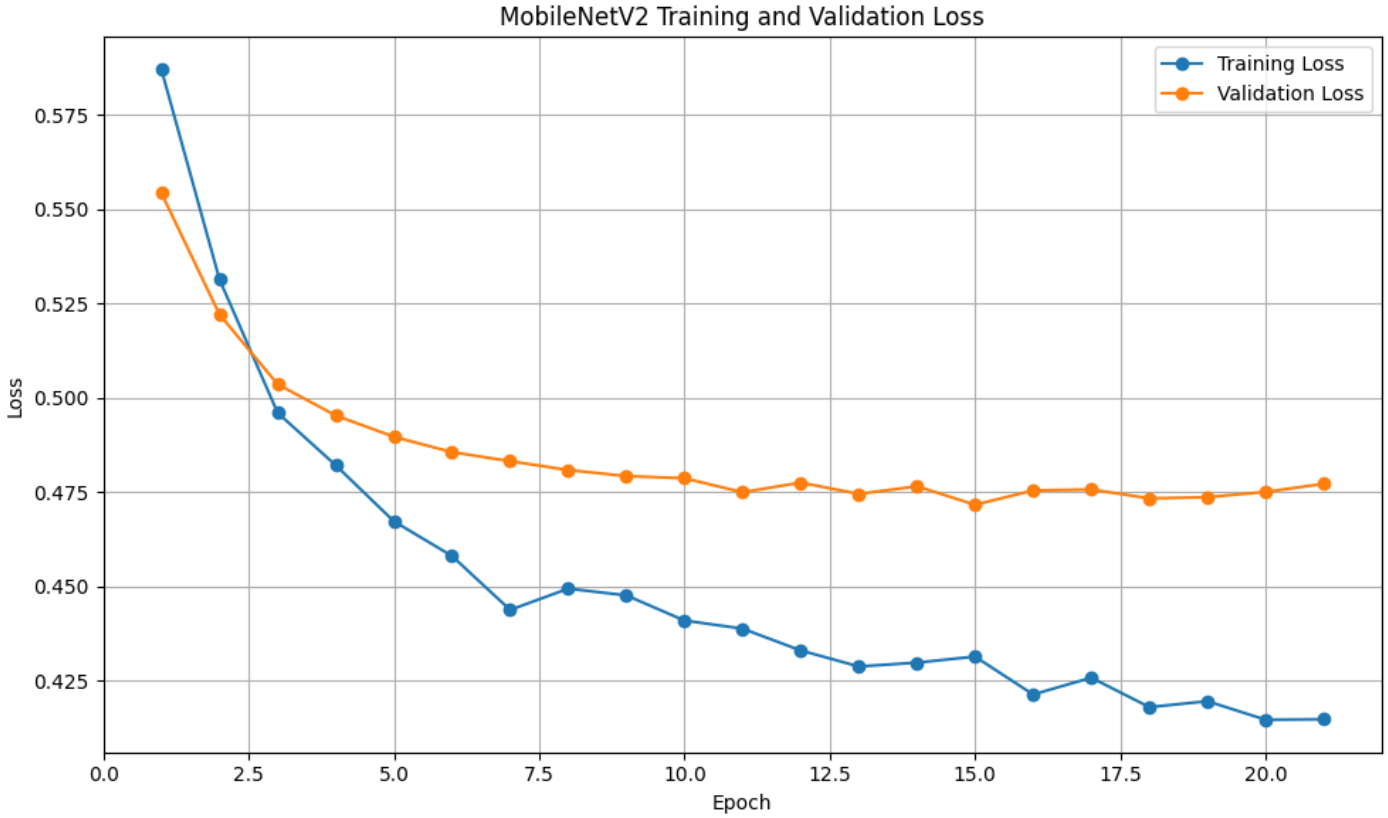
Training / Validation Graphs

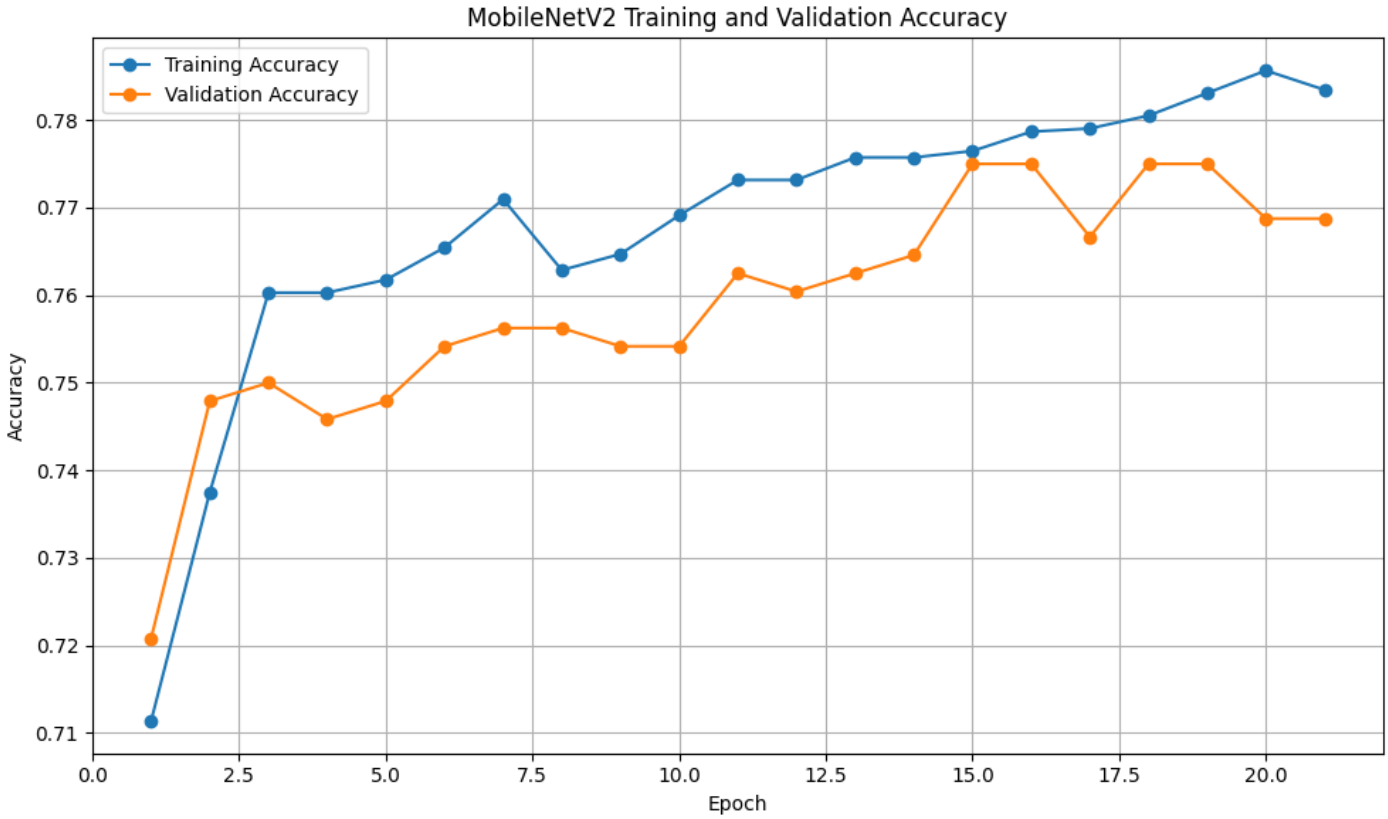




4. MiT-B2 + MobileNetV2 + Gaussian Naive Bayes — graphs (2/2)

Training / Validation Graphs



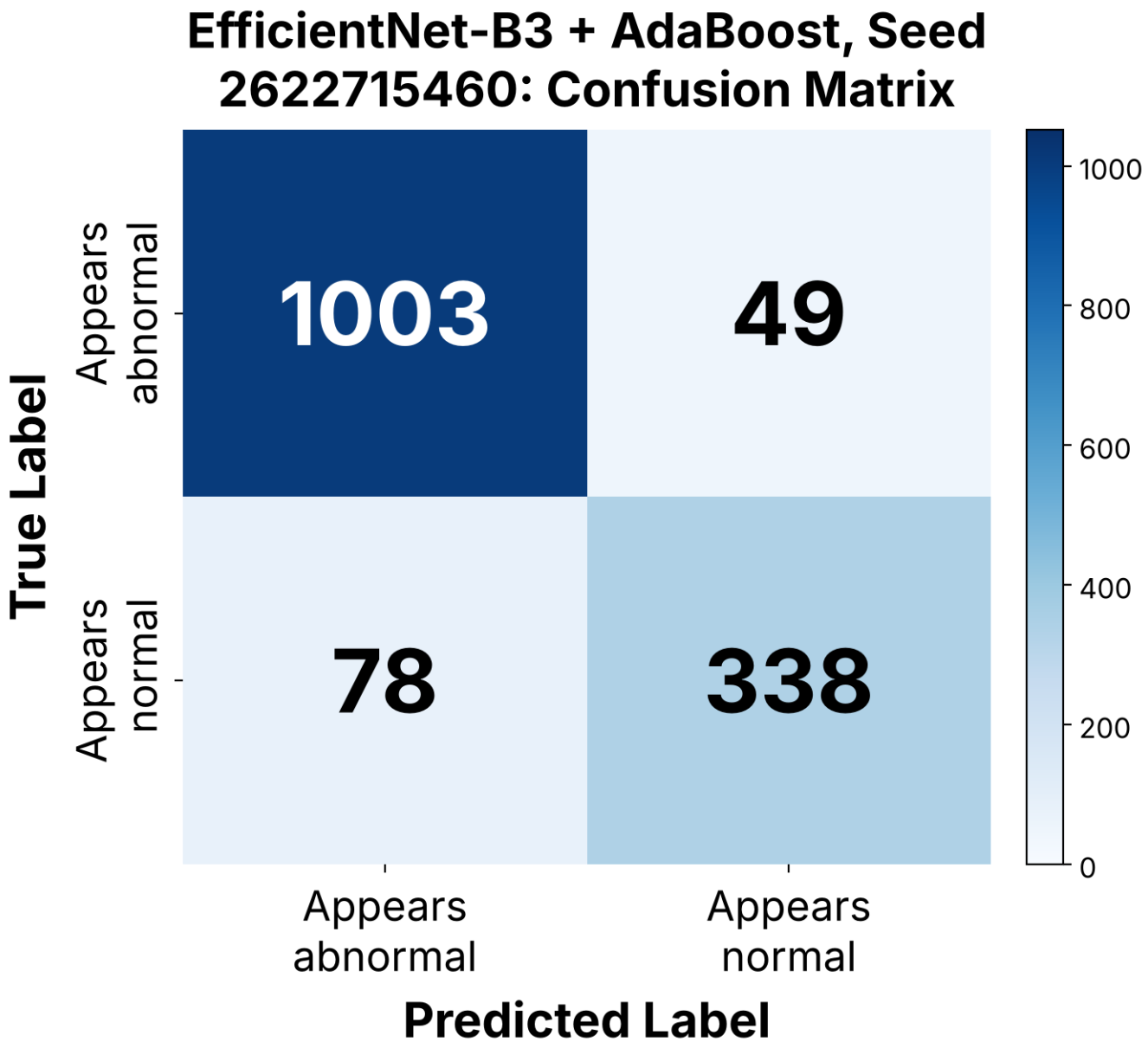


5. EfficientNet-B3 + AdaBoost — Seed 2622715460 (seed 1 of 3)

Source: 3_Seed_exp.ipynb, cell 8. Re-run independently in *(seeded) EfficientNet B3 with Adaboost.ipynb, cell 4, seed 512* (same numbers for that seed).

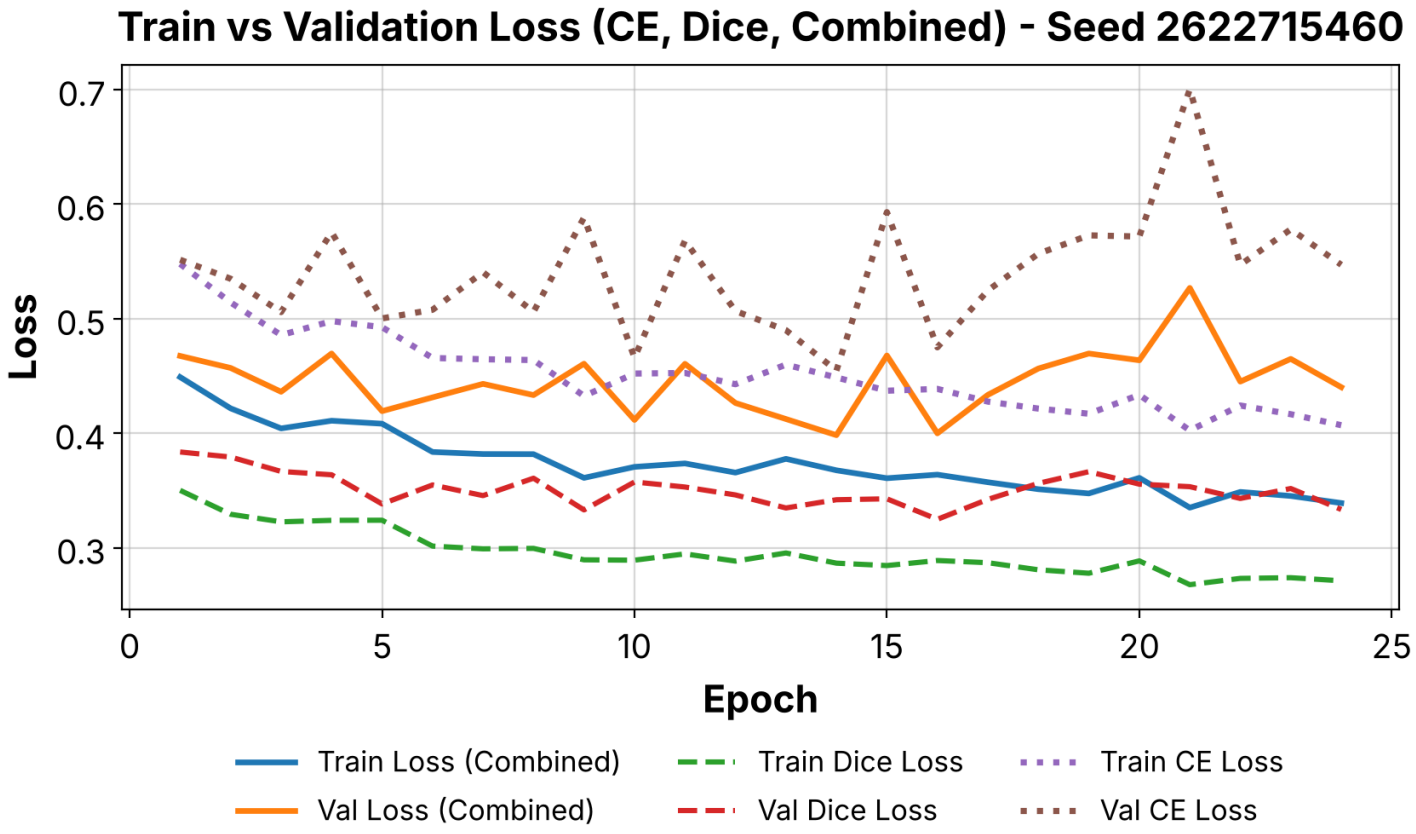
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**91.35%**	92.71%	91.08%	91.71 ± 0.88 %
Precision (weighted)	**91.24%**	92.64%	90.99%	91.62 ± 0.89 %
Recall (weighted)	**91.35%**	92.71%	91.08%	91.71 ± 0.88 %
F1-score (weighted)	**91.25%**	92.64%	90.88%	91.59 ± 0.93 %
AdaBoost validation acc.	69.37%	69.17%	68.33%	68.96 ± 0.55 %
Best validation acc. (CNN)	76.46%	75.62%	72.50%	74.86 ± 2.09 %

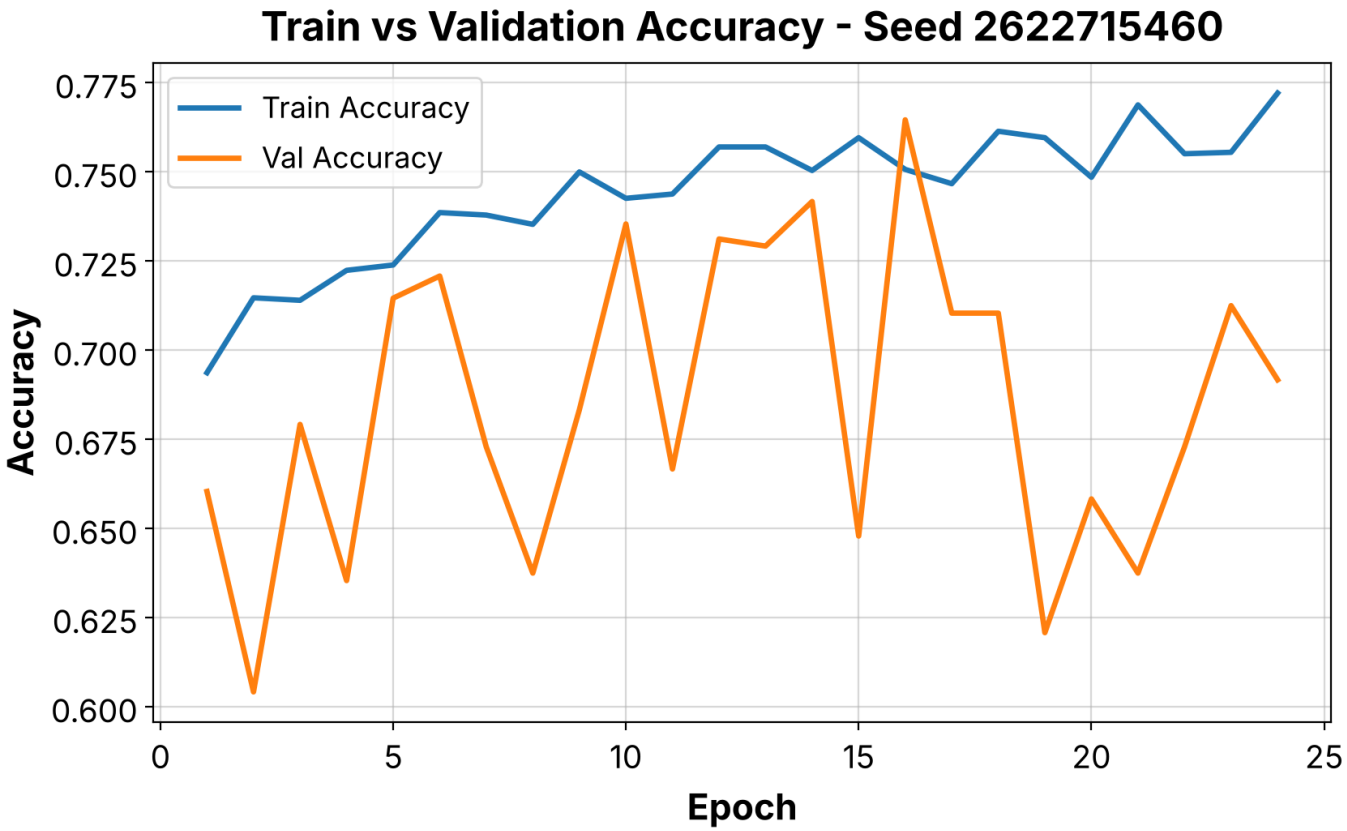
Confusion Matrix



5. EfficientNet-B3 + AdaBoost, Seed 2622715460 — graphs

Training / Validation Graphs



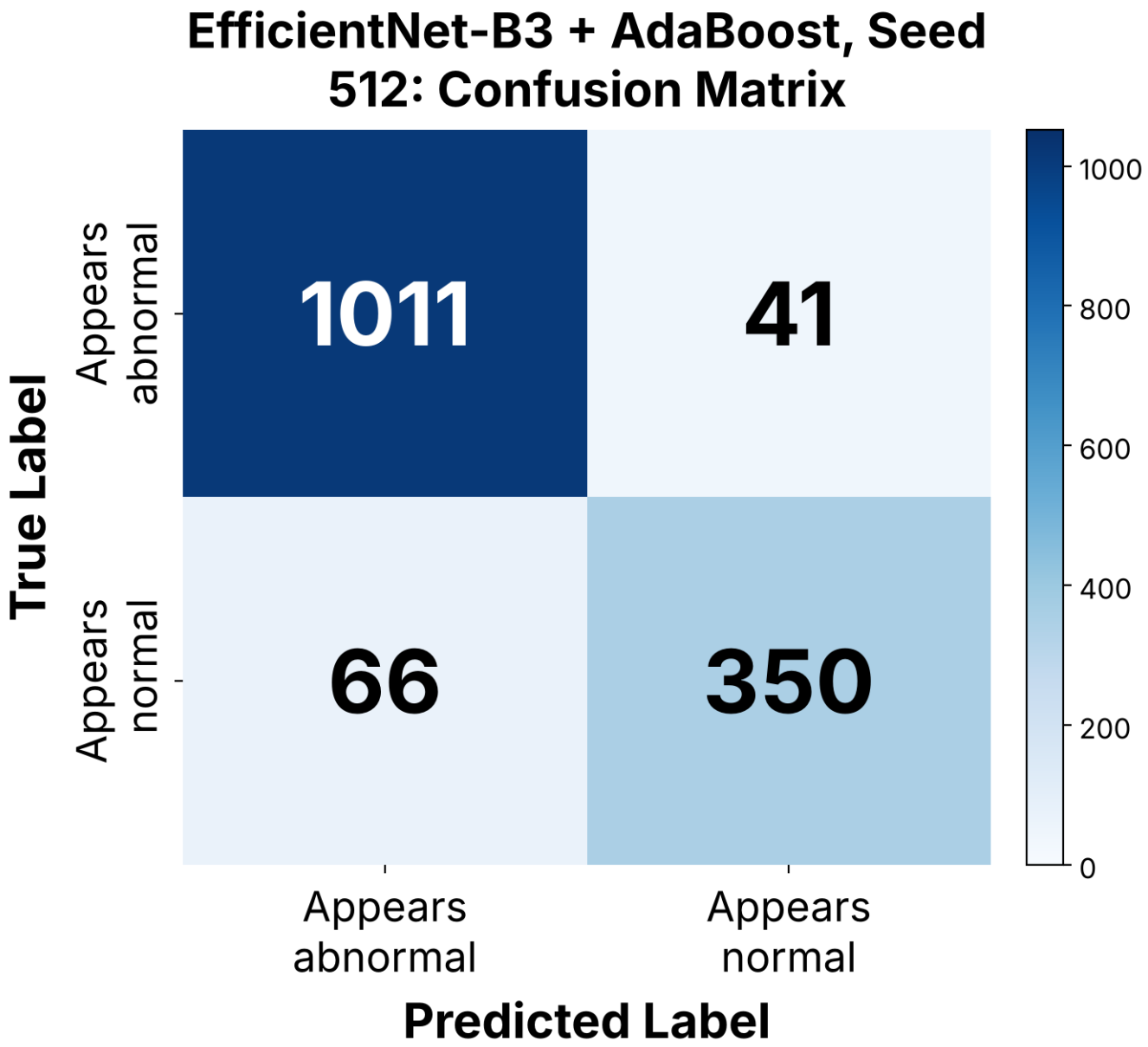


5. EfficientNet-B3 + AdaBoost — Seed 512 (seed 2 of 3)

Source: 3_Seed_exp.ipynb, cell 8. Re-run independently in *(seeded) EfficientNet B3 with Adaboost.ipynb, cell 4, seed 512* (same numbers for that seed).

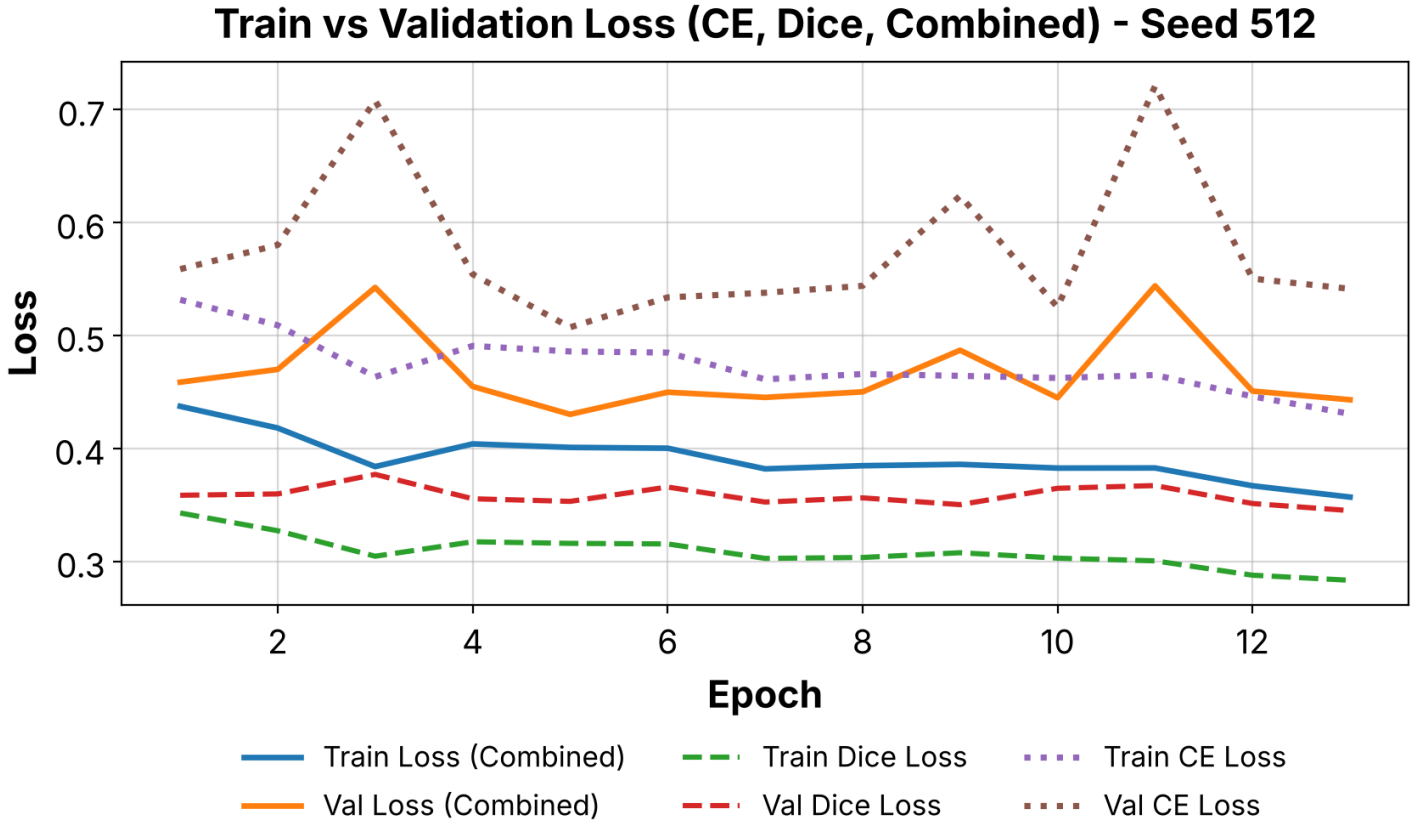
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	91.35%	**92.71%**	91.08%	91.71 ± 0.88 %
Precision (weighted)	91.24%	**92.64%**	90.99%	91.62 ± 0.89 %
Recall (weighted)	91.35%	**92.71%**	91.08%	91.71 ± 0.88 %
F1-score (weighted)	91.25%	**92.64%**	90.88%	91.59 ± 0.93 %
AdaBoost validation acc.	69.37%	69.17%	68.33%	68.96 ± 0.55 %
Best validation acc. (CNN)	76.46%	75.62%	72.50%	74.86 ± 2.09 %

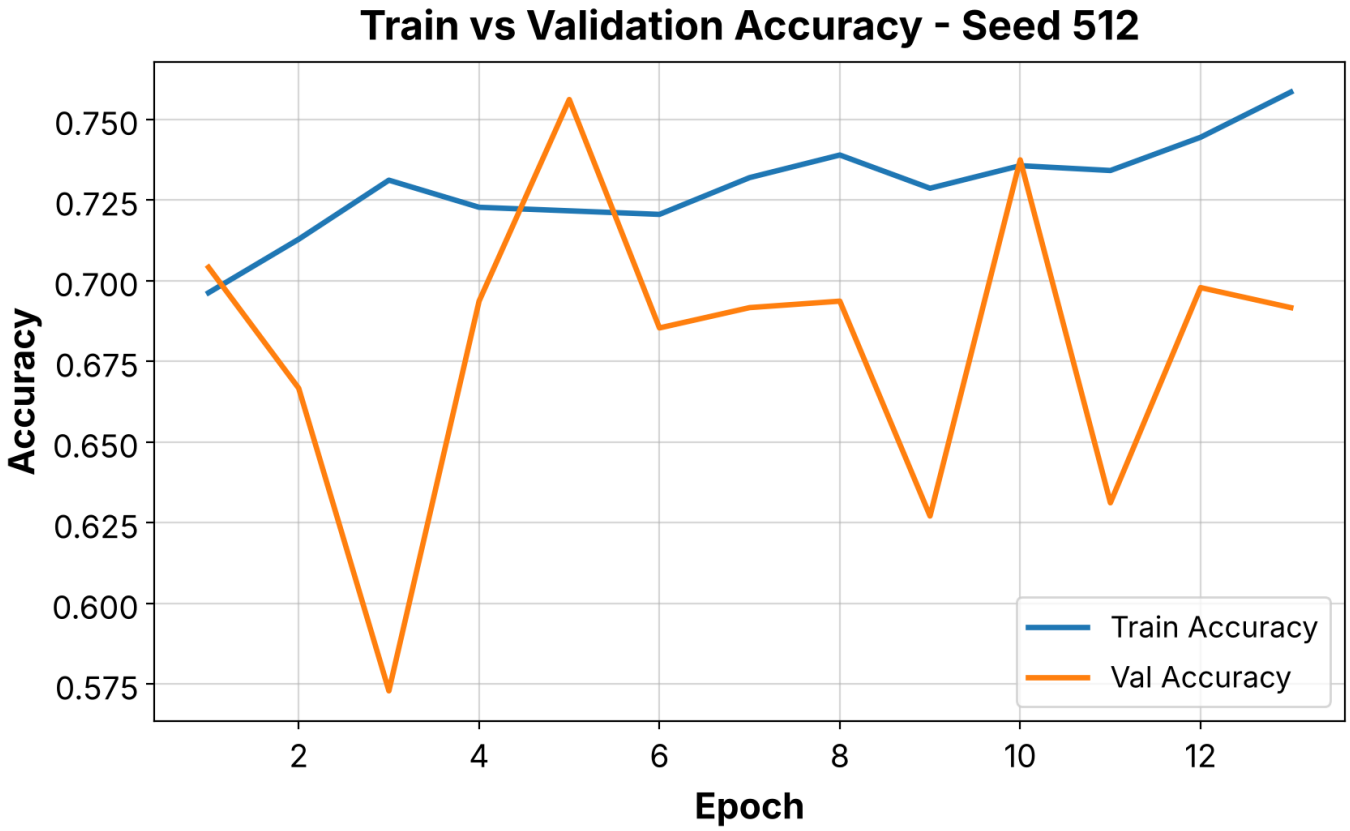
Confusion Matrix



5. EfficientNet-B3 + AdaBoost, Seed 512 — graphs

Training / Validation Graphs



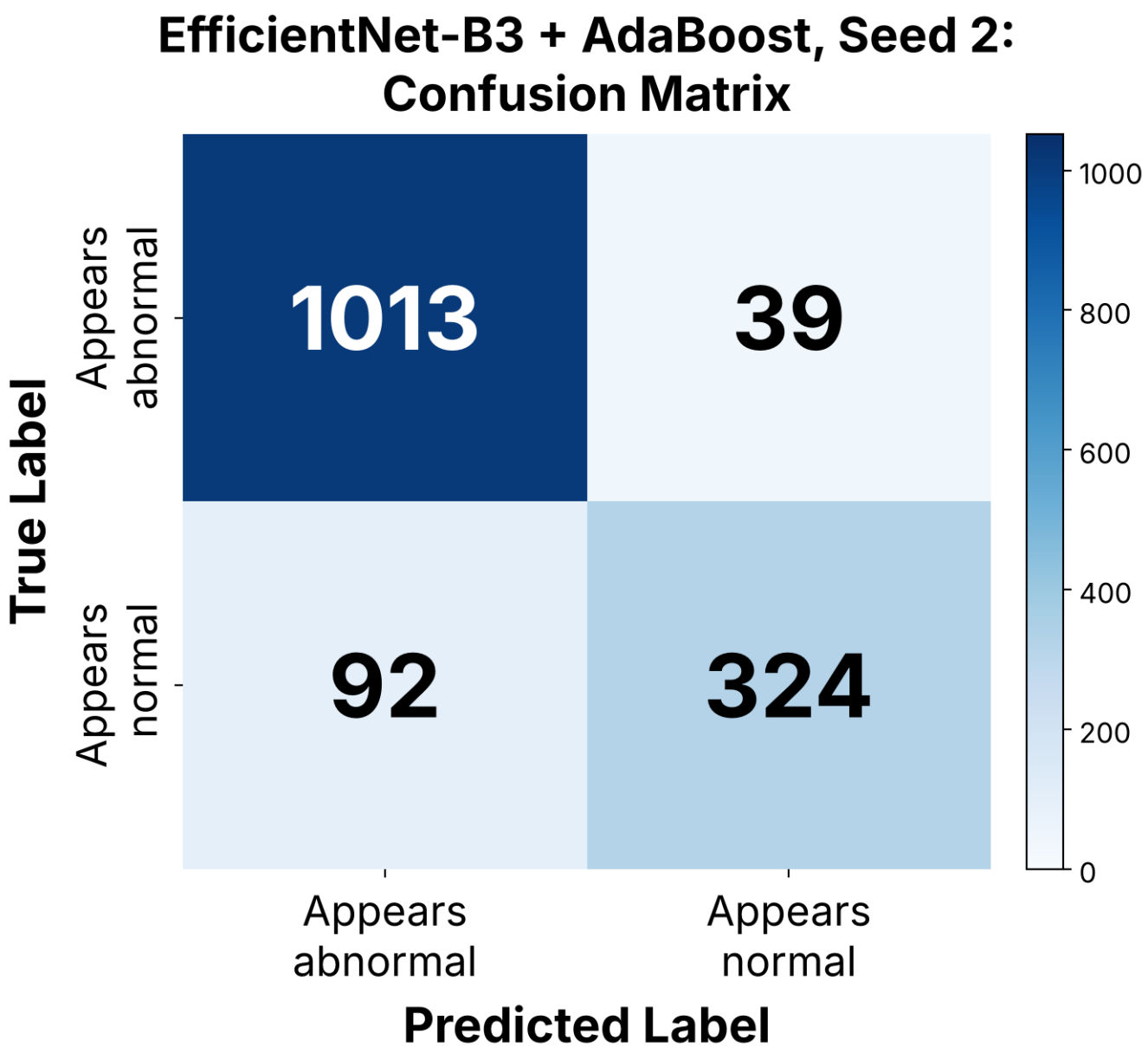


5. EfficientNet-B3 + AdaBoost — Seed 2 (seed 3 of 3)

Source: 3_Seed_exp.ipynb, cell 8. Re-run independently in *(seeded) EfficientNet B3 with Adaboost.ipynb, cell 4, seed 512* (same numbers for that seed).

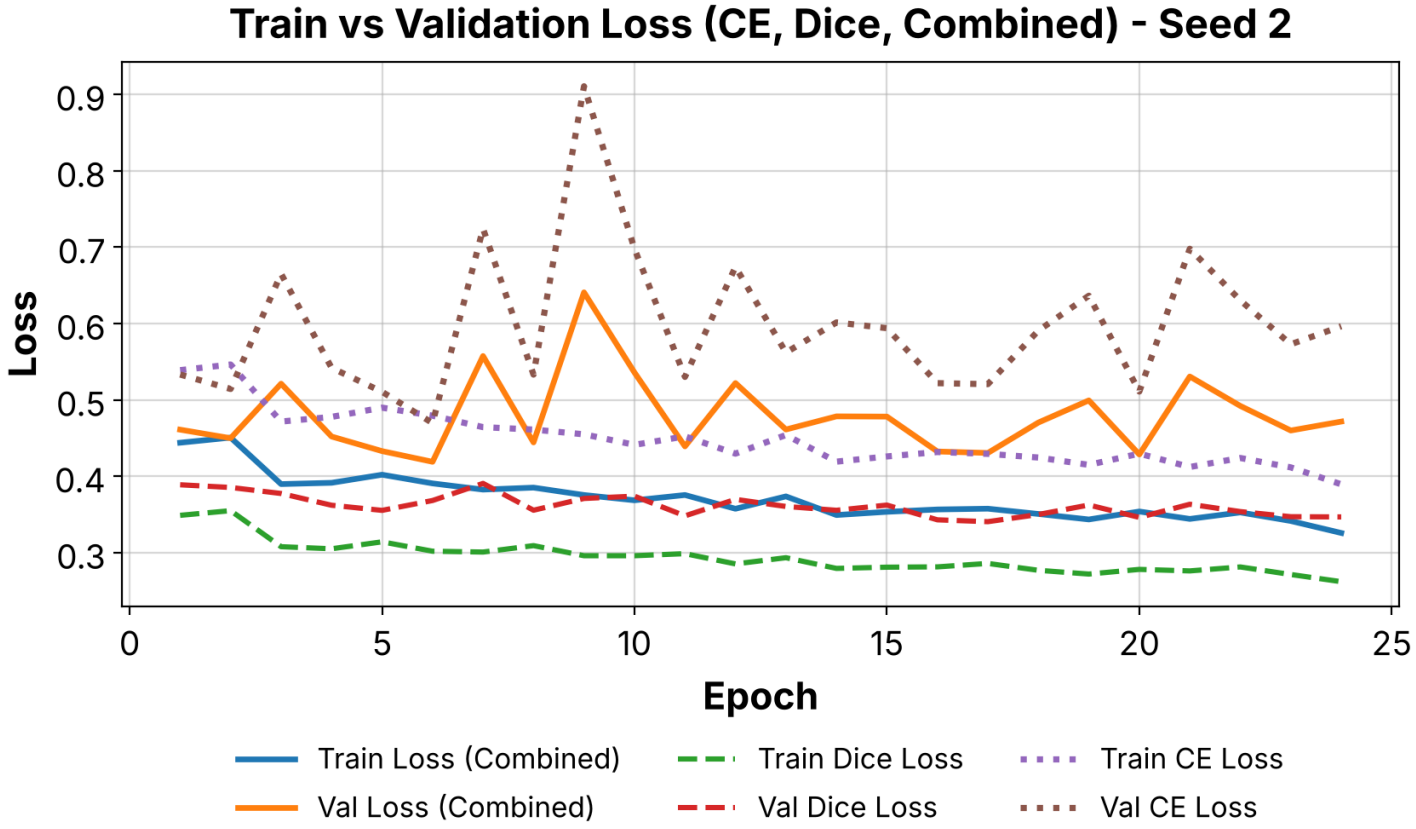
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	91.35%	92.71%	**91.08%**	91.71 ± 0.88 %
Precision (weighted)	91.24%	92.64%	**90.99%**	91.62 ± 0.89 %
Recall (weighted)	91.35%	92.71%	**91.08%**	91.71 ± 0.88 %
F1-score (weighted)	91.25%	92.64%	**90.88%**	91.59 ± 0.93 %
AdaBoost validation acc.	69.37%	69.17%	68.33%	68.96 ± 0.55 %
Best validation acc. (CNN)	76.46%	75.62%	72.50%	74.86 ± 2.09 %

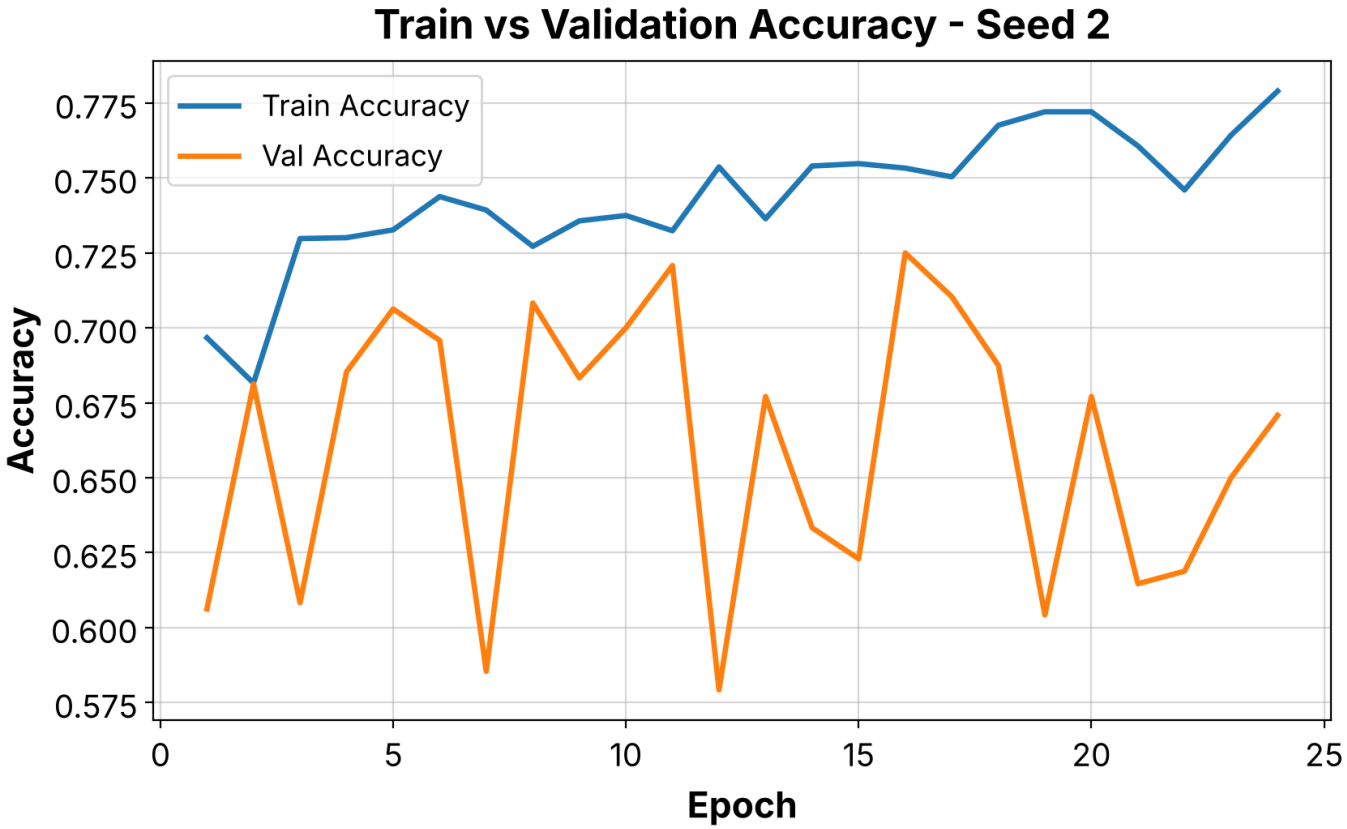
Confusion Matrix



5. EfficientNet-B3 + AdaBoost, Seed 2 — graphs

Training / Validation Graphs



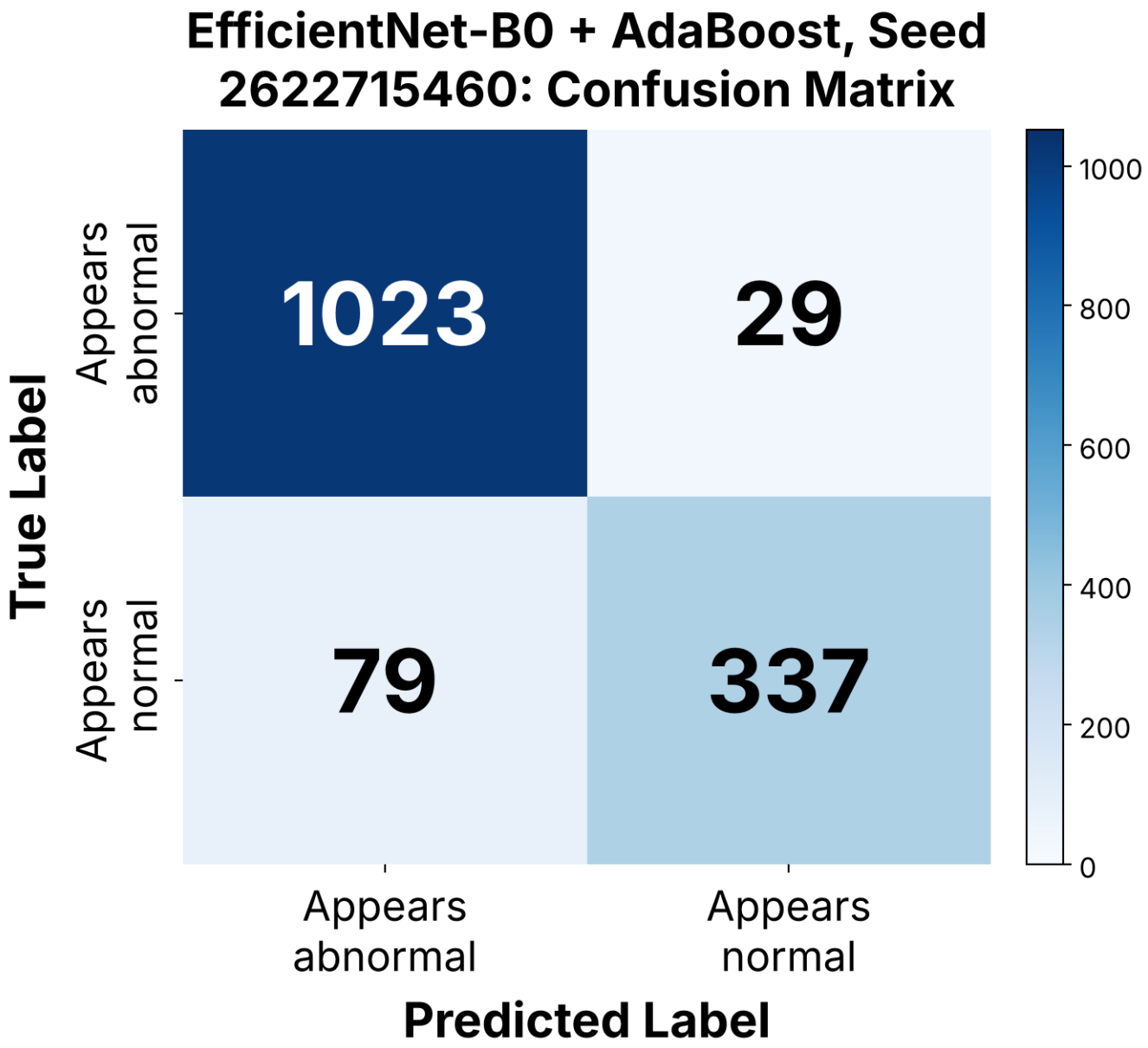


6. EfficientNet-B0 + AdaBoost — Seed 2622715460 (seed 1 of 3)

Source: 3_Seed_exp.ipynb, cell 12. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 8, seed 2622715460* (same numbers for that seed).

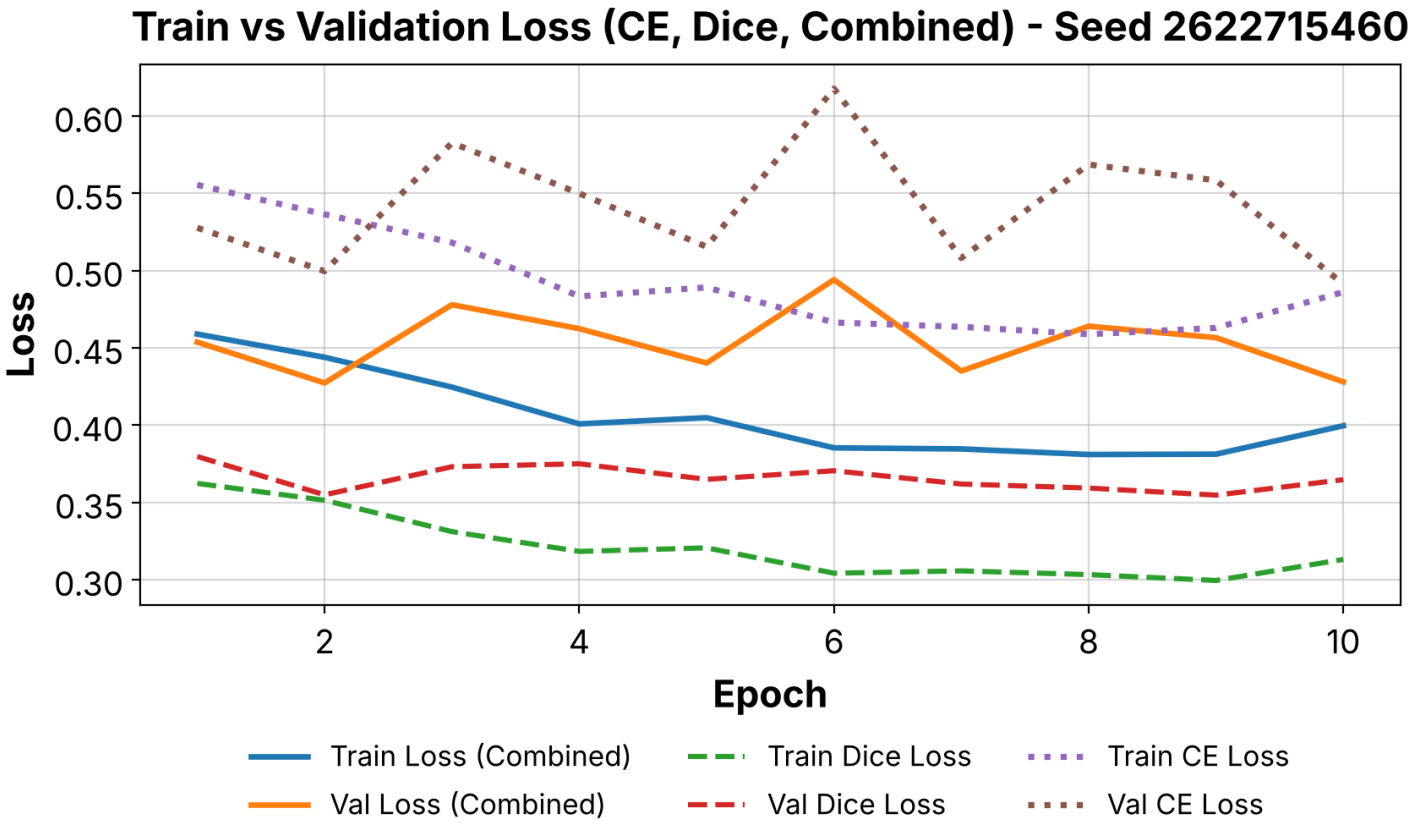
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**92.64%**	91.83%	91.76%	92.08 ± 0.49 %
Precision (weighted)	**92.62%**	91.73%	91.79%	92.05 ± 0.50 %
Recall (weighted)	**92.64%**	91.83%	91.76%	92.08 ± 0.49 %
F1-score (weighted)	**92.49%**	91.72%	91.77%	91.99 ± 0.43 %
AdaBoost validation acc.	71.46%	67.50%	60.00%	66.32 ± 5.82 %
Best validation acc. (CNN)	70.42%	75.62%	75.62%	73.89 ± 3.01 %

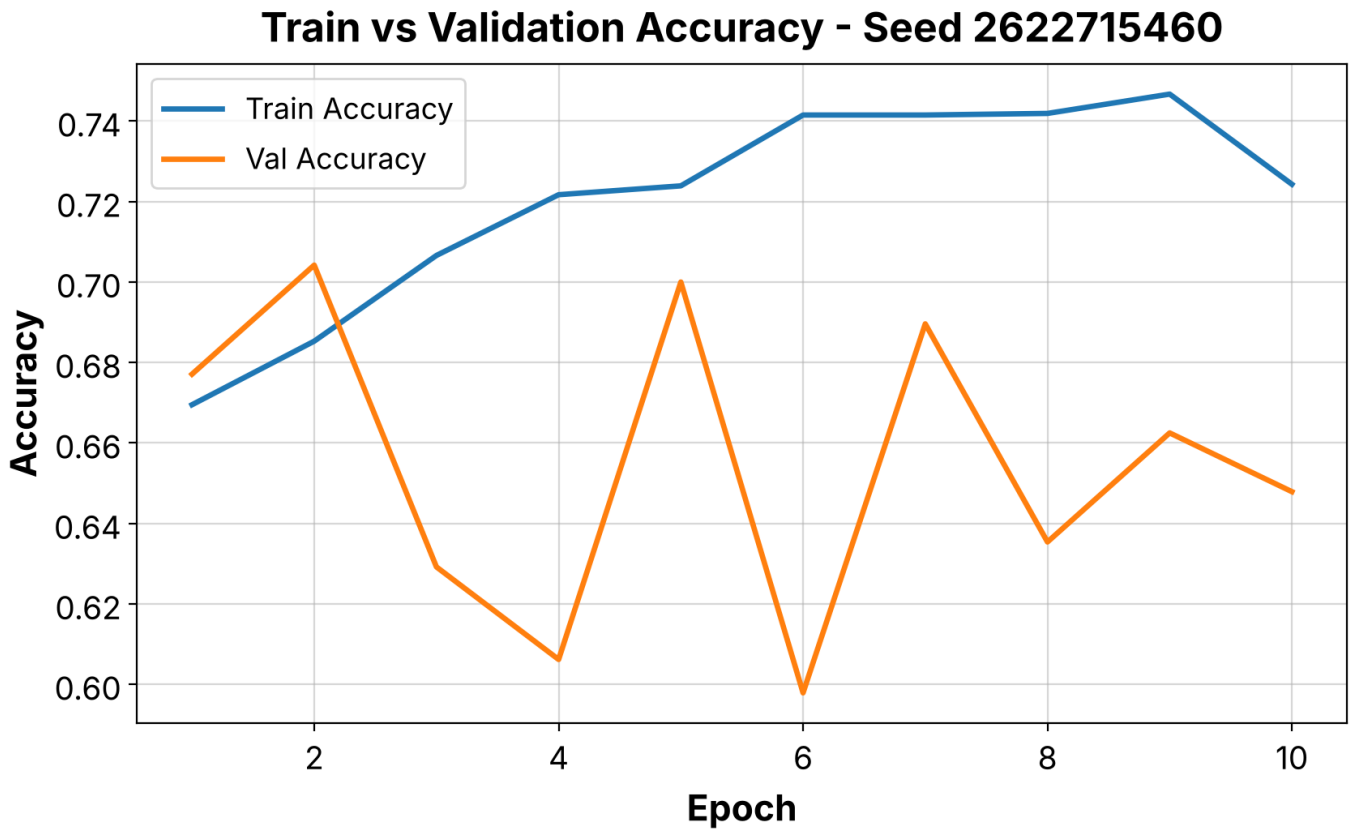
Confusion Matrix



6. EfficientNet-B0 + AdaBoost, Seed 2622715460 — graphs

Training / Validation Graphs



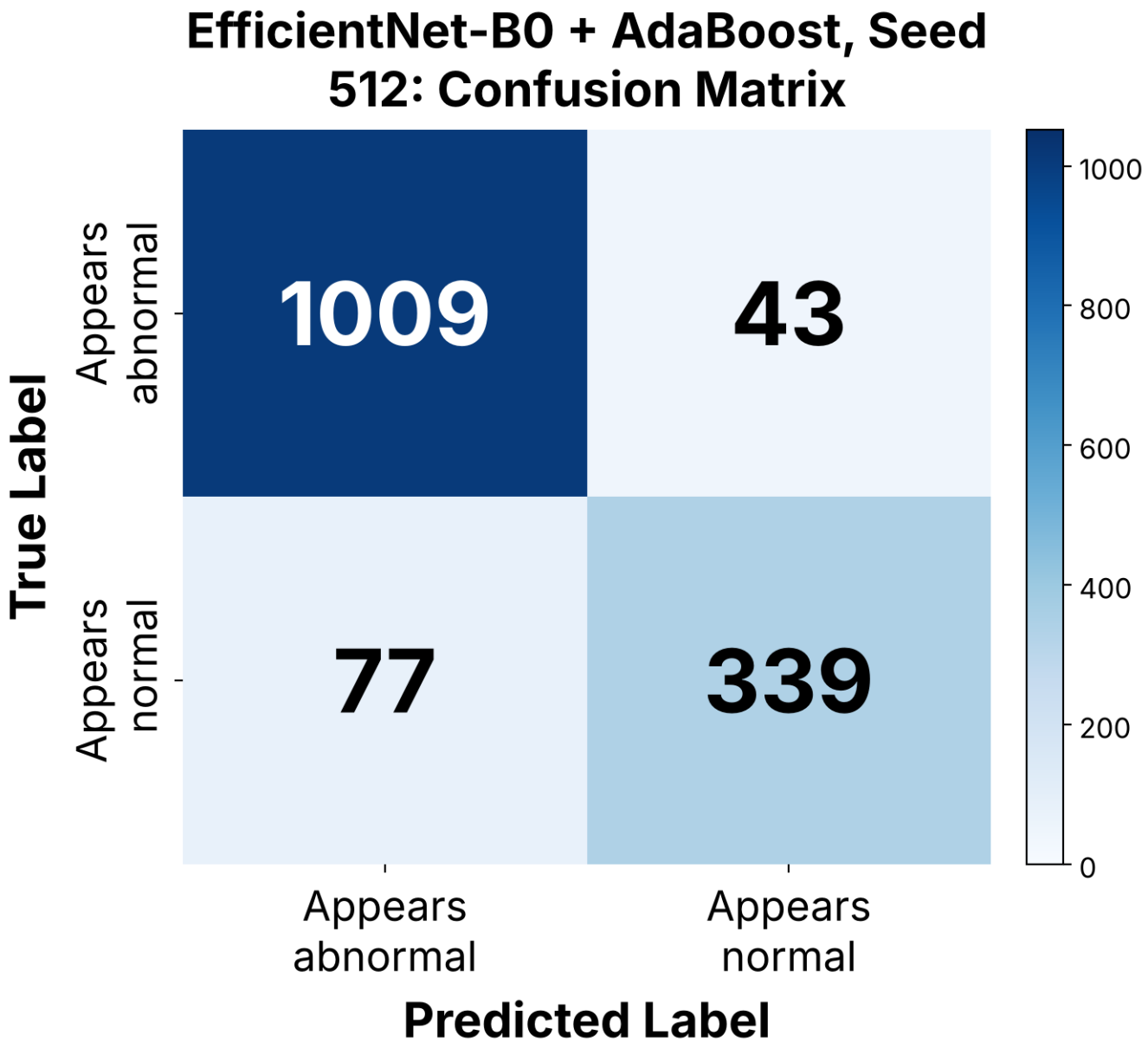


6. EfficientNet-B0 + AdaBoost — Seed 512 (seed 2 of 3)

Source: 3_Seed_exp.ipynb, cell 12. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 8, seed 2622715460* (same numbers for that seed).

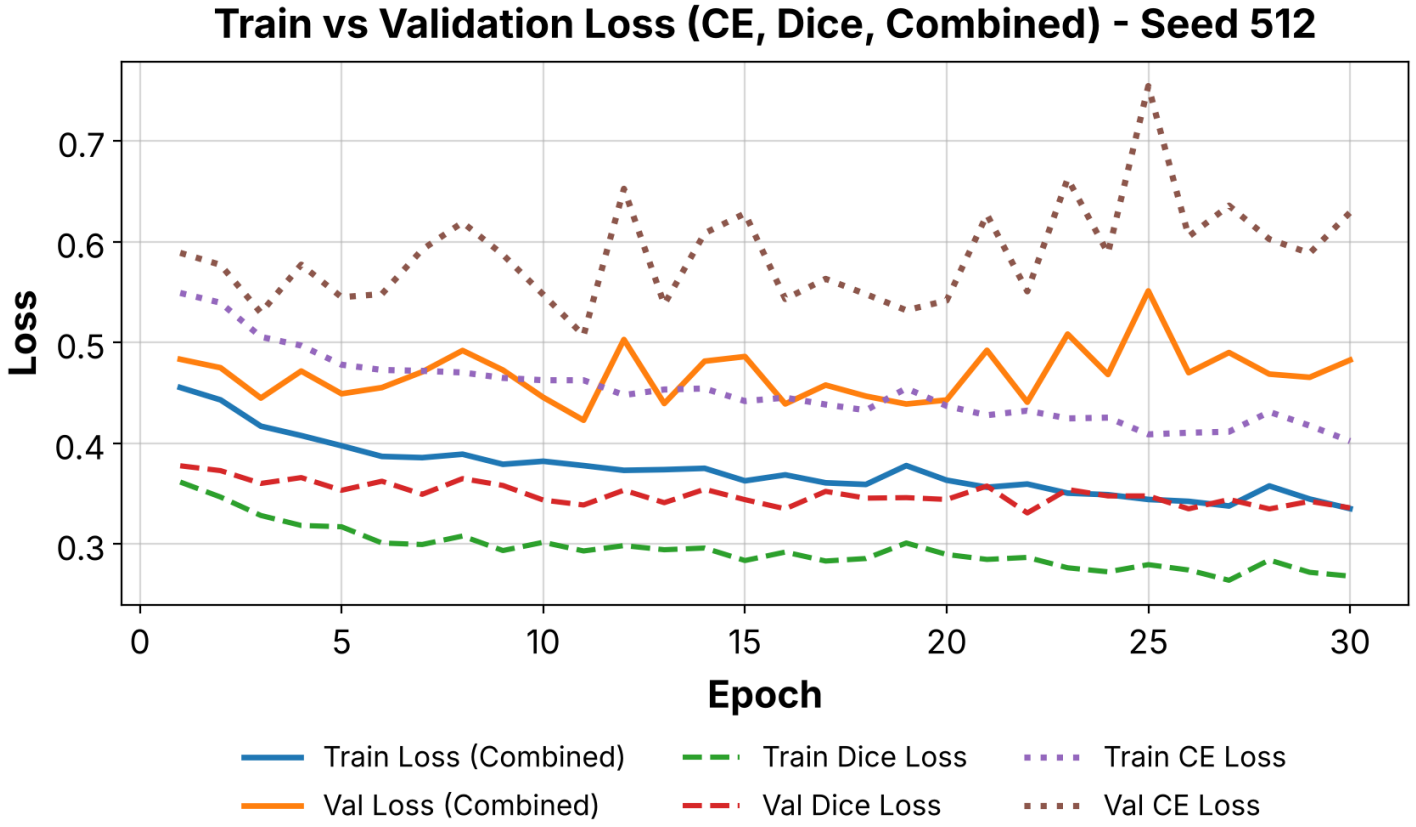
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	92.64%	**91.83%**	91.76%	92.08 ± 0.49 %
Precision (weighted)	92.62%	**91.73%**	91.79%	92.05 ± 0.50 %
Recall (weighted)	92.64%	**91.83%**	91.76%	92.08 ± 0.49 %
F1-score (weighted)	92.49%	**91.72%**	91.77%	91.99 ± 0.43 %
AdaBoost validation acc.	71.46%	67.50%	60.00%	66.32 ± 5.82 %
Best validation acc. (CNN)	70.42%	75.62%	75.62%	73.89 ± 3.01 %

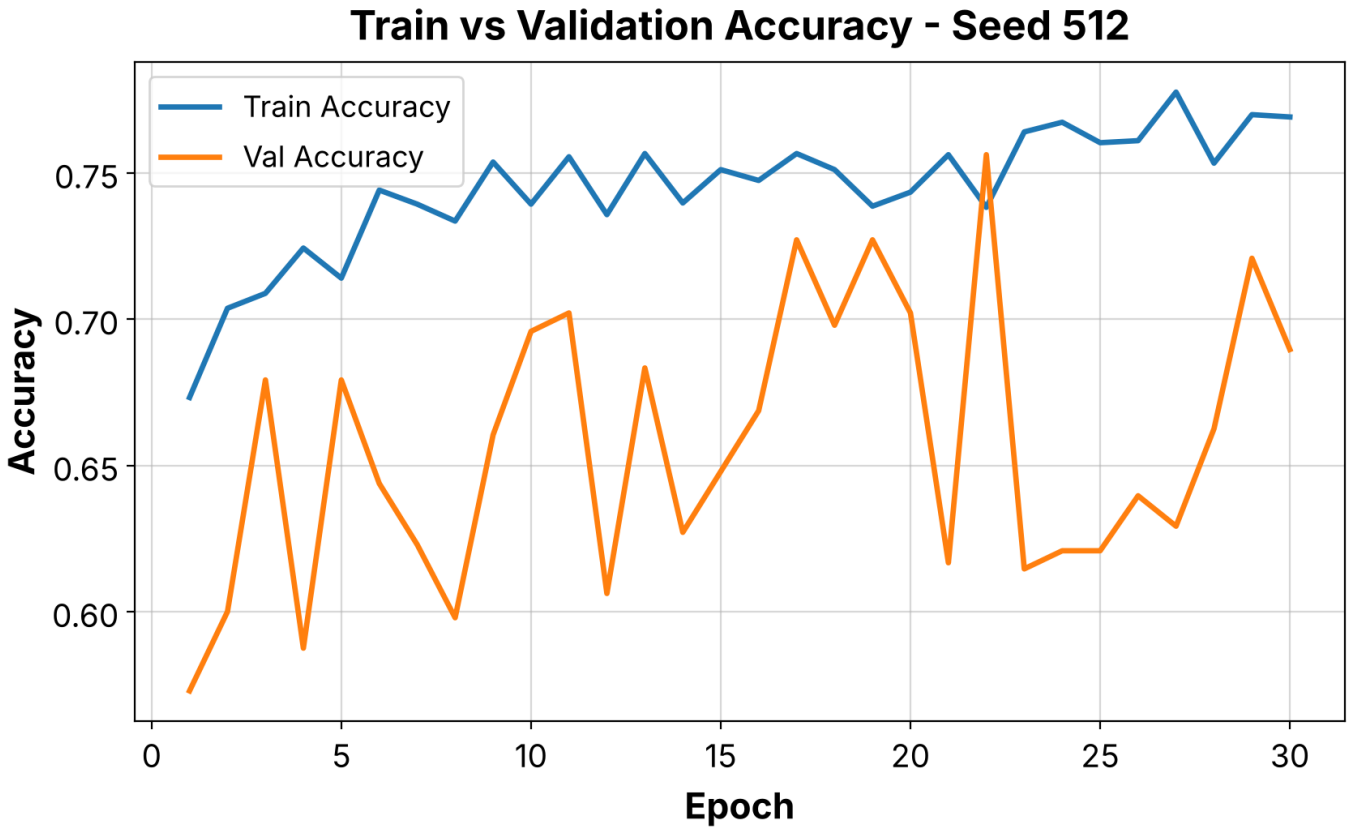
Confusion Matrix



6. EfficientNet-B0 + AdaBoost, Seed 512 — graphs

Training / Validation Graphs



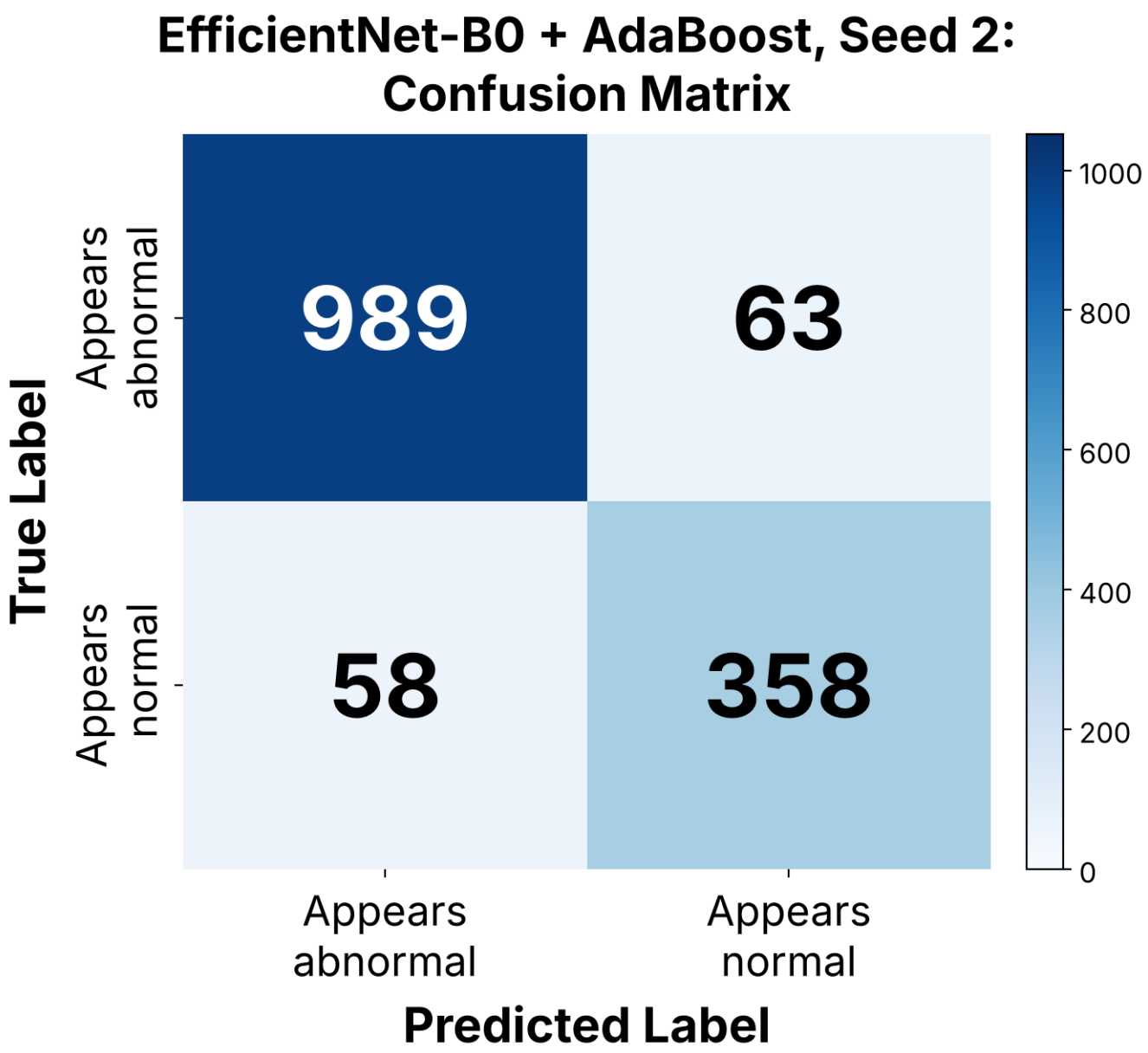


6. EfficientNet-B0 + AdaBoost — Seed 2 (seed 3 of 3)

Source: 3_Seed_exp.ipynb, cell 12. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 8, seed 2622715460* (same numbers for that seed).

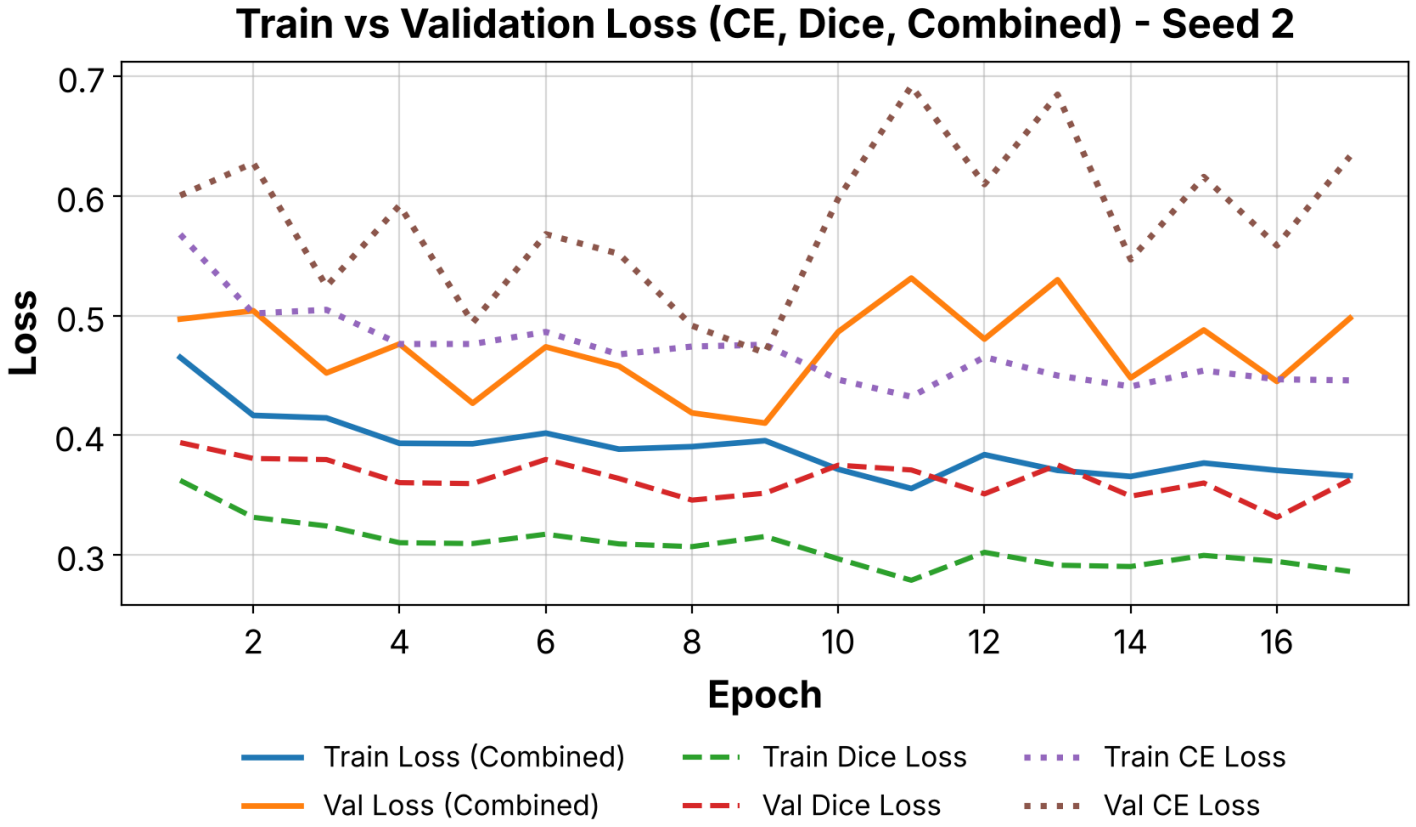
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	92.64%	91.83%	**91.76%**	92.08 ± 0.49 %
Precision (weighted)	92.62%	91.73%	**91.79%**	92.05 ± 0.50 %
Recall (weighted)	92.64%	91.83%	**91.76%**	92.08 ± 0.49 %
F1-score (weighted)	92.49%	91.72%	**91.77%**	91.99 ± 0.43 %
AdaBoost validation acc.	71.46%	67.50%	60.00%	66.32 ± 5.82 %
Best validation acc. (CNN)	70.42%	75.62%	75.62%	73.89 ± 3.01 %

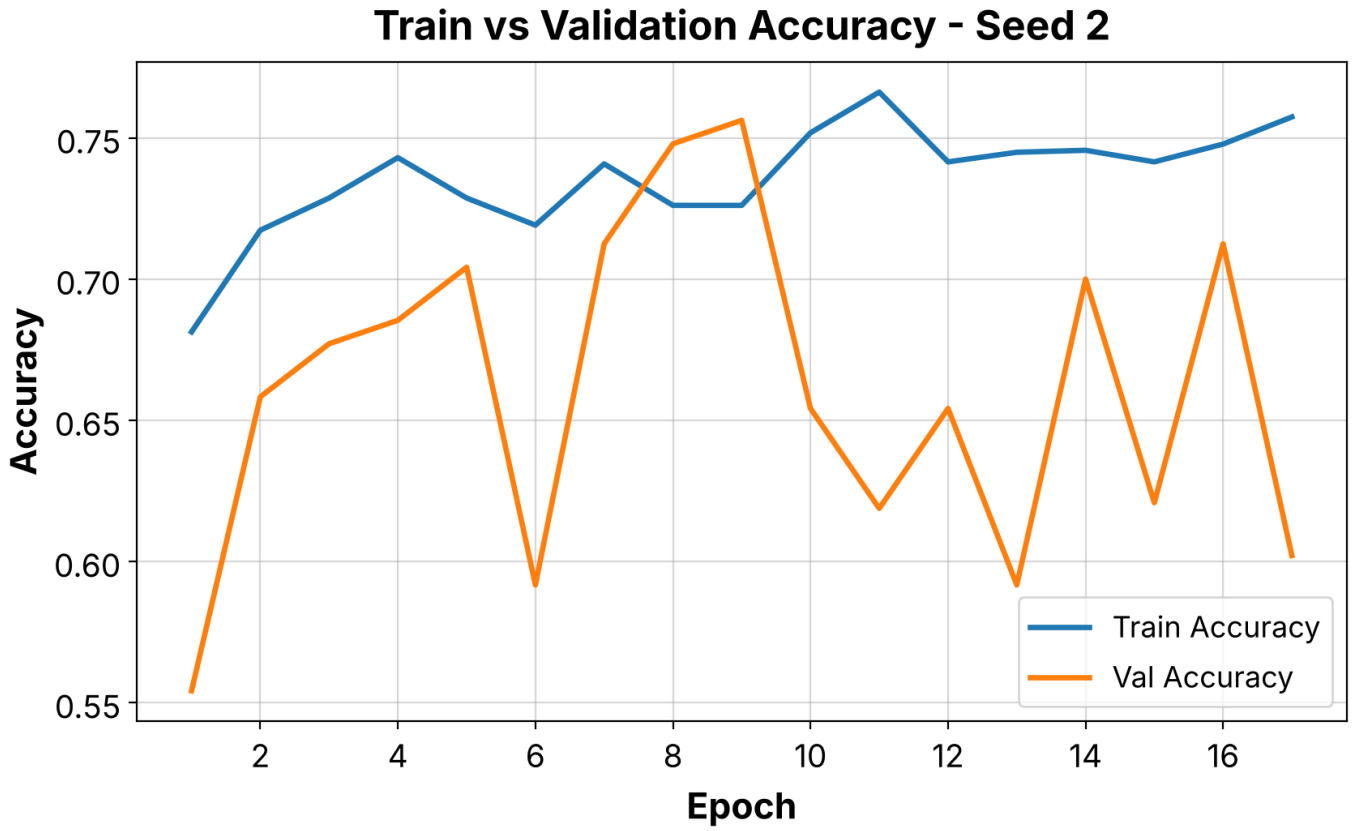
Confusion Matrix



6. EfficientNet-B0 + AdaBoost, Seed 2 — graphs

Training / Validation Graphs





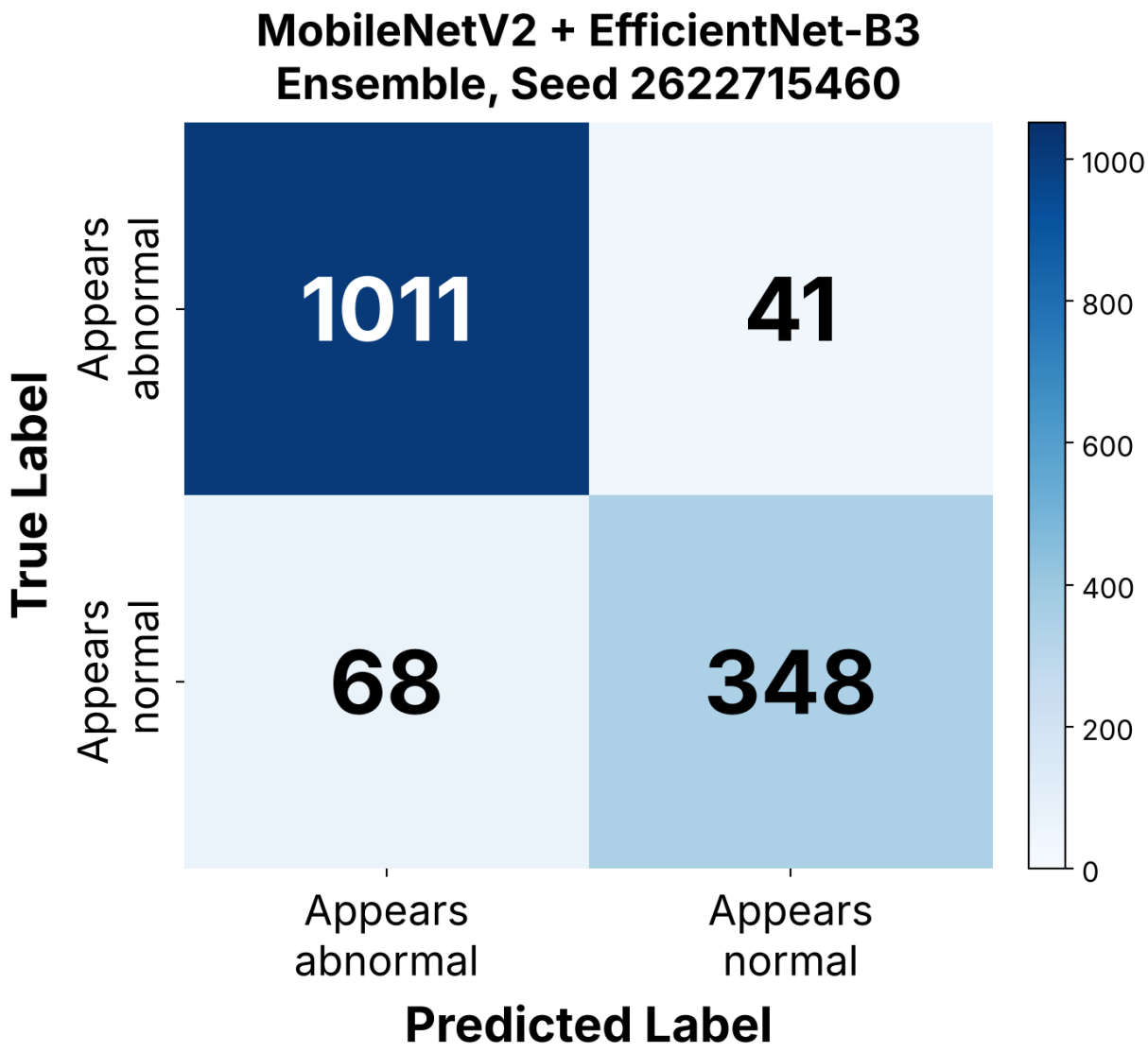
7. Ensemble — MobileNetV2 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 2622715460 (seed 1 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**92.57%**	92.51%	91.28%	92.12 ± 0.73 %
Precision (weighted)	**92.50%**	92.45%	91.26%	92.07 ± 0.70 %
Recall (weighted)	**92.57%**	92.51%	91.28%	92.12 ± 0.73 %
F1-score (weighted)	**92.50%**	92.37%	91.05%	91.97 ± 0.80 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 1, EfficientNet-B3 = row 5.

Confusion Matrix



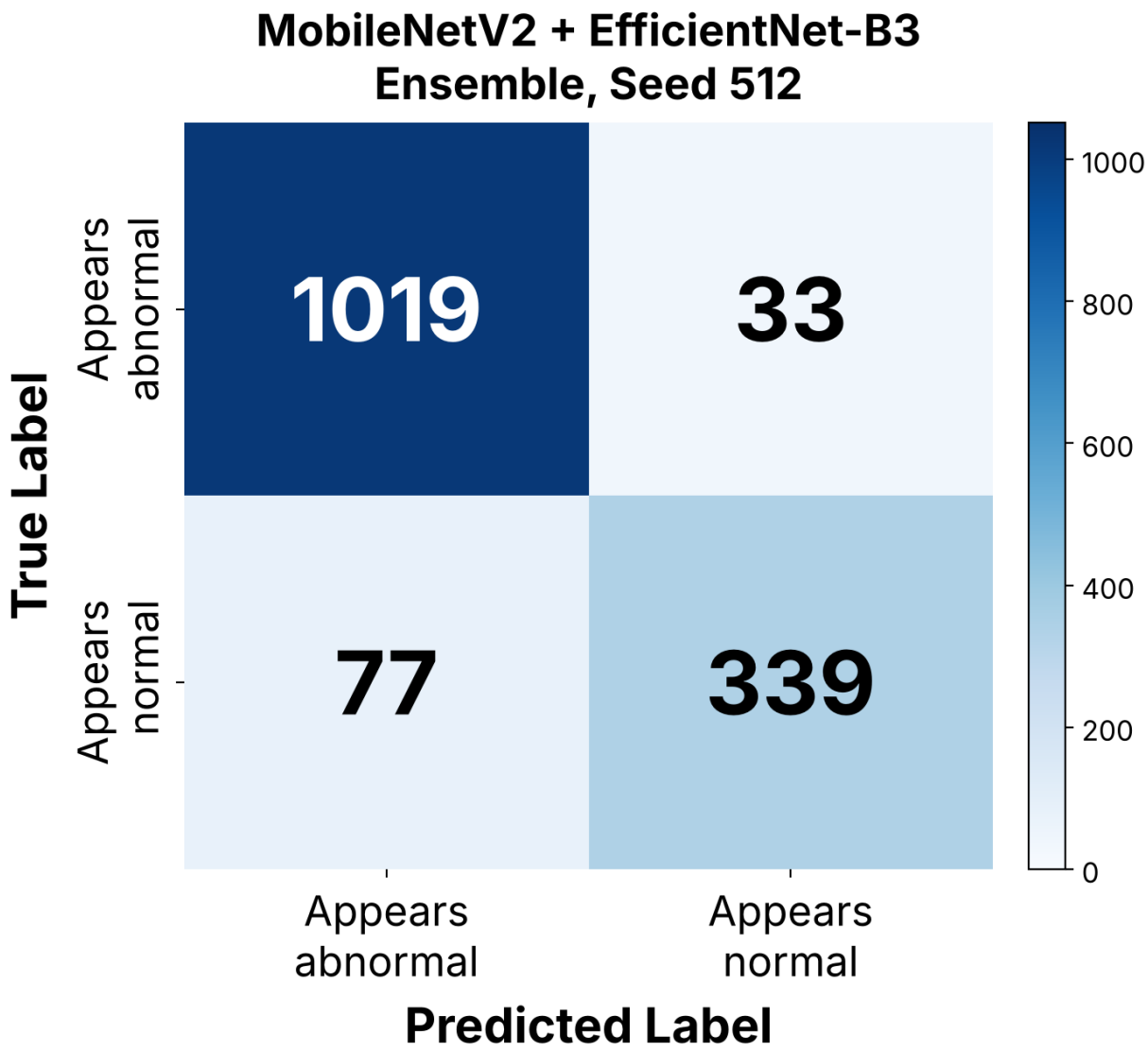
7. Ensemble — MobileNetV2 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 512 (seed 2 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	92.57%	**92.51%**	91.28%	92.12 ± 0.73 %
Precision (weighted)	92.50%	**92.45%**	91.26%	92.07 ± 0.70 %
Recall (weighted)	92.57%	**92.51%**	91.28%	92.12 ± 0.73 %
F1-score (weighted)	92.50%	**92.37%**	91.05%	91.97 ± 0.80 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 1, EfficientNet-B3 = row 5.

Confusion Matrix



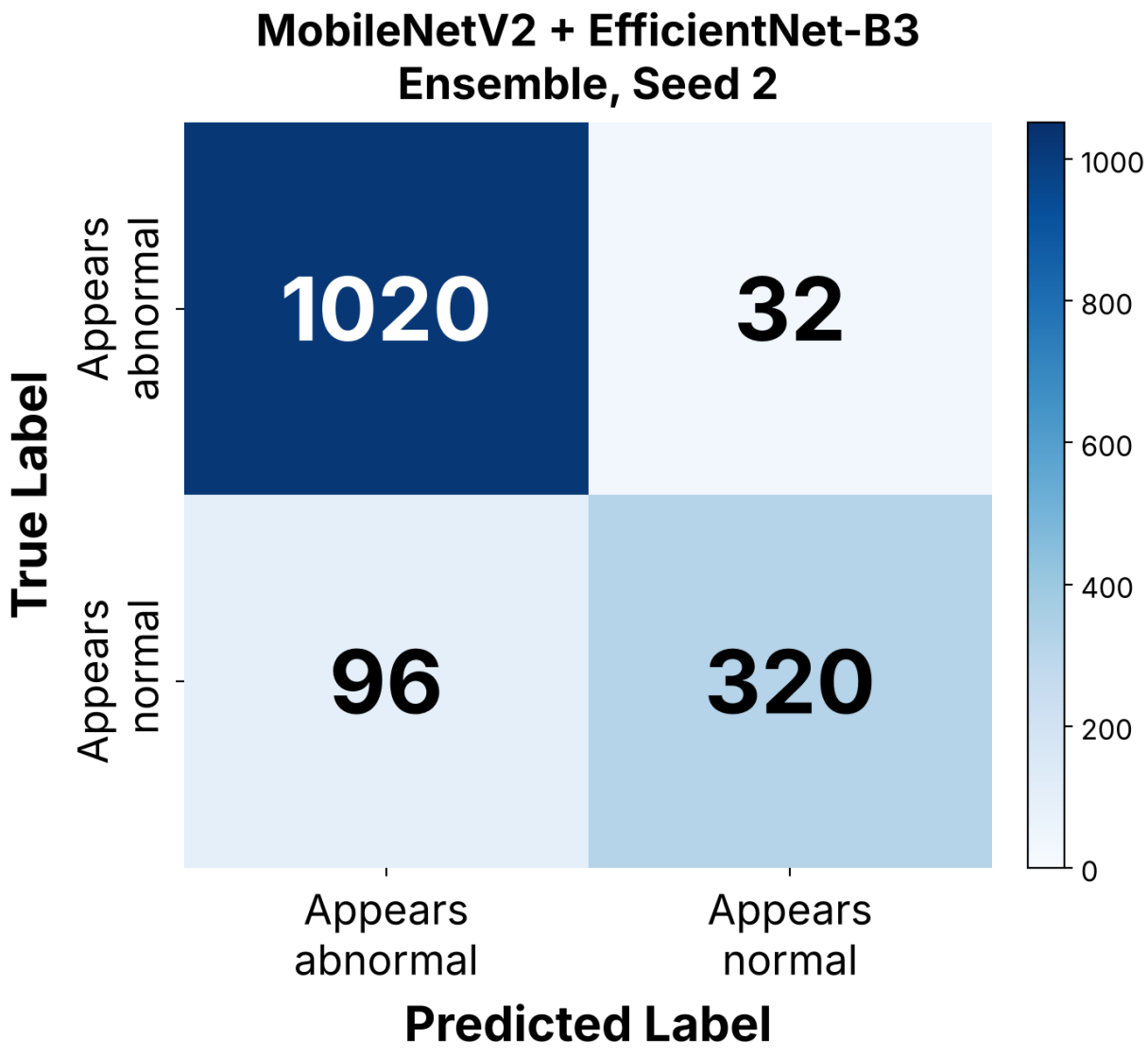
7. Ensemble — MobileNetV2 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 2 (seed 3 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	92.57%	92.51%	**91.28%**	92.12 ± 0.73 %
Precision (weighted)	92.50%	92.45%	**91.26%**	92.07 ± 0.70 %
Recall (weighted)	92.57%	92.51%	**91.28%**	92.12 ± 0.73 %
F1-score (weighted)	92.50%	92.37%	**91.05%**	91.97 ± 0.80 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 1, EfficientNet-B3 = row 5.

Confusion Matrix

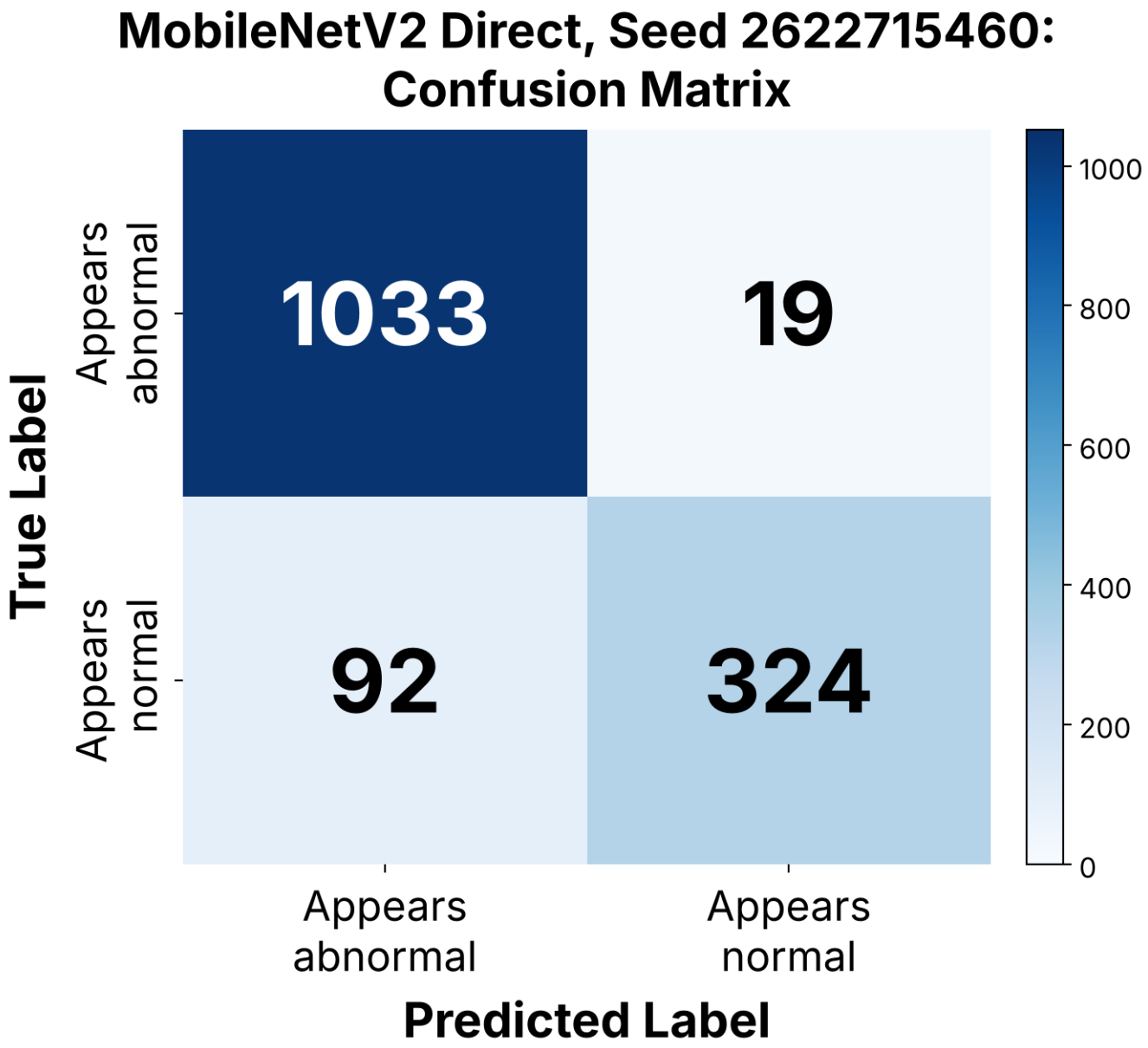


8. MobileNetV2 — Direct (fine-tuned CNN) — Seed 2622715460 (seed 1 of 3)

Source: 3_Seed_exp.ipynb, cell 4. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 4, seed 2622715460* (same numbers for that seed).

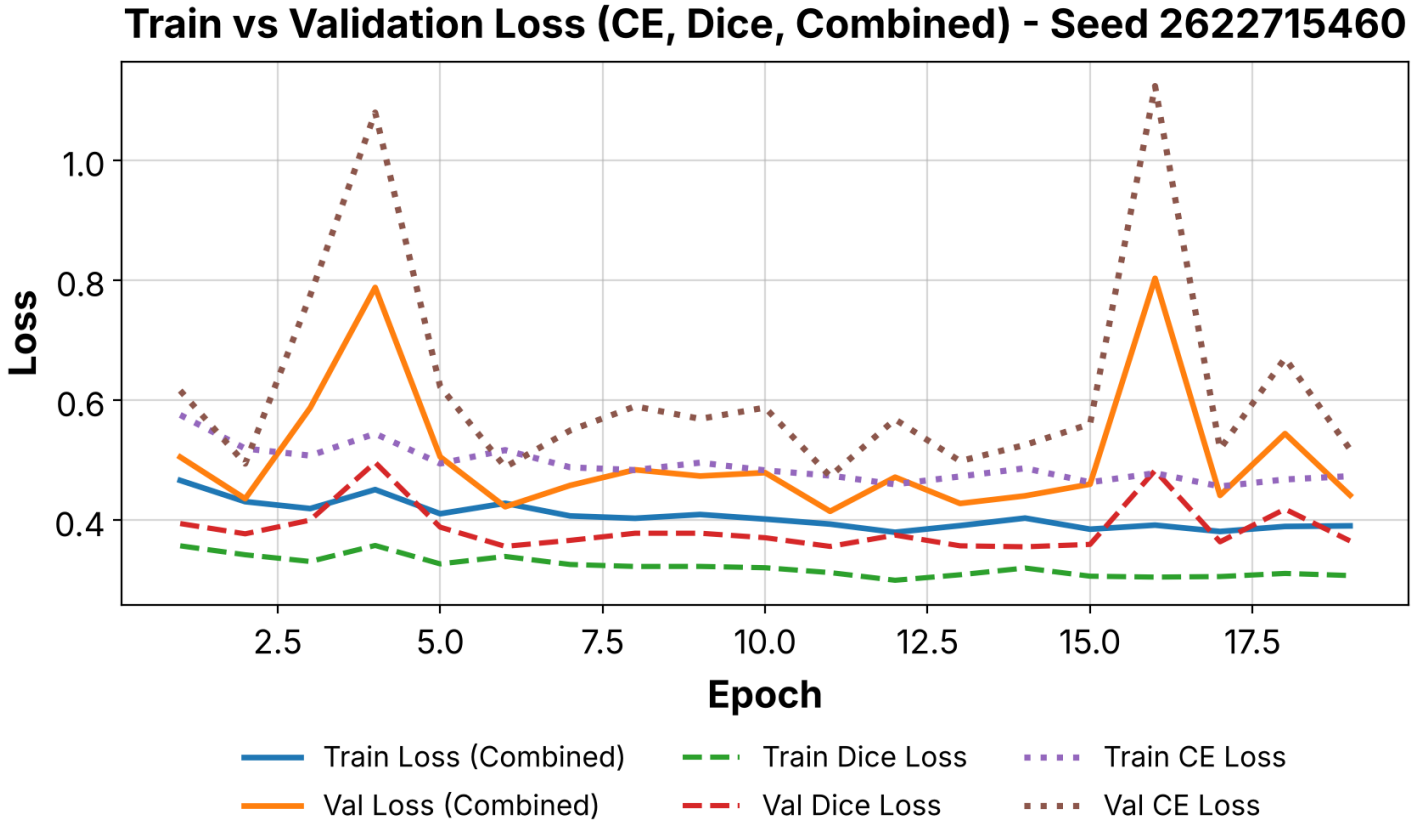
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**92.44%**	85.83%	88.62%	88.96 ± 3.32 %
Precision (weighted)	**92.57%**	85.76%	89.07%	89.13 ± 3.41 %
Recall (weighted)	**92.44%**	85.83%	88.62%	88.96 ± 3.32 %
F1-score (weighted)	**92.20%**	85.02%	87.94%	88.39 ± 3.61 %
Test Dice loss	0.5267	0.5283	0.5245	0.5265 ± 0.0019
Best validation acc. (CNN)	72.08%	71.25%	77.50%	73.61 ± 3.39 %

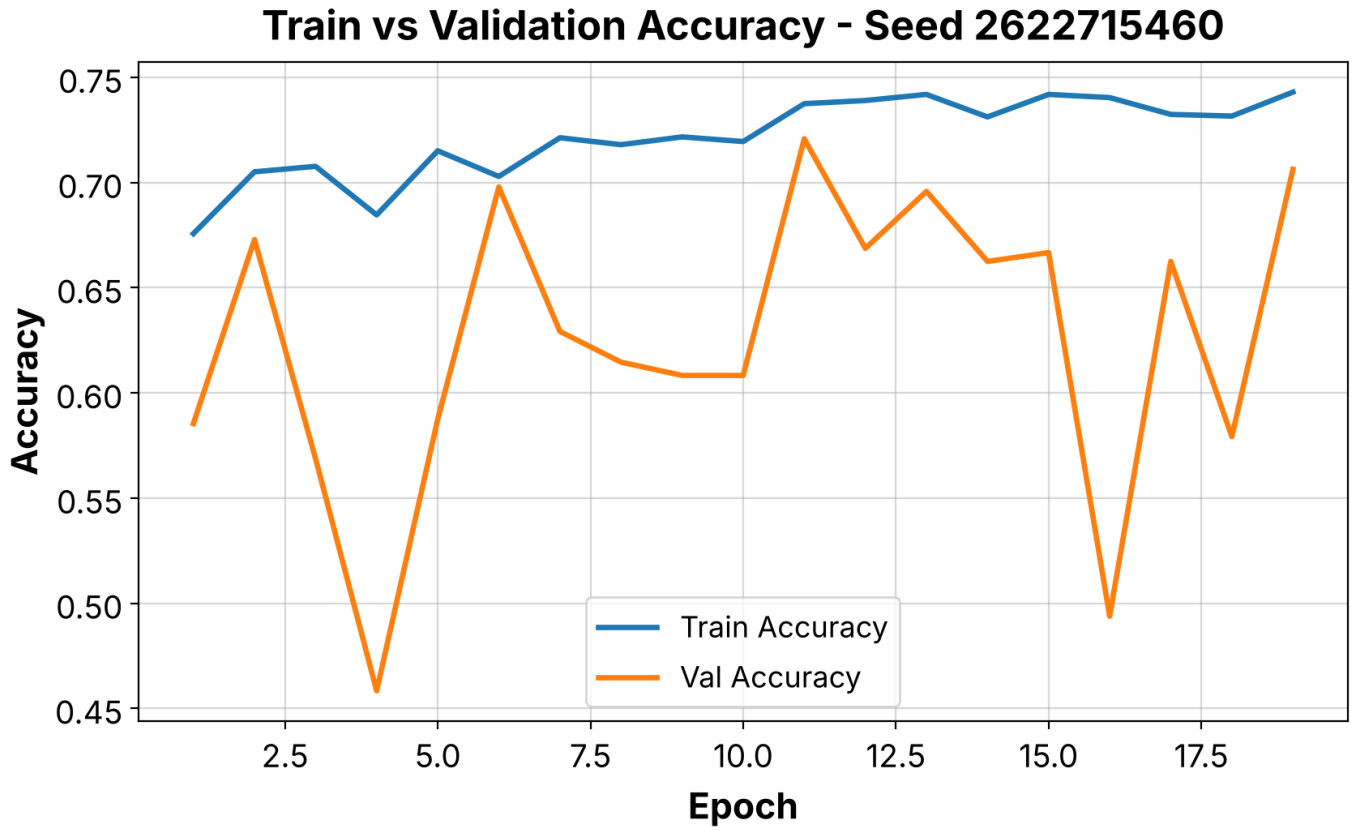
Confusion Matrix



8. MobileNetV2 — Direct (fine-tuned CNN), Seed 2622715460 — graphs

Training / Validation Graphs



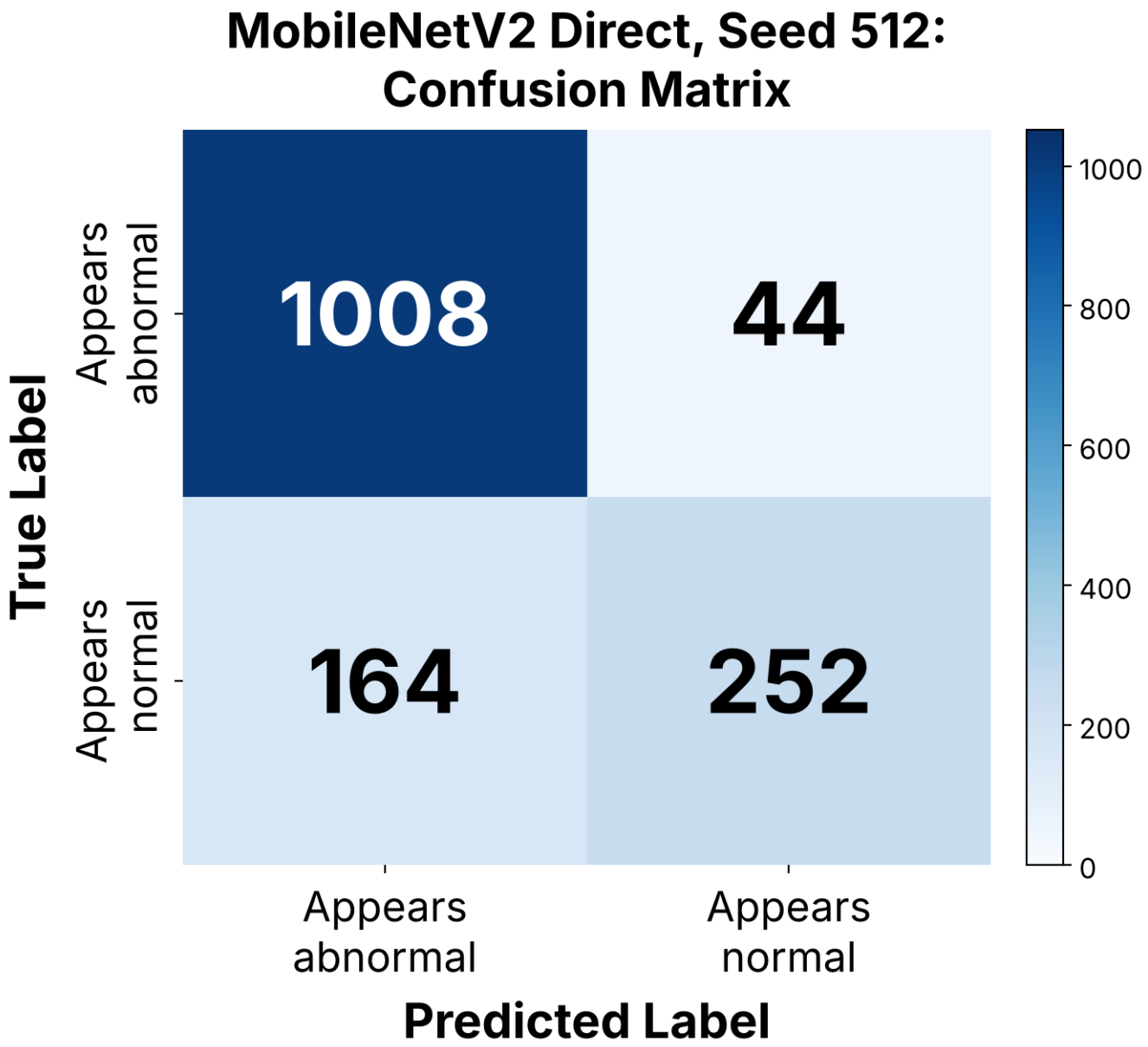


8. MobileNetV2 — Direct (fine-tuned CNN) — Seed 512 (seed 2 of 3)

Source: 3_Seed_exp.ipynb, cell 4. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 4, seed 2622715460* (same numbers for that seed).

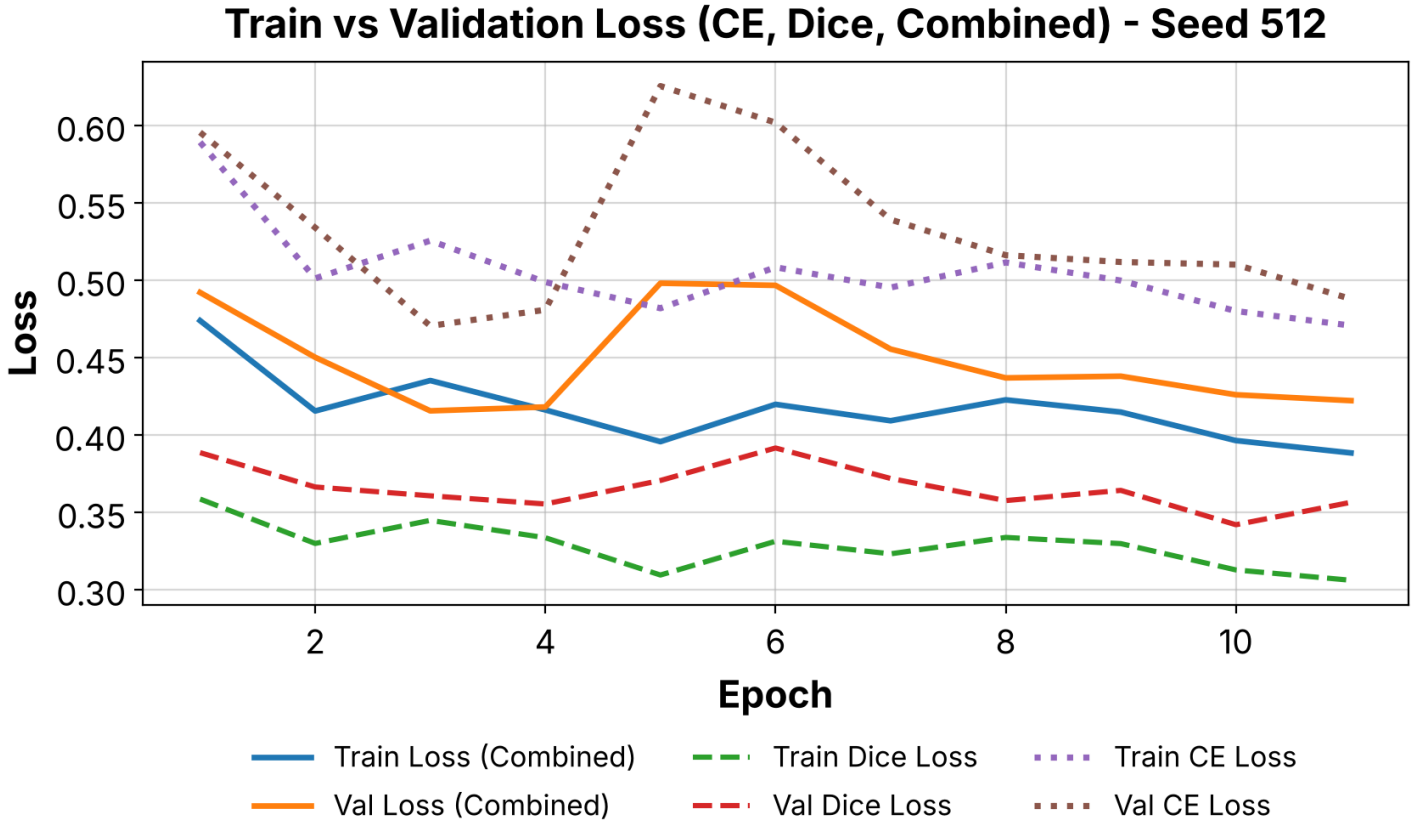
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	92.44%	**85.83%**	88.62%	88.96 ± 3.32 %
Precision (weighted)	92.57%	**85.76%**	89.07%	89.13 ± 3.41 %
Recall (weighted)	92.44%	**85.83%**	88.62%	88.96 ± 3.32 %
F1-score (weighted)	92.20%	**85.02%**	87.94%	88.39 ± 3.61 %
Test Dice loss	0.5267	0.5283	0.5245	0.5265 ± 0.0019
Best validation acc. (CNN)	72.08%	71.25%	77.50%	73.61 ± 3.39 %

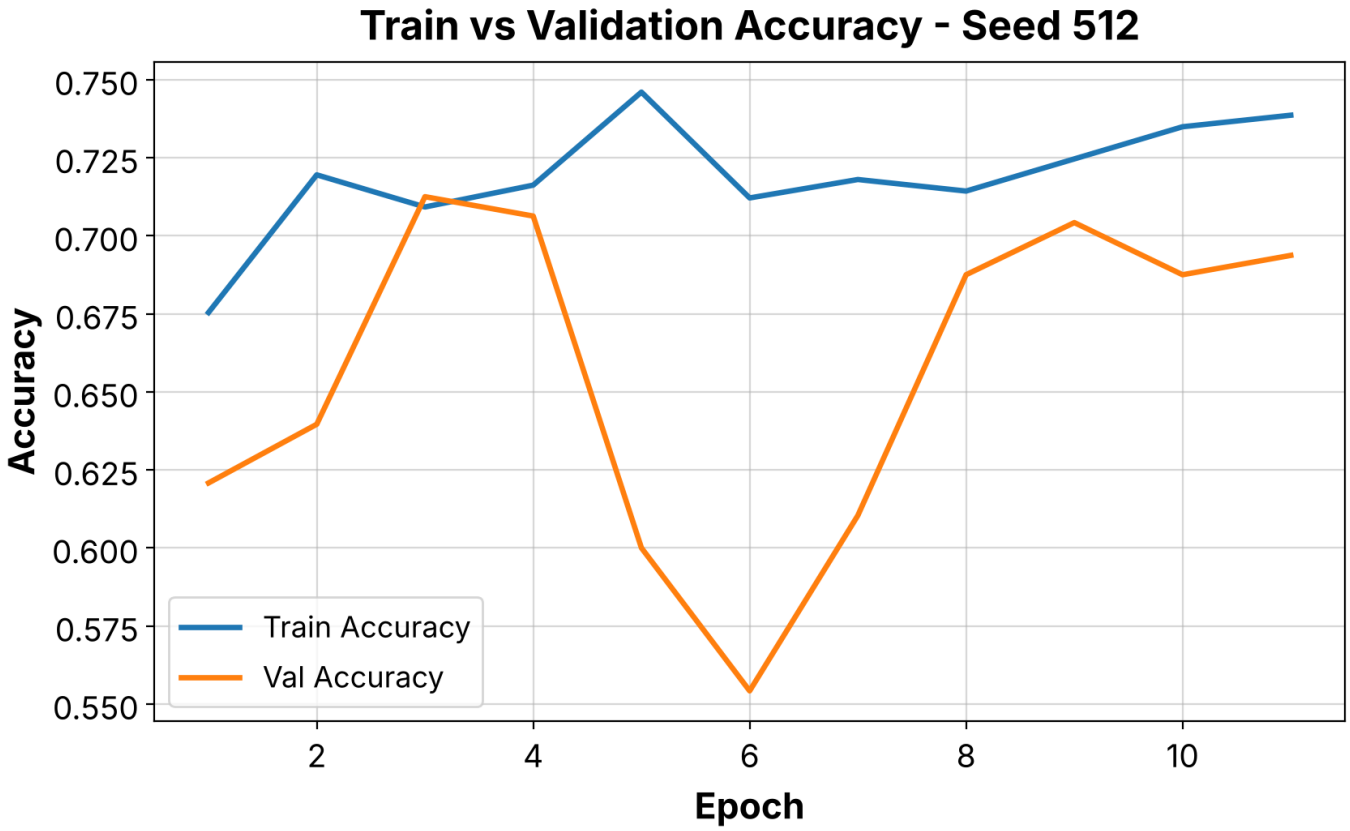
Confusion Matrix



8. MobileNetV2 — Direct (fine-tuned CNN), Seed 512 — graphs

Training / Validation Graphs



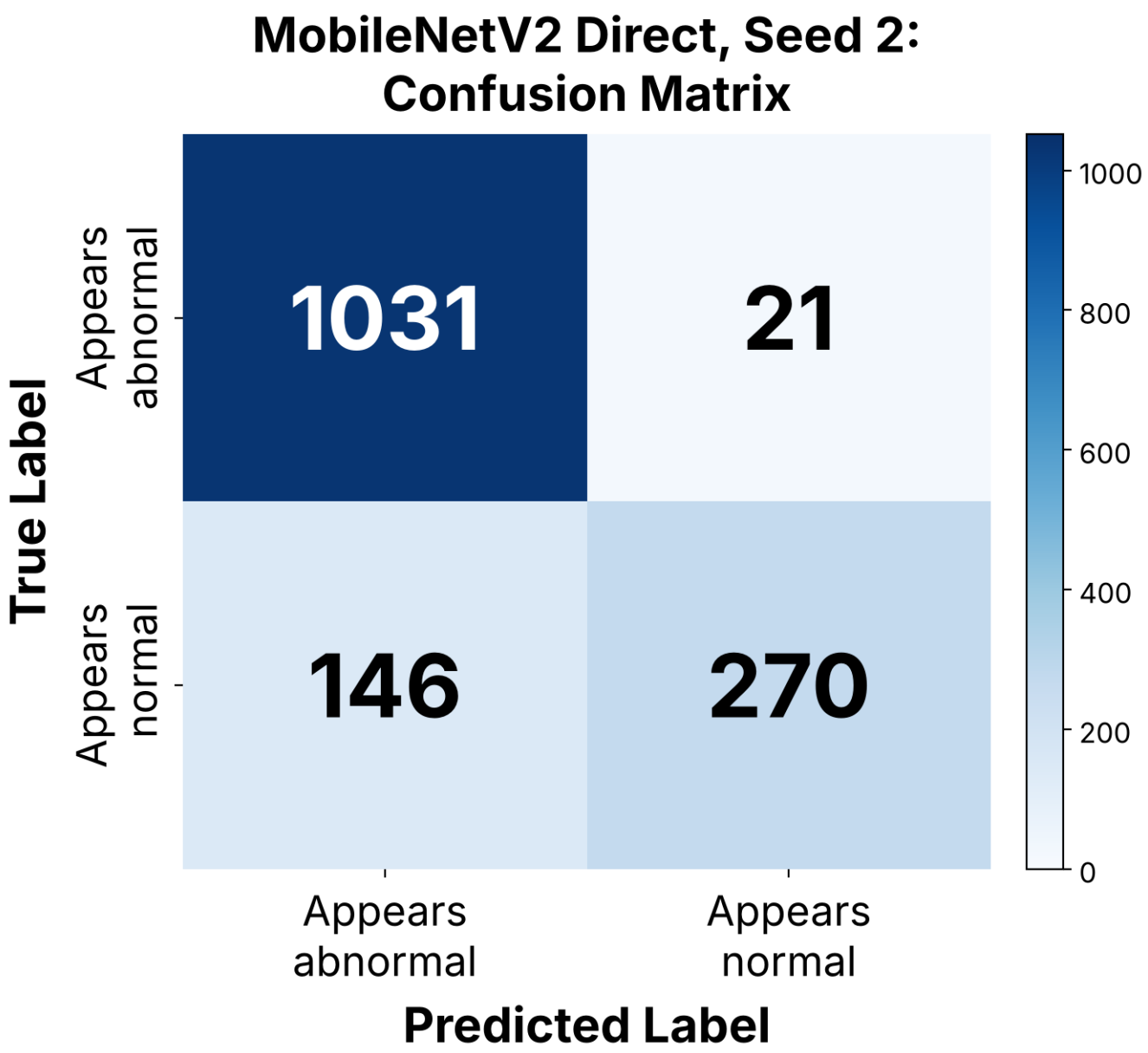


8. MobileNetV2 — Direct (fine-tuned CNN) — Seed 2 (seed 3 of 3)

Source: 3_Seed_exp.ipynb, cell 4. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 4, seed 2622715460* (same numbers for that seed).

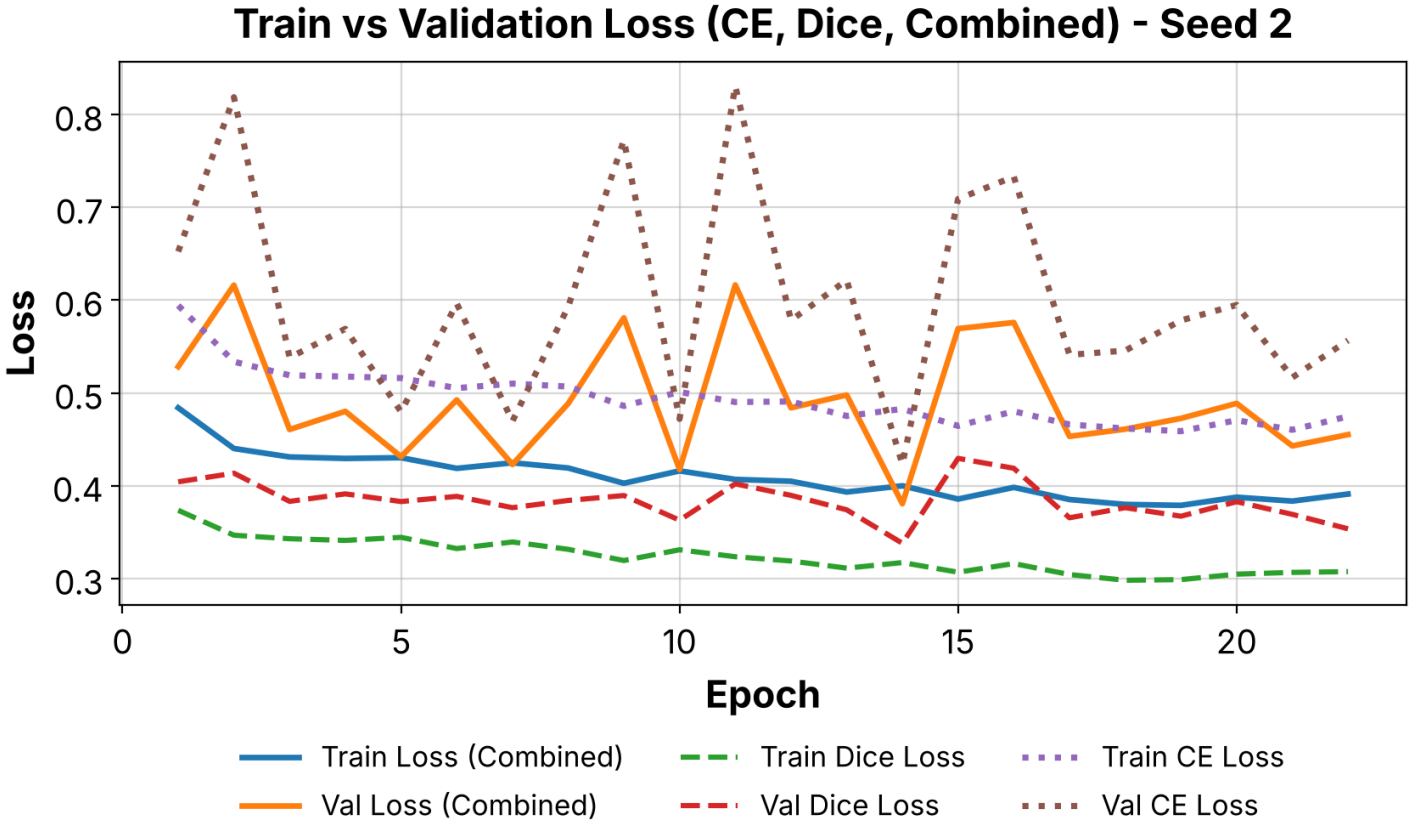
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	92.44%	85.83%	**88.62%**	88.96 ± 3.32 %
Precision (weighted)	92.57%	85.76%	**89.07%**	89.13 ± 3.41 %
Recall (weighted)	92.44%	85.83%	**88.62%**	88.96 ± 3.32 %
F1-score (weighted)	92.20%	85.02%	**87.94%**	88.39 ± 3.61 %
Test Dice loss	0.5267	0.5283	0.5245	0.5265 ± 0.0019
Best validation acc. (CNN)	72.08%	71.25%	77.50%	73.61 ± 3.39 %

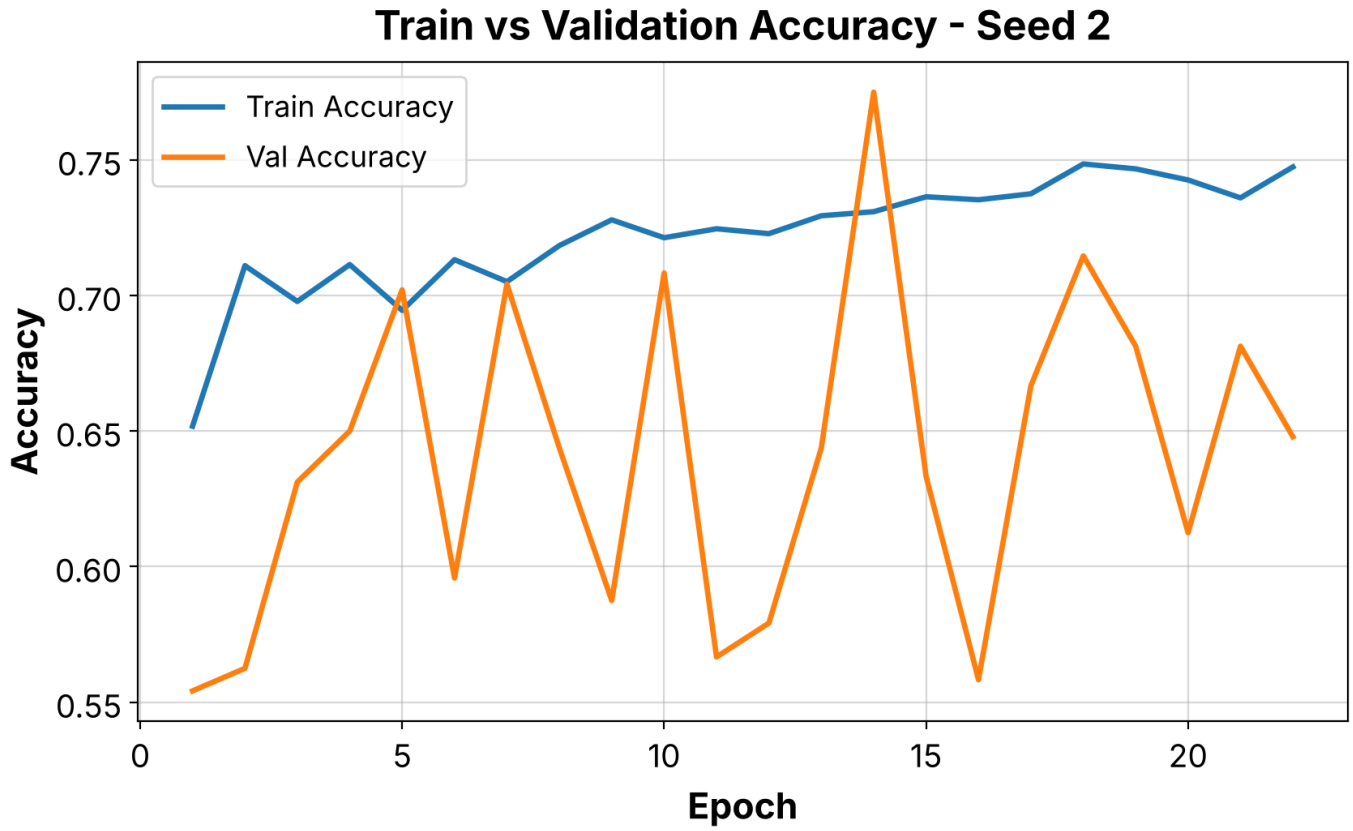
Confusion Matrix



8. MobileNetV2 — Direct (fine-tuned CNN), Seed 2 — graphs

Training / Validation Graphs





9. EfficientNet-B3 + SVM

Source notebook: efficinetnet b3 baseline and ml.ipynb (inside Models_DL.zip)

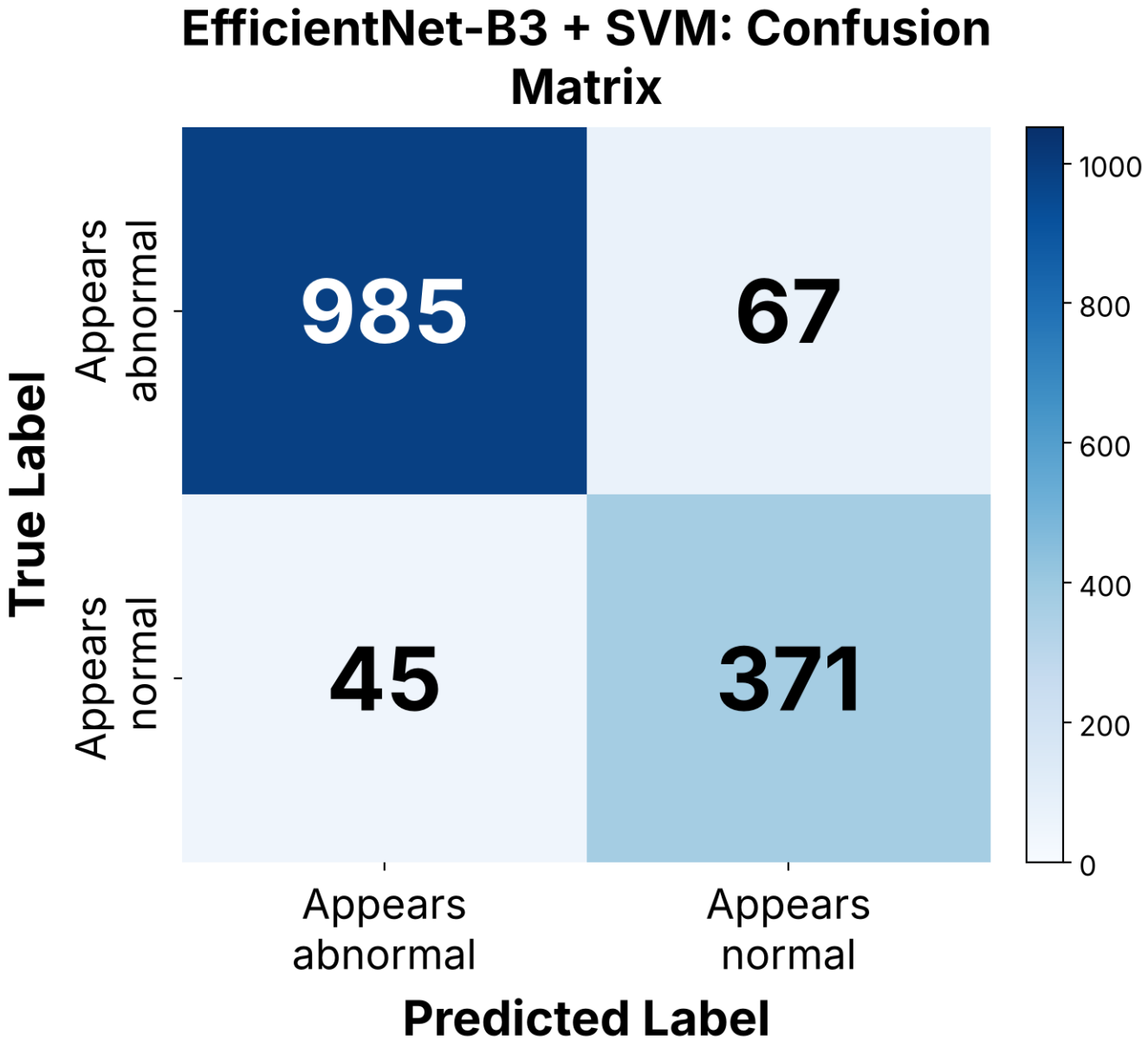
Accuracy: 92.37%

Precision: 92.53%

Recall: 92.37%

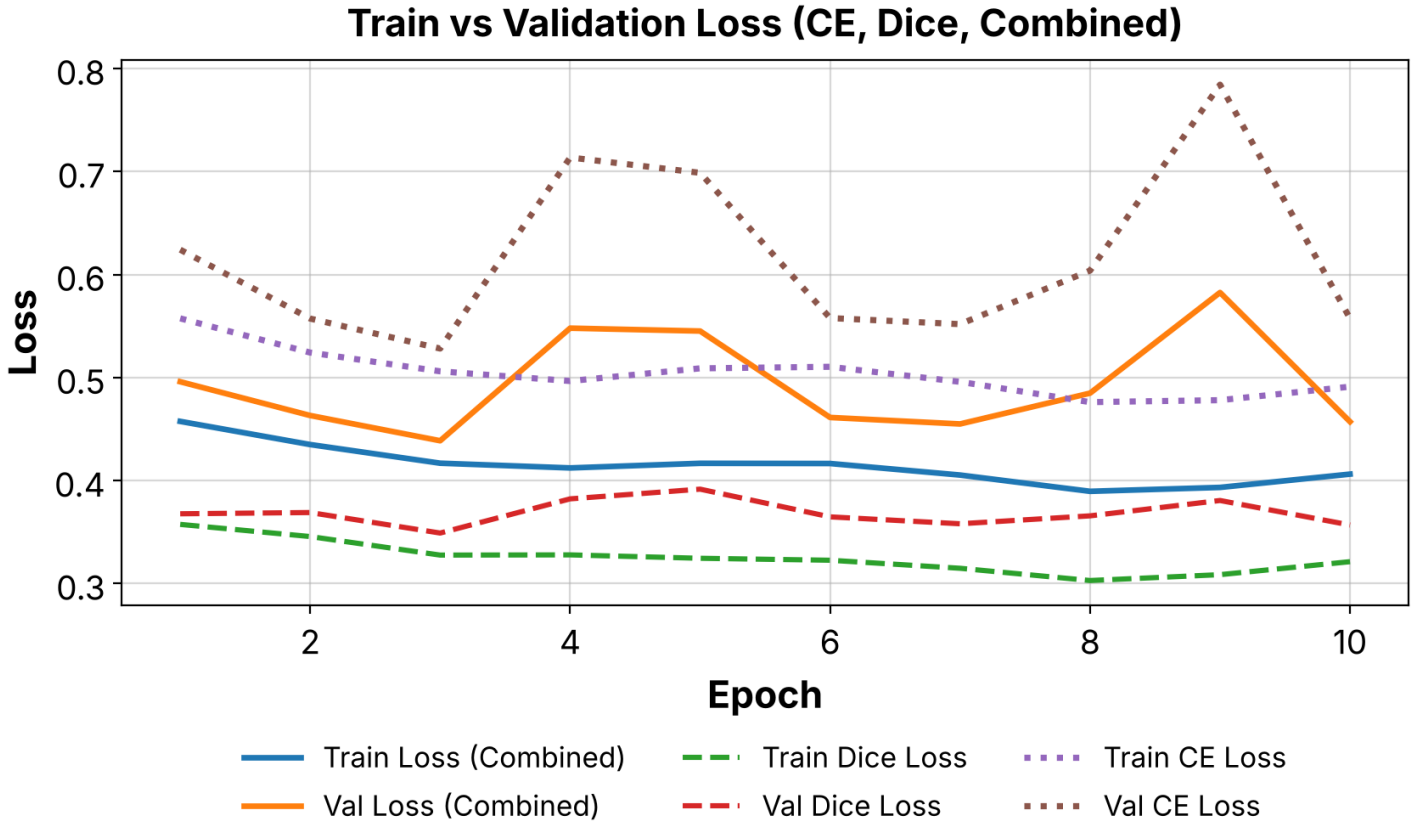
F1-score: 92.43%

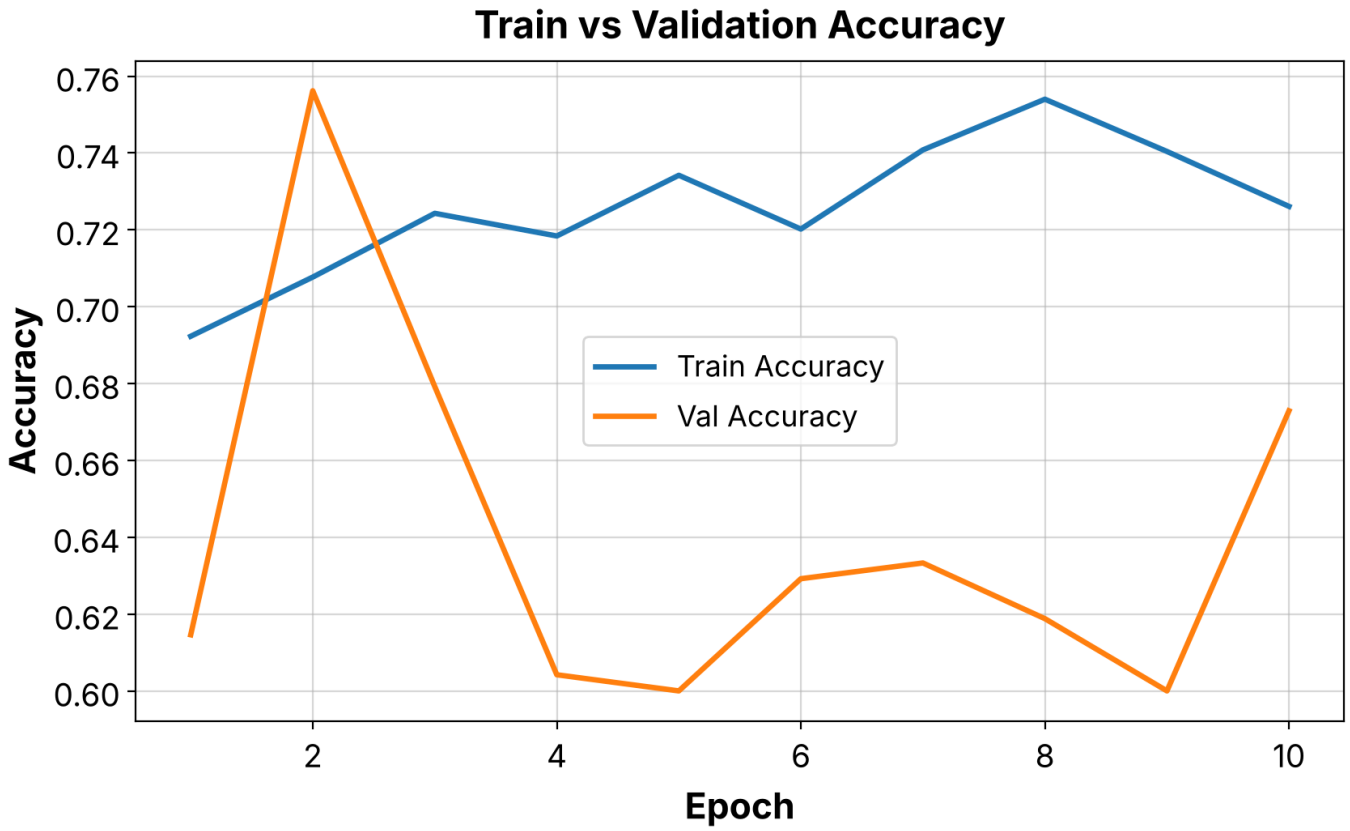
Confusion Matrix



9. EfficientNet-B3 + SVM — graphs

Training / Validation Graphs





10. EfficientNet-B3 + Random Forest

Source notebook: efficinetnet b3 baseline and ml.ipynb (inside Models_DL.zip)

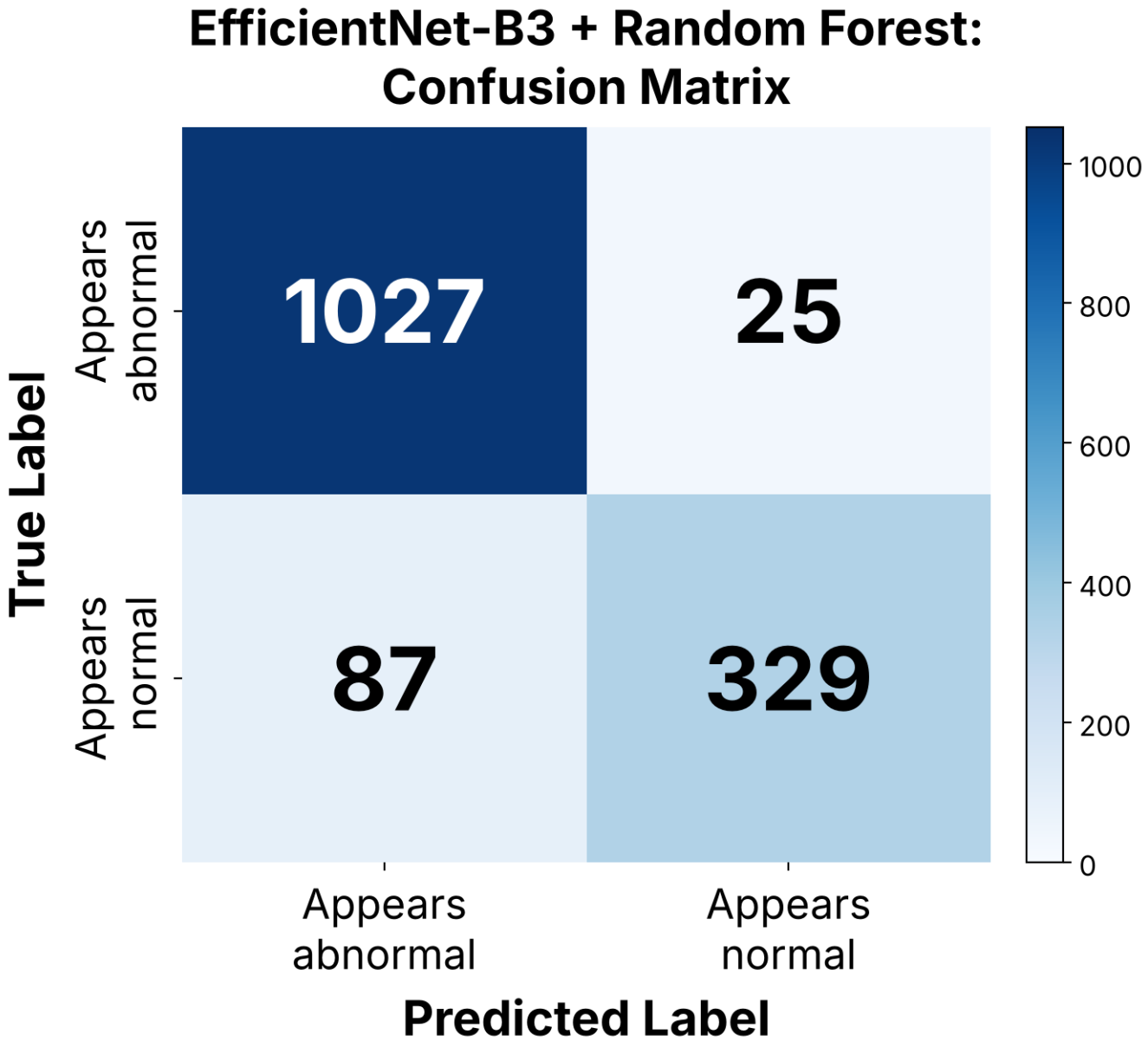
Accuracy: 92.37%

Precision: 92.40%

Recall: 92.37%

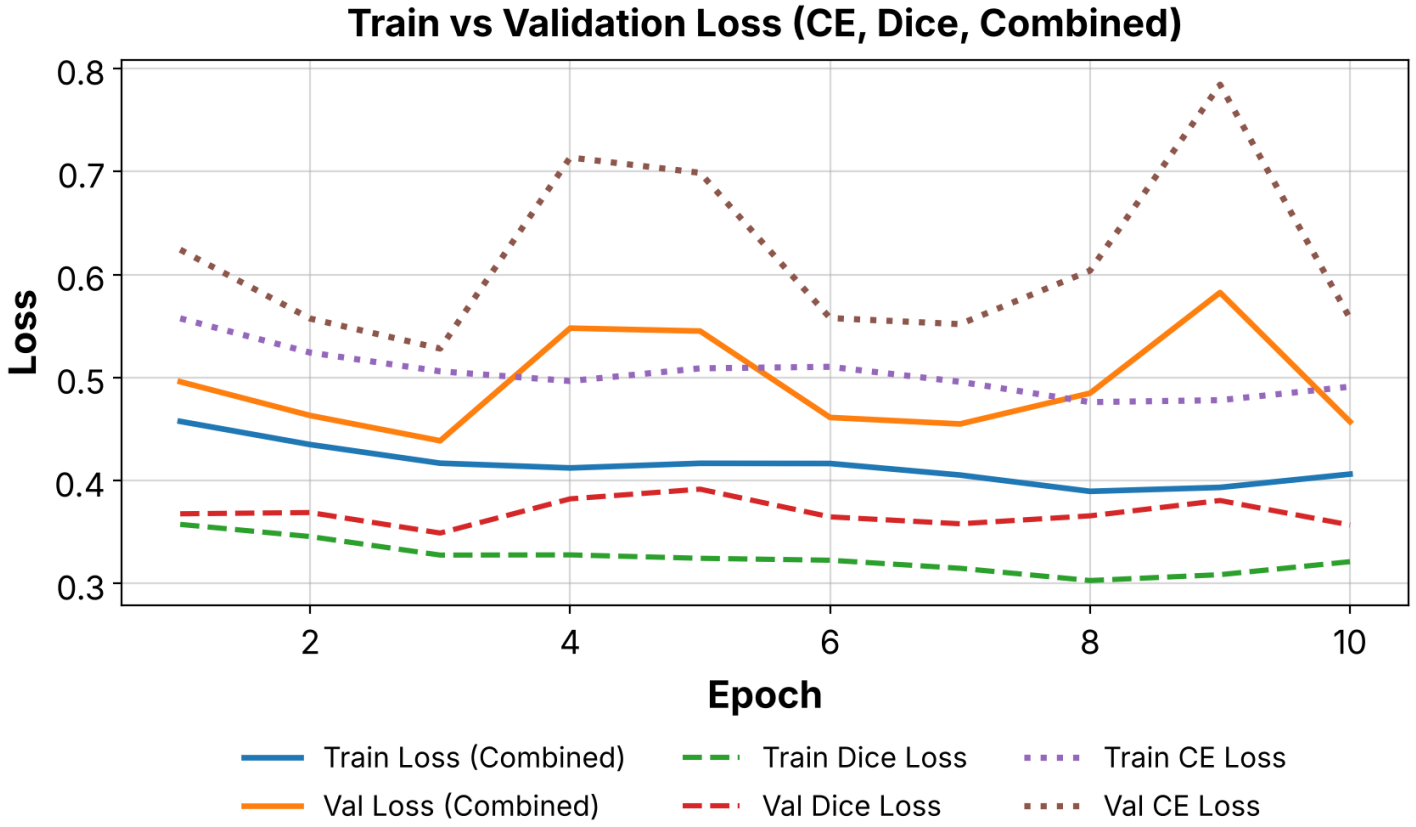
F1-score: 92.17%

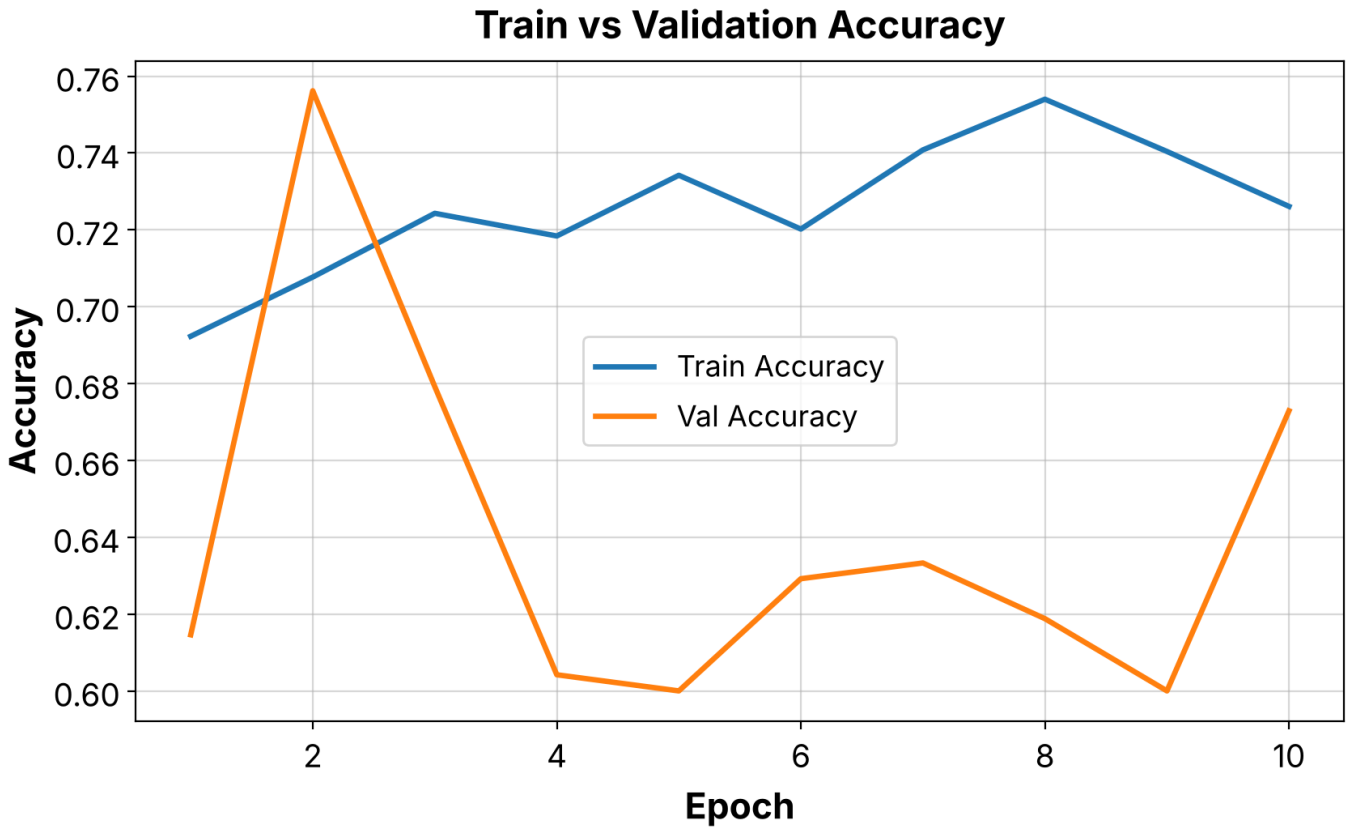
Confusion Matrix



10. EfficientNet-B3 + Random Forest — graphs

Training / Validation Graphs





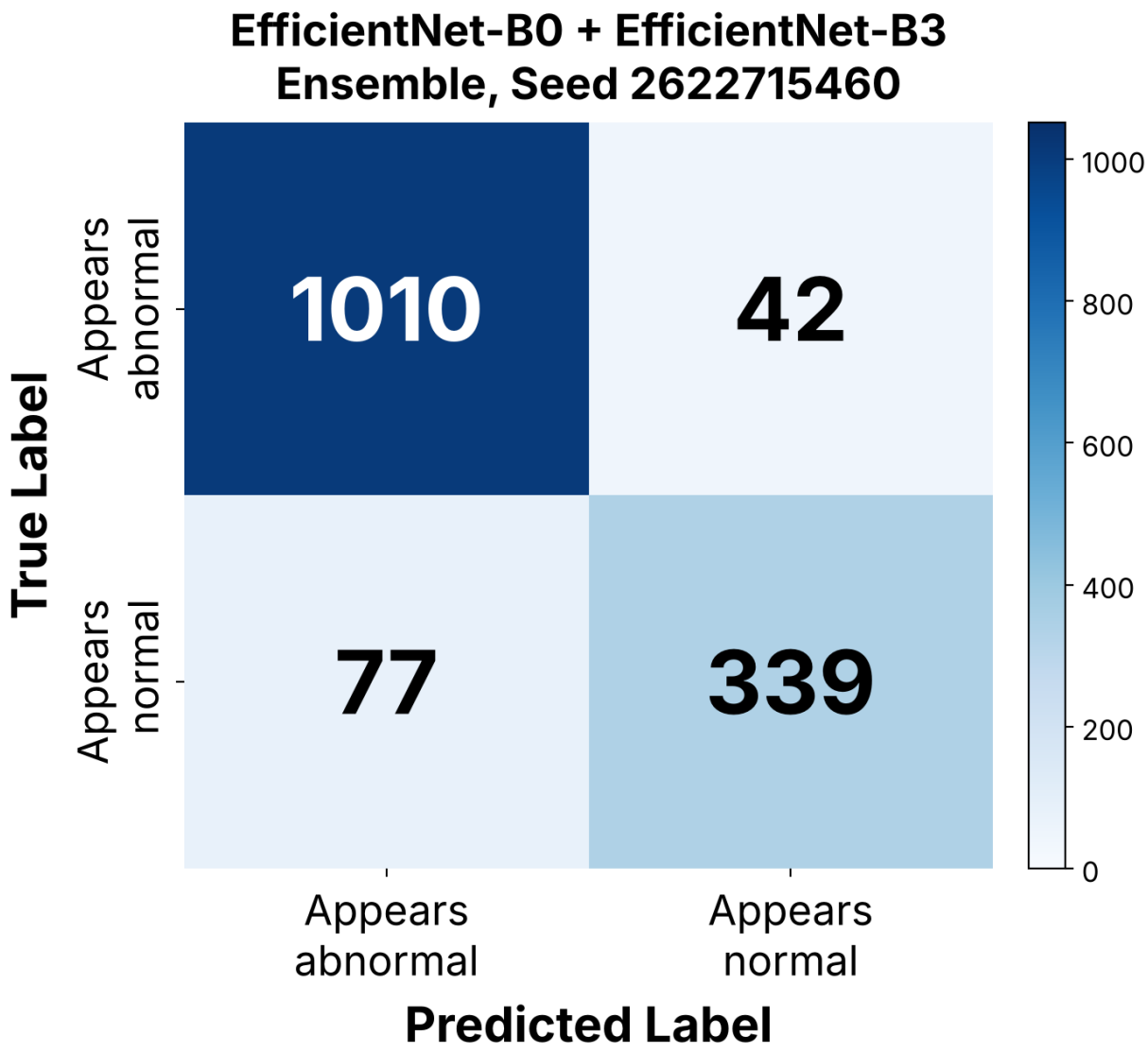
11. Ensemble — EfficientNet-B0 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 2622715460 (seed 1 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**91.89%**	92.30%	91.08%	91.76 ± 0.62 %
Precision (weighted)	**91.80%**	92.27%	90.96%	91.67 ± 0.66 %
Recall (weighted)	**91.89%**	92.30%	91.08%	91.76 ± 0.62 %
F1-score (weighted)	**91.78%**	92.14%	90.94%	91.62 ± 0.62 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: EfficientNet-B0 = row 6, EfficientNet-B3 = row 5.

Confusion Matrix



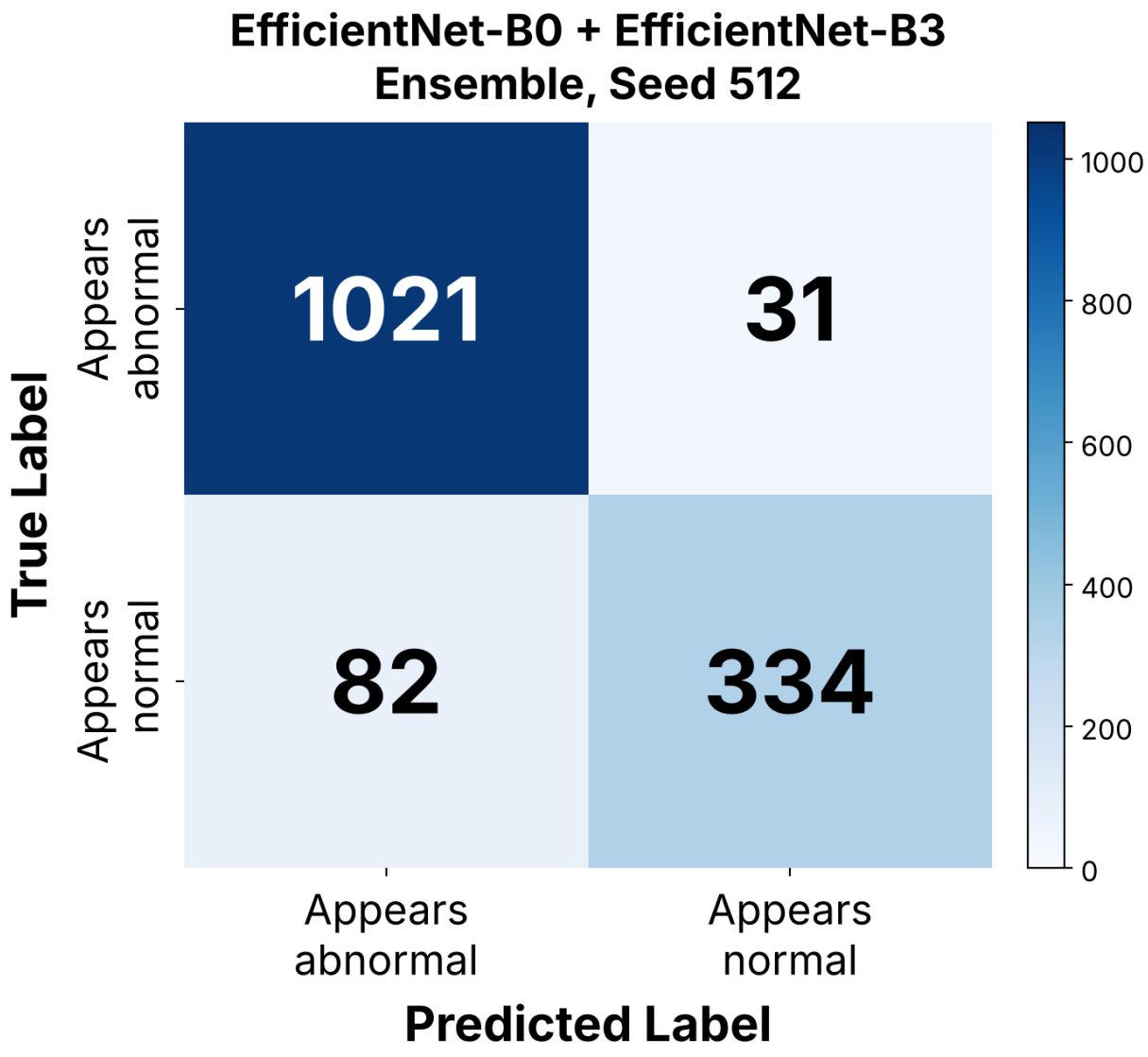
11. Ensemble — EfficientNet-B0 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 512 (seed 2 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	91.89%	**92.30%**	91.08%	91.76 ± 0.62 %
Precision (weighted)	91.80%	**92.27%**	90.96%	91.67 ± 0.66 %
Recall (weighted)	91.89%	**92.30%**	91.08%	91.76 ± 0.62 %
F1-score (weighted)	91.78%	**92.14%**	90.94%	91.62 ± 0.62 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: EfficientNet-B0 = row 6, EfficientNet-B3 = row 5.

Confusion Matrix



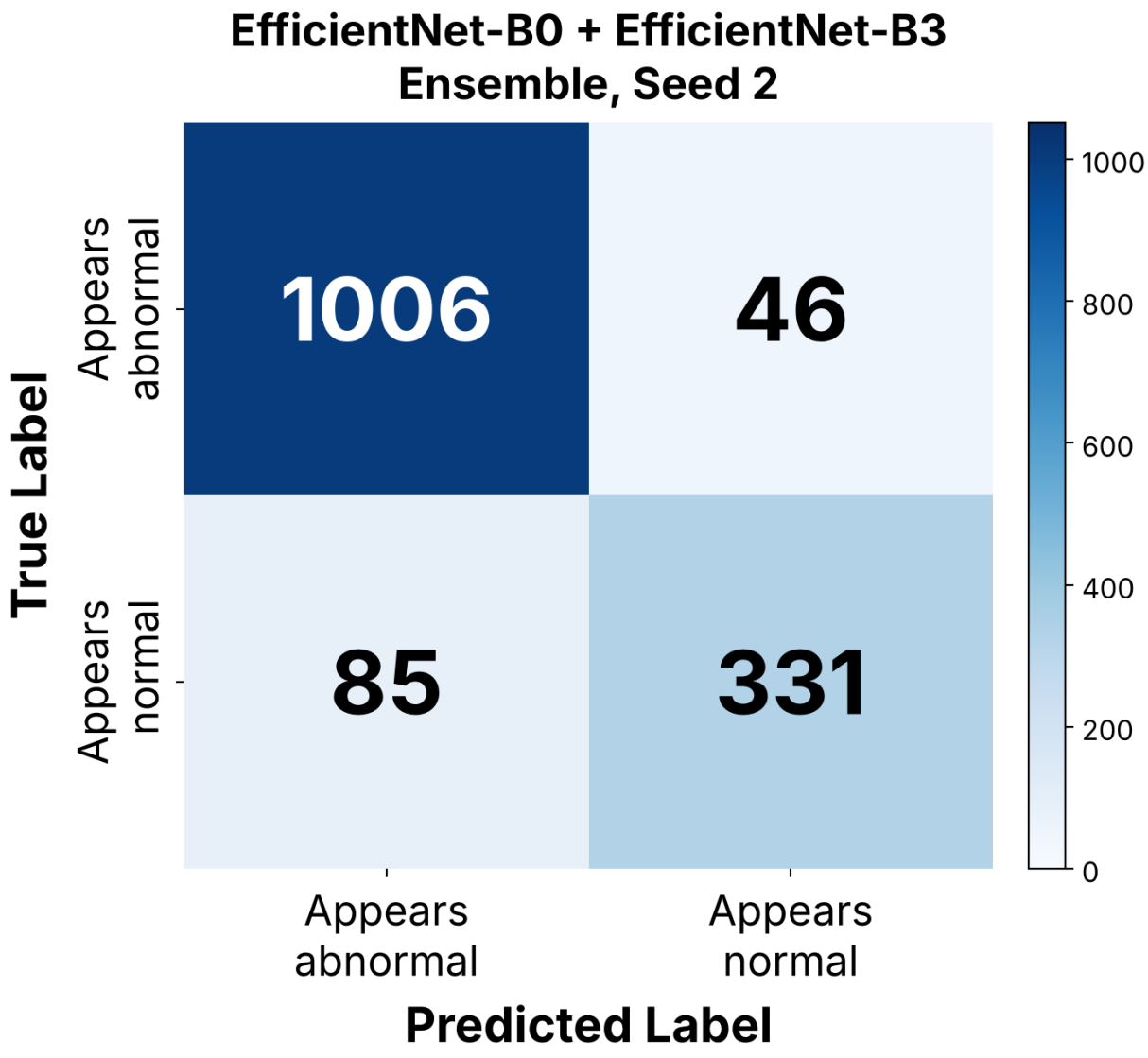
11. Ensemble — EfficientNet-B0 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 2 (seed 3 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	91.89%	92.30%	**91.08%**	91.76 ± 0.62 %
Precision (weighted)	91.80%	92.27%	**90.96%**	91.67 ± 0.66 %
Recall (weighted)	91.89%	92.30%	**91.08%**	91.76 ± 0.62 %
F1-score (weighted)	91.78%	92.14%	**90.94%**	91.62 ± 0.62 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: EfficientNet-B0 = row 6, EfficientNet-B3 = row 5.

Confusion Matrix



12. EfficientNet-B3 + XGBoost

Source notebook: efficinetnet b3 baseline and ml.ipynb (inside Models_DL.zip)

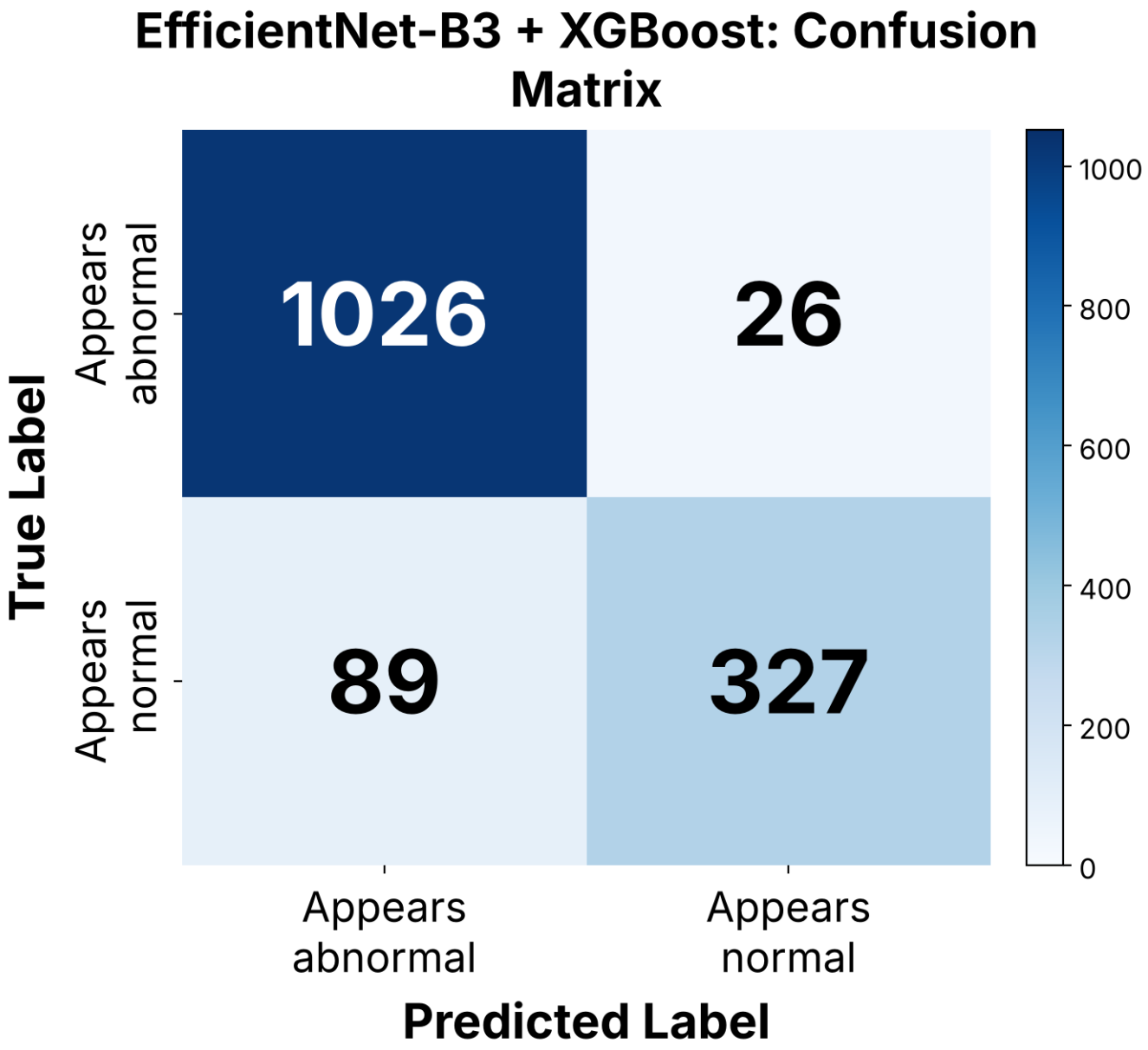
Accuracy: 92.17%

Precision: 92.19%

Recall: 92.17%

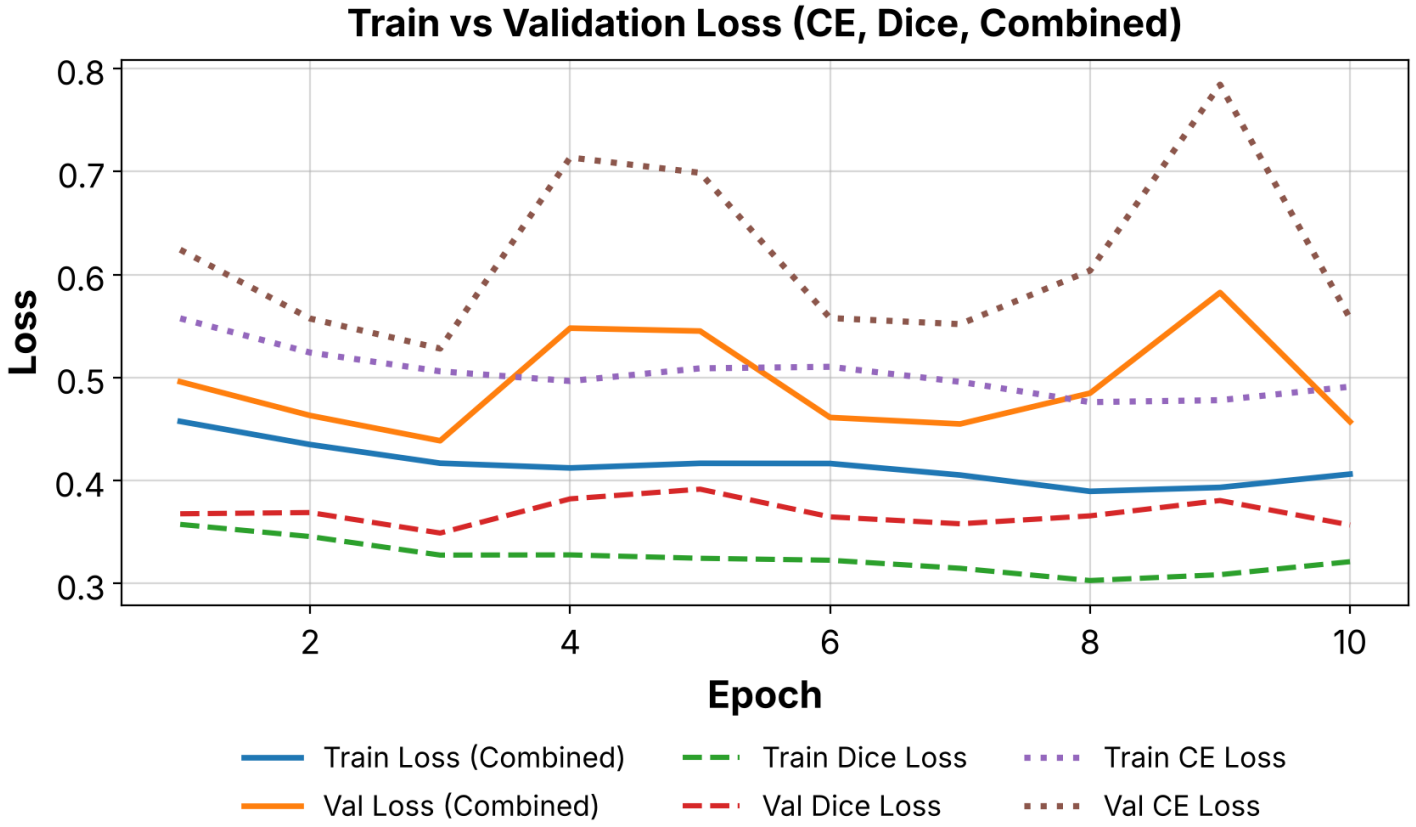
F1-score: 91.96%

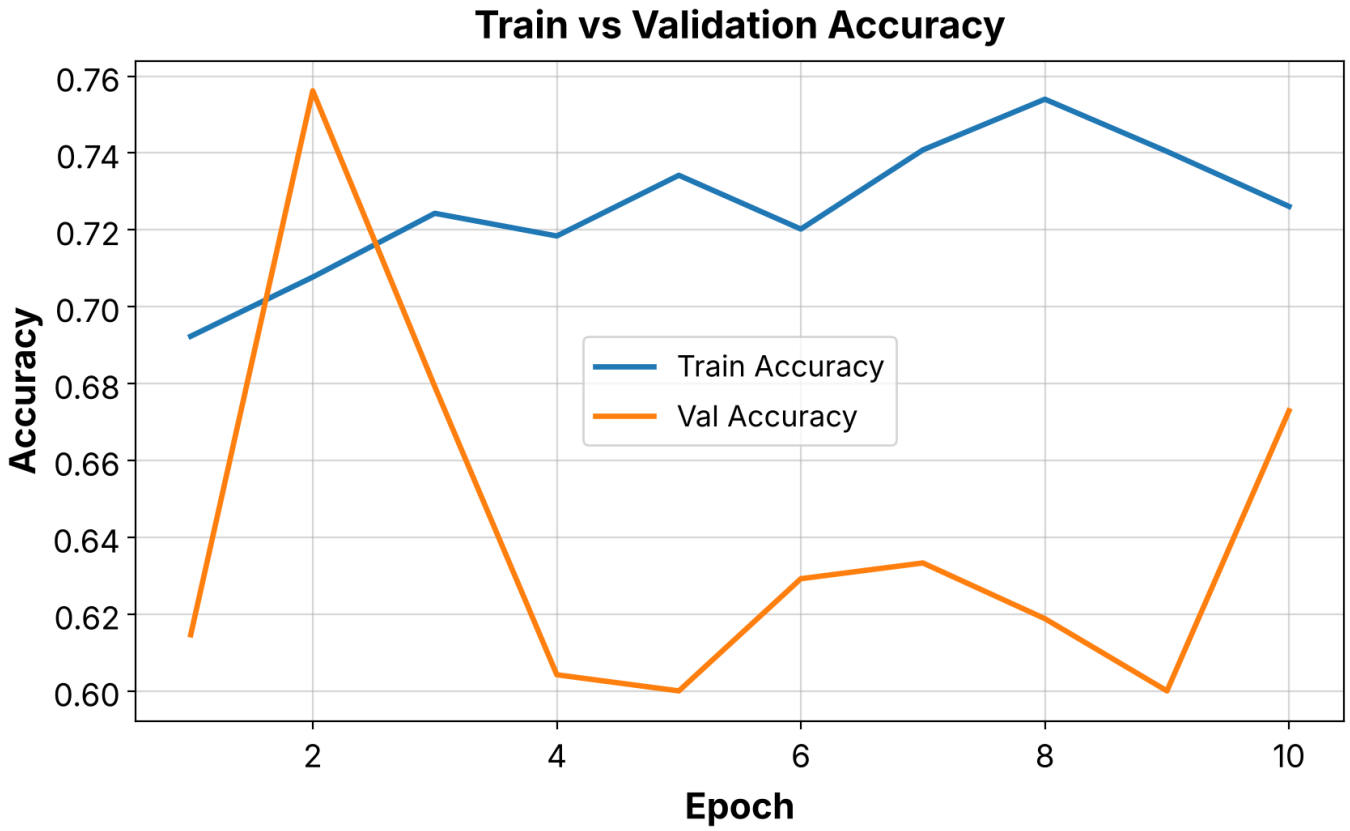
Confusion Matrix



12. EfficientNet-B3 + XGBoost — graphs

Training / Validation Graphs





13. InceptionV3 — Direct (fine-tuned CNN)

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

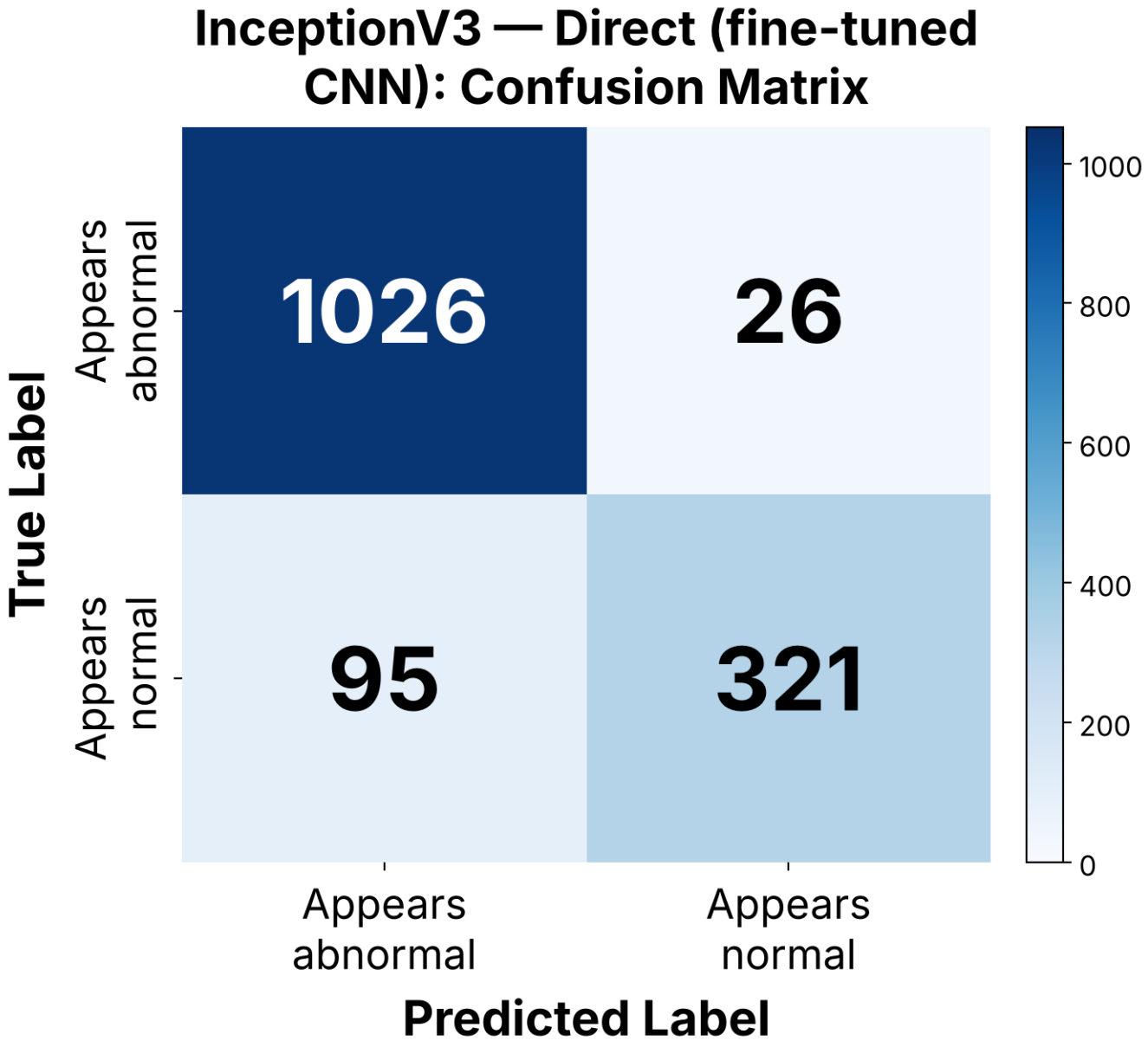
Accuracy: 91.76%

Precision: 91.80%

Recall: 91.76%

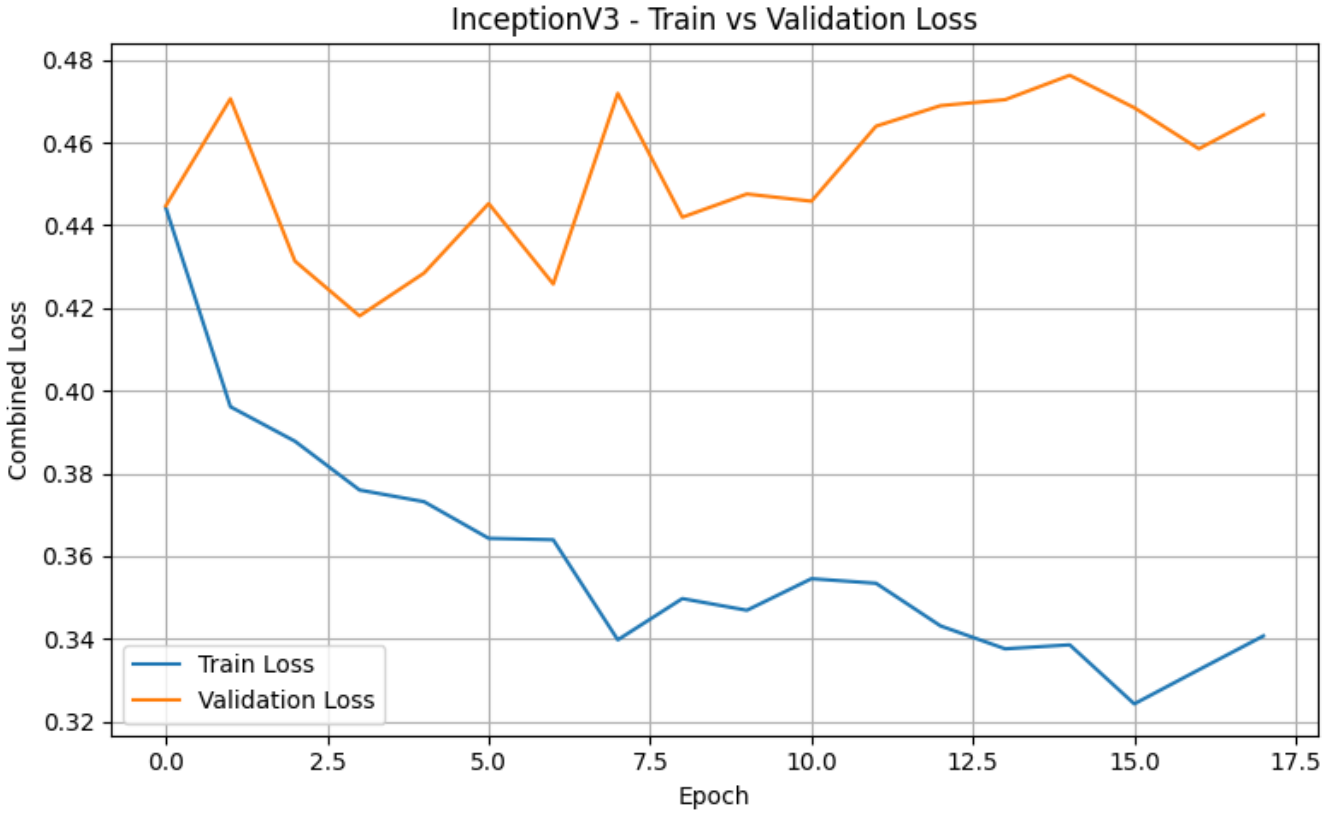
F1-score: 91.52%

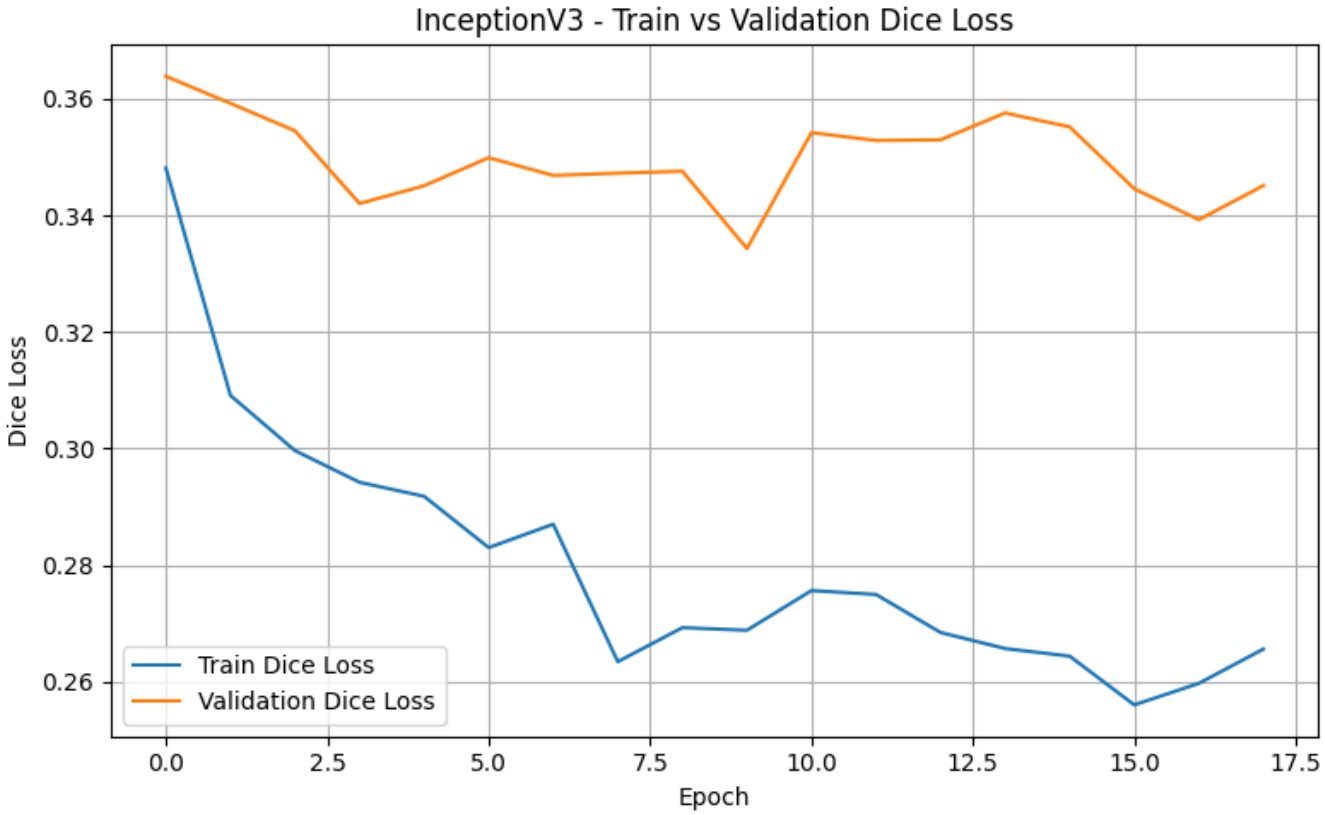
Confusion Matrix



13. InceptionV3 — Direct (fine-tuned CNN) — graphs (1/2)

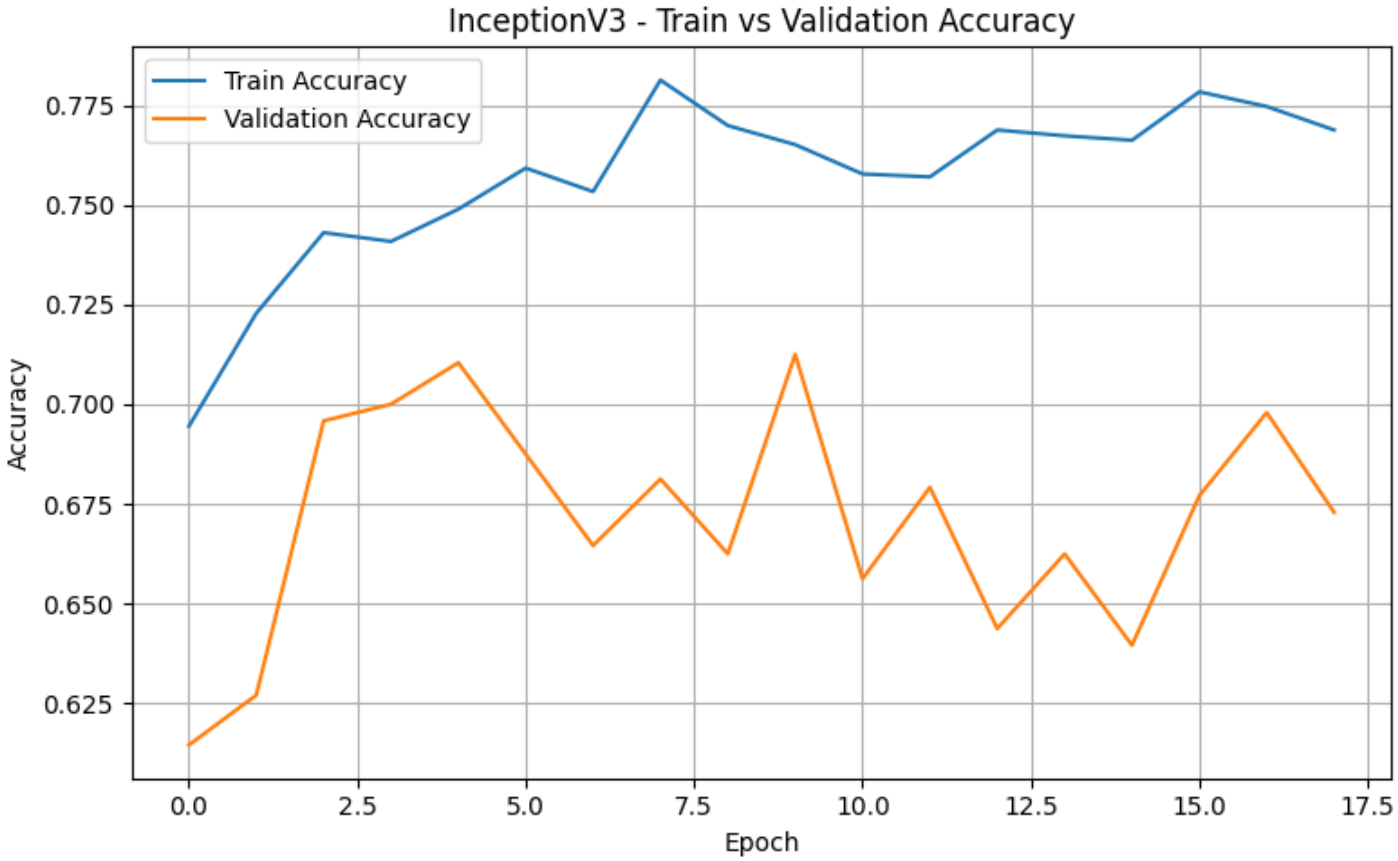
Training / Validation Graphs





13. InceptionV3 — Direct (fine-tuned CNN) — graphs (2/2)

Training / Validation Graphs



14. MobileNetV2 + Gaussian Naive Bayes

Source notebook: MobileNet with ML.ipynb (inside Models_DL.zip)

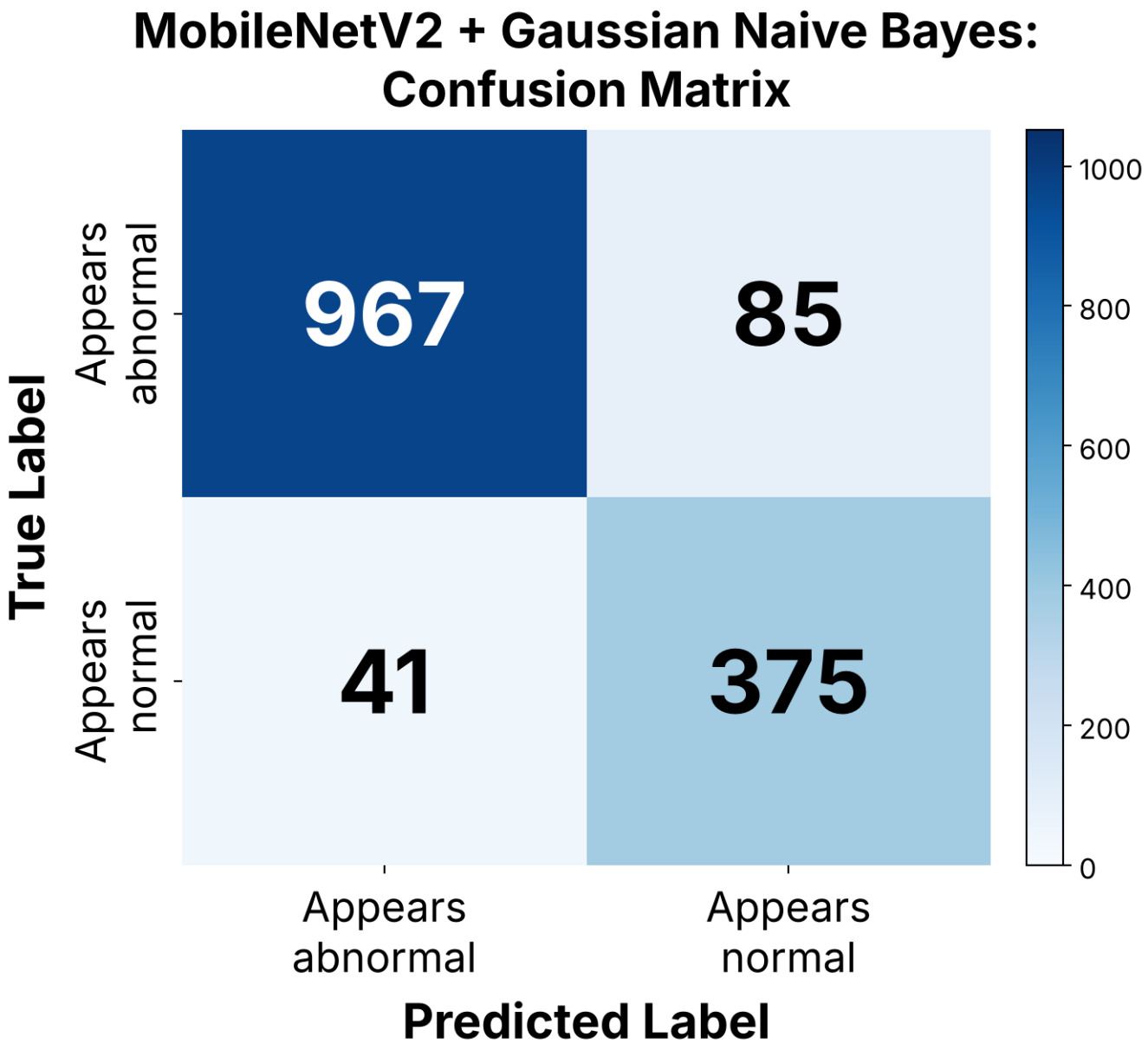
Accuracy: 91.42%

Precision: 91.85%

Recall: 91.42%

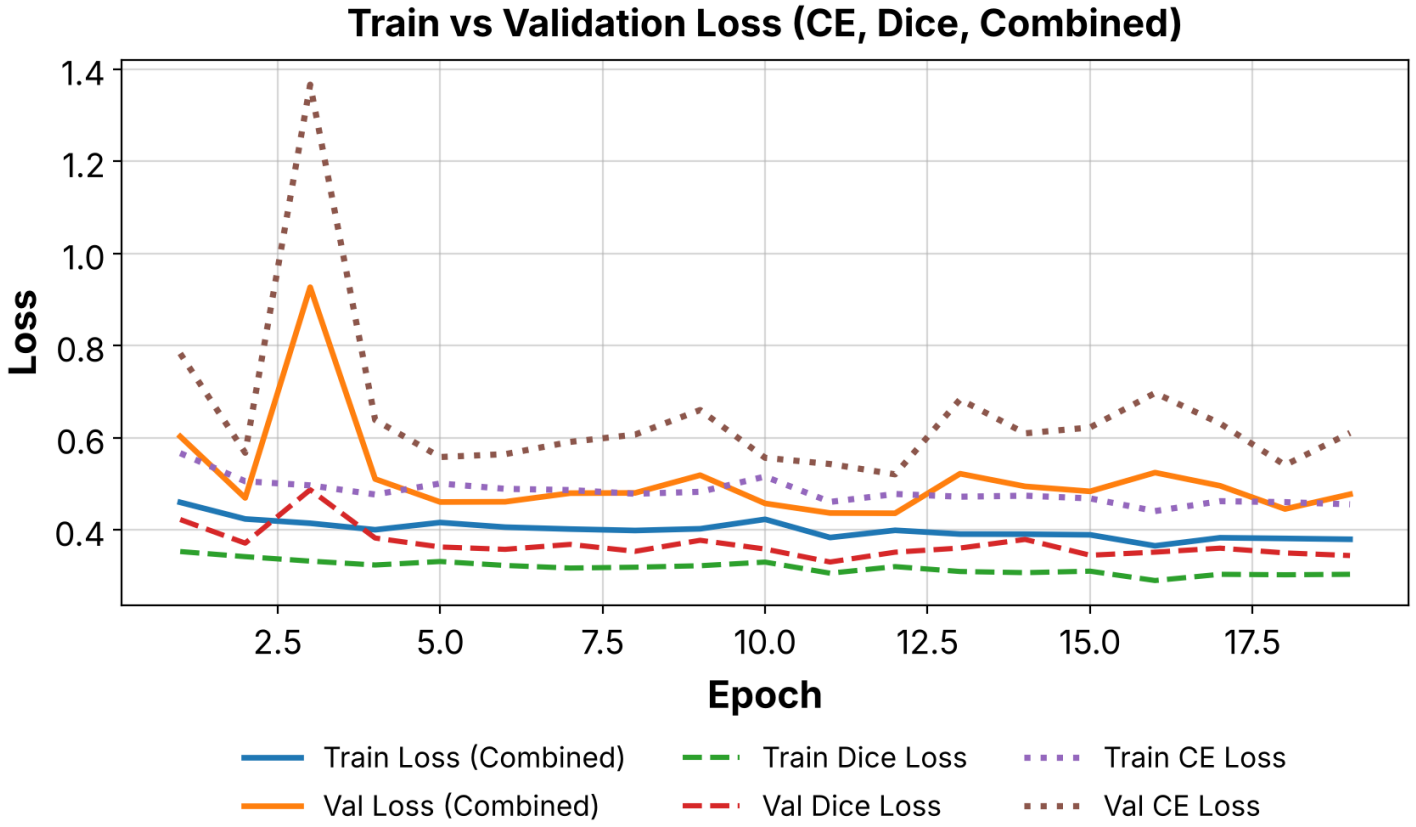
F1-score: 91.54%

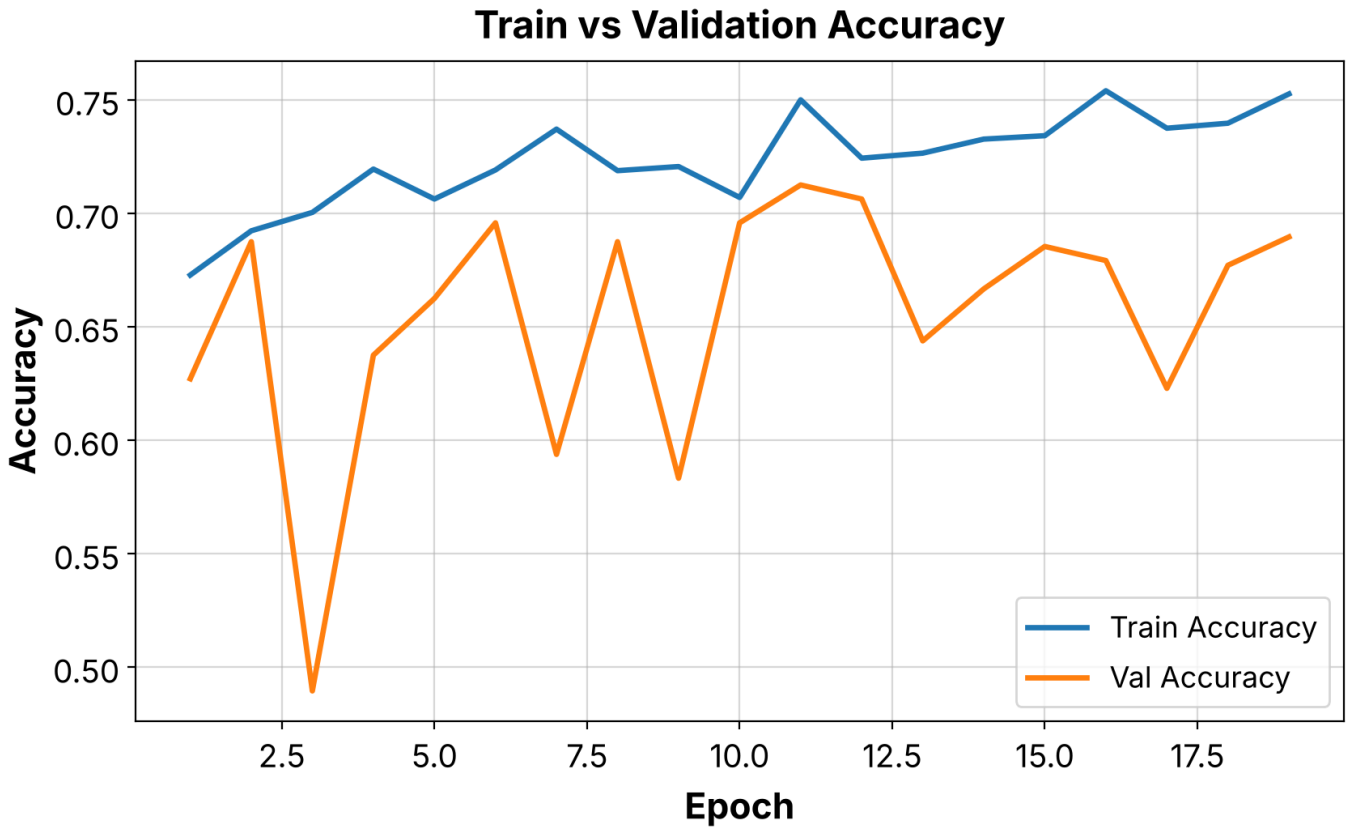
Confusion Matrix



14. MobileNetV2 + Gaussian Naive Bayes — graphs

Training / Validation Graphs





15. ResNet18 + Random Forest

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

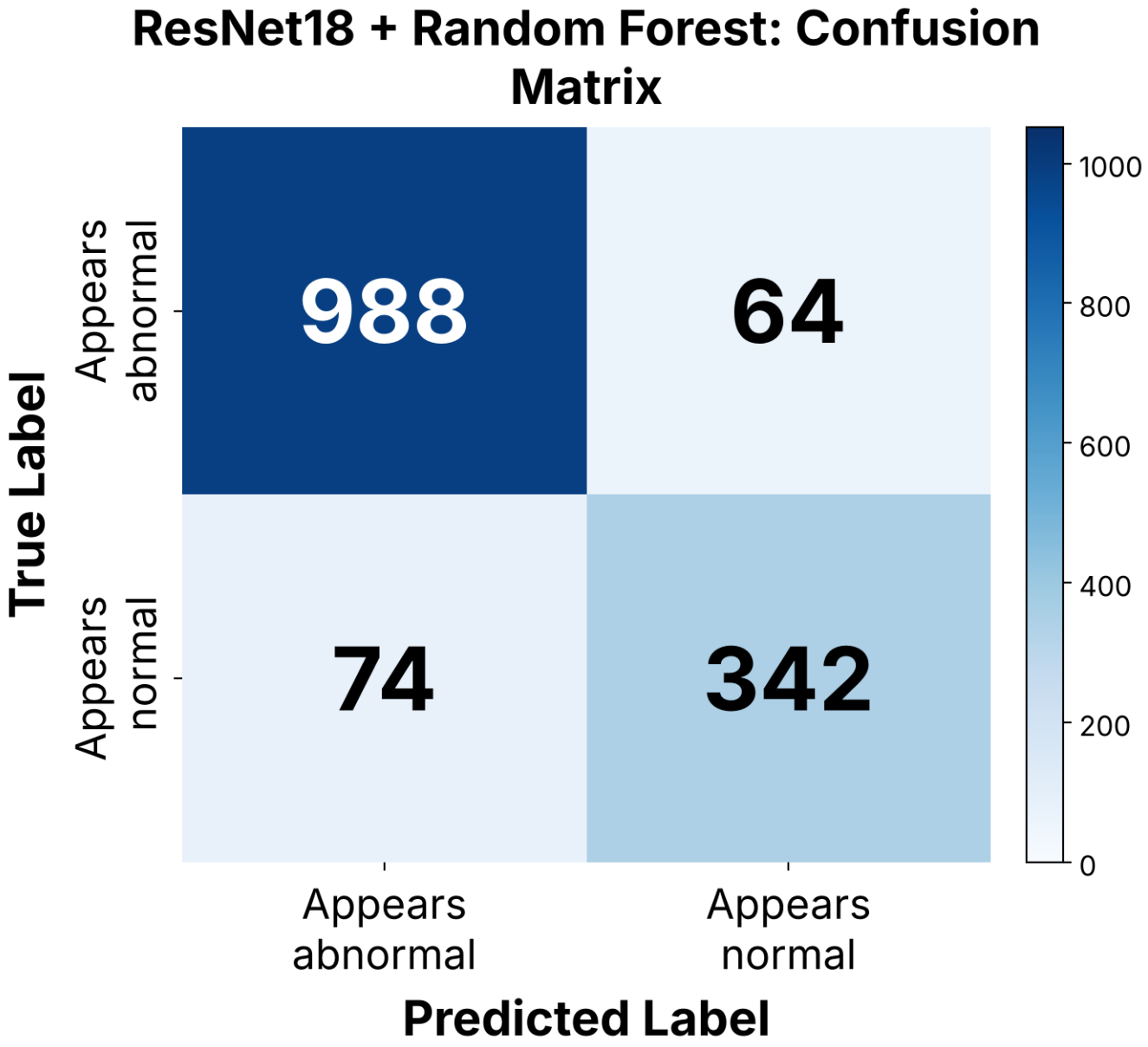
Accuracy: 90.60%

Precision: 90.54%

Recall: 90.60%

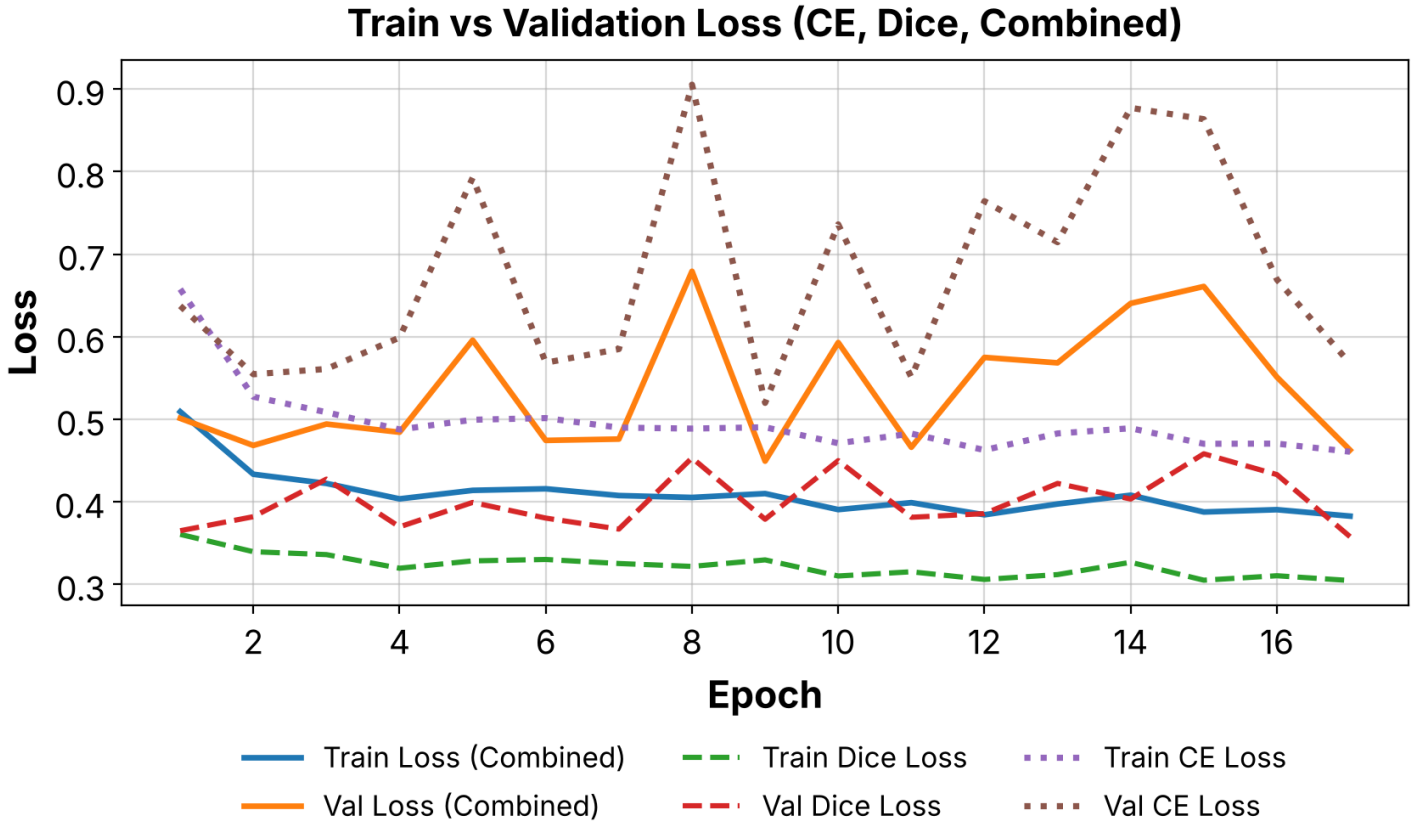
F1-score: 90.56%

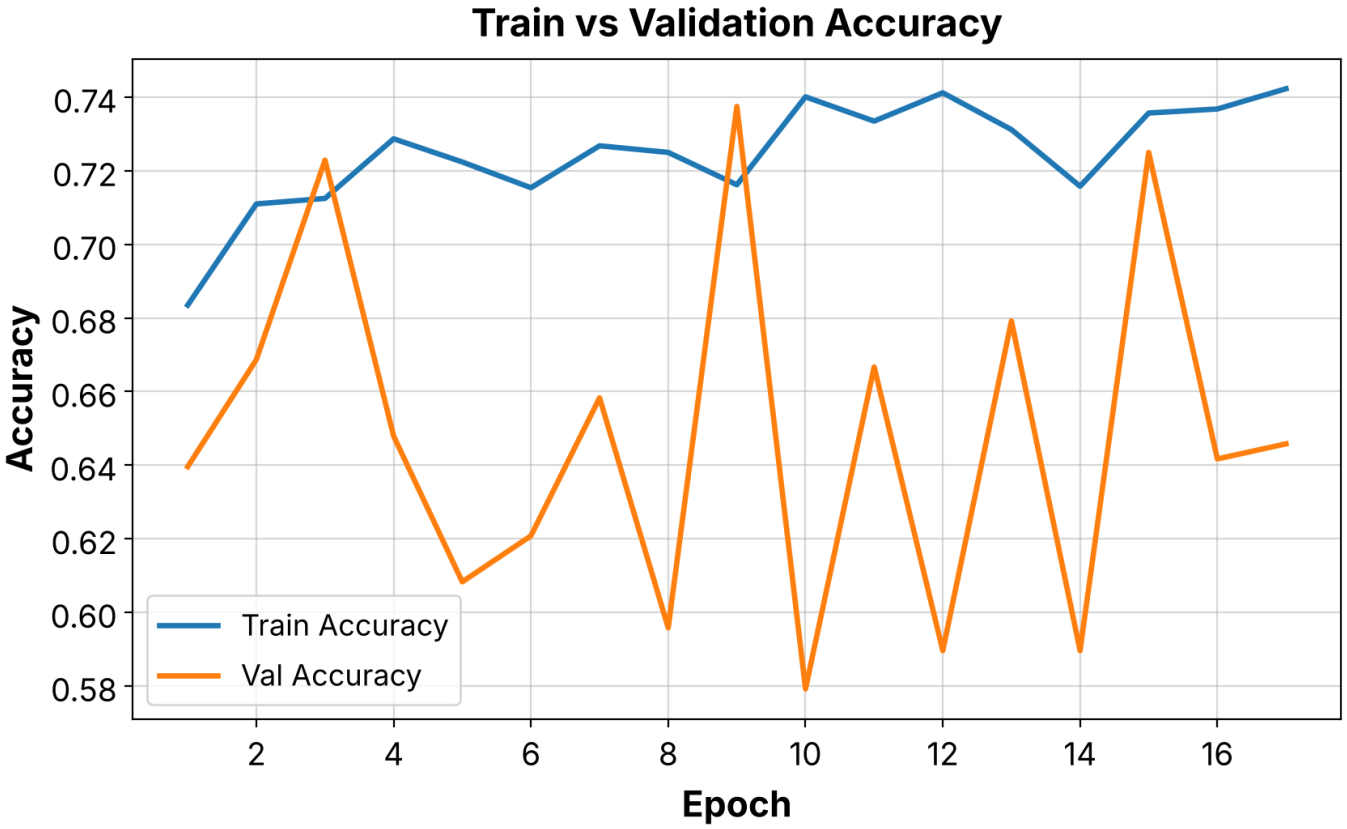
Confusion Matrix



15. ResNet18 + Random Forest — graphs

Training / Validation Graphs





16. MobileNetV2 + Random Forest

Source notebook: MobileNet with ML.ipynb (inside Models_DL.zip)

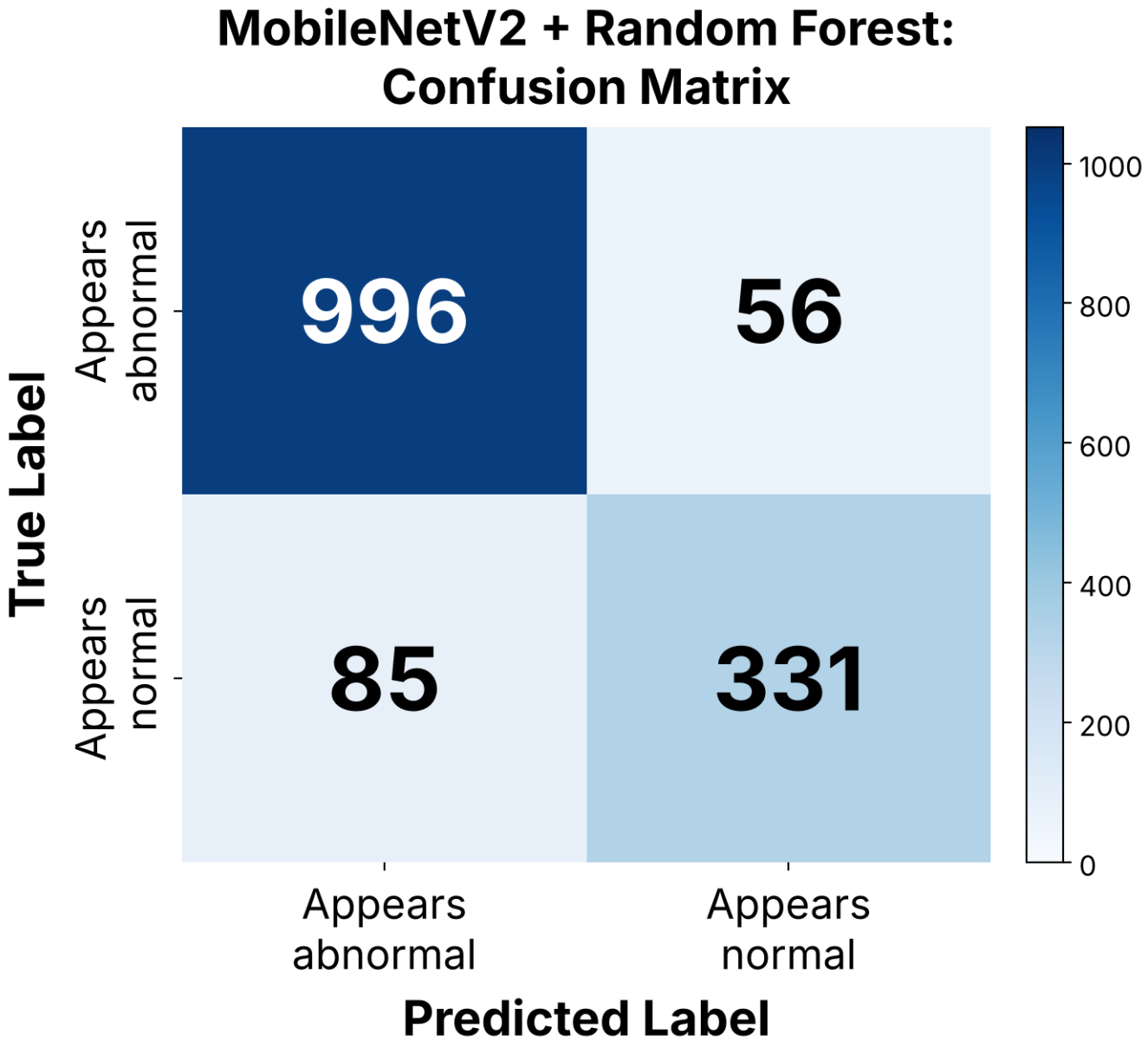
Accuracy: 90.40%

Precision: 90.26%

Recall: 90.40%

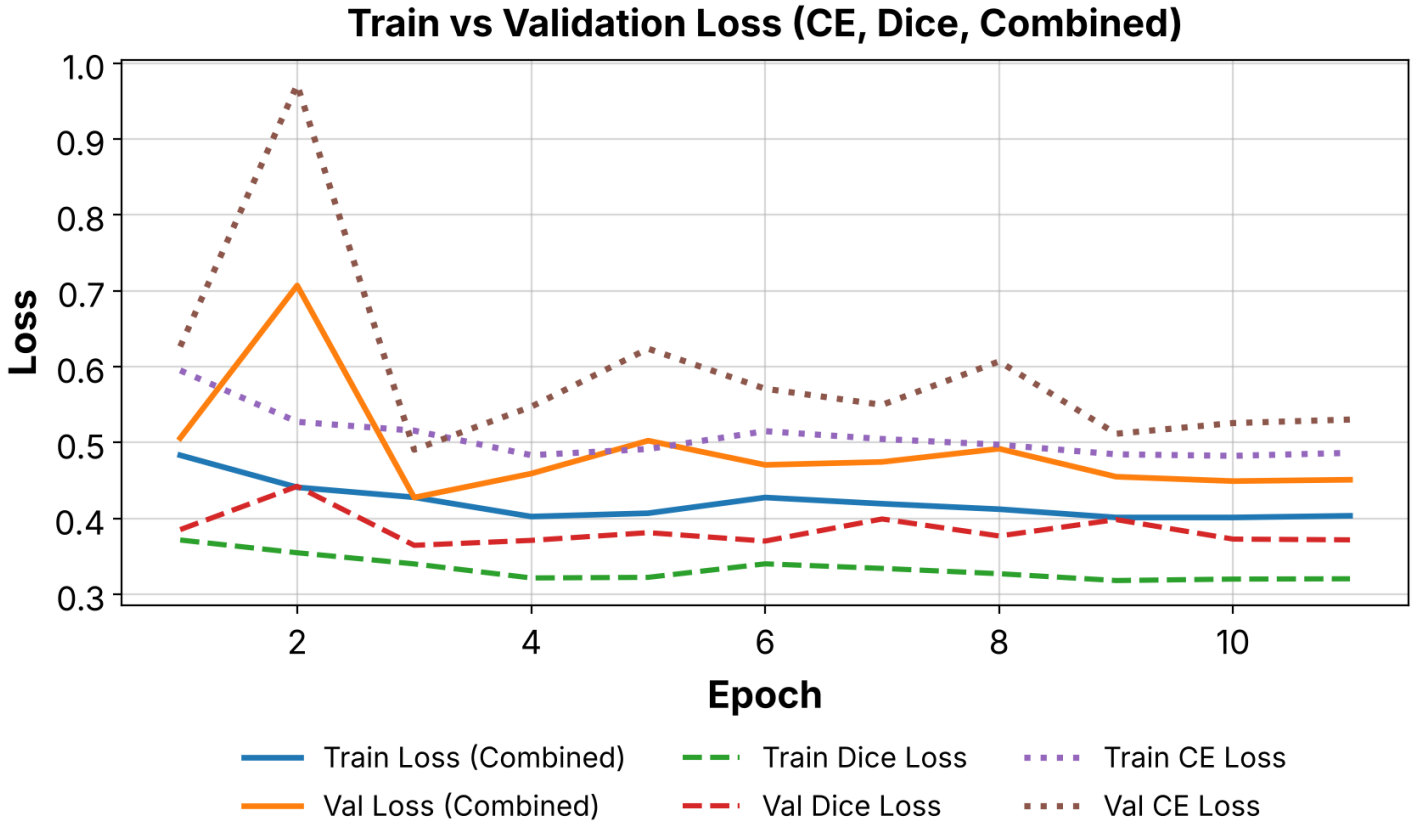
F1-score: 90.29%

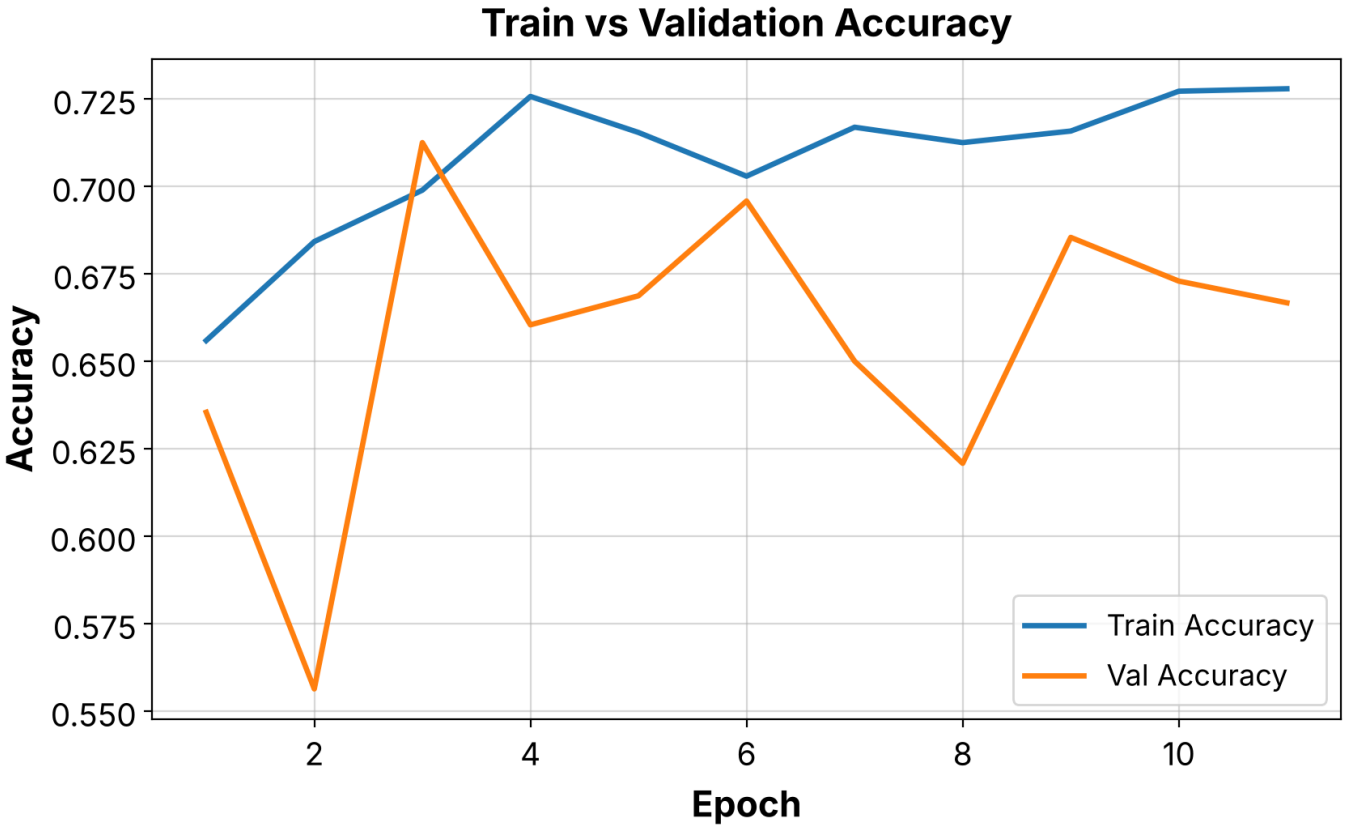
Confusion Matrix



16. MobileNetV2 + Random Forest — graphs

Training / Validation Graphs





17. VGG16 — Direct (fine-tuned CNN)

Source notebook: Vgg with Dice Loss (Baseline).ipynb (inside Models_DL.zip)

Accuracy: 90.33%

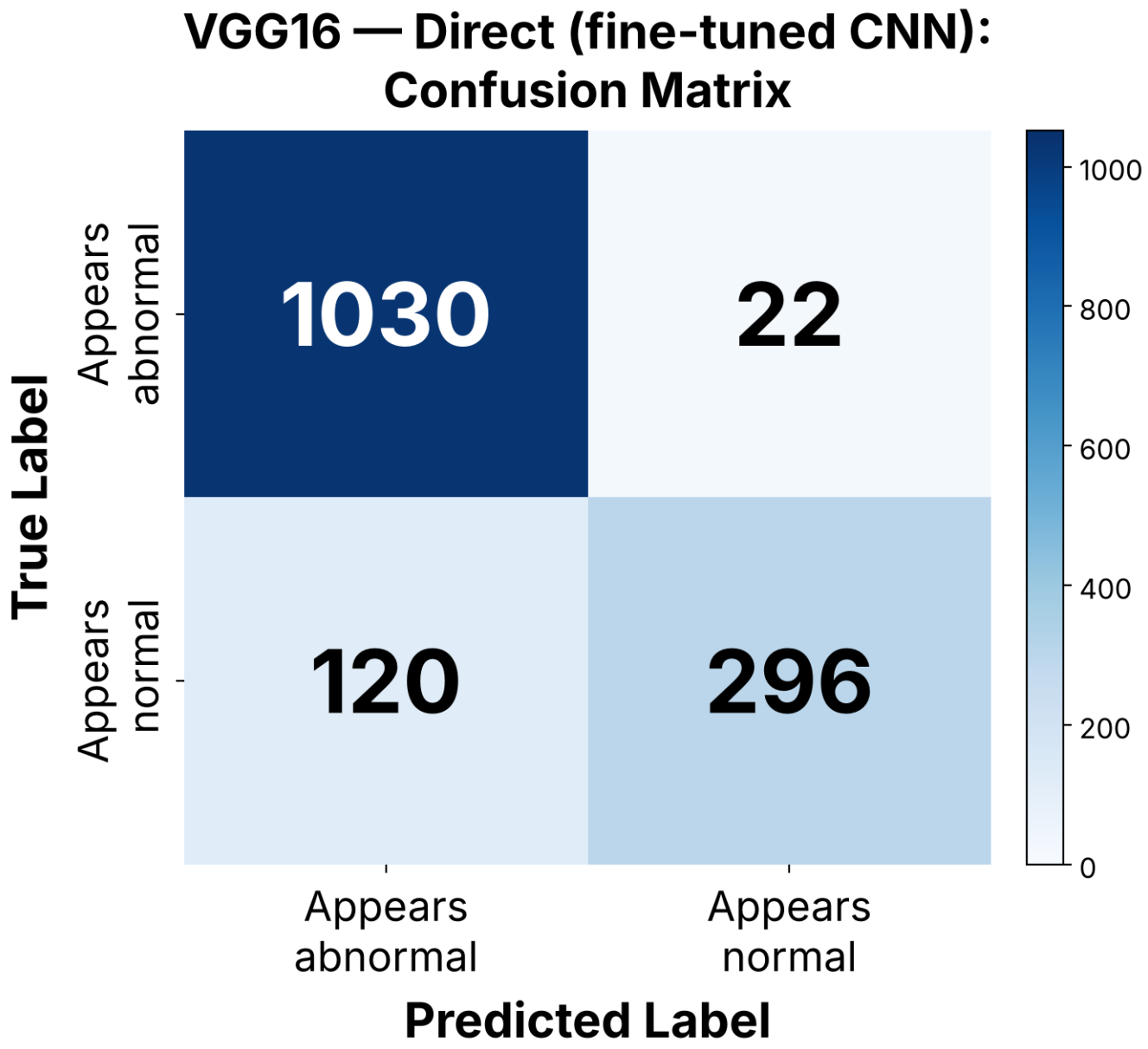
Precision: 90.56%

Recall: 90.33%

F1-score: 89.90%

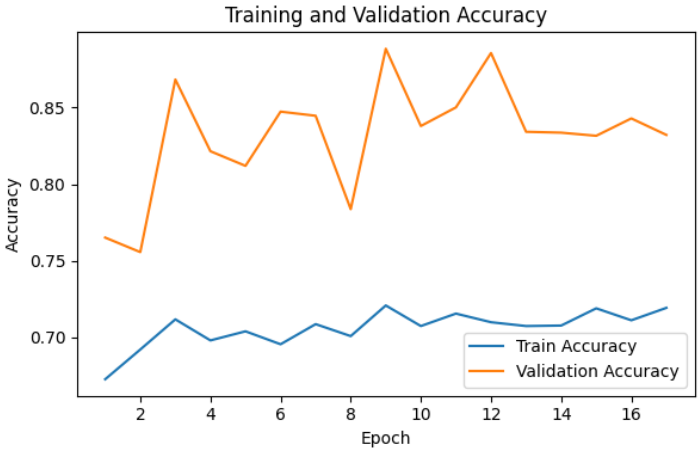
Note: the notebook prints 88.84% as its test / “Final Test Accuracy”, which does not agree with the notebook’s own printed confusion matrix and weighted recall (in the Dice-loss notebook this number is simply the best *validation* accuracy). The test accuracy shown here is the one implied by the confusion matrix and equals the weighted recall.

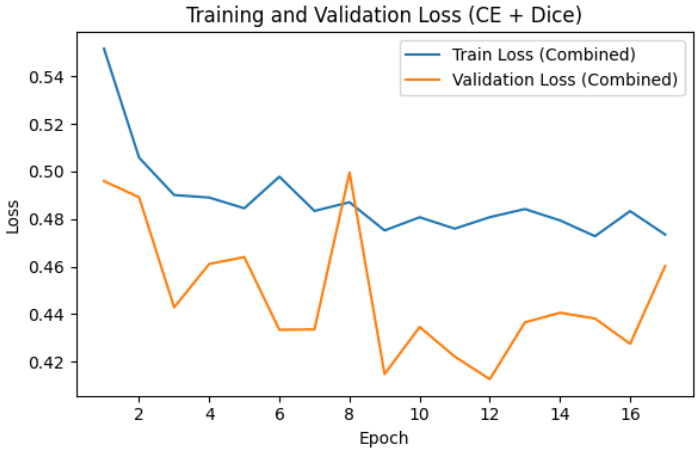
Confusion Matrix

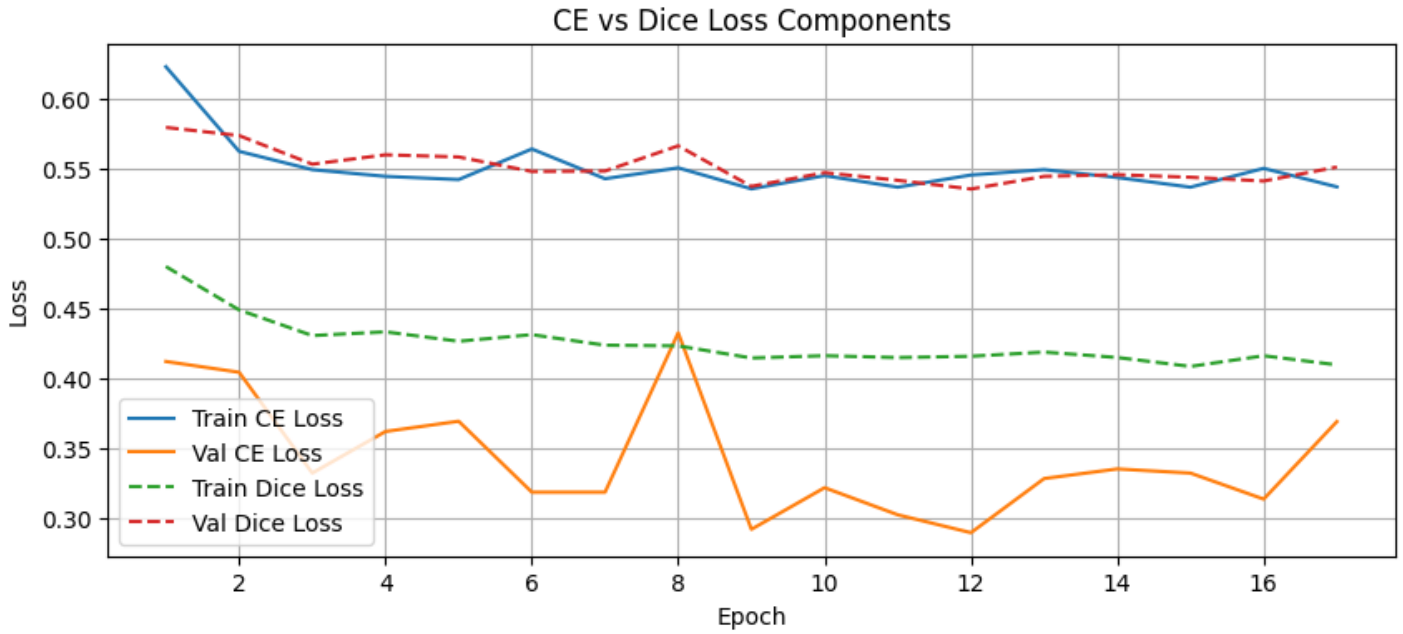


17. VGG16 — Direct (fine-tuned CNN) — graphs

Training / Validation Graphs







18. VGG16 + SVM

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

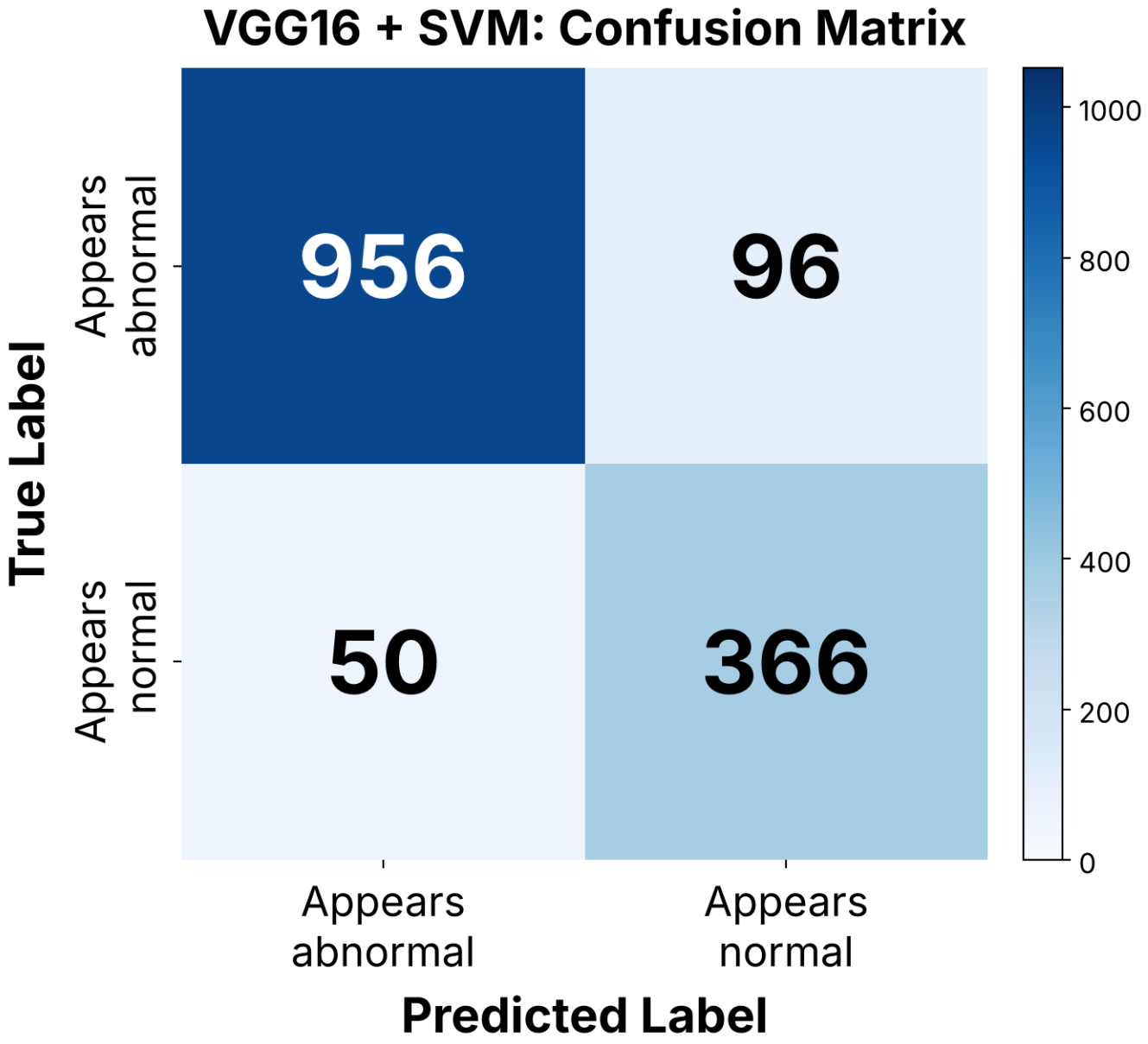
Accuracy: 90.05%

Precision: 90.55%

Recall: 90.05%

F1-score: 90.20%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

19. EfficientNet-B3 + KNN

Source notebook: efficinetnet b3 baseline and ml.ipynb (inside Models_DL.zip)

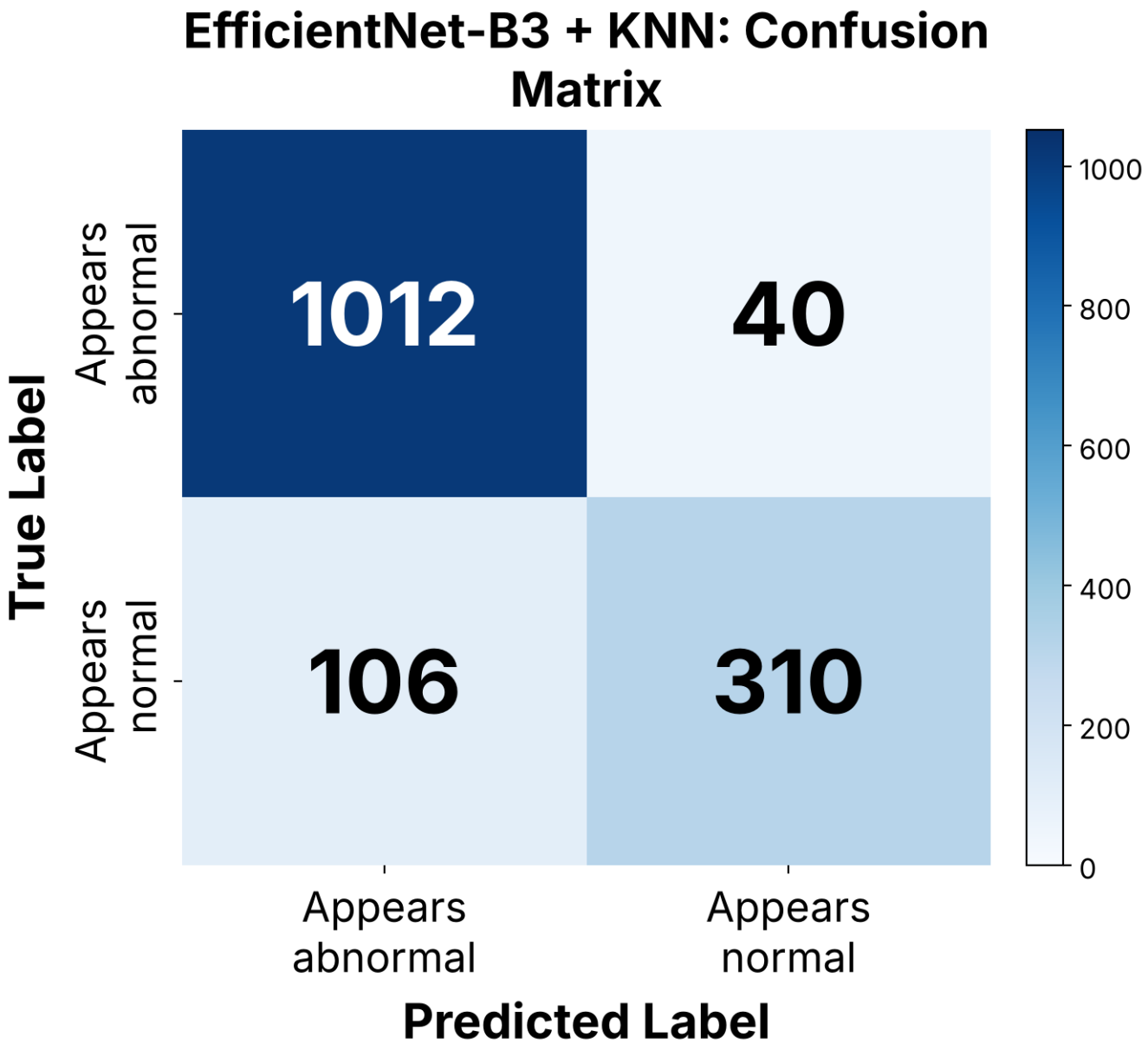
Accuracy: 90.05%

Precision: 89.97%

Recall: 90.05%

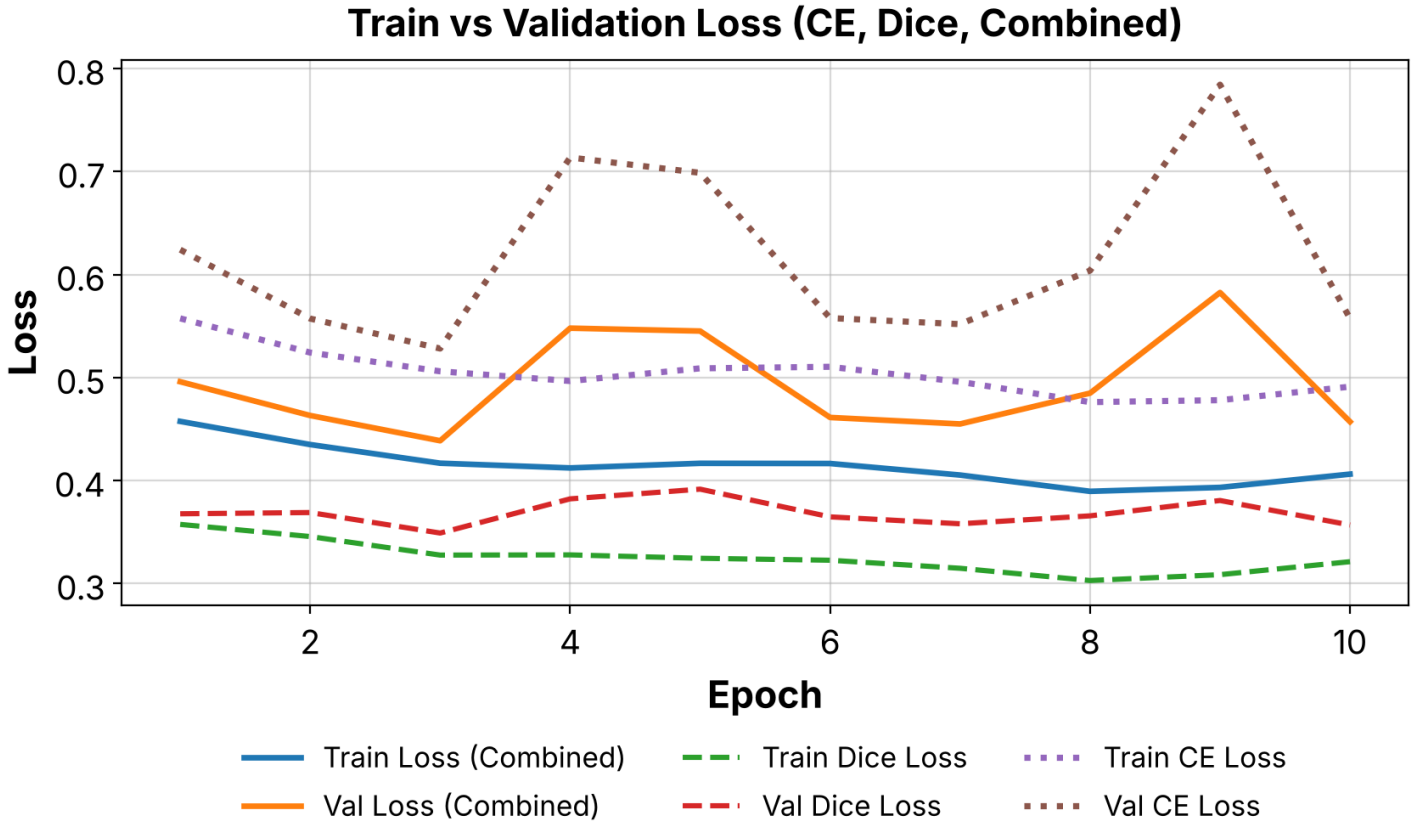
F1-score: 89.78%

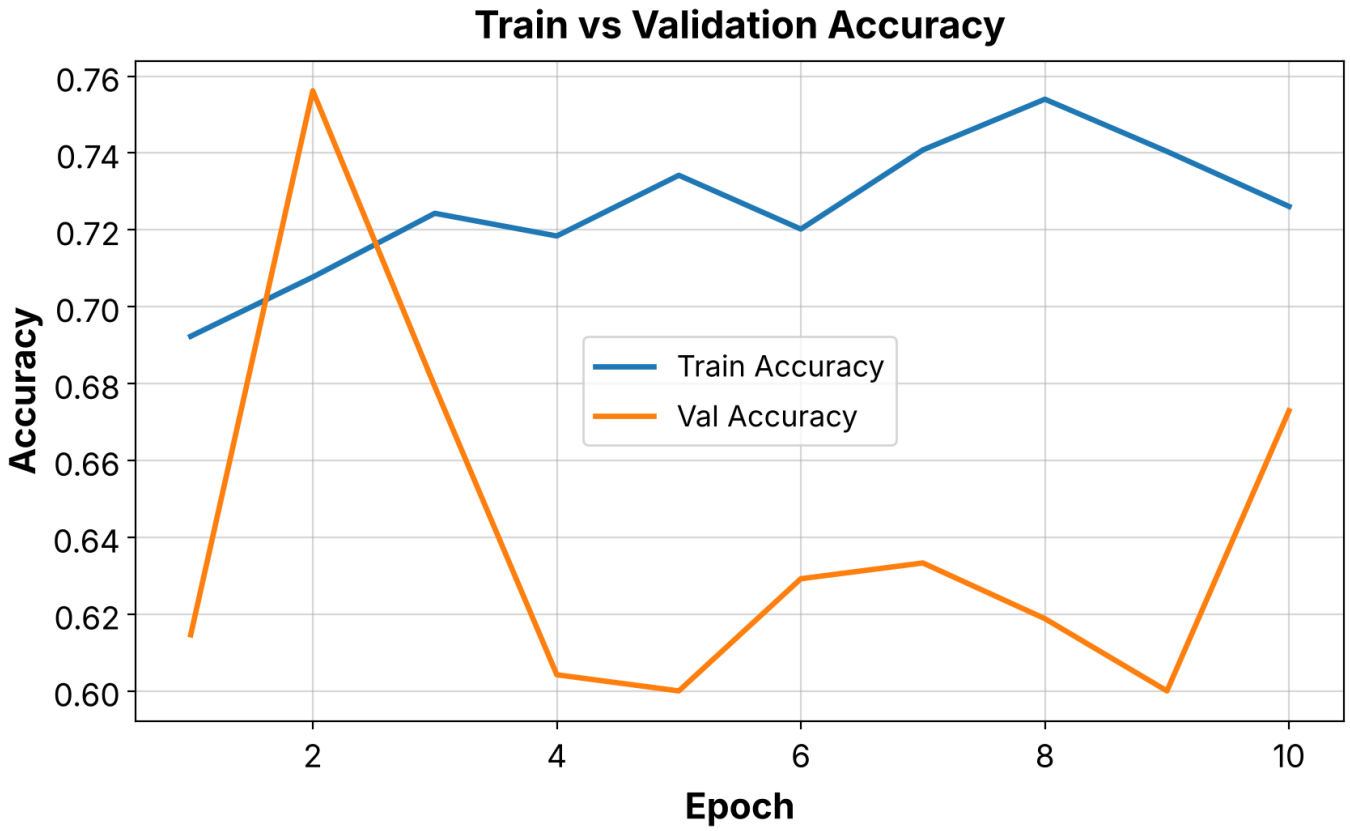
Confusion Matrix



19. EfficientNet-B3 + KNN — graphs

Training / Validation Graphs



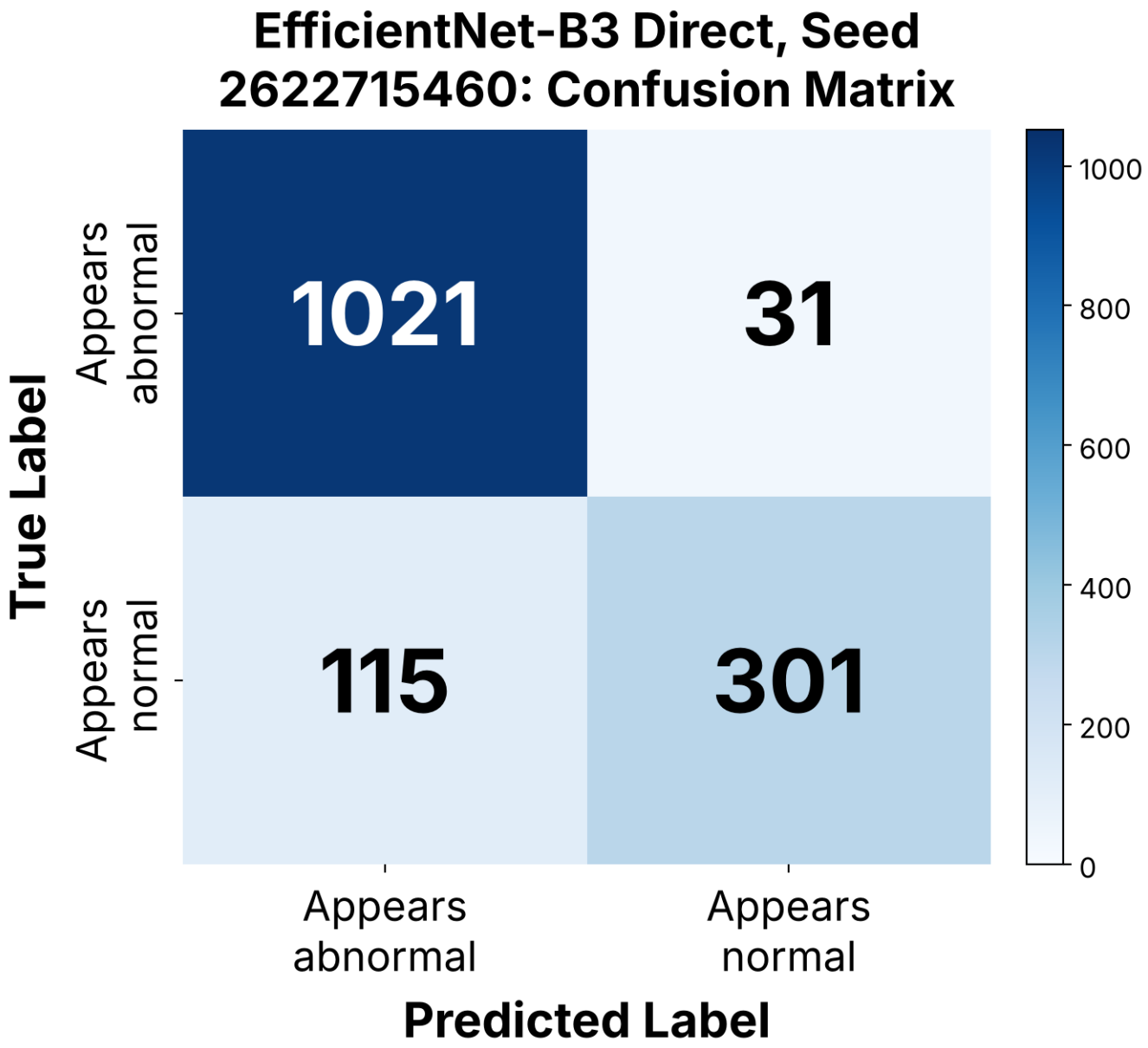


20. EfficientNet-B3 — Direct (fine-tuned CNN) — Seed 2622715460 (seed 1 of 3)

Source: 3_Seed_exp.ipynb, cell 8. Re-run independently in *(seeded) EfficientNet B3 with Adaboost.ipynb, cell 4, seed 512* (same numbers for that seed).

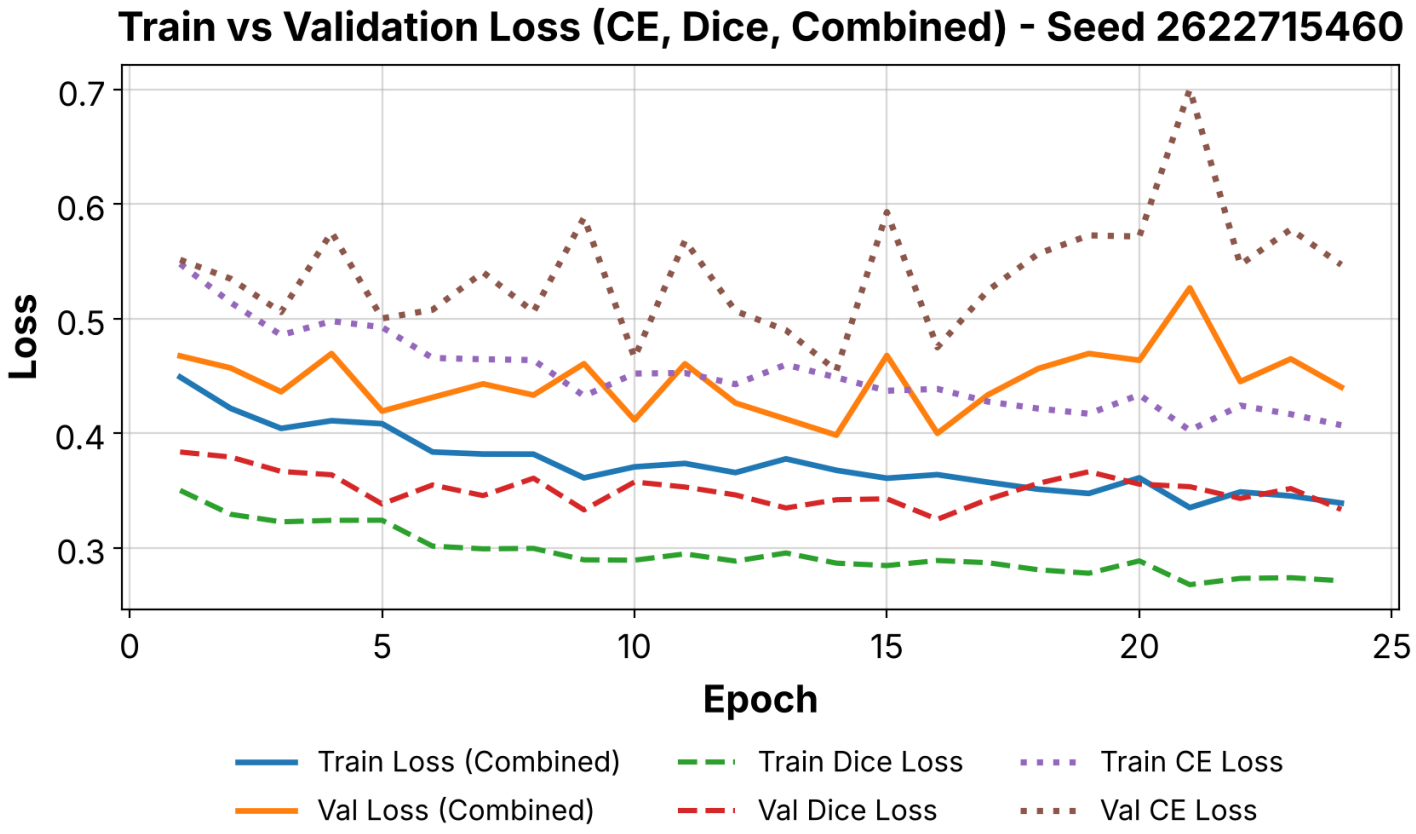
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**90.05%**	87.60%	89.71%	89.12 ± 1.33 %
Precision (weighted)	**90.10%**	87.90%	89.84%	89.28 ± 1.21 %
Recall (weighted)	**90.05%**	87.60%	89.71%	89.12 ± 1.33 %
F1-score (weighted)	**89.69%**	86.85%	89.27%	88.60 ± 1.53 %
Test Dice loss	0.5196	0.5218	0.5247	0.5220 ± 0.0025
Best validation acc. (CNN)	76.46%	75.62%	72.50%	74.86 ± 2.09 %

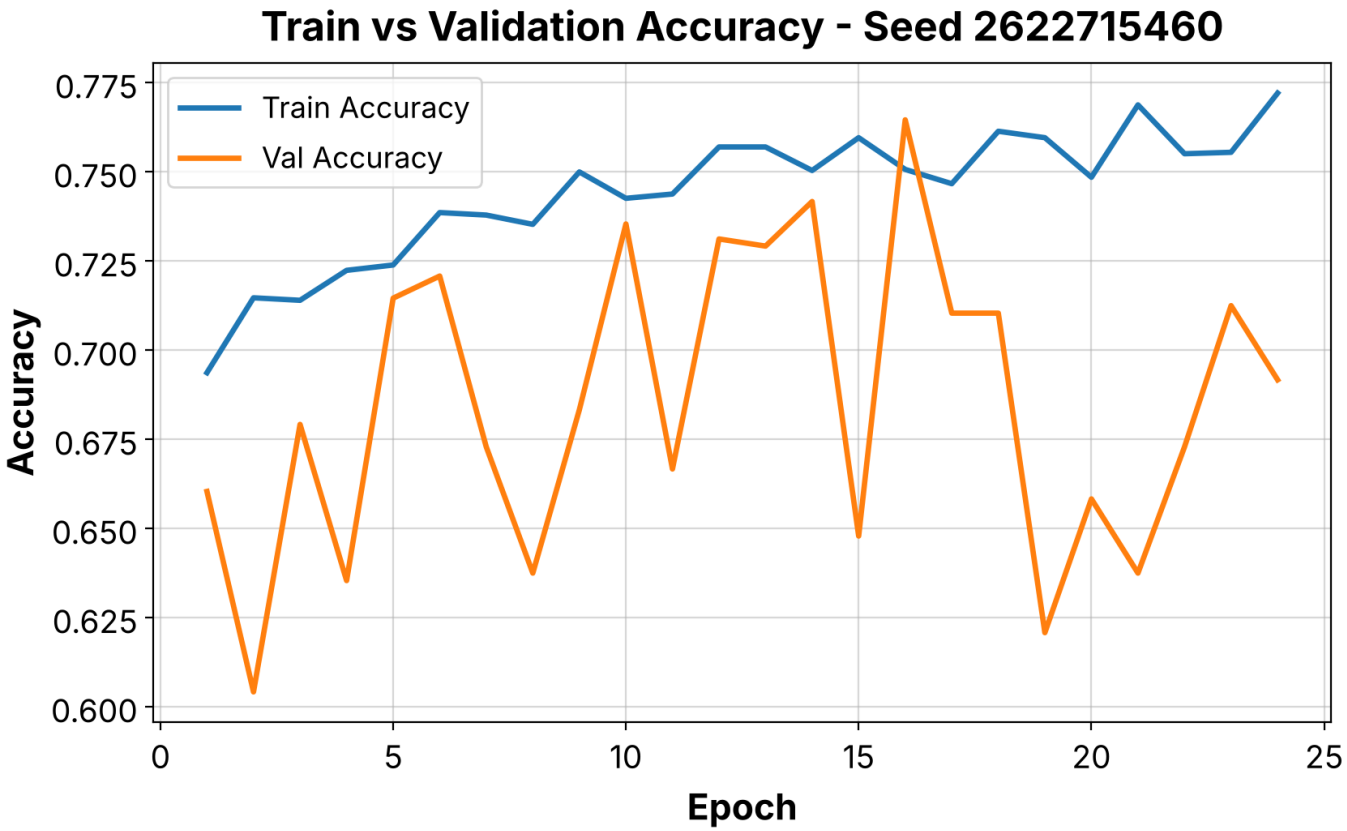
Confusion Matrix



20. EfficientNet-B3 — Direct (fine-tuned CNN), Seed 2622715460 — graphs

Training / Validation Graphs



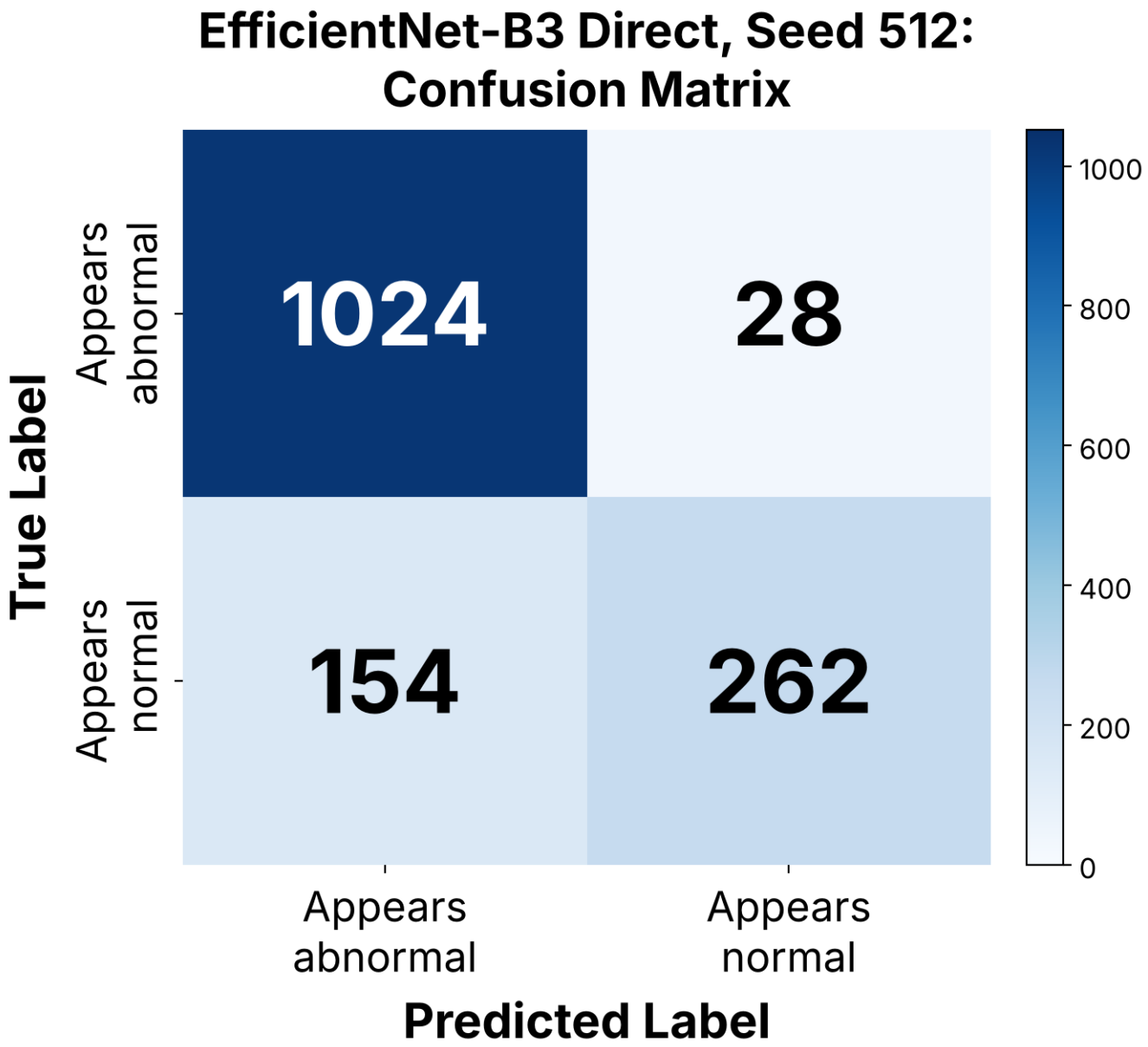


20. EfficientNet-B3 — Direct (fine-tuned CNN) — Seed 512 (seed 2 of 3)

Source: 3_Seed_exp.ipynb, cell 8. Re-run independently in *(seeded) EfficientNet B3 with Adaboost.ipynb, cell 4, seed 512* (same numbers for that seed).

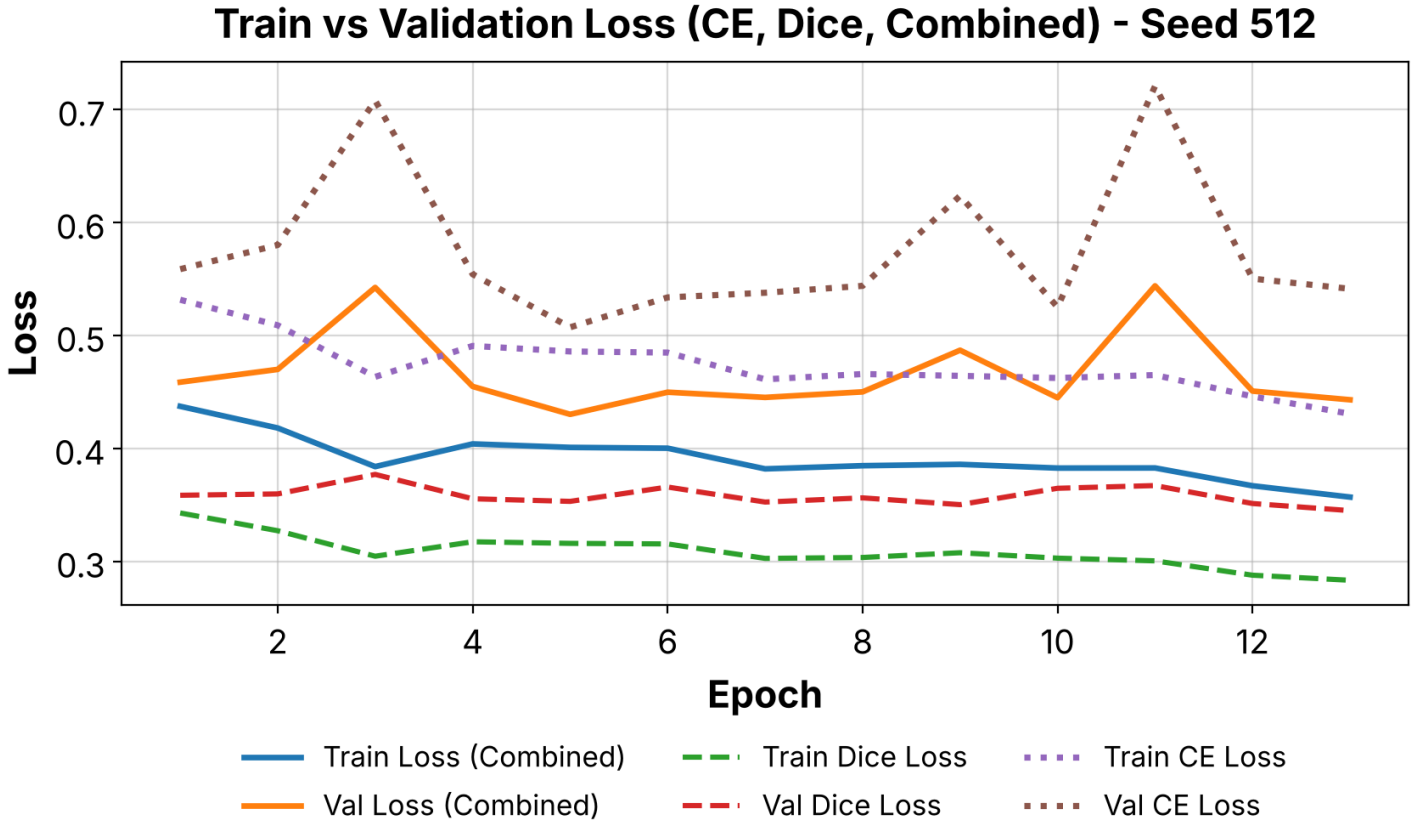
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	90.05%	**87.60%**	89.71%	89.12 ± 1.33 %
Precision (weighted)	90.10%	**87.90%**	89.84%	89.28 ± 1.21 %
Recall (weighted)	90.05%	**87.60%**	89.71%	89.12 ± 1.33 %
F1-score (weighted)	89.69%	**86.85%**	89.27%	88.60 ± 1.53 %
Test Dice loss	0.5196	0.5218	0.5247	0.5220 ± 0.0025
Best validation acc. (CNN)	76.46%	75.62%	72.50%	74.86 ± 2.09 %

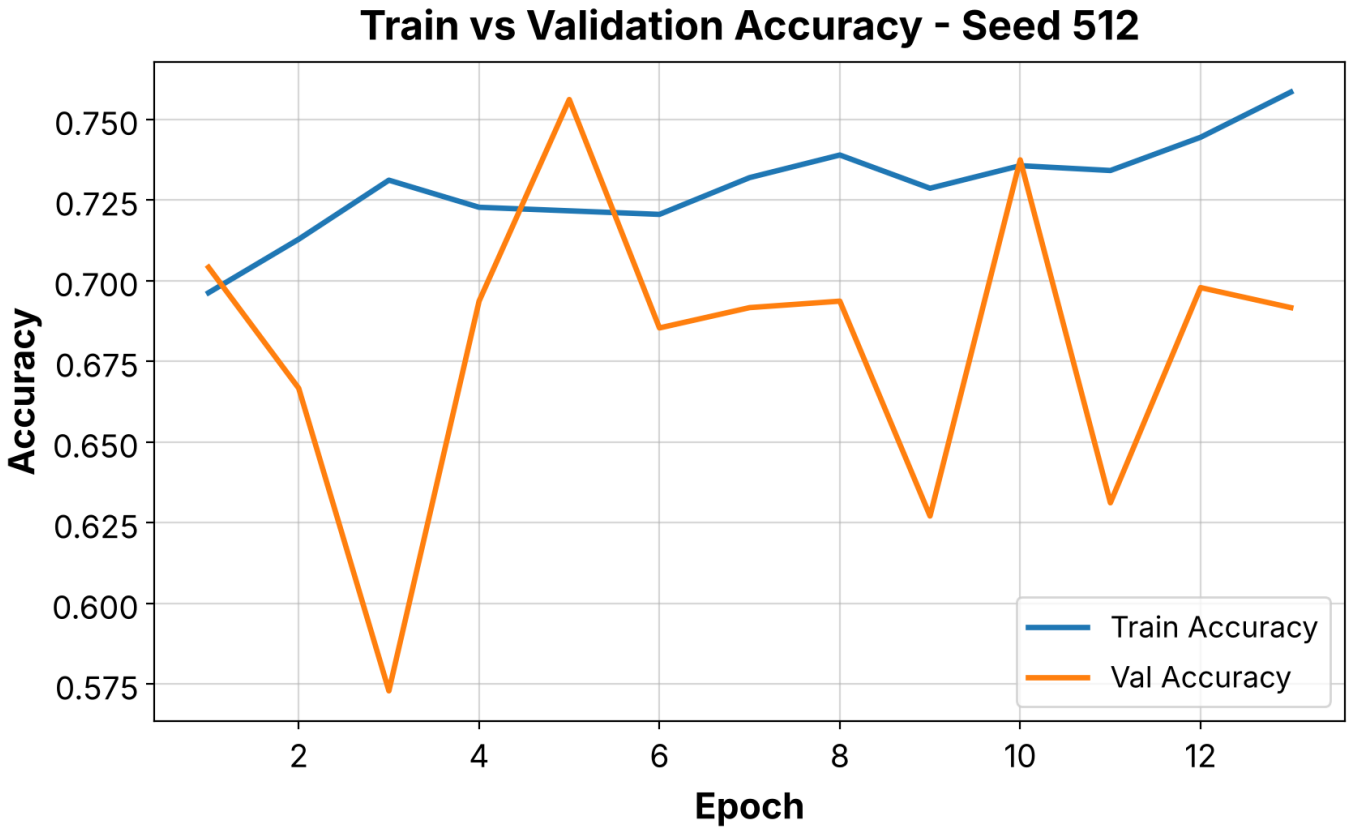
Confusion Matrix



20. EfficientNet-B3 — Direct (fine-tuned CNN), Seed 512 — graphs

Training / Validation Graphs



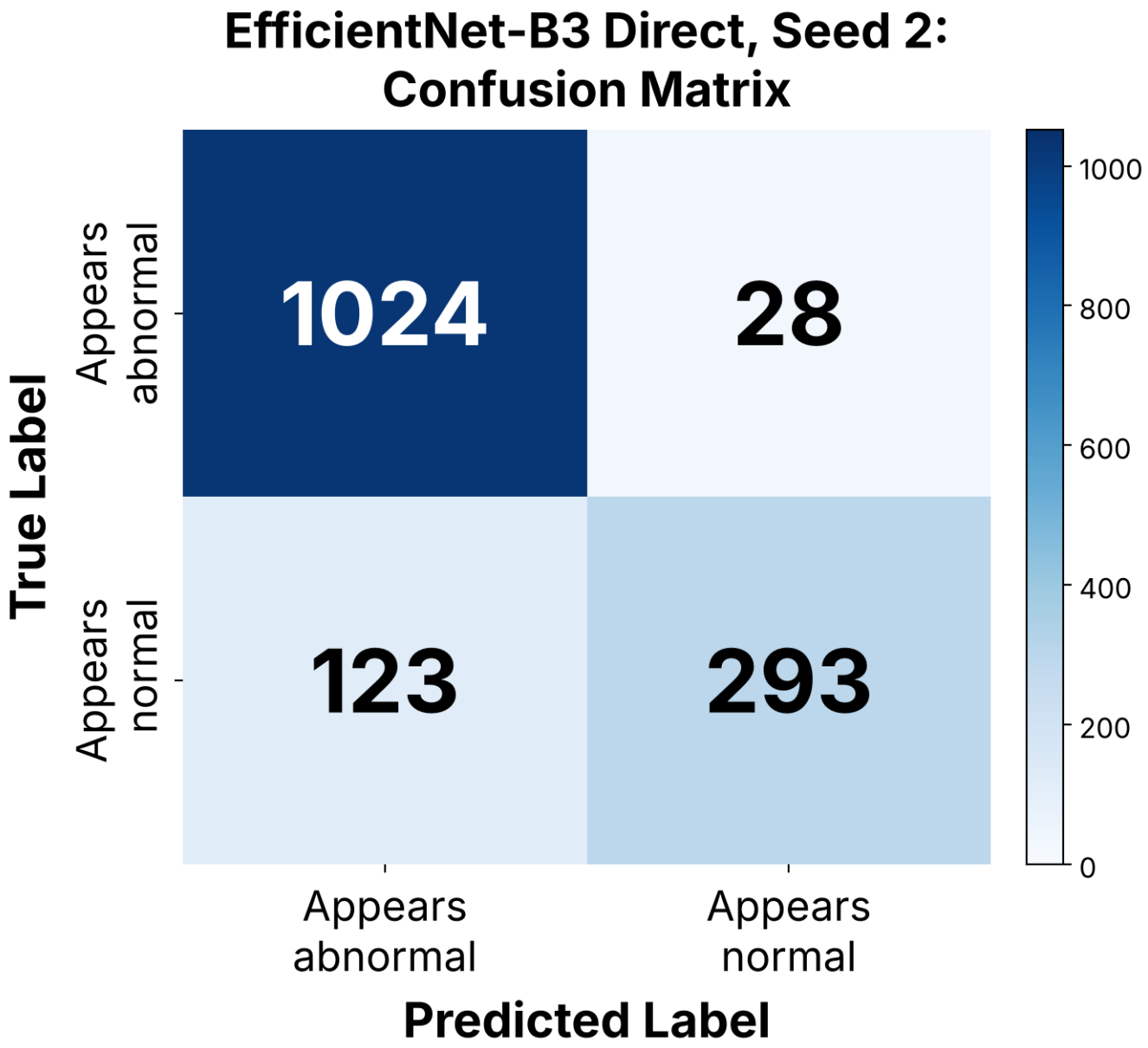


20. EfficientNet-B3 — Direct (fine-tuned CNN) — Seed 2 (seed 3 of 3)

Source: 3_Seed_exp.ipynb, cell 8. Re-run independently in *(seeded) EfficientNet B3 with Adaboost.ipynb, cell 4, seed 512* (same numbers for that seed).

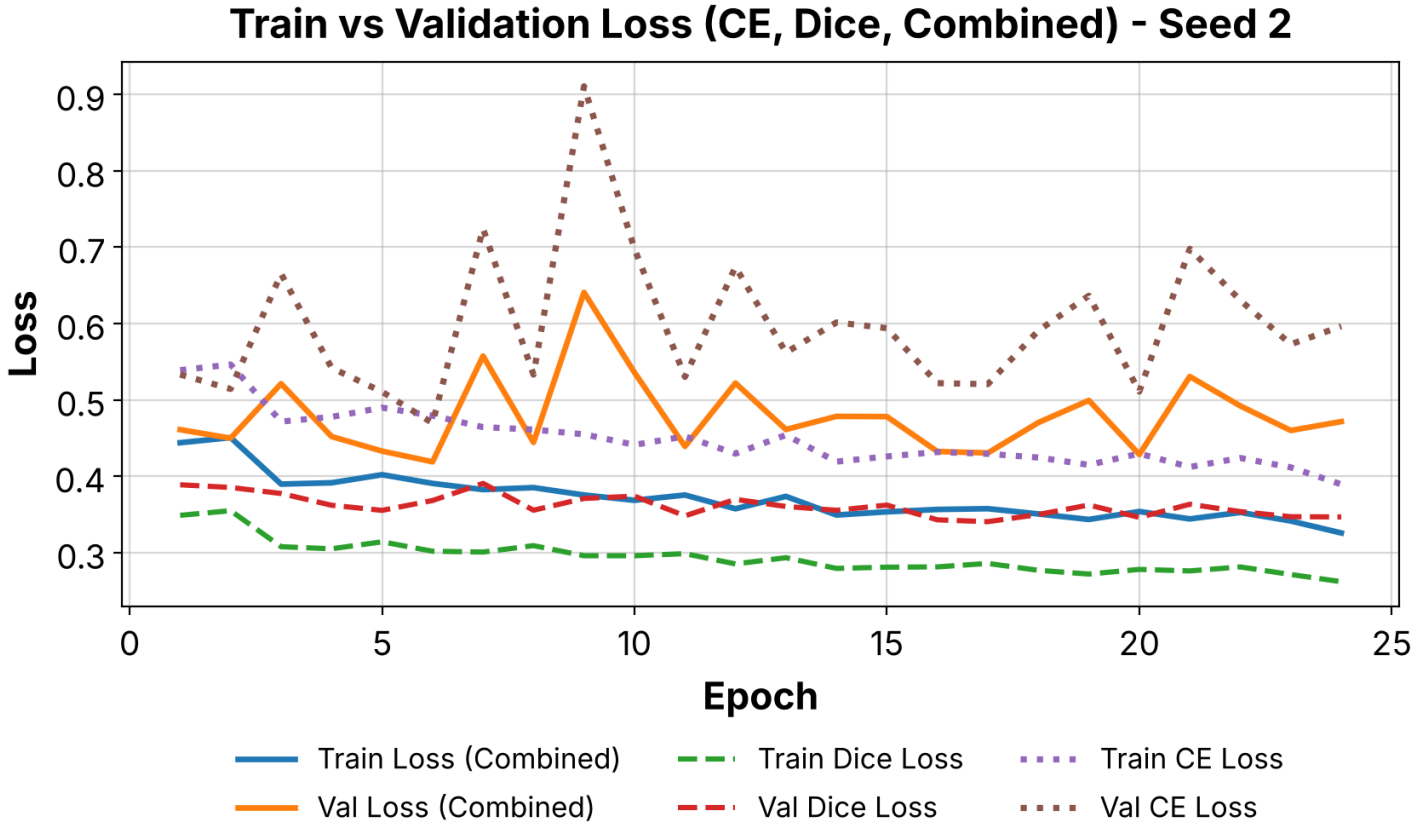
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	90.05%	87.60%	**89.71%**	89.12 ± 1.33 %
Precision (weighted)	90.10%	87.90%	**89.84%**	89.28 ± 1.21 %
Recall (weighted)	90.05%	87.60%	**89.71%**	89.12 ± 1.33 %
F1-score (weighted)	89.69%	86.85%	**89.27%**	88.60 ± 1.53 %
Test Dice loss	0.5196	0.5218	0.5247	0.5220 ± 0.0025
Best validation acc. (CNN)	76.46%	75.62%	72.50%	74.86 ± 2.09 %

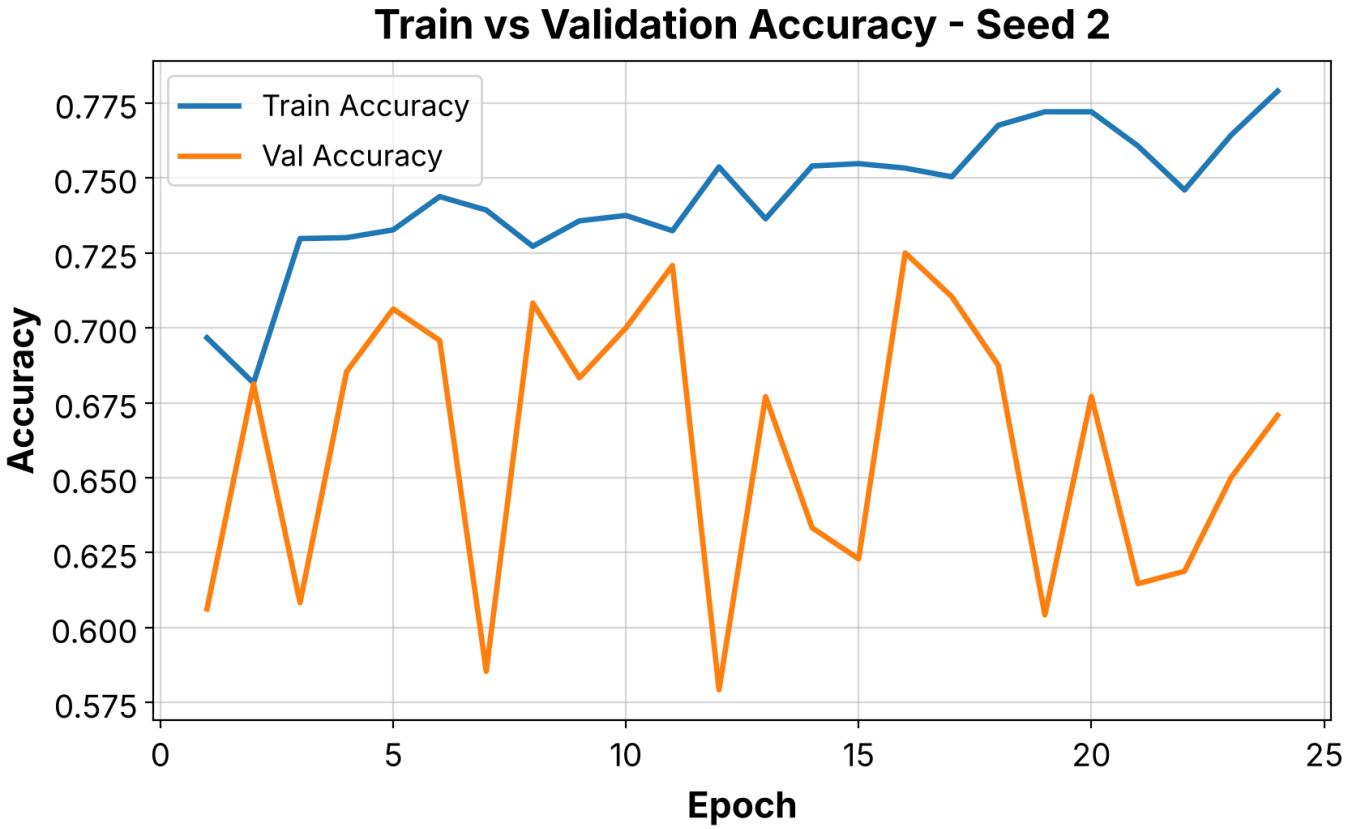
Confusion Matrix



20. EfficientNet-B3 — Direct (fine-tuned CNN), Seed 2 — graphs

Training / Validation Graphs



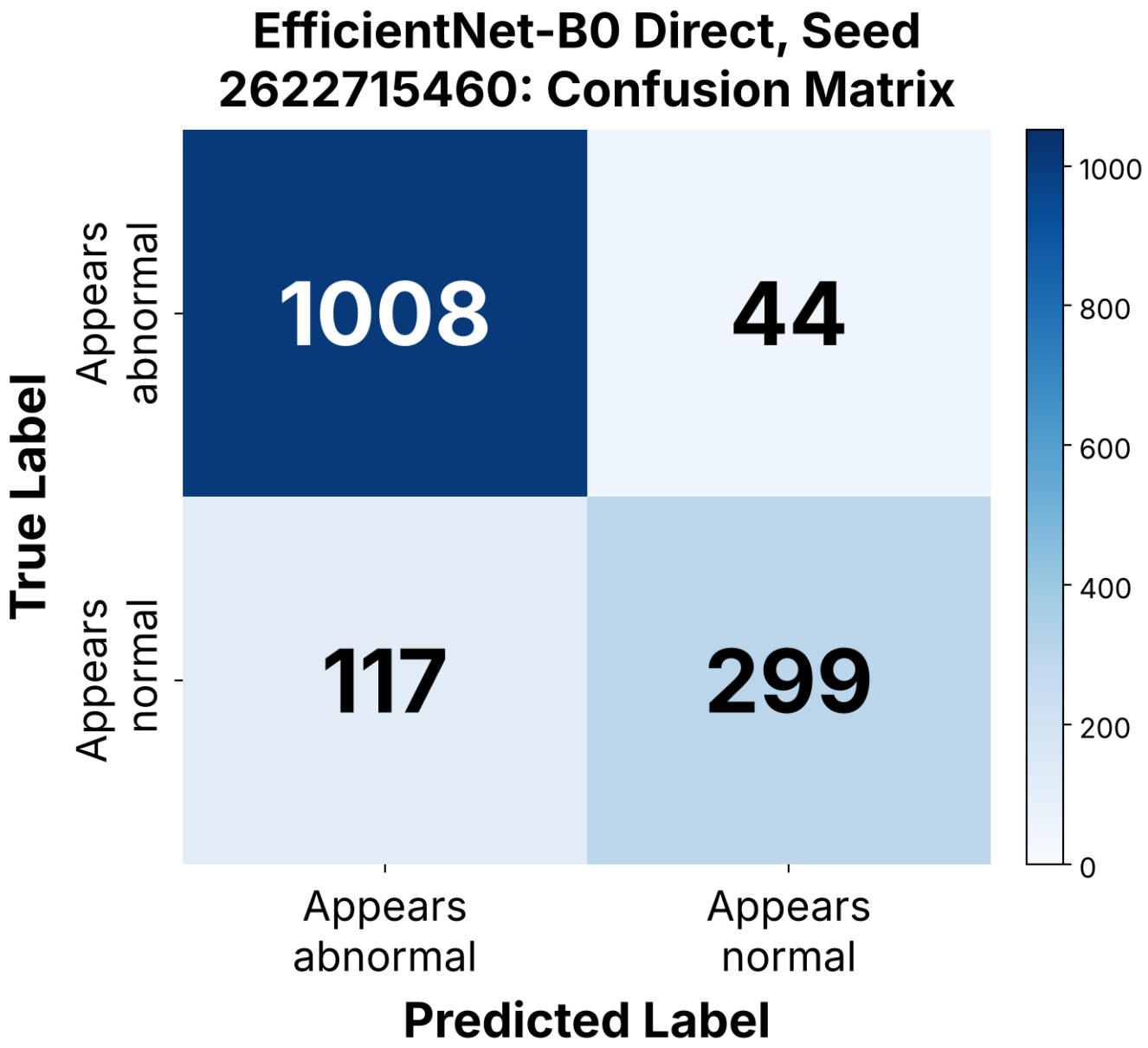


21. EfficientNet-B0 — Direct (fine-tuned CNN) — Seed 2622715460 (seed 1 of 3)

Source: 3_Seed_exp.ipynb, cell 12. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 8, seed 2622715460* (same numbers for that seed).

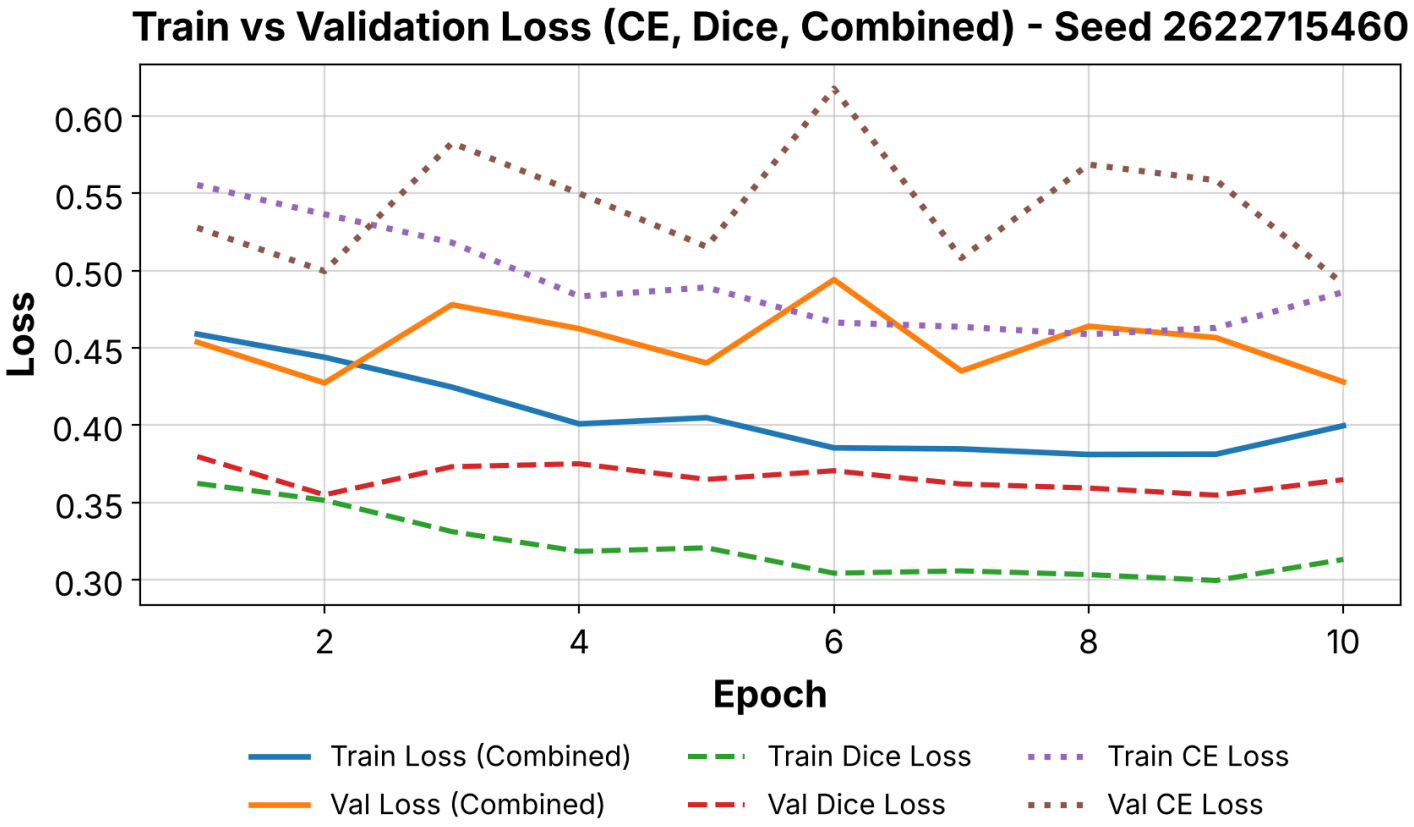
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**89.03%**	89.58%	88.42%	89.01 ± 0.58 %
Precision (weighted)	**88.91%**	89.52%	88.68%	89.04 ± 0.44 %
Recall (weighted)	**89.03%**	89.58%	88.42%	89.01 ± 0.58 %
F1-score (weighted)	**88.69%**	89.23%	87.78%	88.57 ± 0.73 %
Test Dice loss	0.5193	0.5218	0.5244	0.5218 ± 0.0026
Best validation acc. (CNN)	70.42%	75.62%	75.62%	73.89 ± 3.01 %

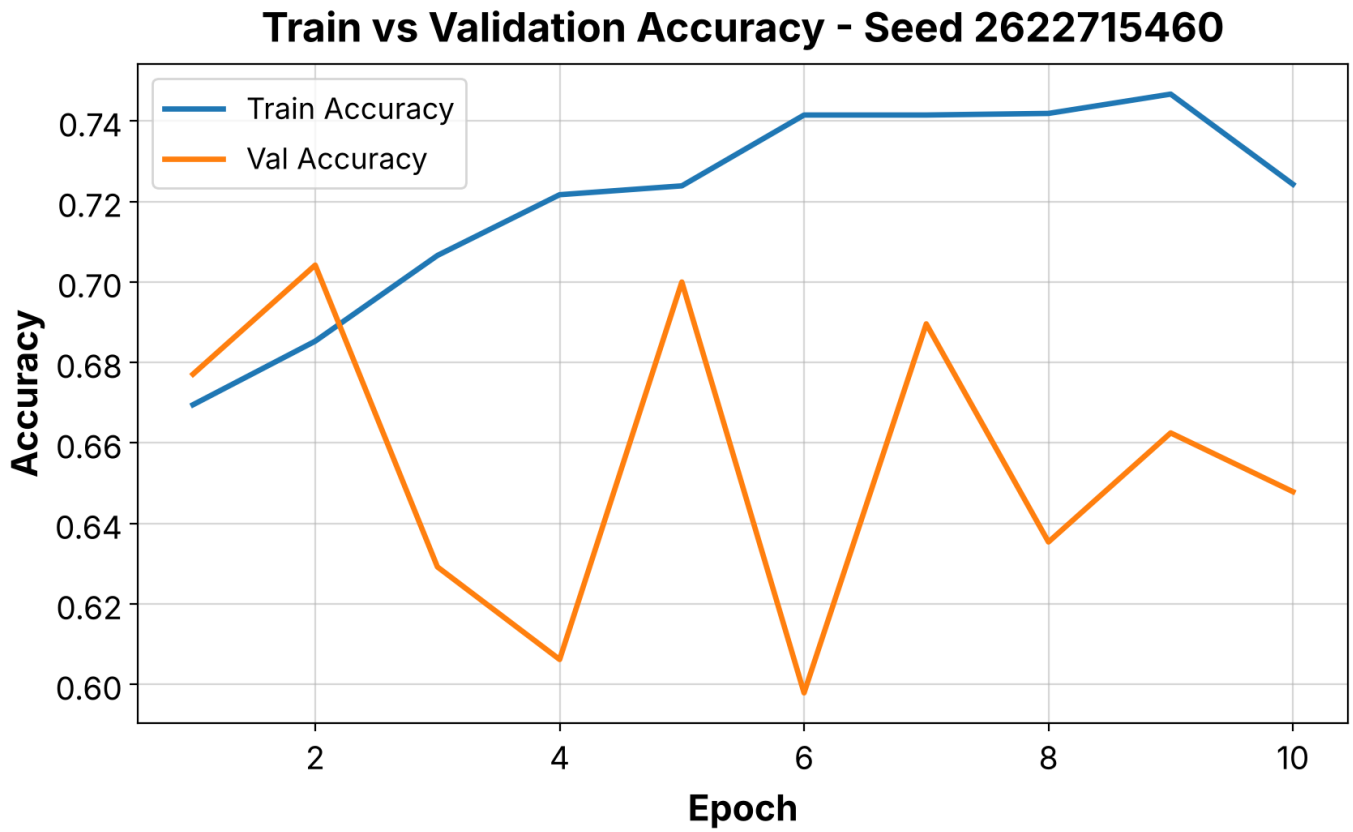
Confusion Matrix



21. EfficientNet-B0 — Direct (fine-tuned CNN), Seed 2622715460 — graphs

Training / Validation Graphs



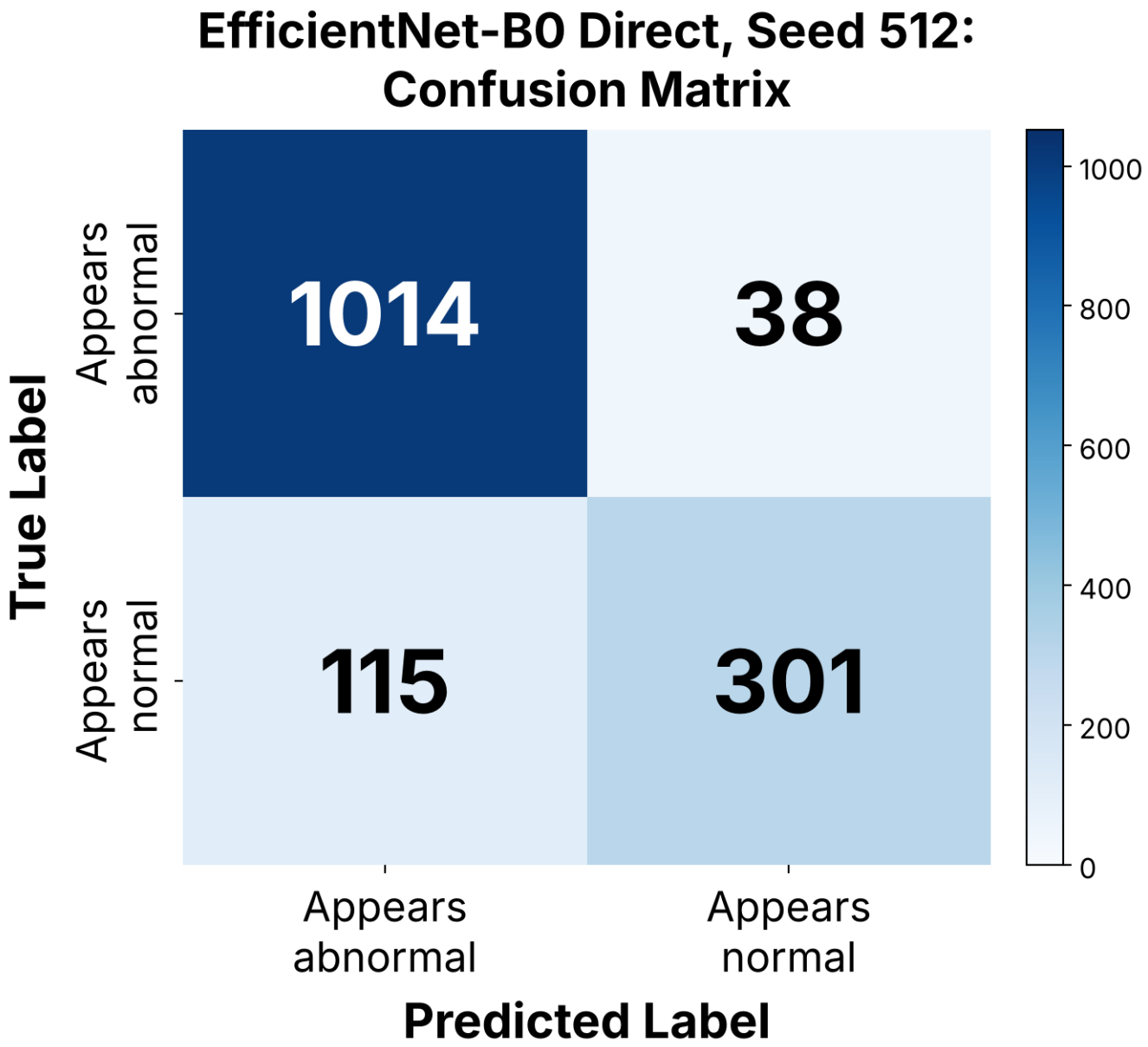


21. EfficientNet-B0 — Direct (fine-tuned CNN) — Seed 512 (seed 2 of 3)

Source: 3_Seed_exp.ipynb, cell 12. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 8, seed 2622715460* (same numbers for that seed).

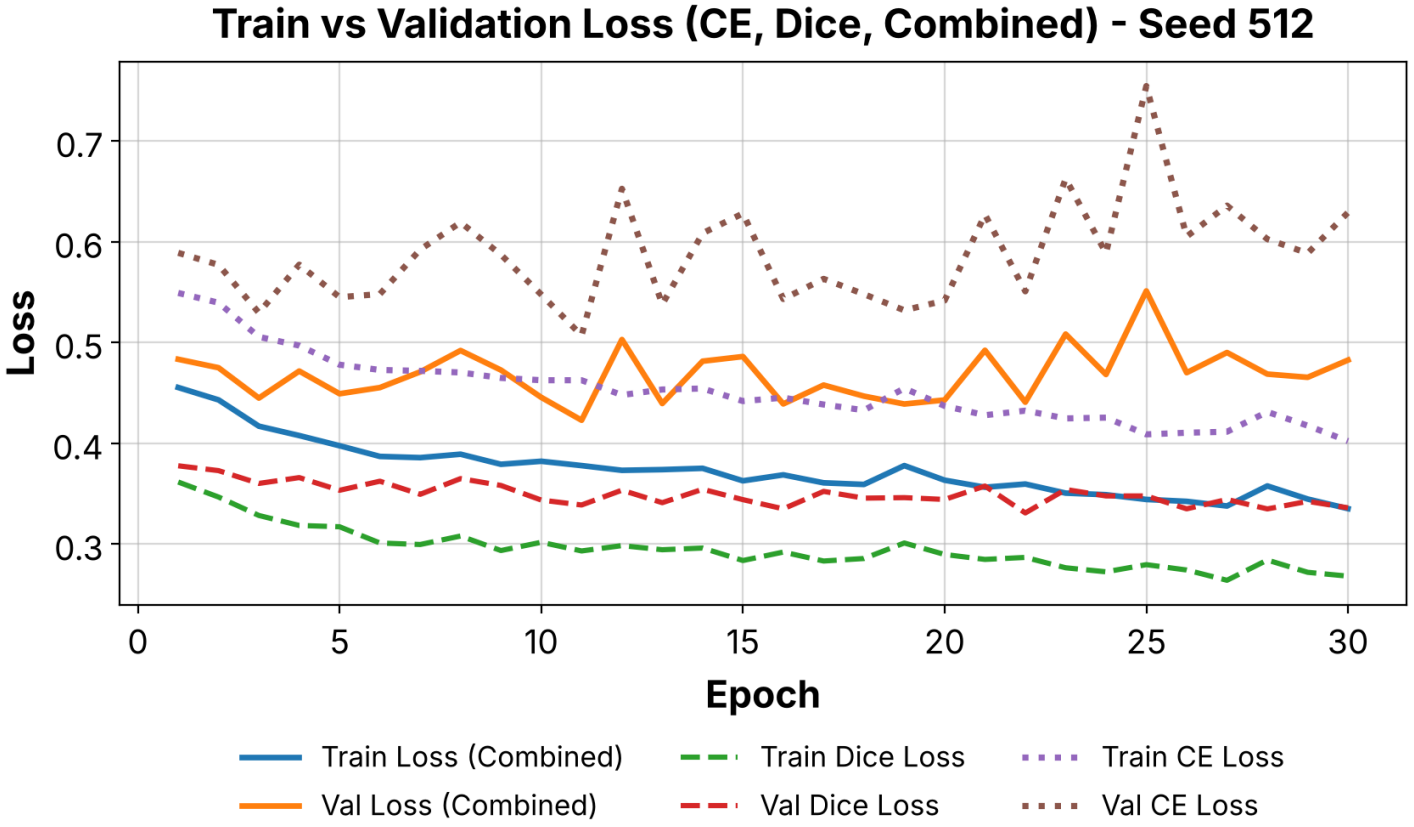
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	89.03%	**89.58%**	88.42%	89.01 ± 0.58 %
Precision (weighted)	88.91%	**89.52%**	88.68%	89.04 ± 0.44 %
Recall (weighted)	89.03%	**89.58%**	88.42%	89.01 ± 0.58 %
F1-score (weighted)	88.69%	**89.23%**	87.78%	88.57 ± 0.73 %
Test Dice loss	0.5193	0.5218	0.5244	0.5218 ± 0.0026
Best validation acc. (CNN)	70.42%	75.62%	75.62%	73.89 ± 3.01 %

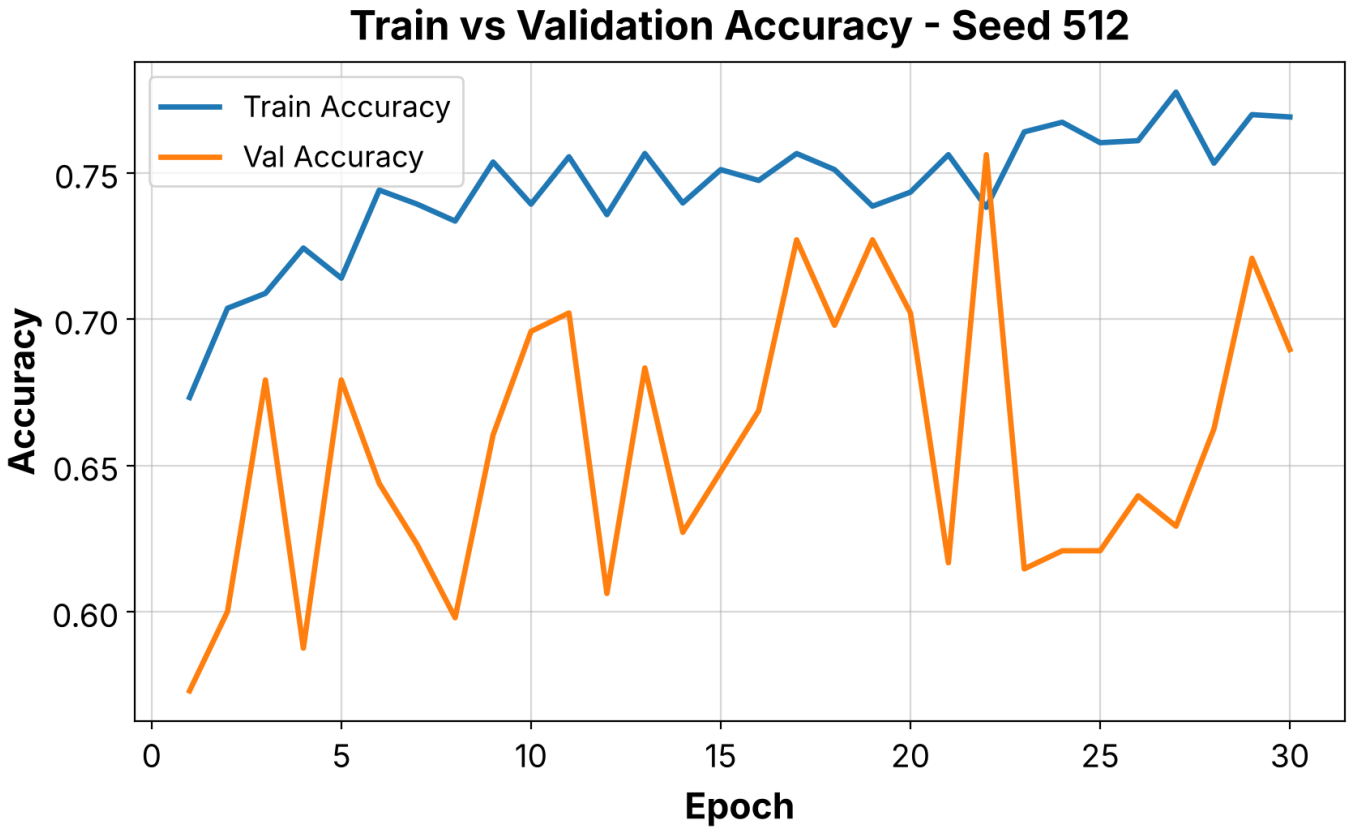
Confusion Matrix



21. EfficientNet-B0 — Direct (fine-tuned CNN), Seed 512 — graphs

Training / Validation Graphs



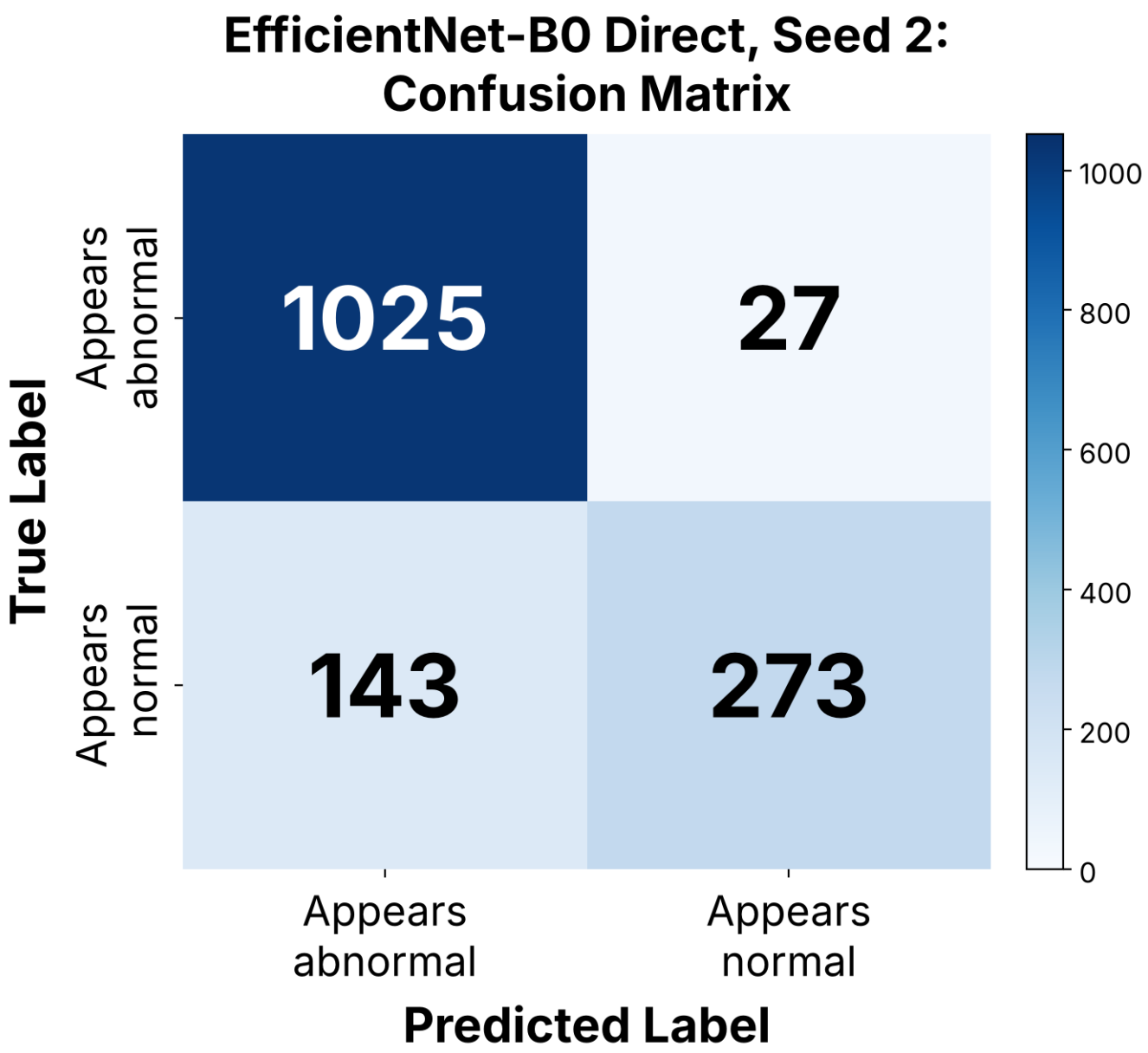


21. EfficientNet-B0 — Direct (fine-tuned CNN) — Seed 2 (seed 3 of 3)

Source: 3_Seed_exp.ipynb, cell 12. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 8, seed 2622715460* (same numbers for that seed).

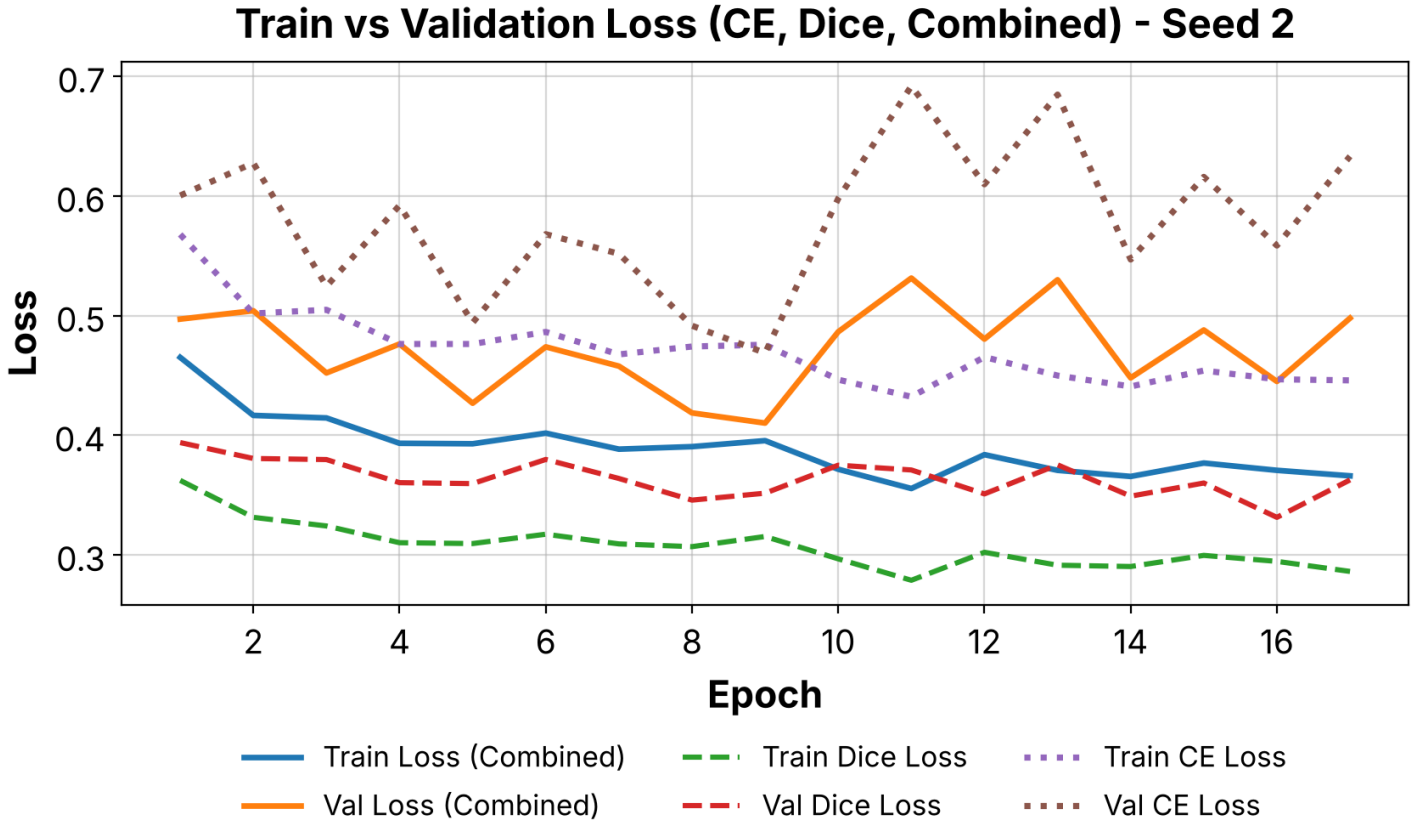
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	89.03%	89.58%	**88.42%**	89.01 ± 0.58 %
Precision (weighted)	88.91%	89.52%	**88.68%**	89.04 ± 0.44 %
Recall (weighted)	89.03%	89.58%	**88.42%**	89.01 ± 0.58 %
F1-score (weighted)	88.69%	89.23%	**87.78%**	88.57 ± 0.73 %
Test Dice loss	0.5193	0.5218	0.5244	0.5218 ± 0.0026
Best validation acc. (CNN)	70.42%	75.62%	75.62%	73.89 ± 3.01 %

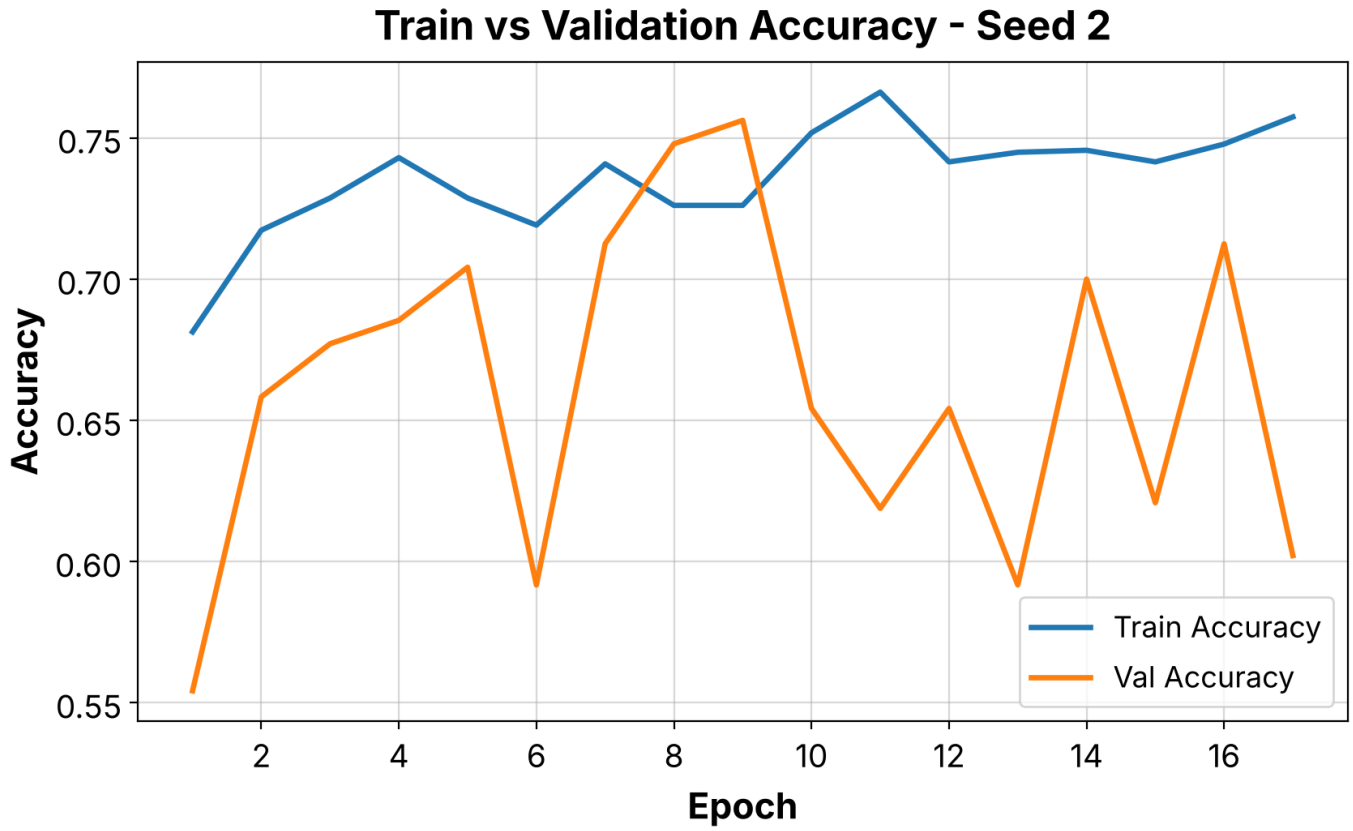
Confusion Matrix



21. EfficientNet-B0 — Direct (fine-tuned CNN), Seed 2 — graphs

Training / Validation Graphs





22. MobileNetV2 + KNN

Source notebook: MobileNet with ML.ipynb (inside Models_DL.zip)

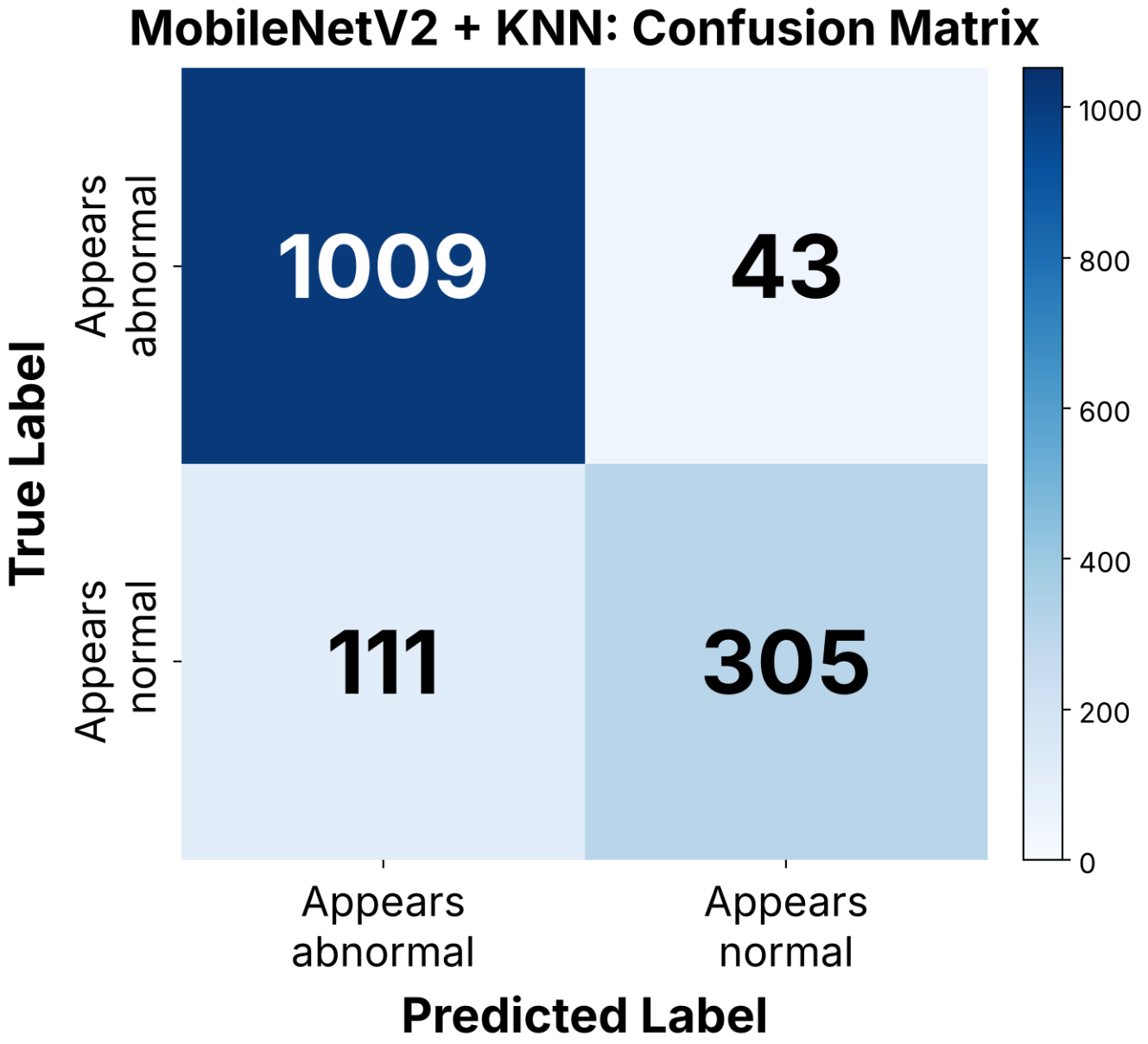
Accuracy: 89.51%

Precision: 89.40%

Recall: 89.51%

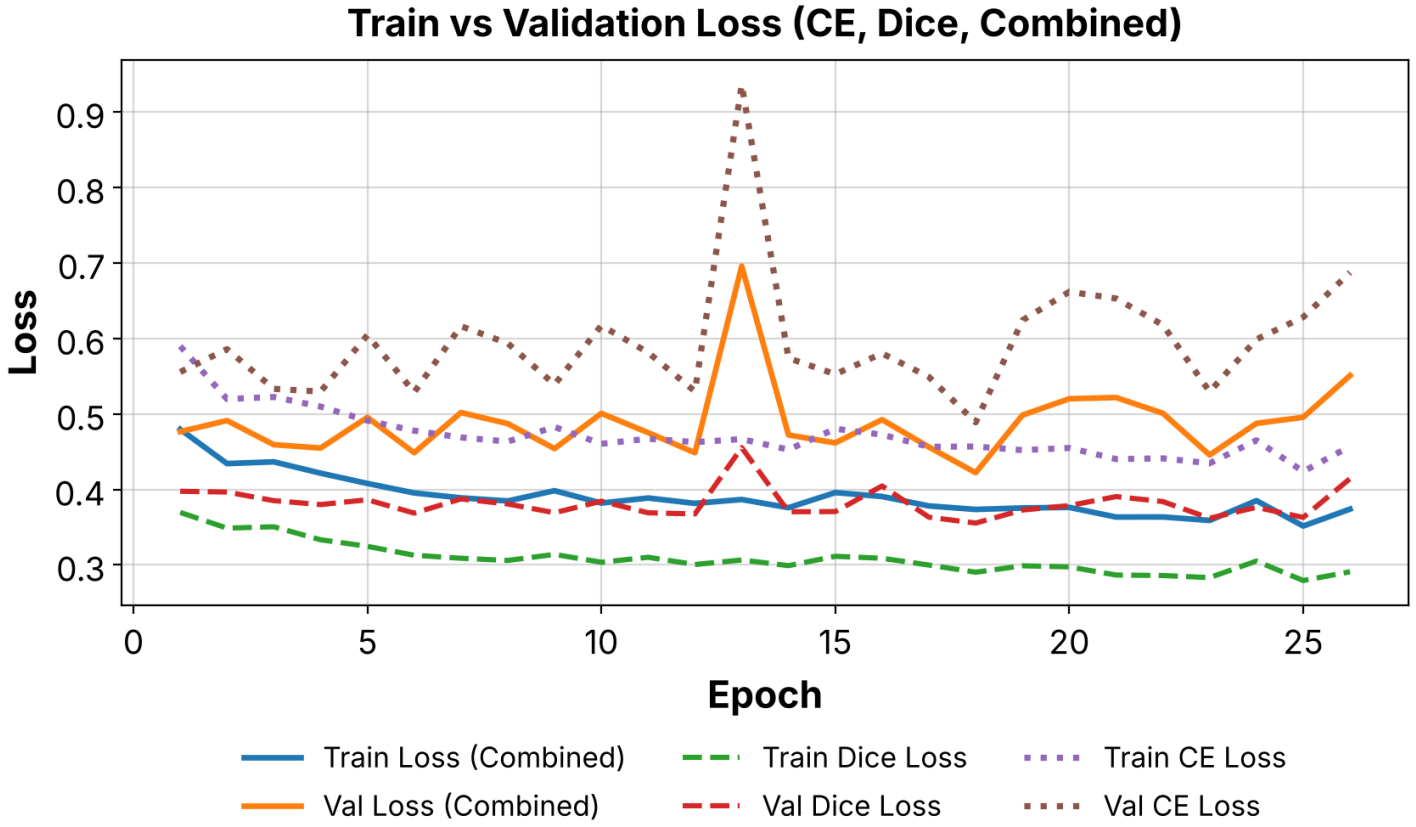
F1-score: 89.21%

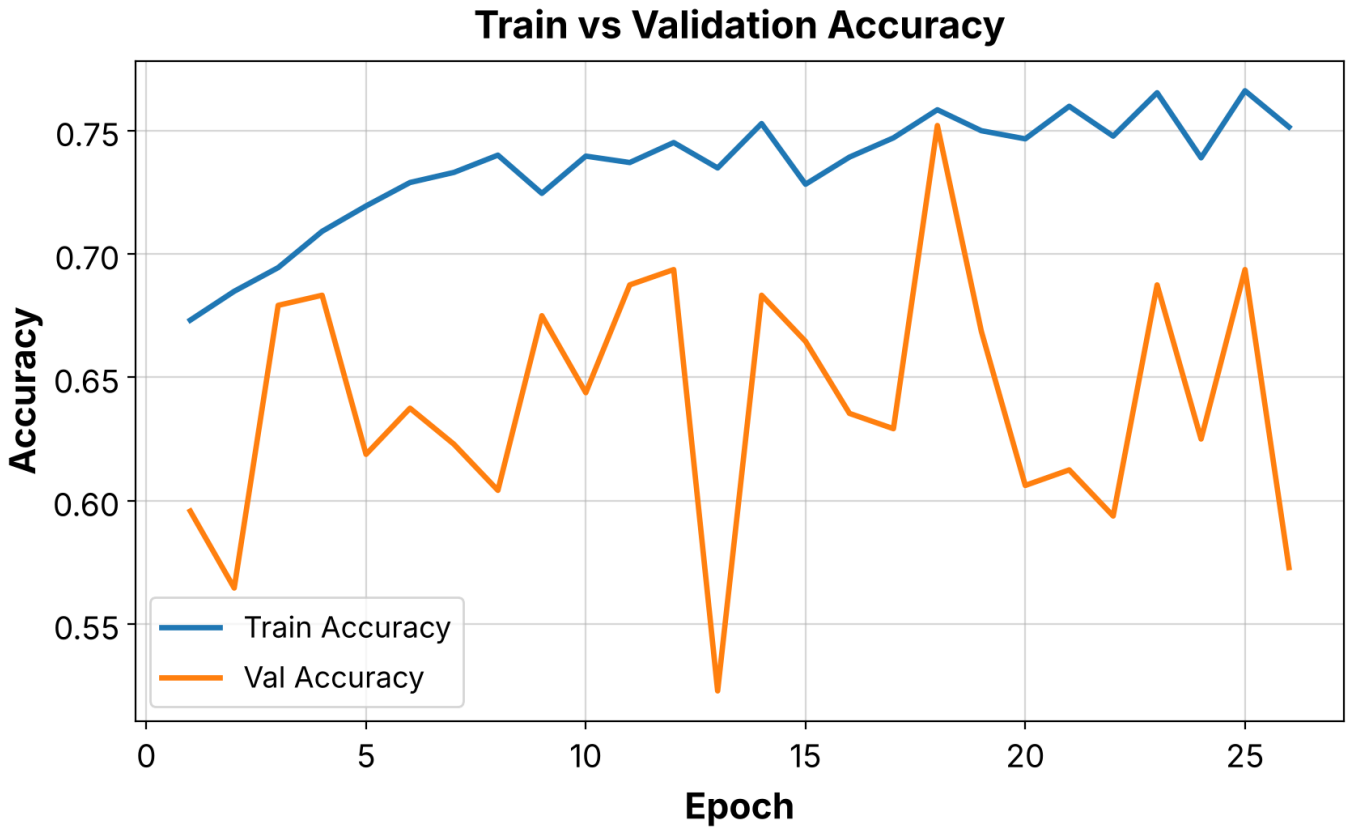
Confusion Matrix



22. MobileNetV2 + KNN — graphs

Training / Validation Graphs





23. InceptionV3 + SVM

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

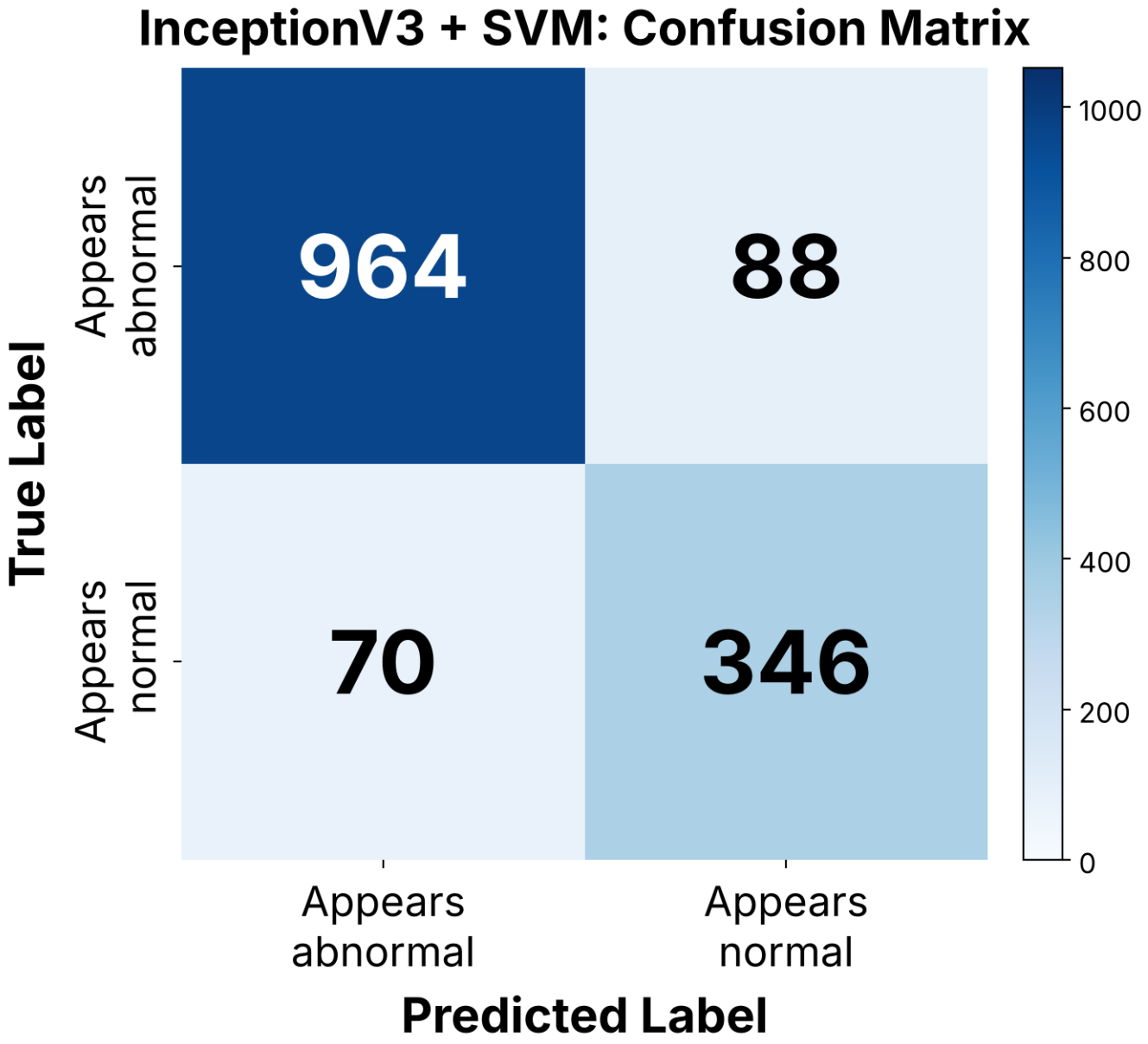
Accuracy: 89.24%

Precision: 89.40%

Recall: 89.24%

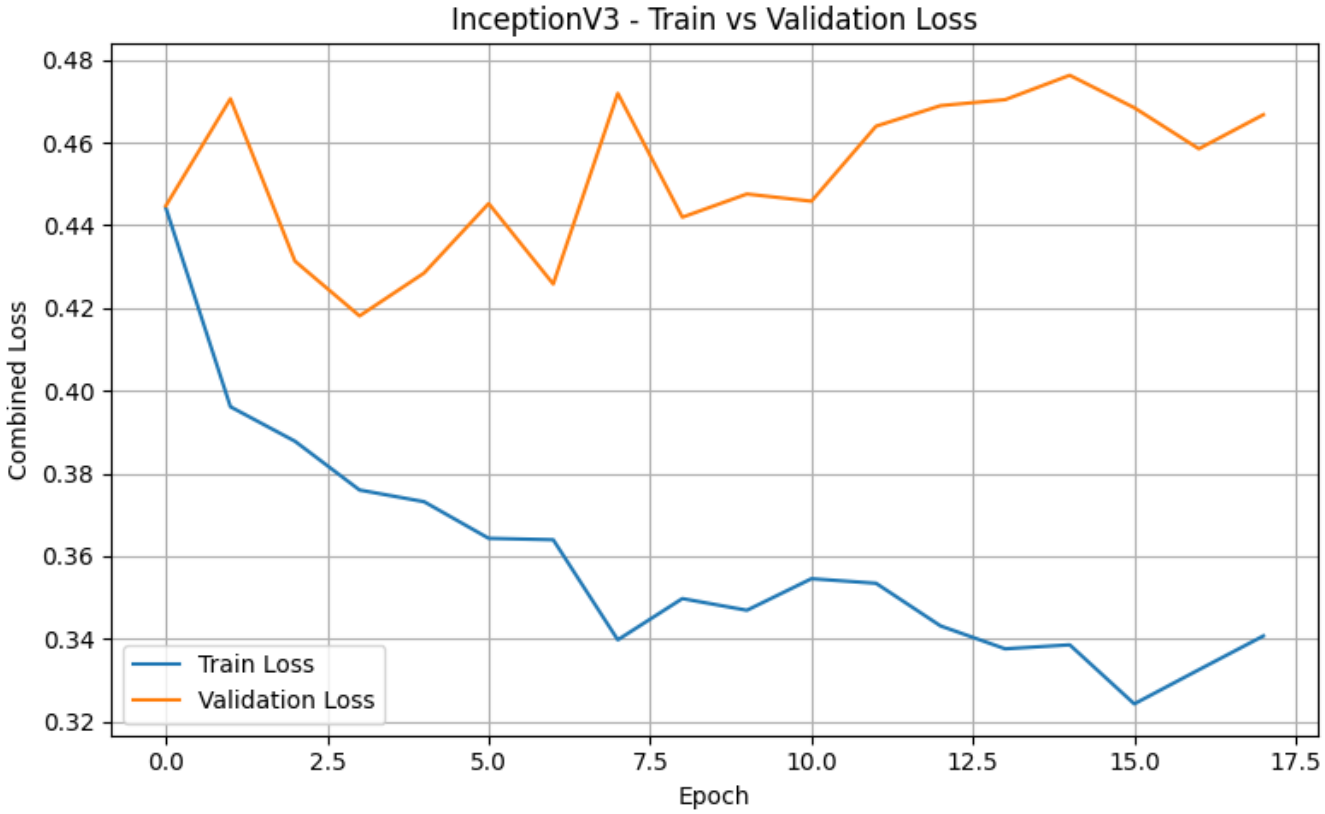
F1-score: 89.30%

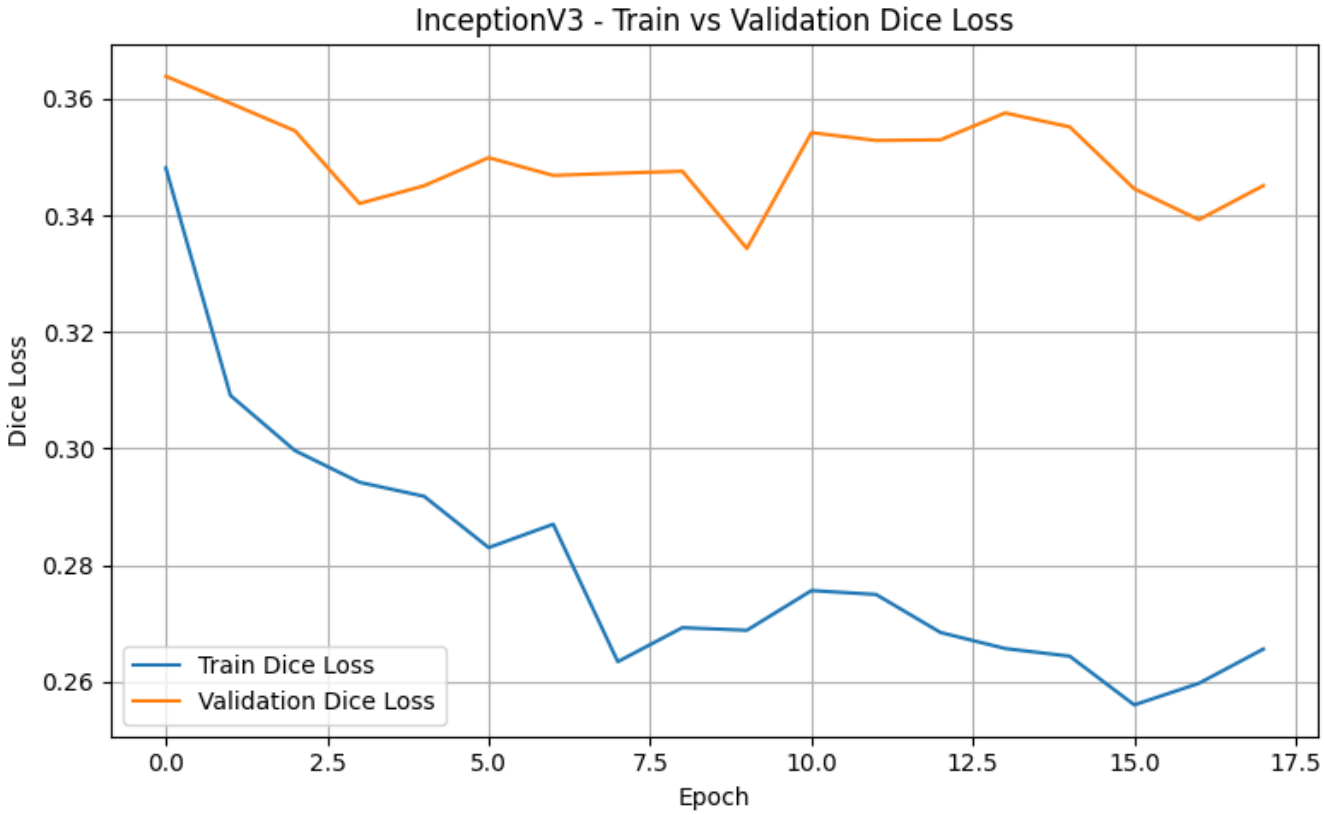
Confusion Matrix



23. InceptionV3 + SVM — graphs (1/2)

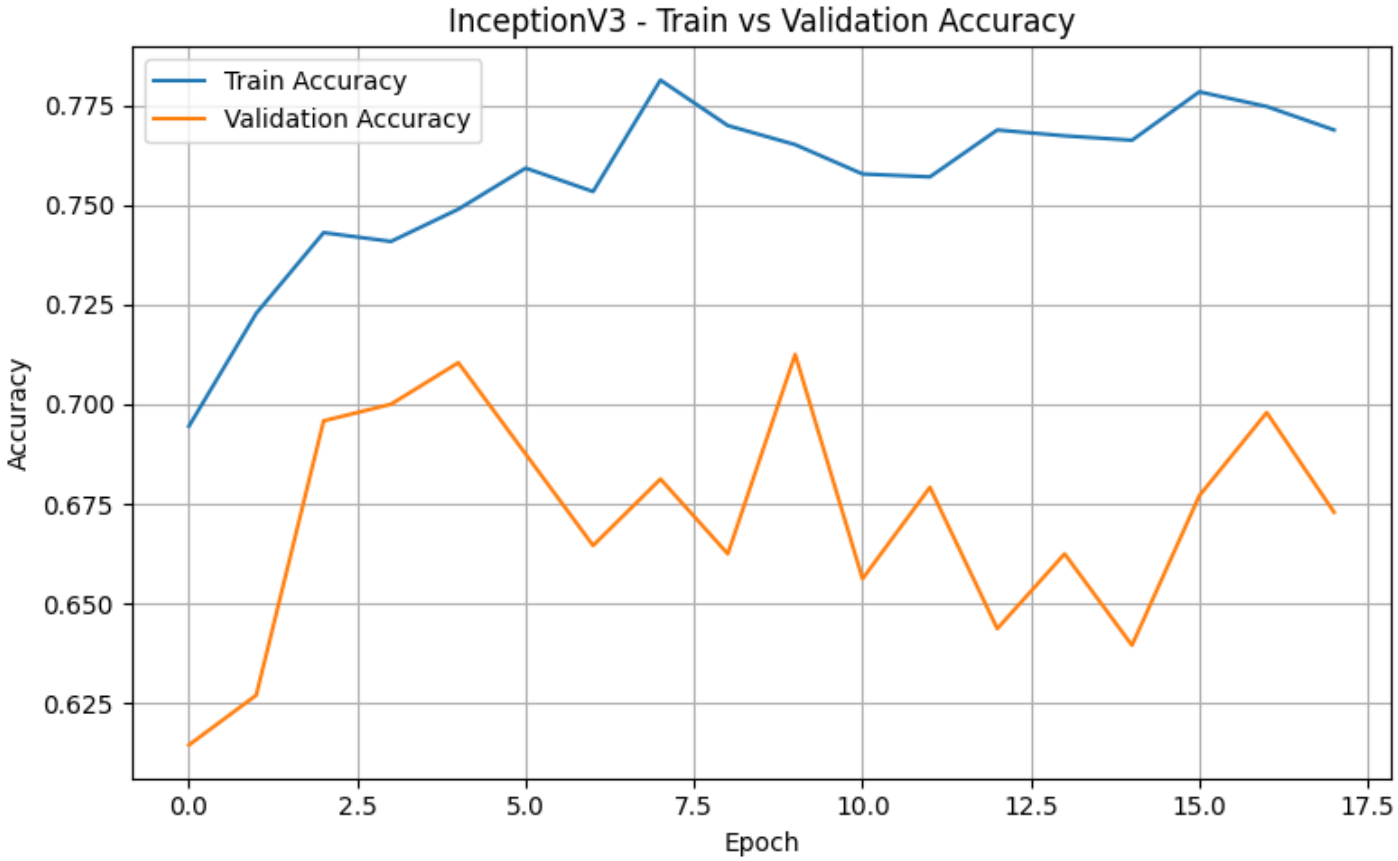
Training / Validation Graphs





23. InceptionV3 + SVM — graphs (2/2)

Training / Validation Graphs



24. MobileNetV2 + SVM

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

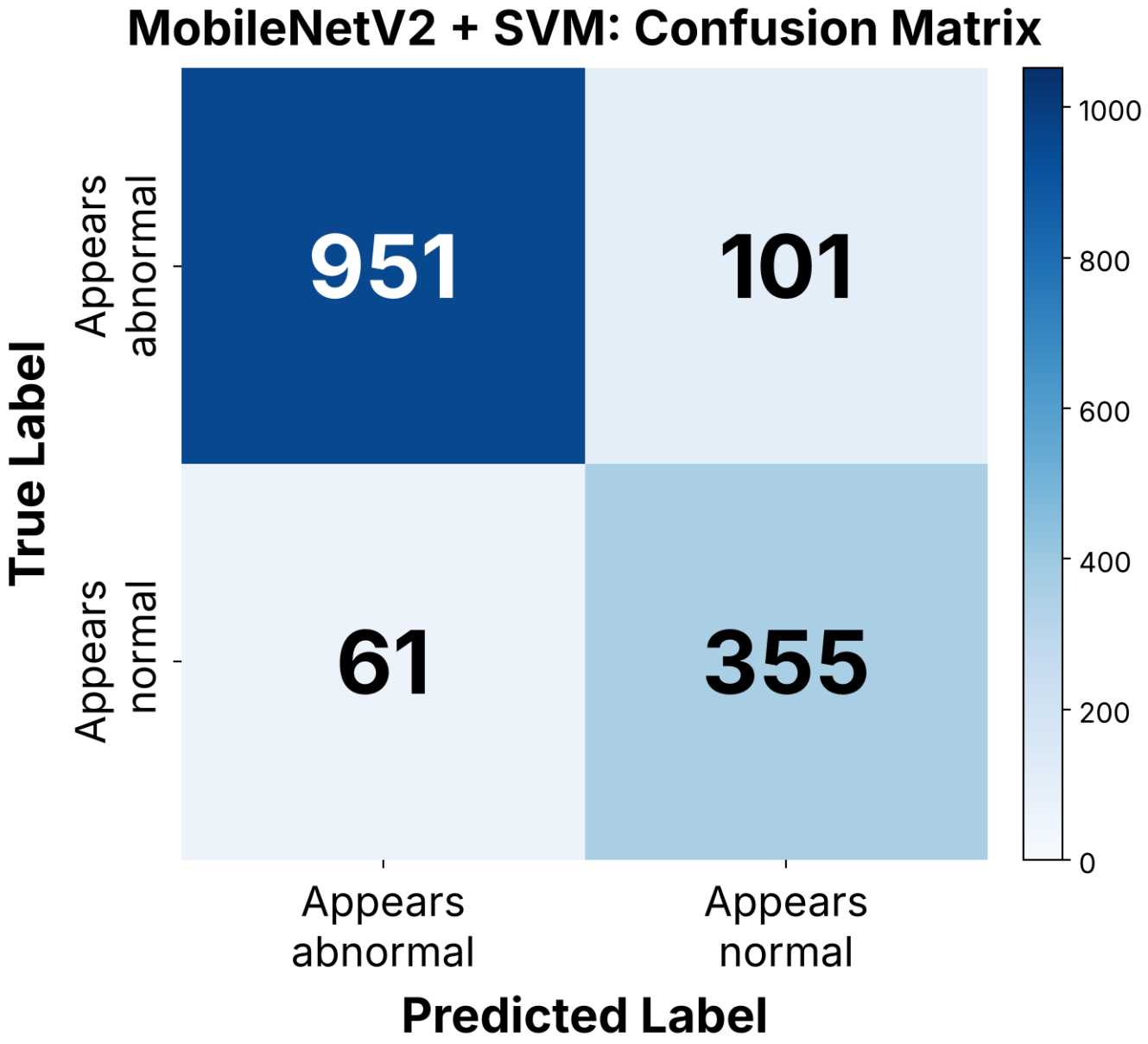
Accuracy: 88.96%

Precision: 89.40%

Recall: 88.96%

F1-score: 89.11%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

25. MiT-B2 + MobileNetV2 + KNN

Source notebook: MIT more epochs.ipynb (inside MIT + MobileNet.zip)

Accuracy: 88.96%

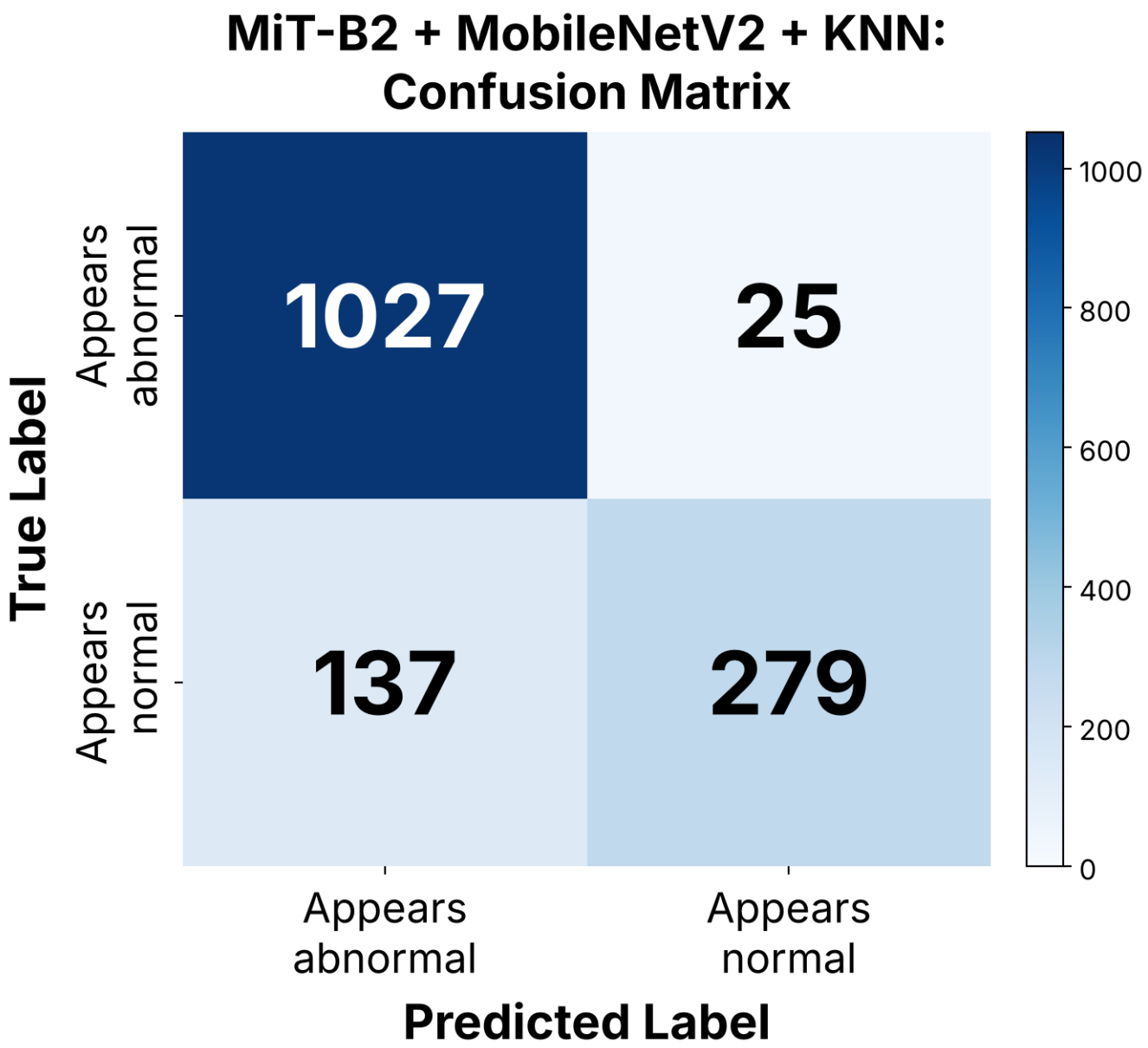
Precision: 89.24%

Recall: 88.96%

F1-score: 88.39%

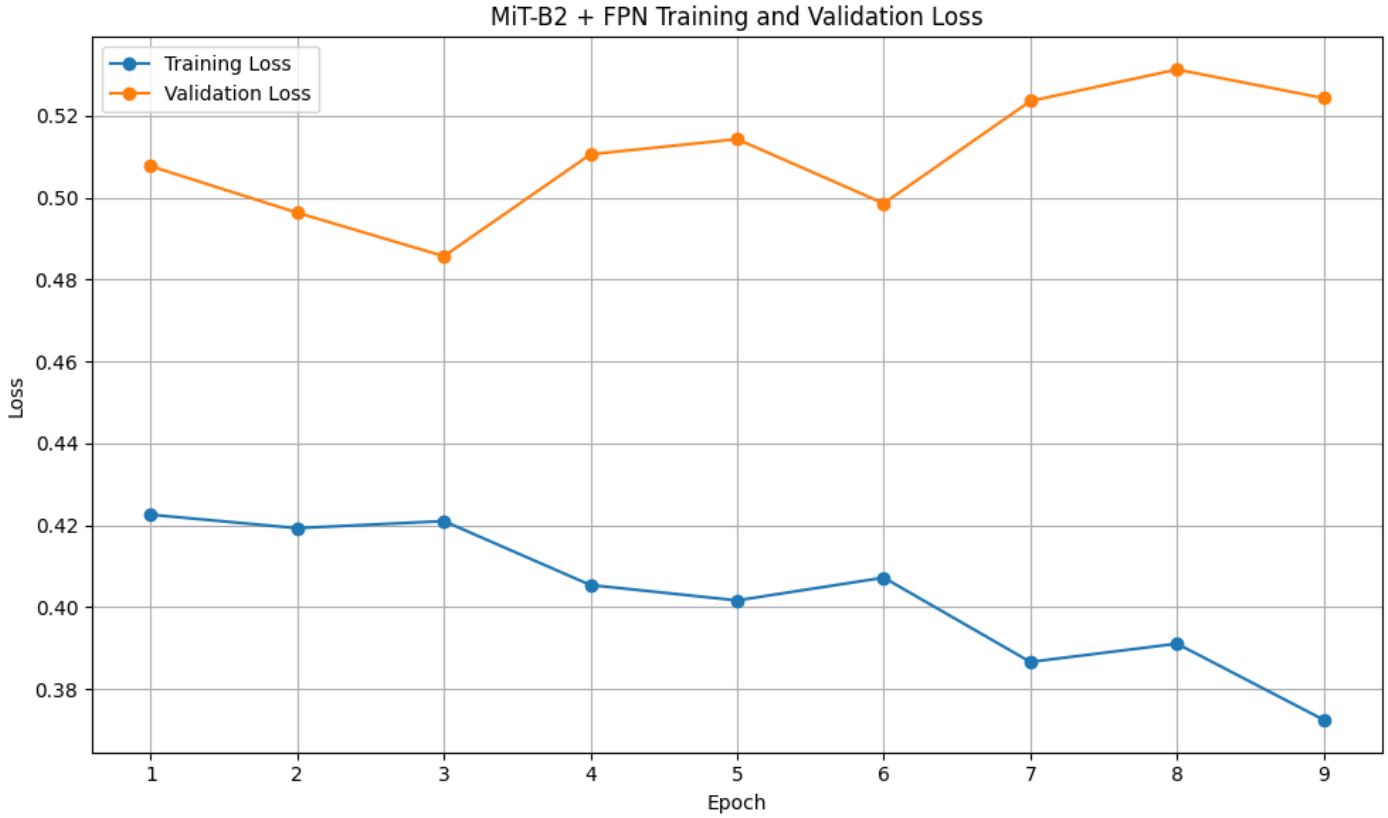
Same pair also run elsewhere with lower accuracy: 88.49% (not shown).

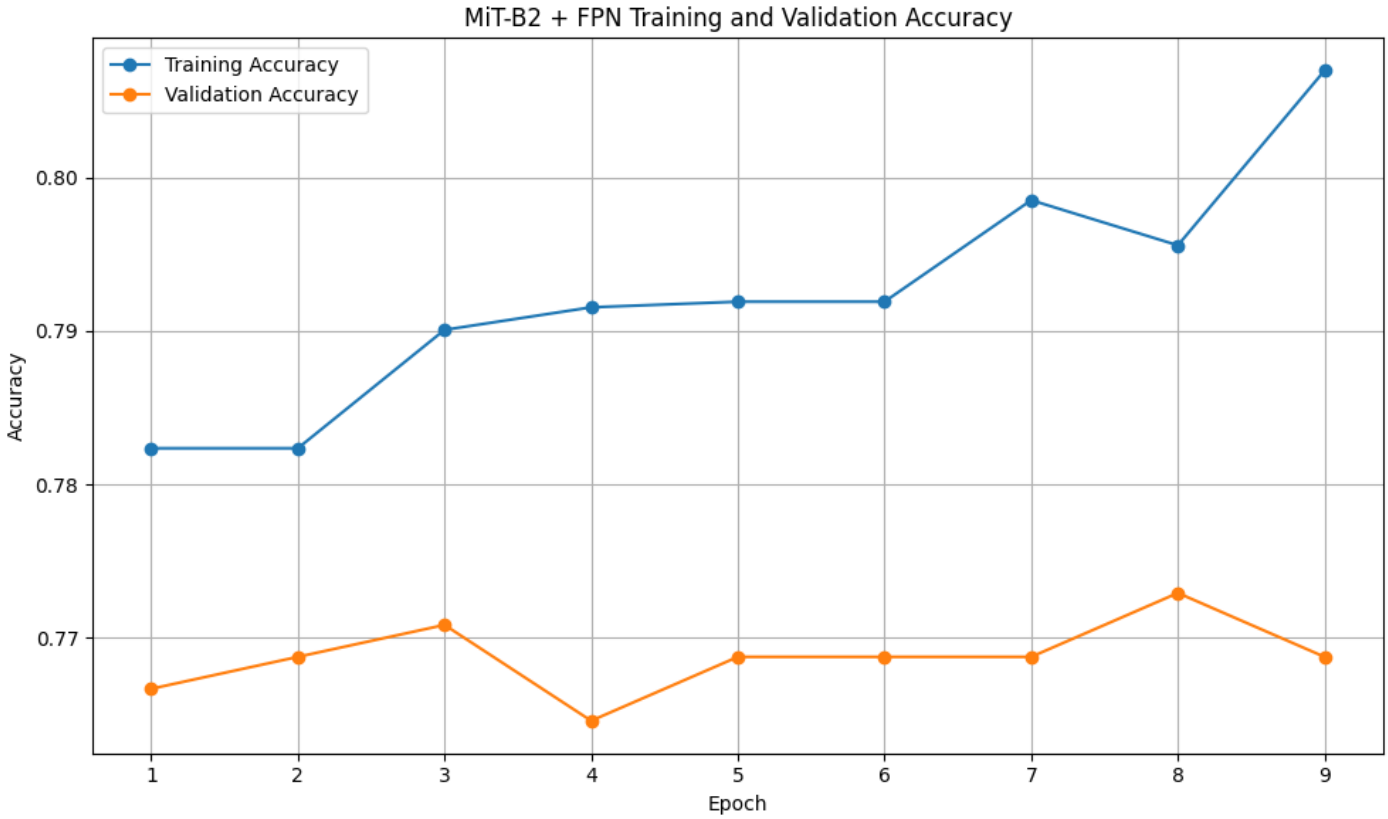
Confusion Matrix



25. MiT-B2 + MobileNetV2 + KNN — graphs (1/2)

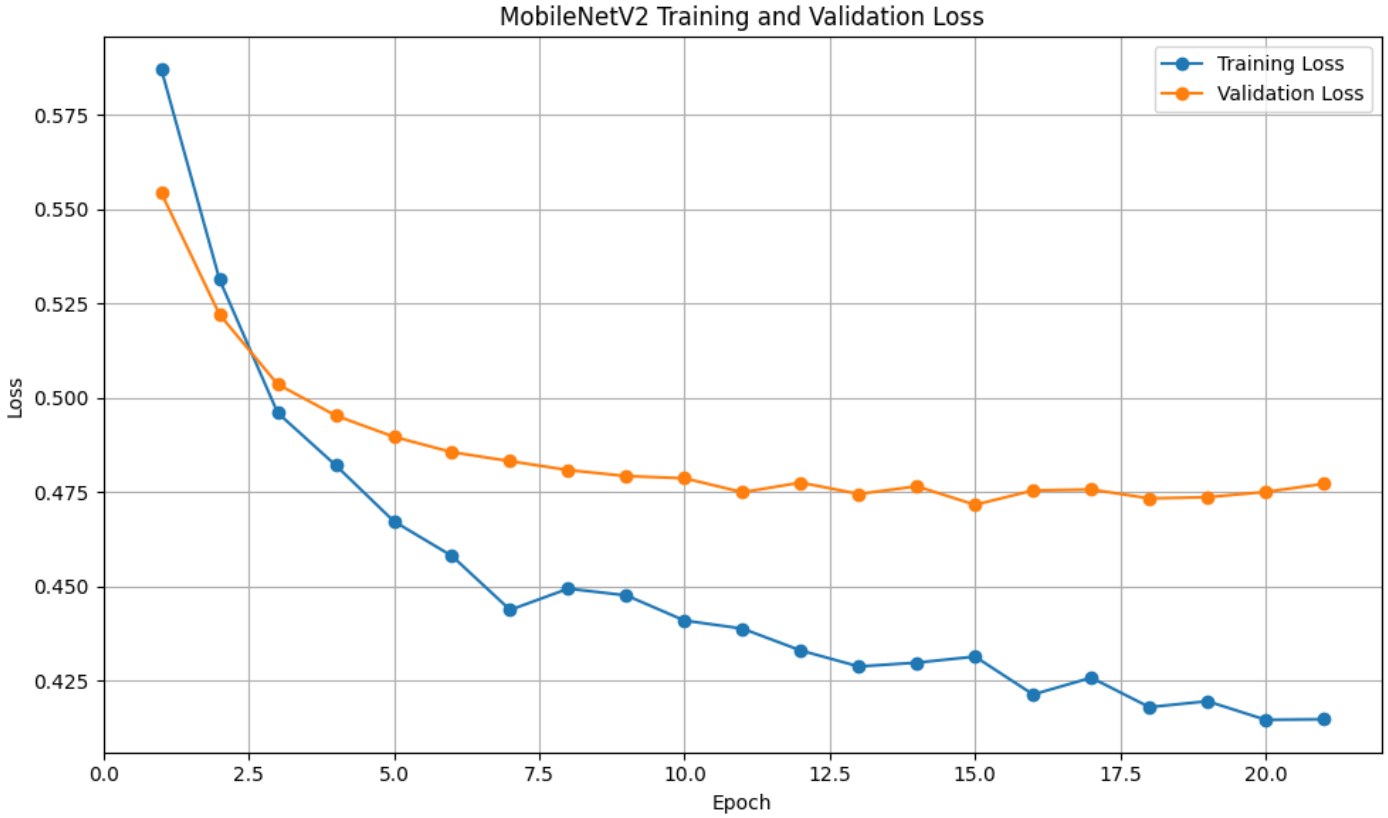
Training / Validation Graphs

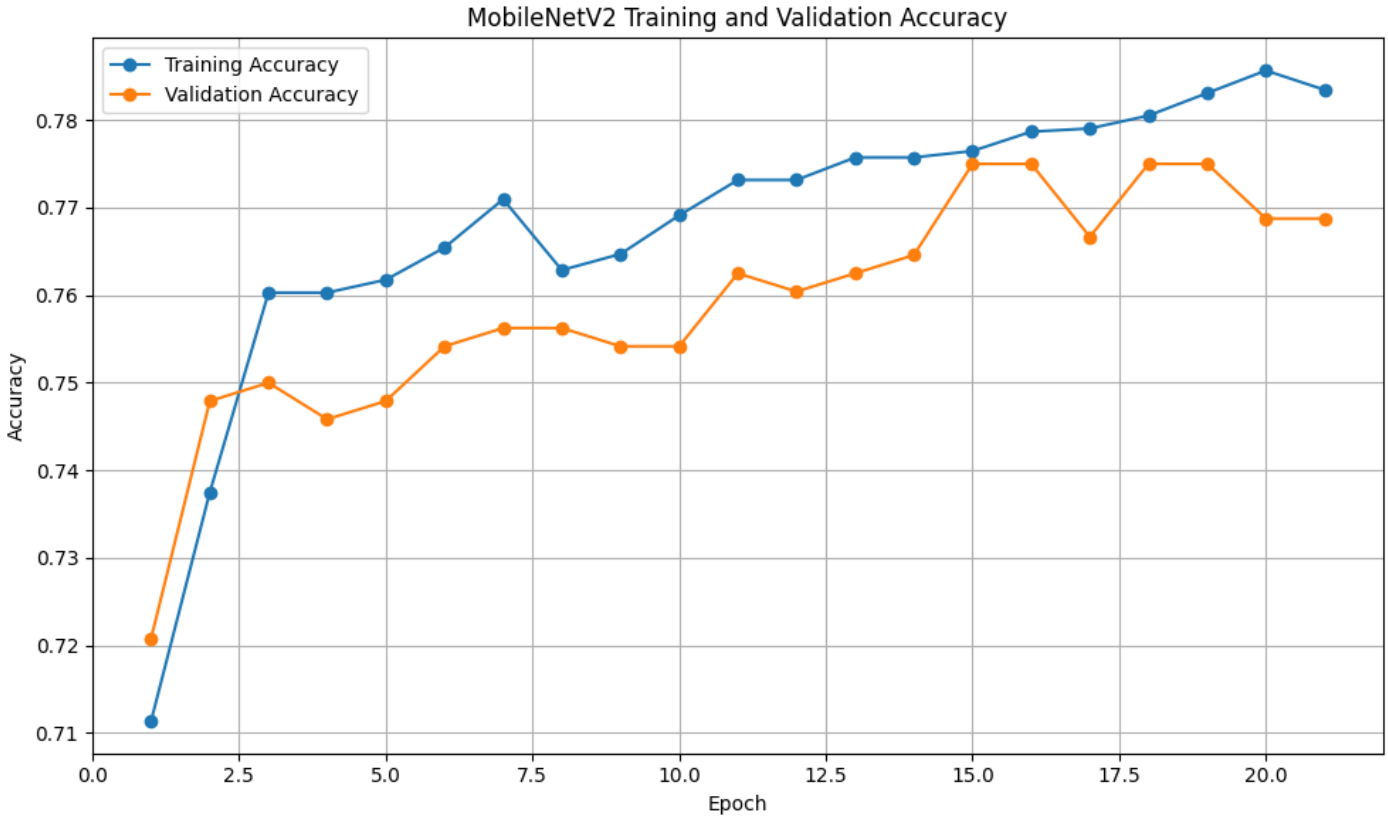




25. MiT-B2 + MobileNetV2 + KNN — graphs (2/2)

Training / Validation Graphs





26. ResNet18 + KNN

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

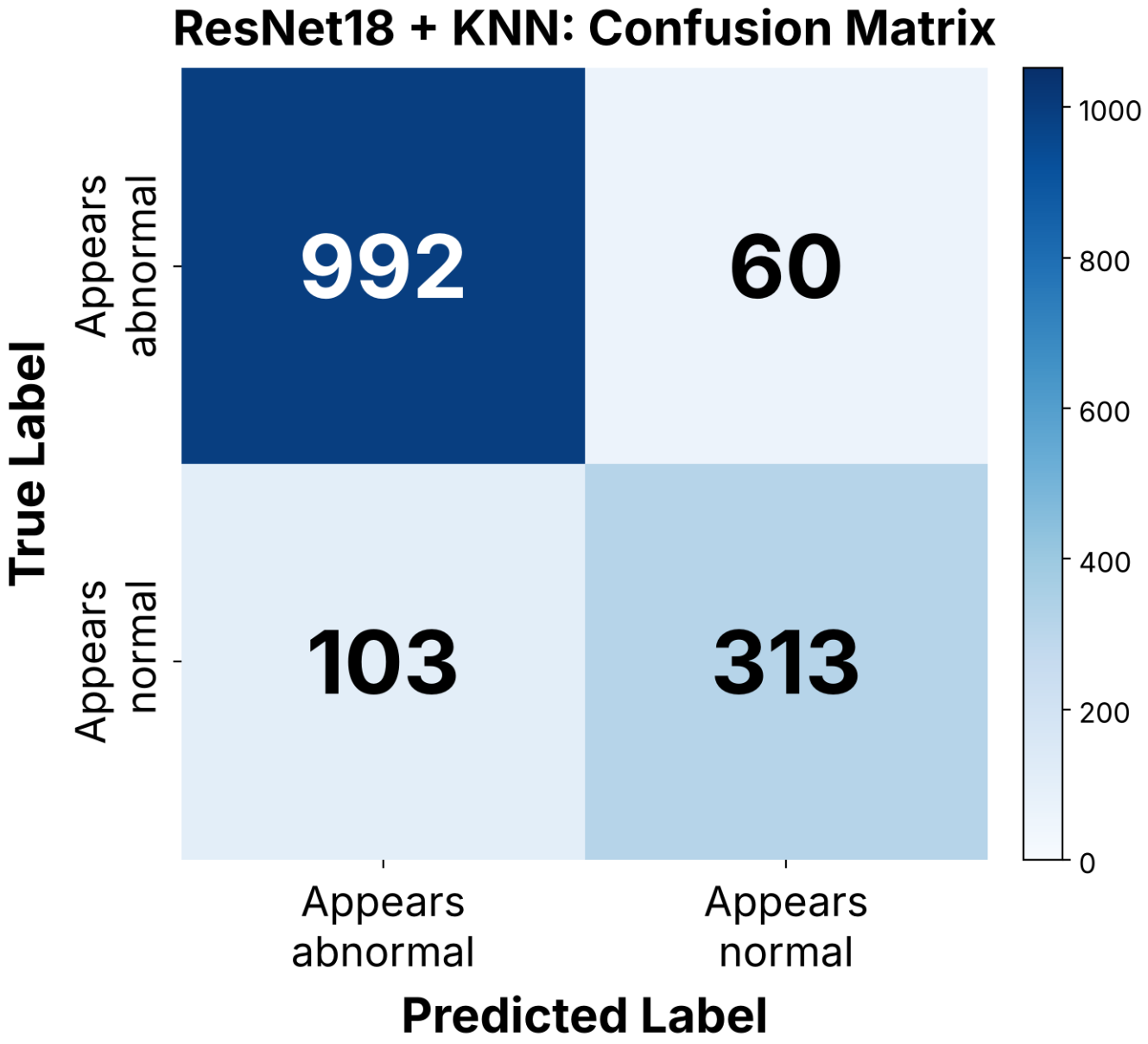
Accuracy: 88.90%

Precision: 88.70%

Recall: 88.90%

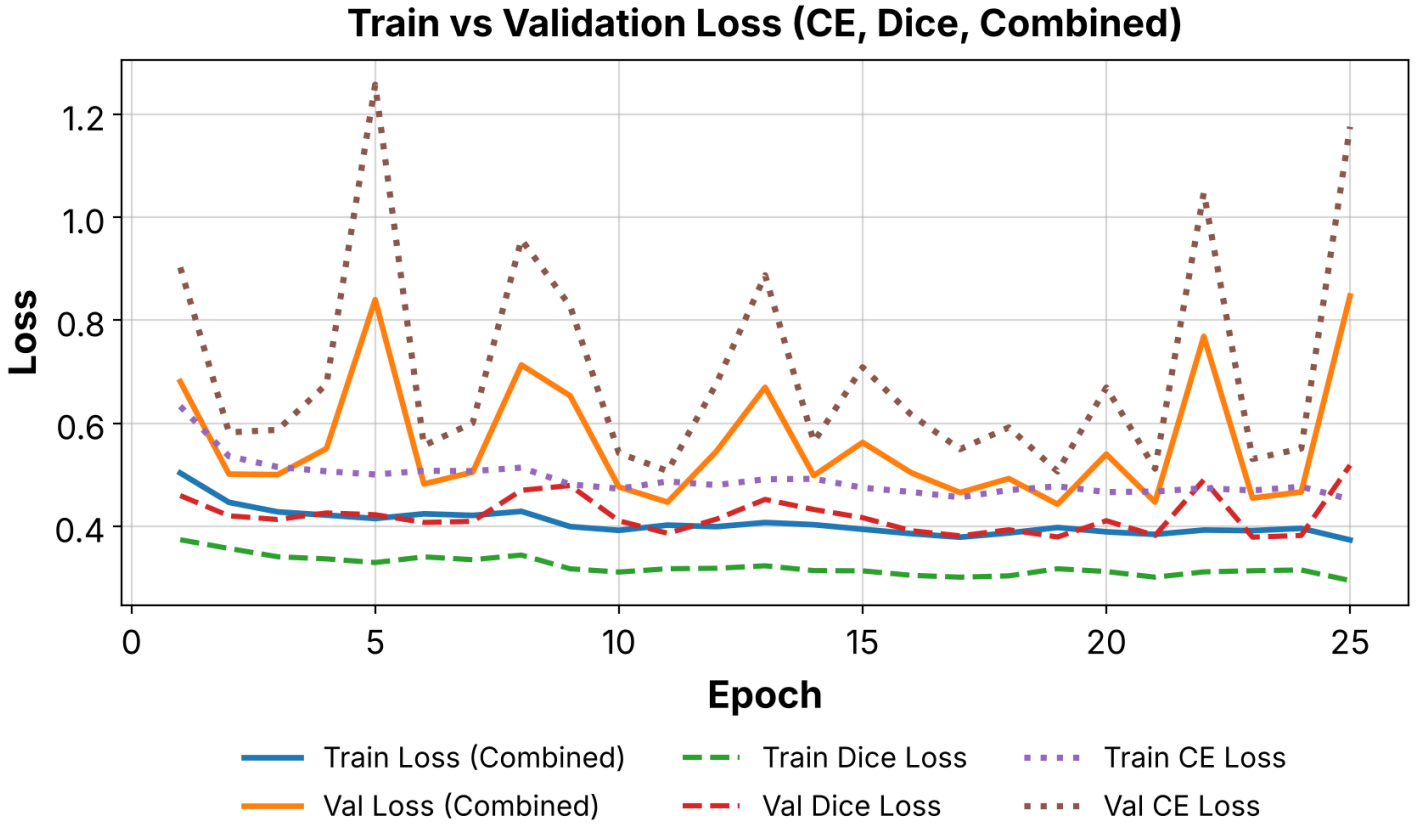
F1-score: 88.71%

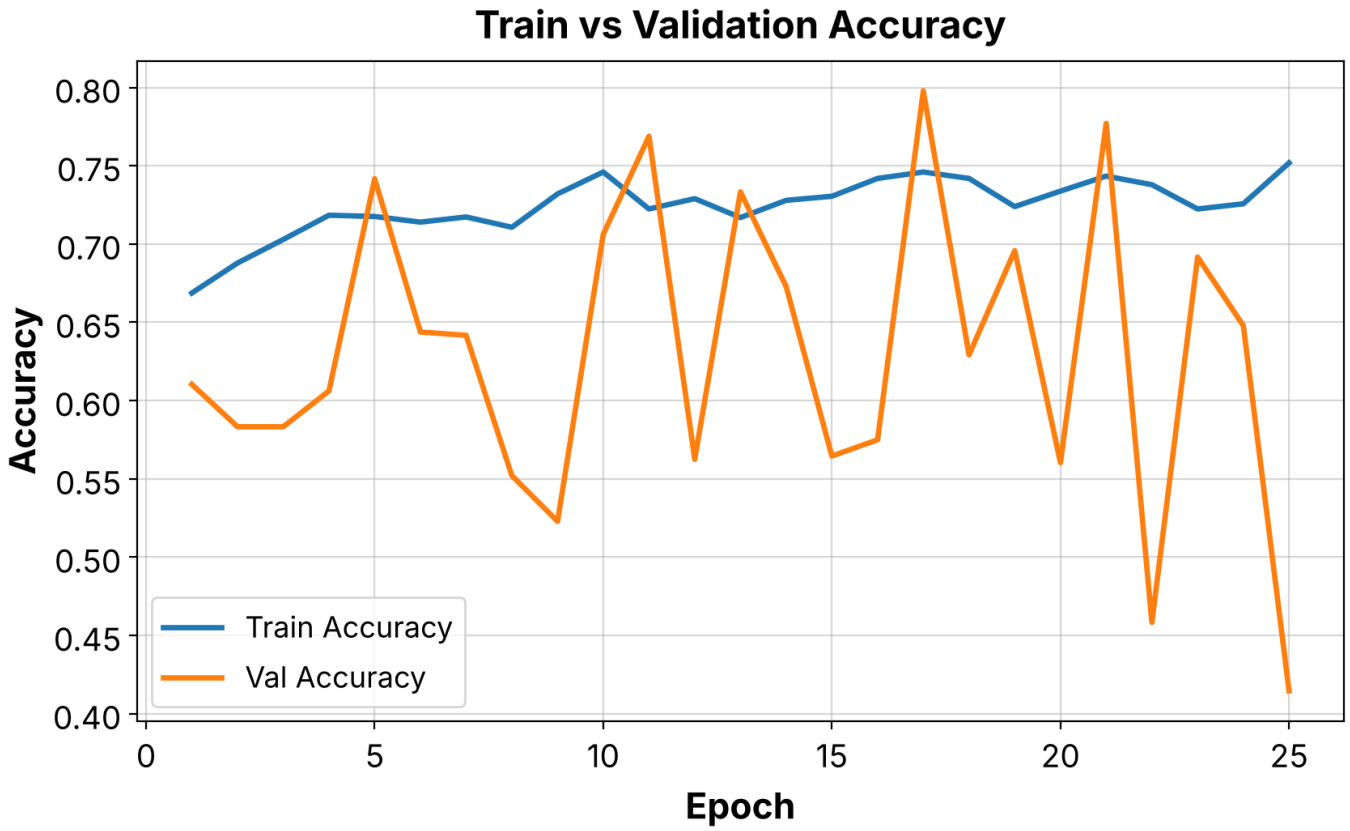
Confusion Matrix



26. ResNet18 + KNN — graphs

Training / Validation Graphs





27. ResNet18 + XGBoost

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

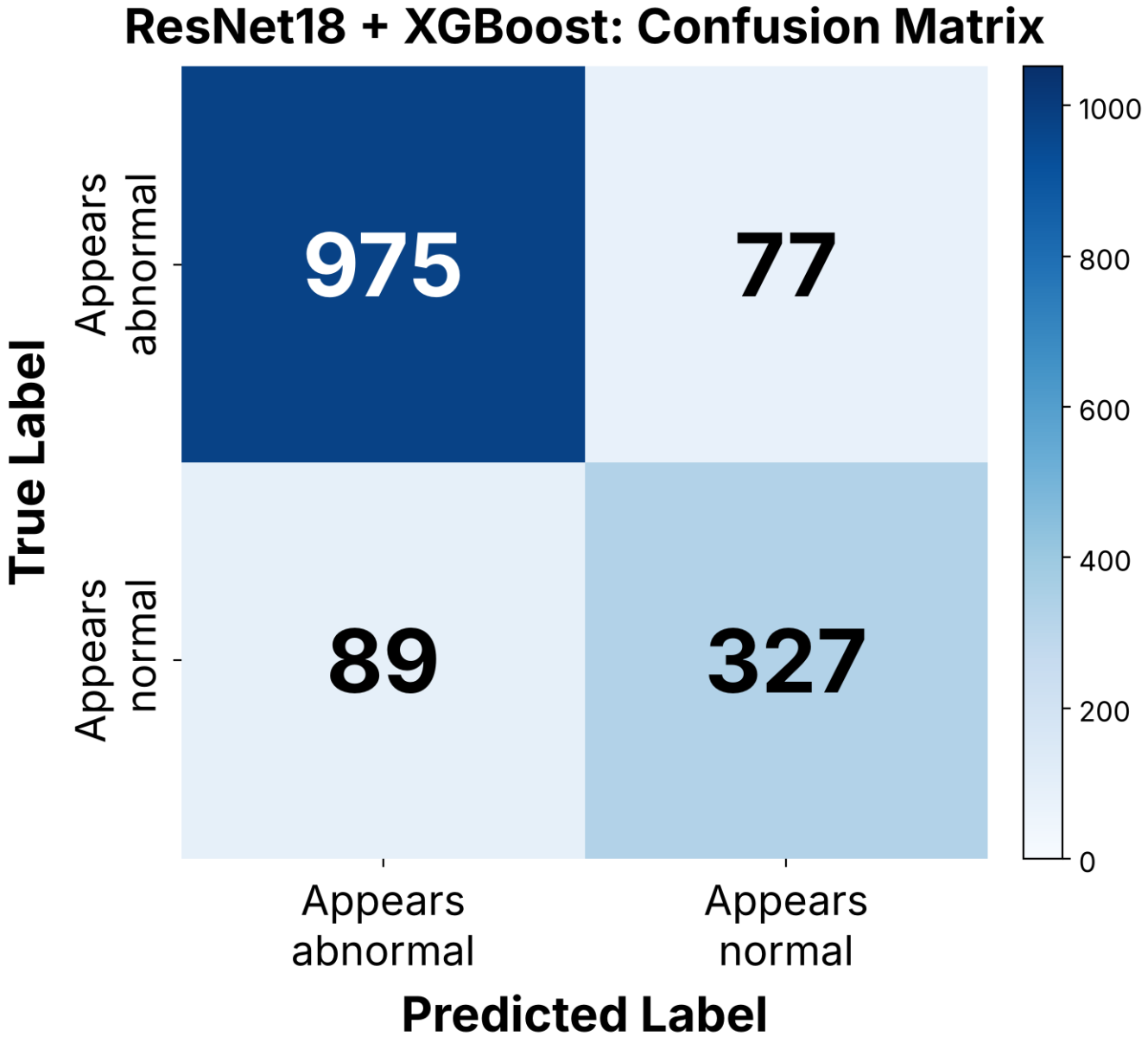
Accuracy: 88.69%

Precision: 88.60%

Recall: 88.69%

F1-score: 88.64%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

28. ResNet18 — Direct (fine-tuned CNN)

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

Accuracy: 88.42%

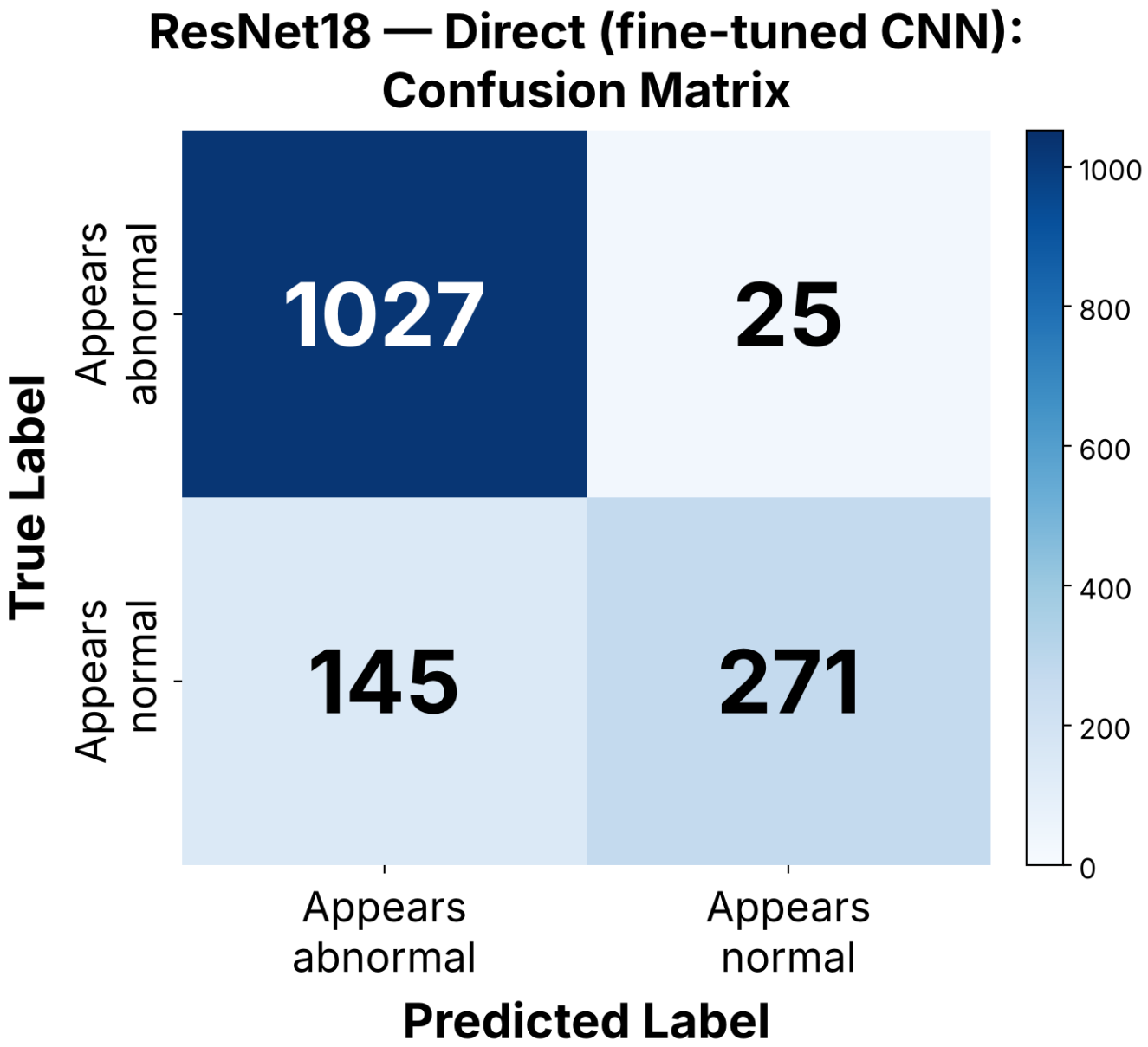
Precision: 88.74%

Recall: 88.42%

F1-score: 87.76%

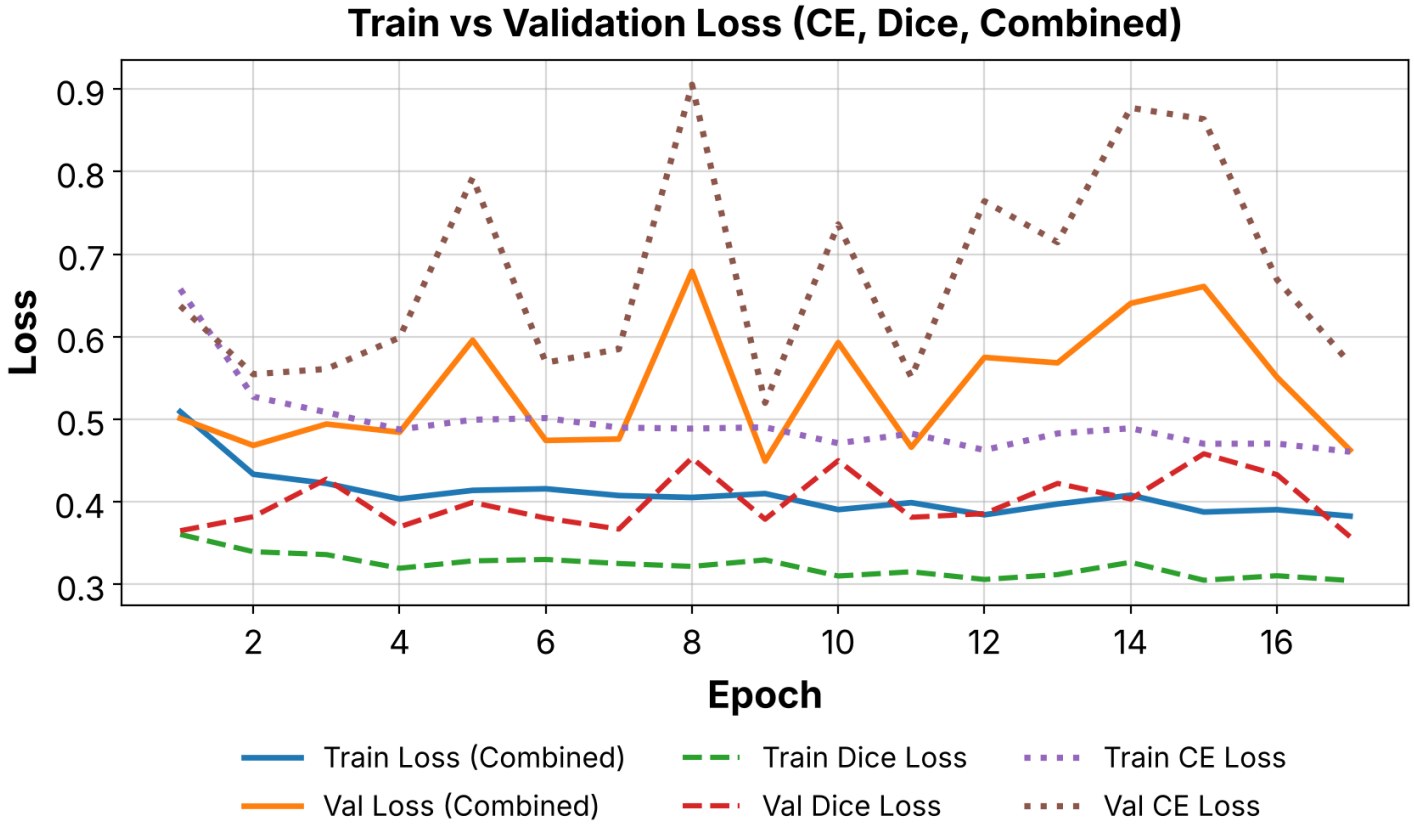
Same pair also run elsewhere with lower accuracy: 88.28%, 86.10%, 83.45%, 82.02% (not shown).

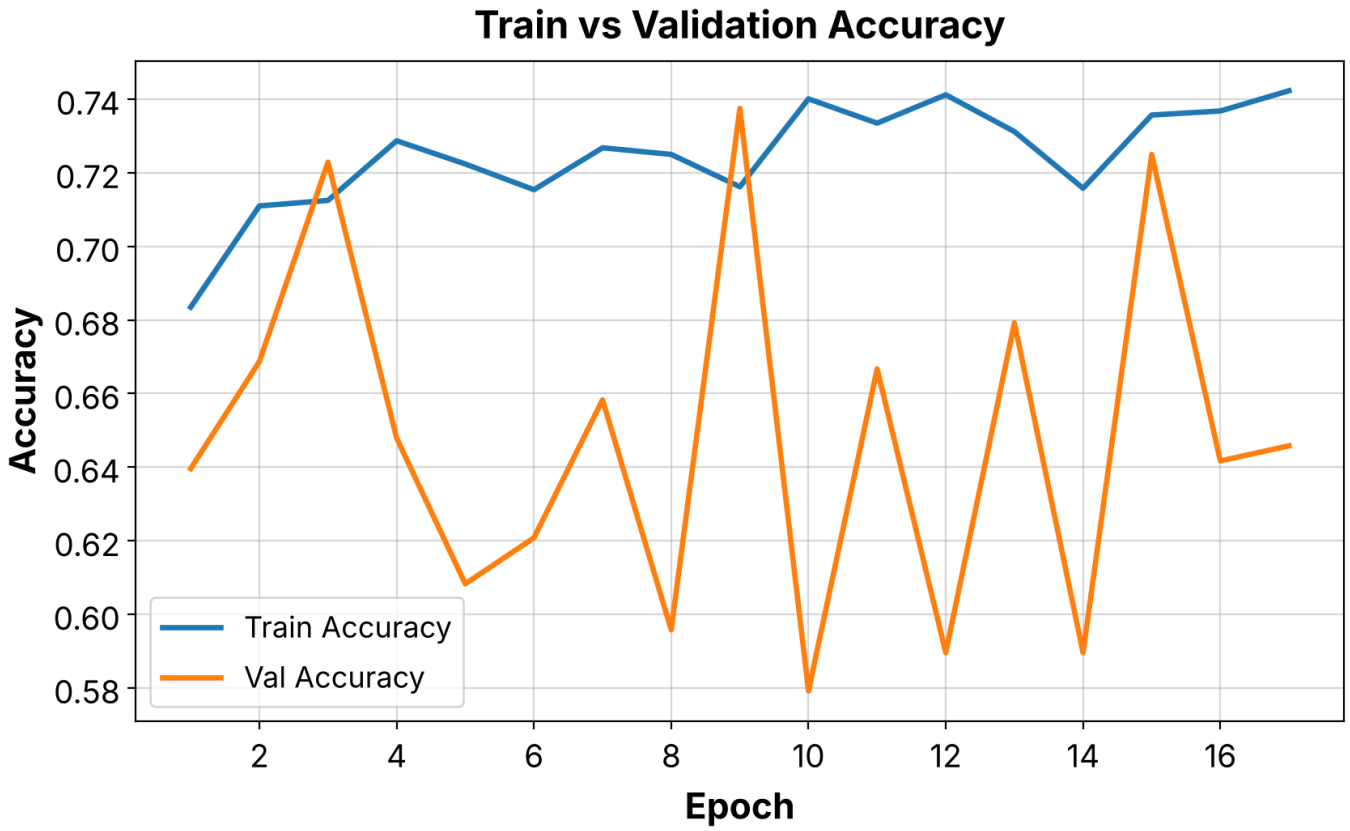
Confusion Matrix



28. ResNet18 — Direct (fine-tuned CNN) — graphs

Training / Validation Graphs





29. InceptionV3 + Random Forest

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

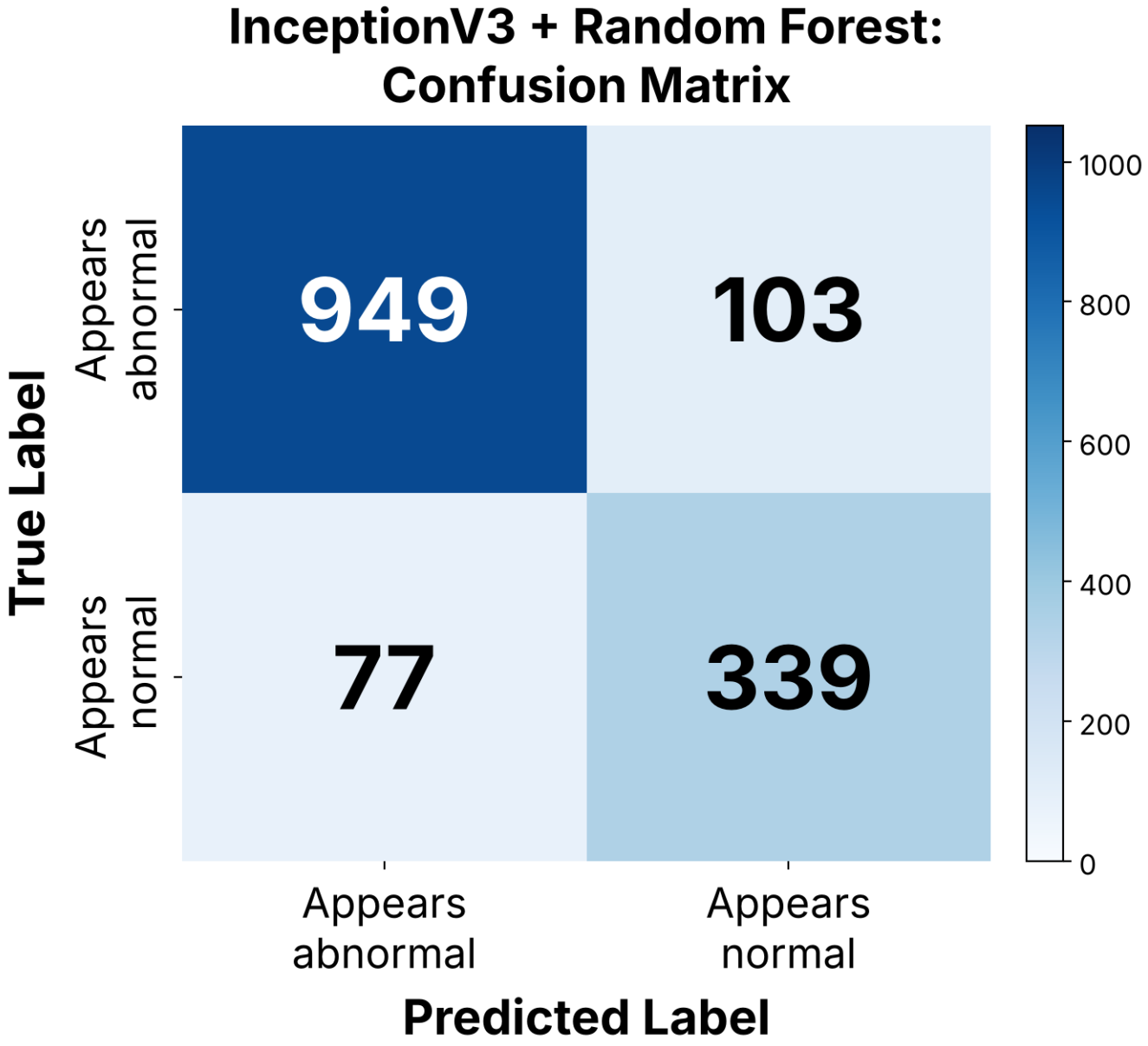
Accuracy: 87.74%

Precision: 88.02%

Recall: 87.74%

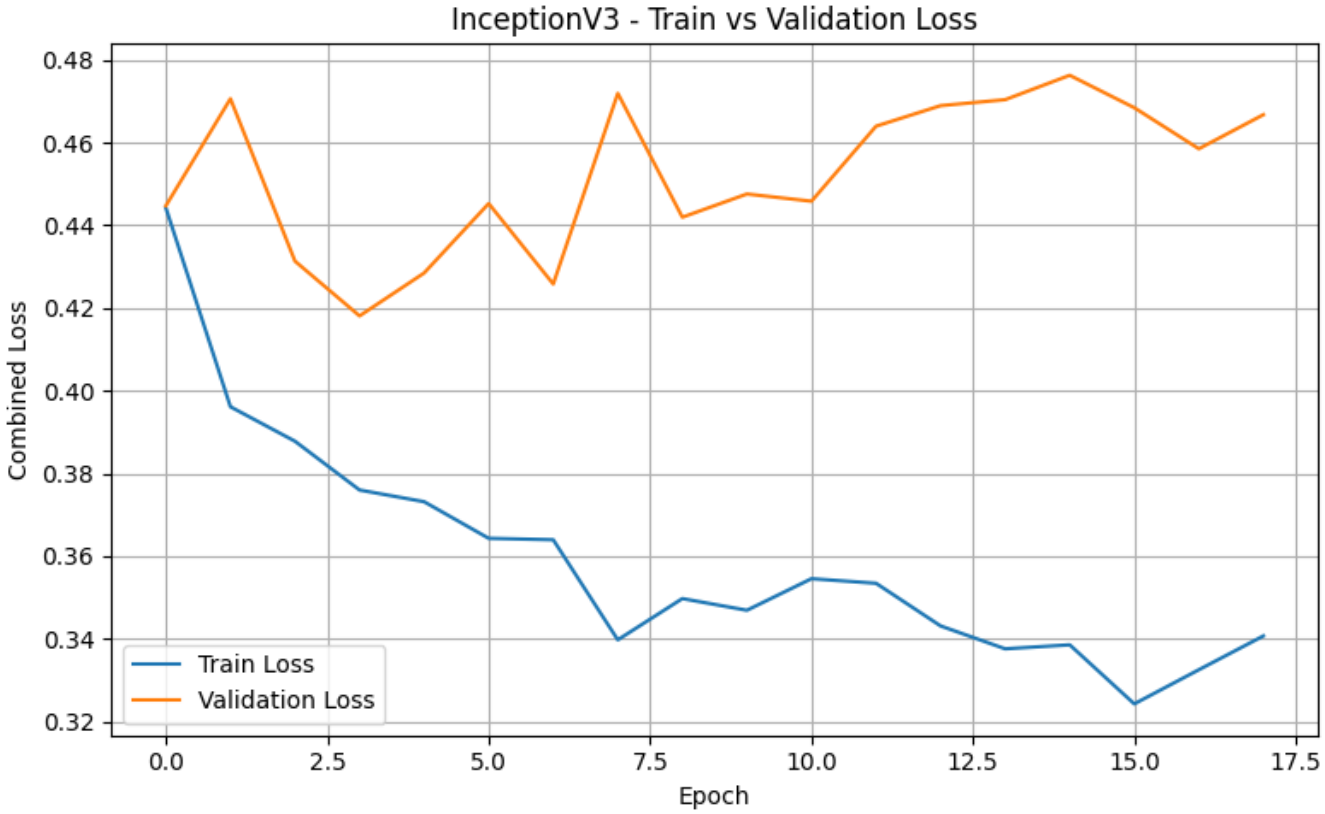
F1-score: 87.85%

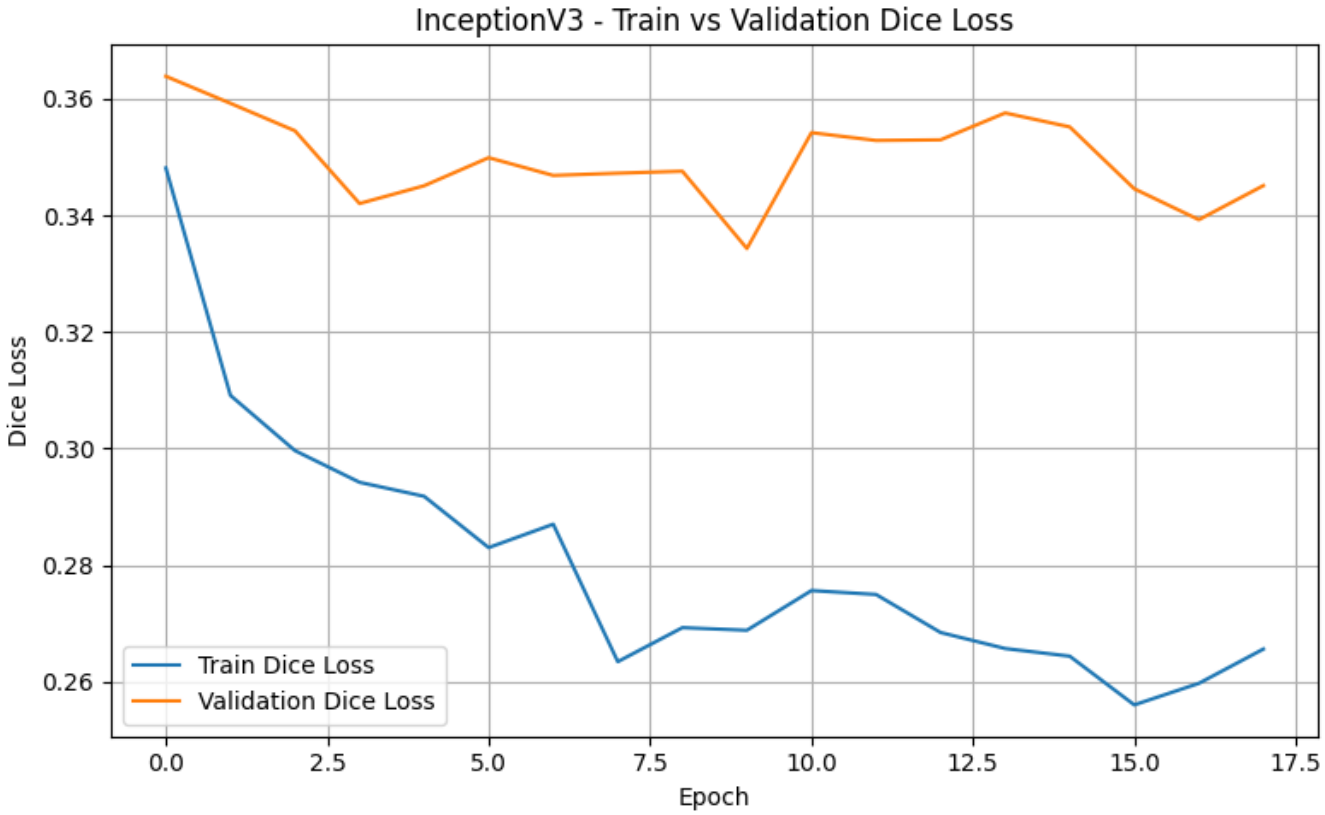
Confusion Matrix



29. InceptionV3 + Random Forest — graphs (1/2)

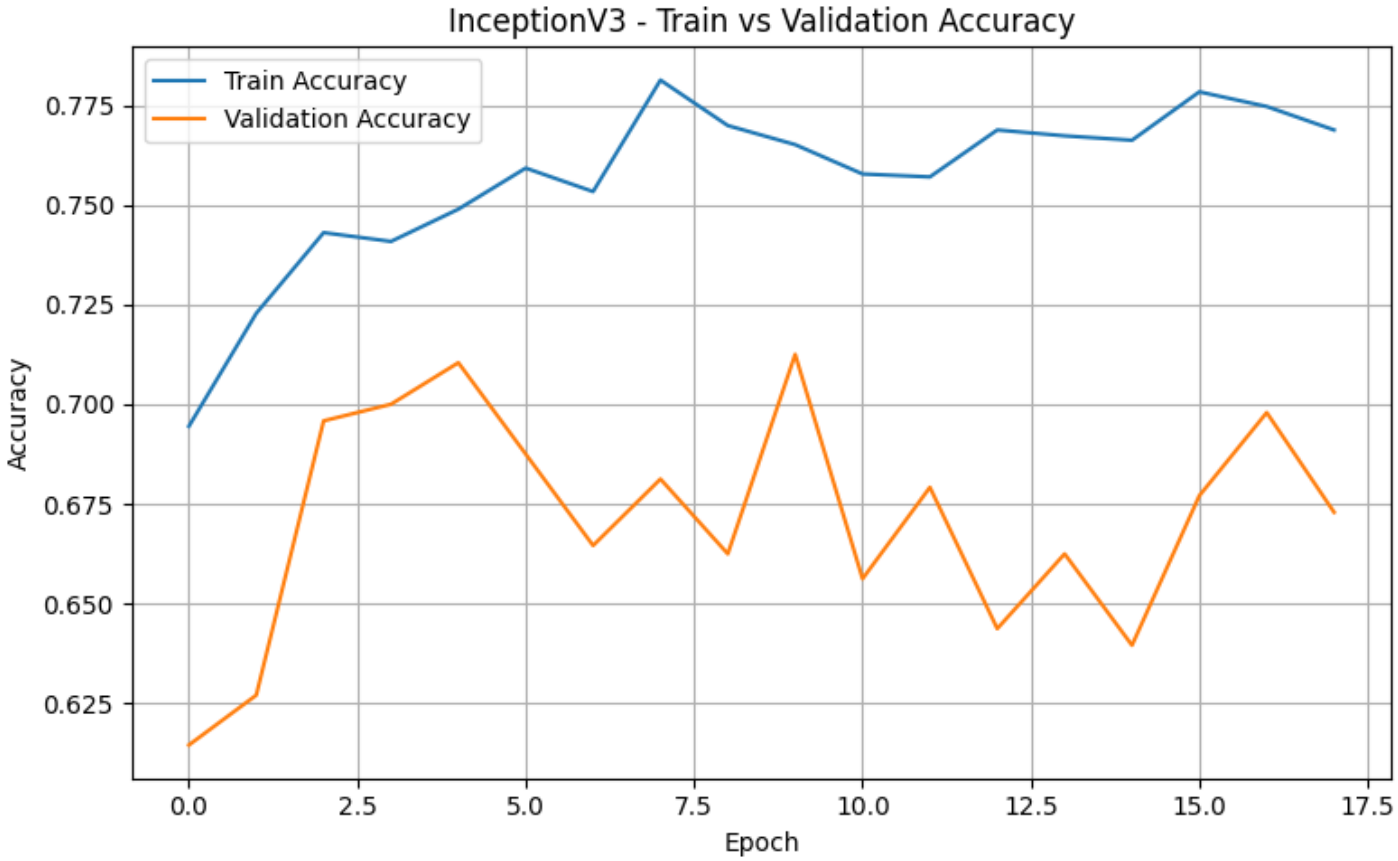
Training / Validation Graphs





29. InceptionV3 + Random Forest — graphs (2/2)

Training / Validation Graphs



30. MiT-B2 + FPN + Random Forest

Source notebook: MIT + RF and MIT only.ipynb (inside MIT + MobileNet.zip)

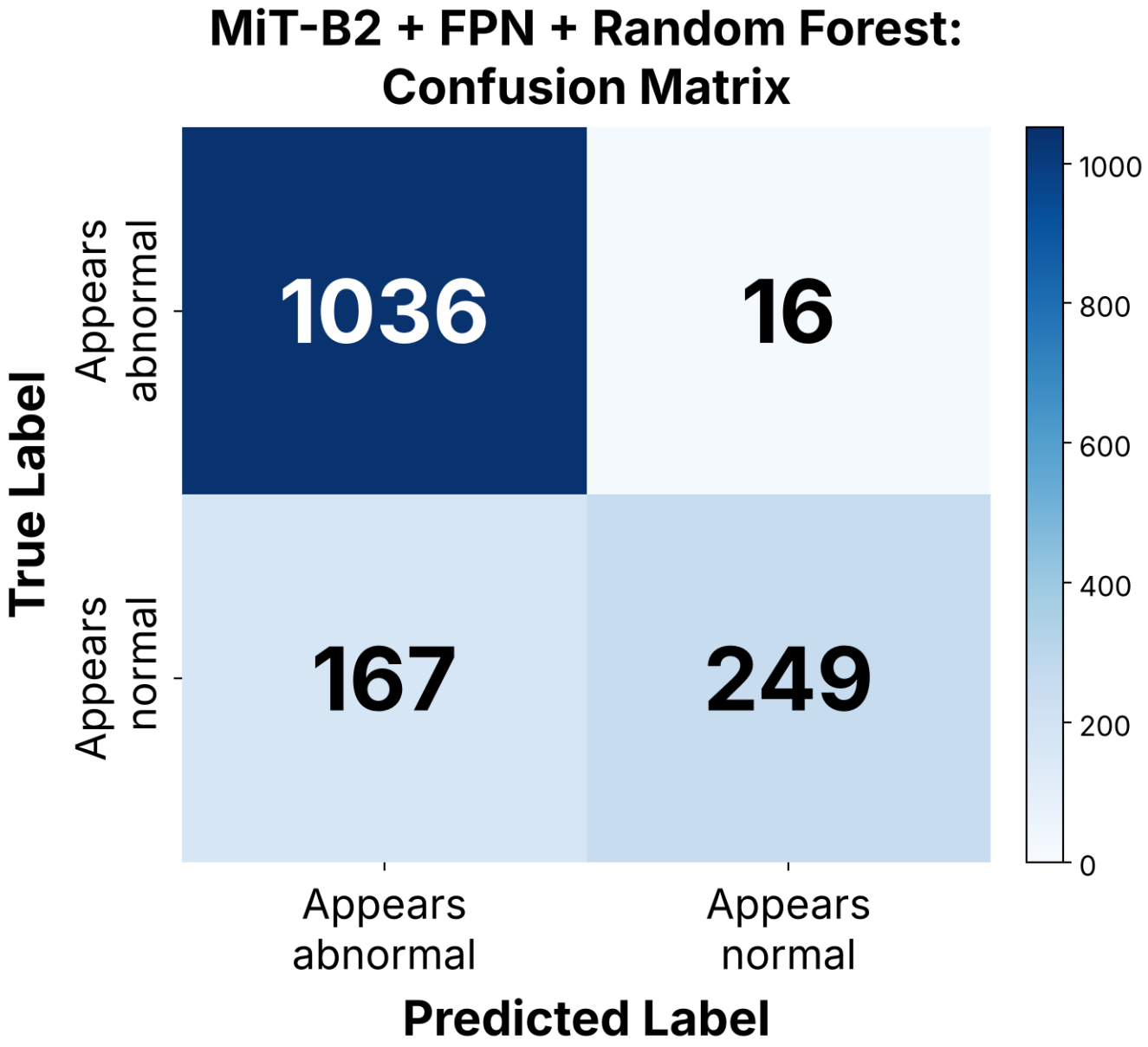
Accuracy: 87.53%

Precision: 88.34%

Recall: 87.53%

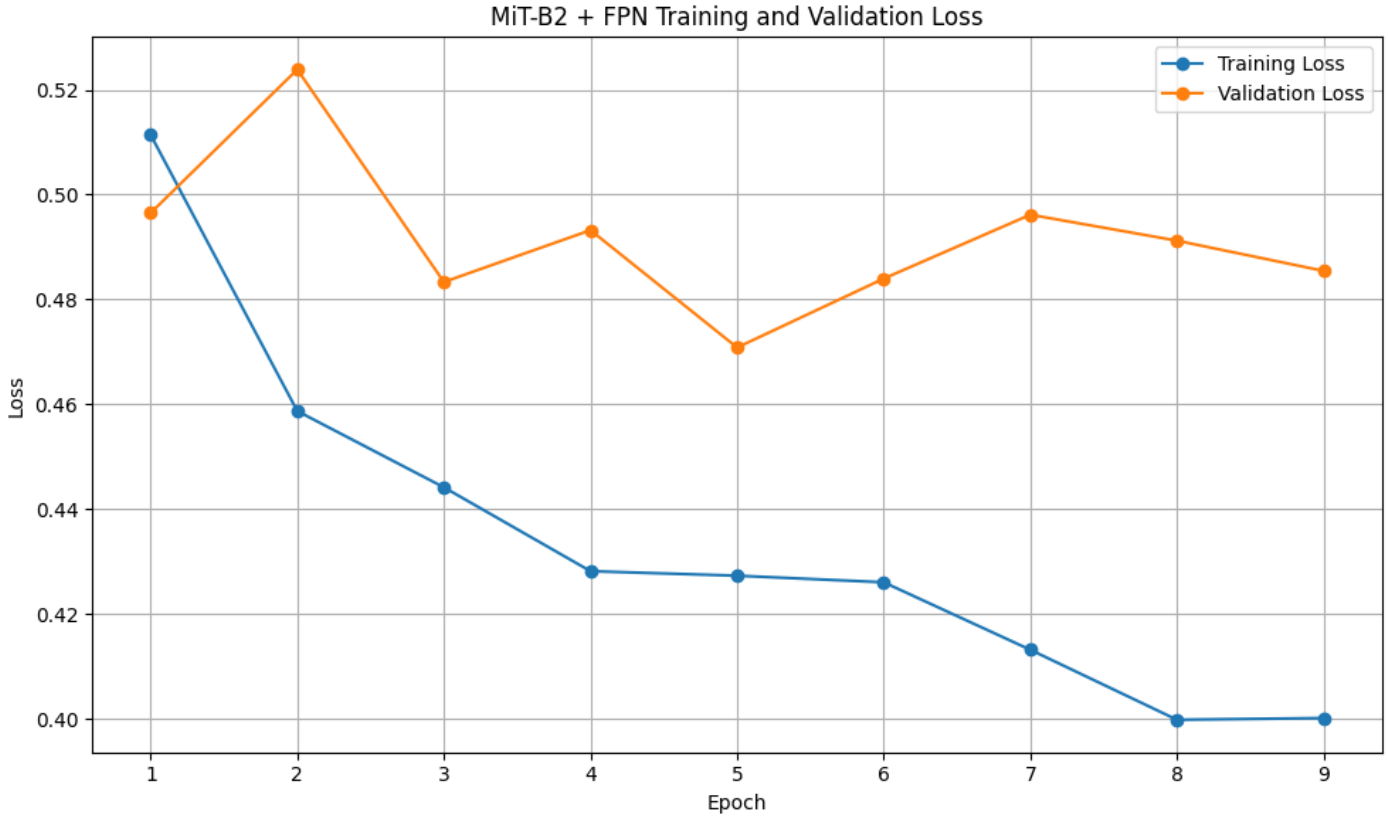
F1-score: 86.57%

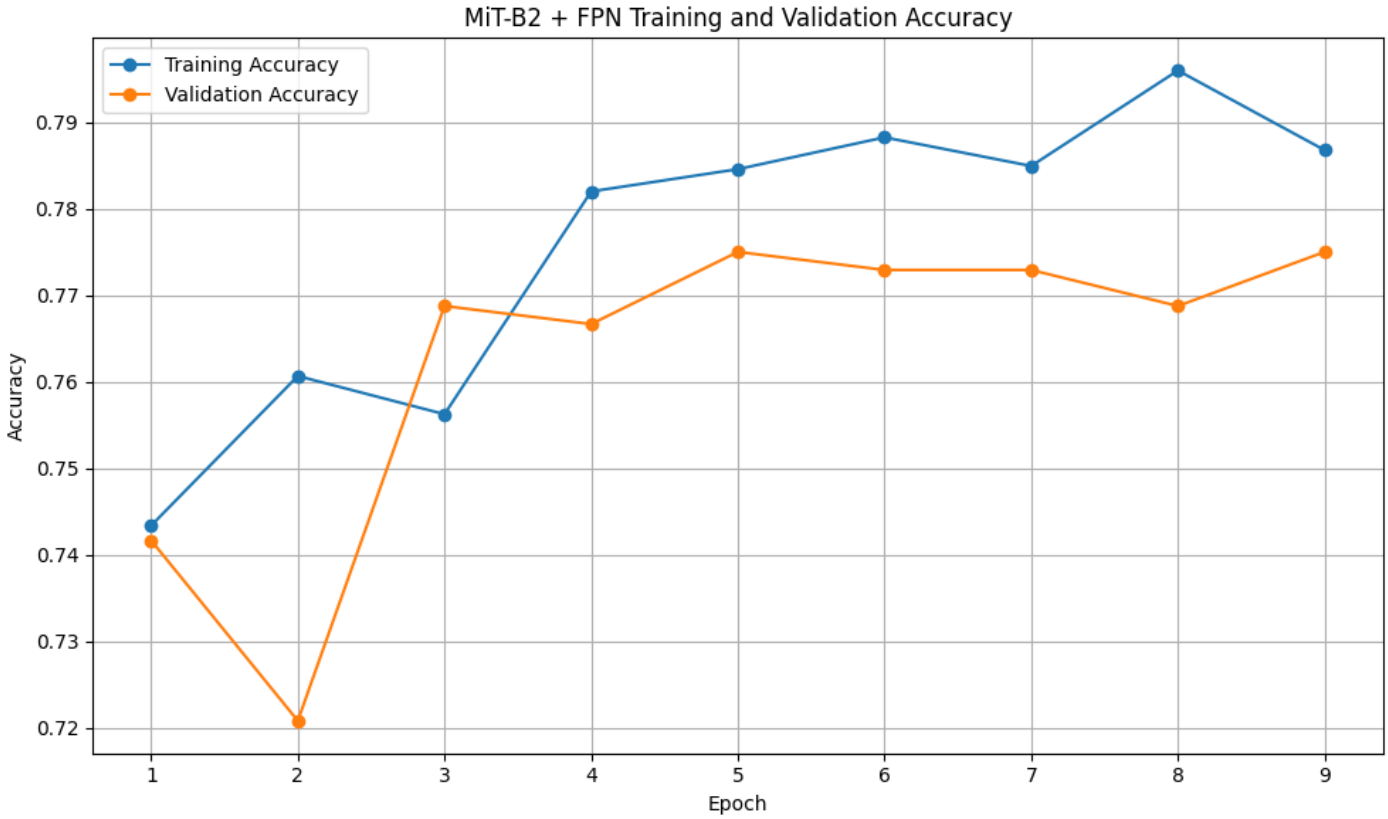
Confusion Matrix



30. MiT-B2 + FPN + Random Forest — graphs

Training / Validation Graphs





31. VGG16 + KNN

Source notebook: Vgg with ML.ipynb (inside Models_DL.zip)

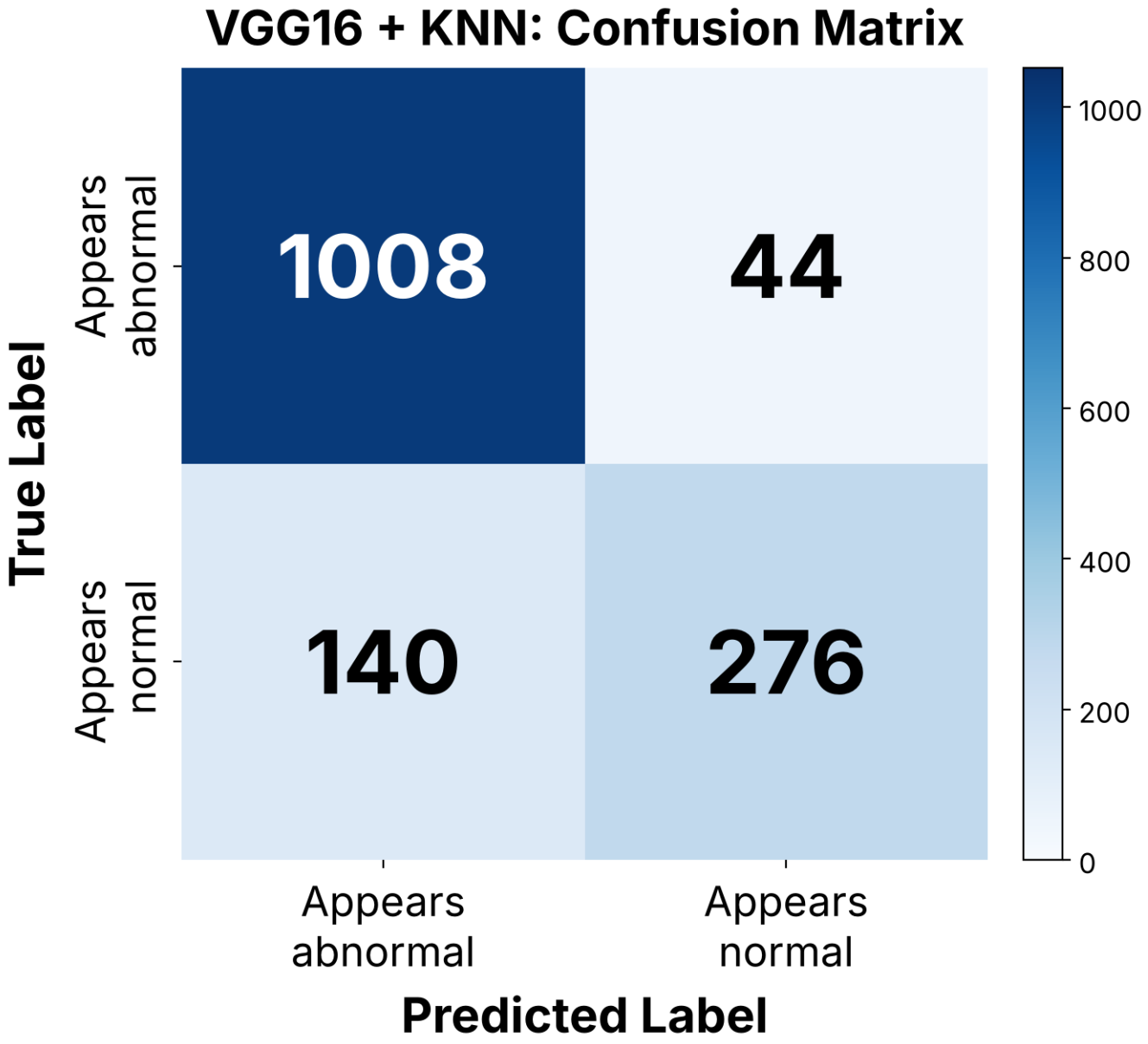
Accuracy: 87.47%

Precision: 87.36%

Recall: 87.47%

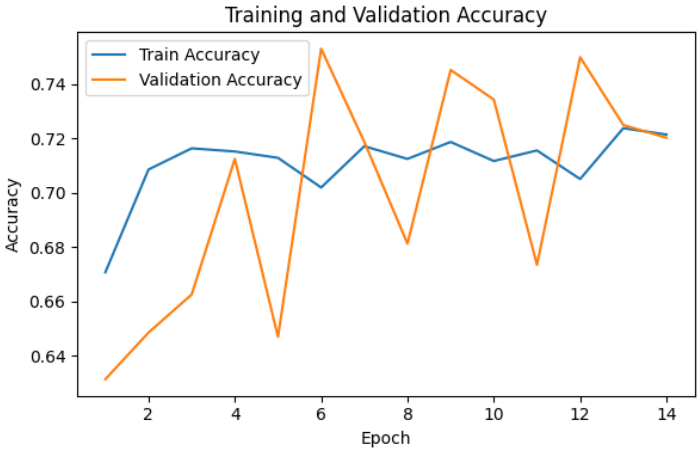
F1-score: 86.92%

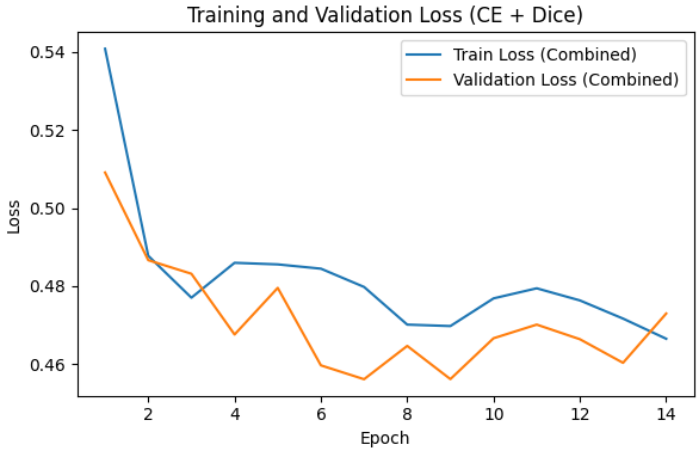
Confusion Matrix

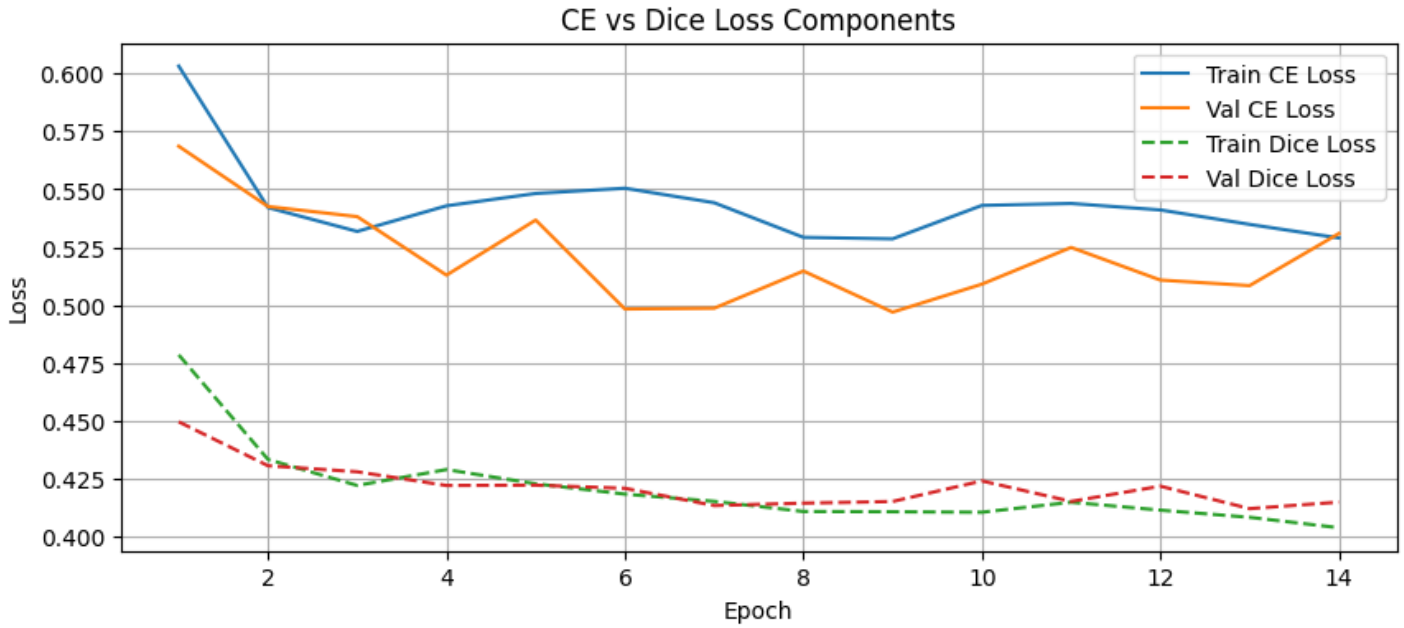


31. VGG16 + KNN — graphs

Training / Validation Graphs







32. ResNet18 + AdaBoost

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

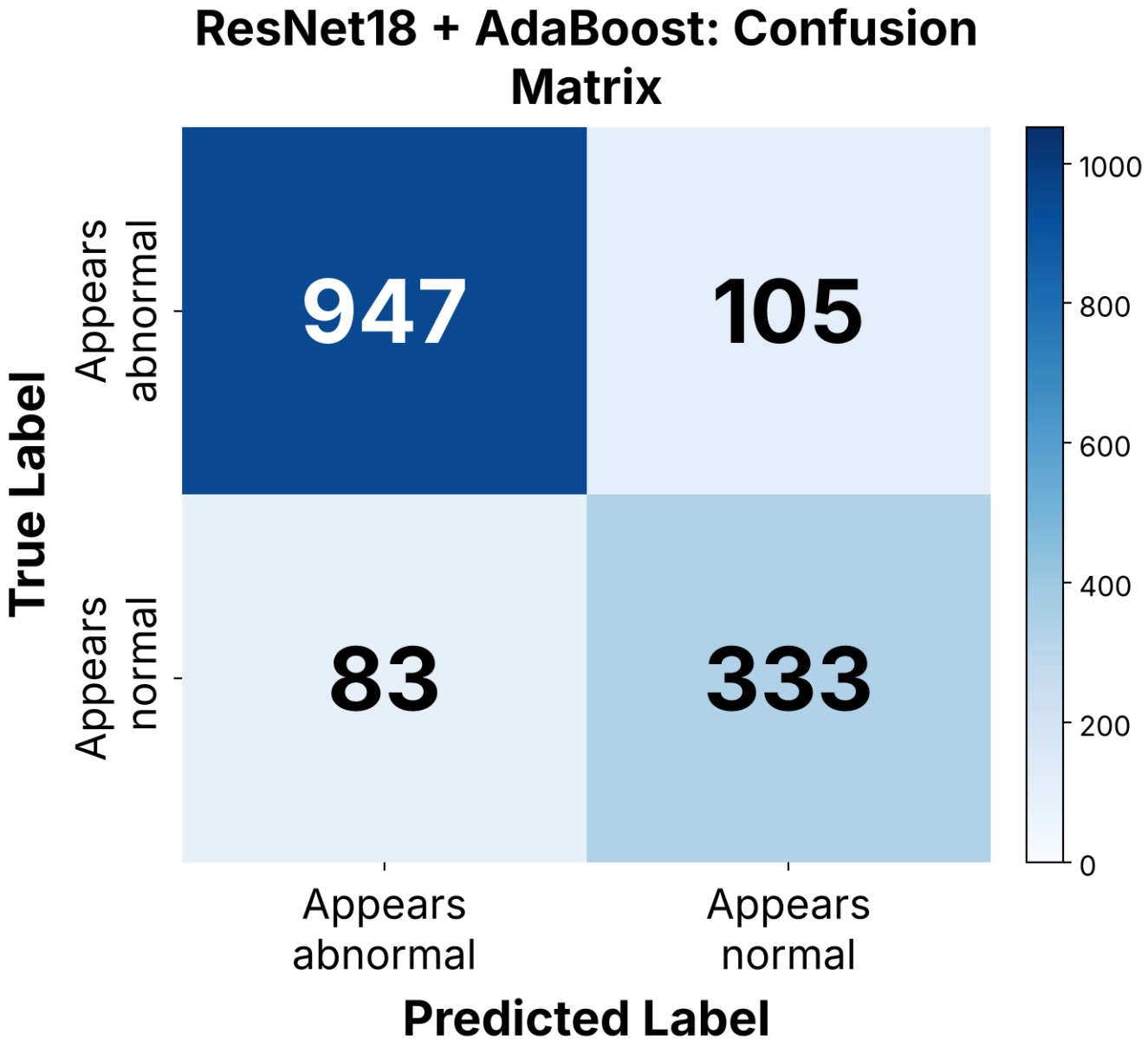
Accuracy: 87.19%

Precision: 87.43%

Recall: 87.19%

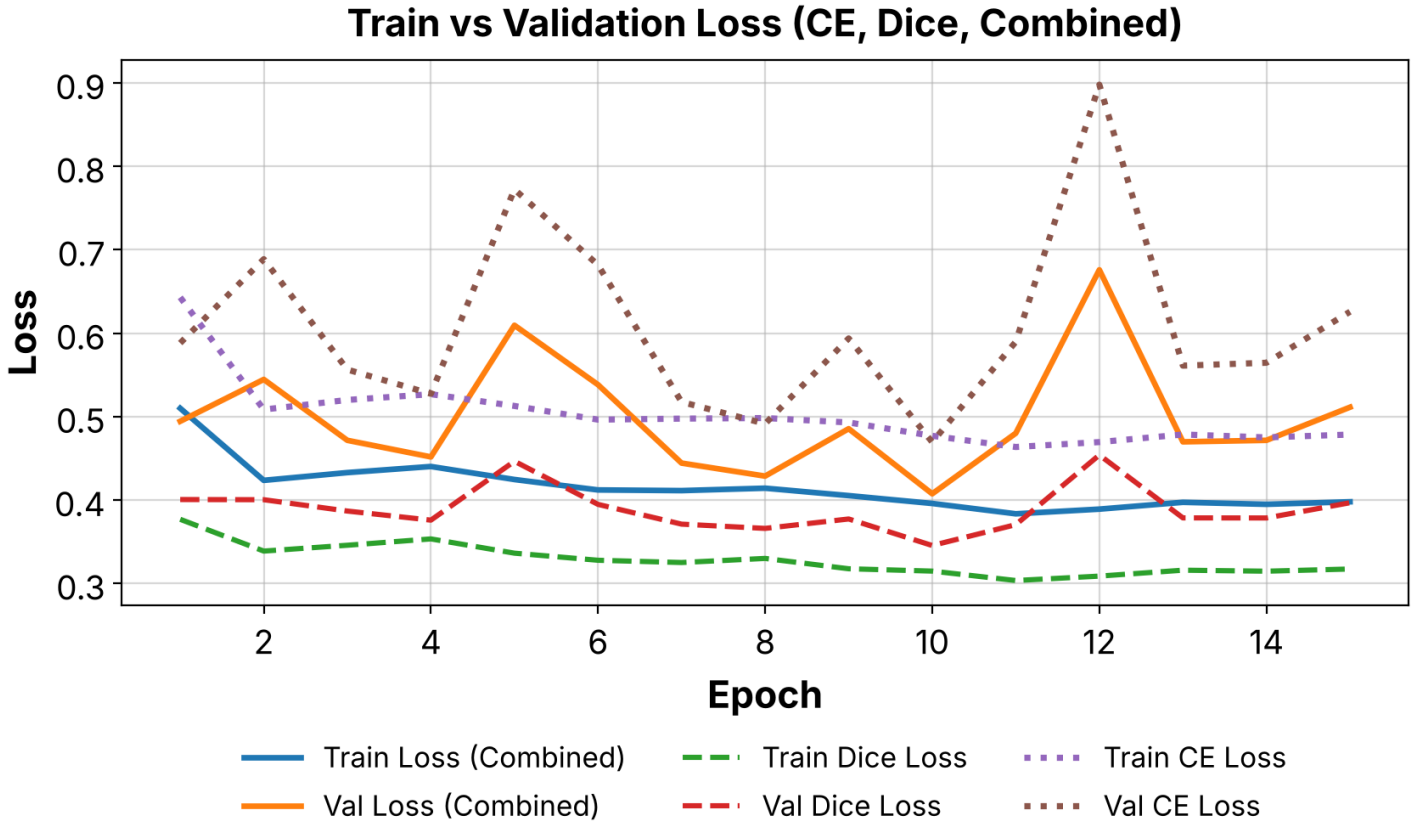
F1-score: 87.29%

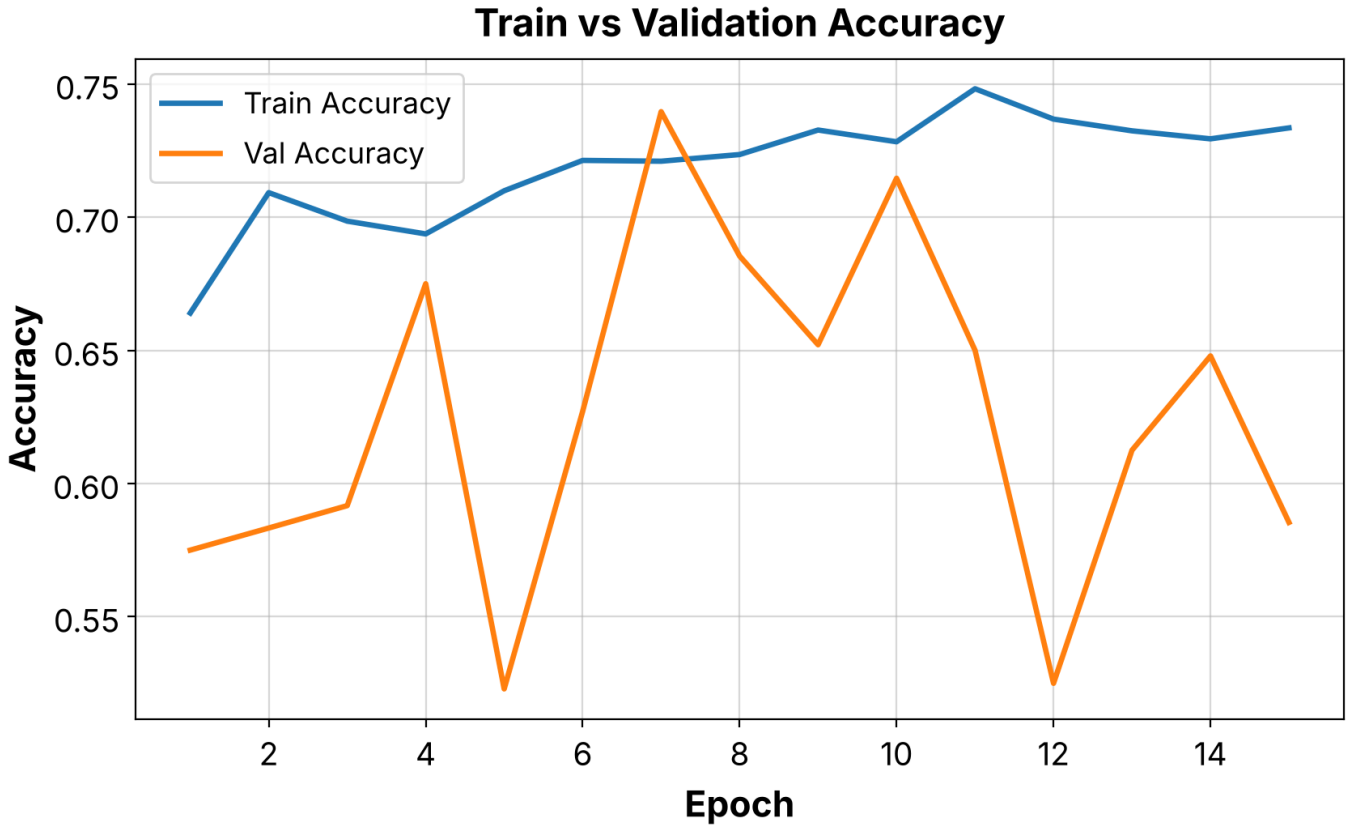
Confusion Matrix



32. ResNet18 + AdaBoost — graphs

Training / Validation Graphs





33. MobileNetV2 — Direct — 5-fold CV

Source notebook: MobileNet with Kfold.ipynb (inside Models_DL.zip)

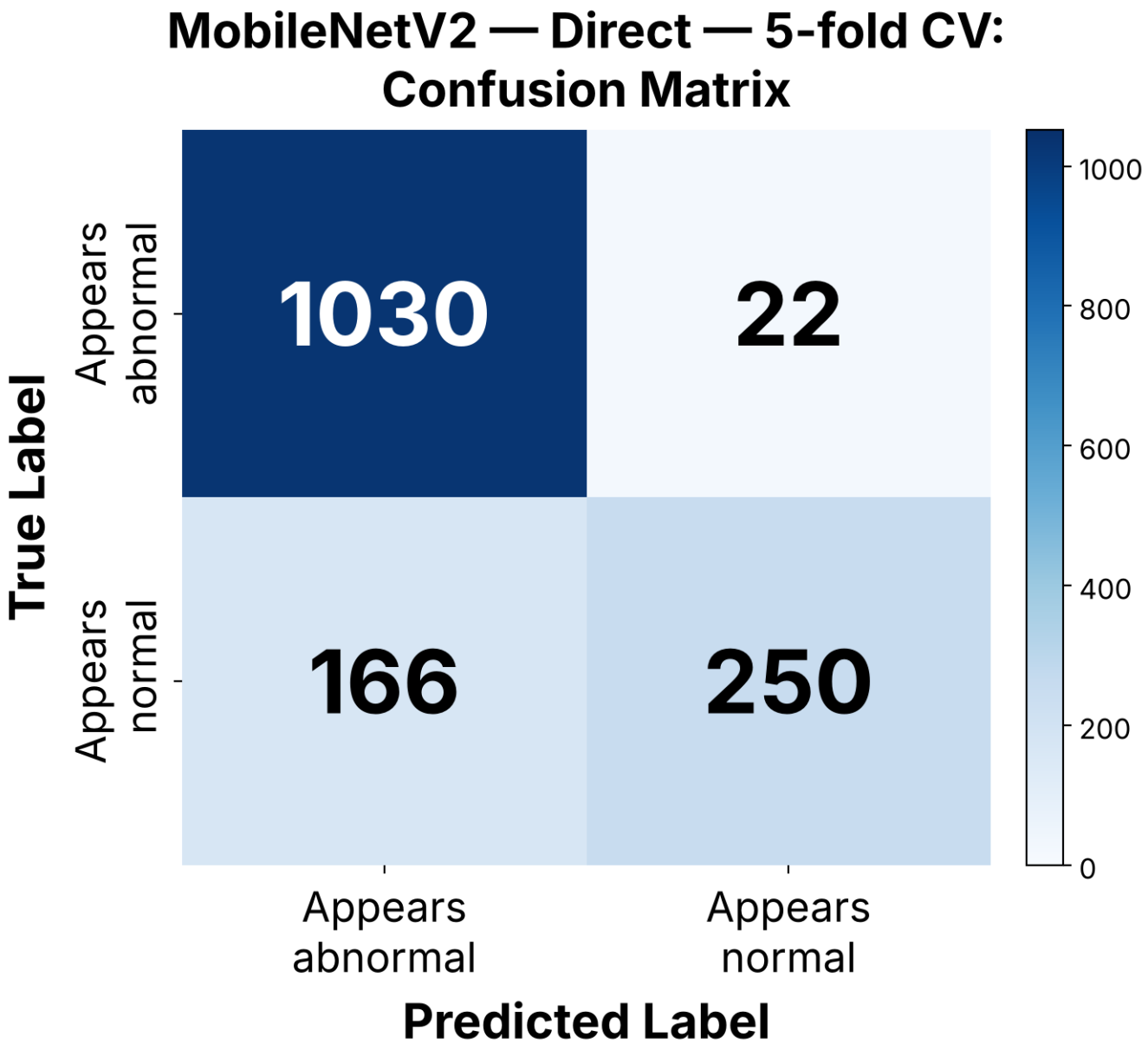
Accuracy: 87.19%

Precision: 87.76%

Recall: 87.19%

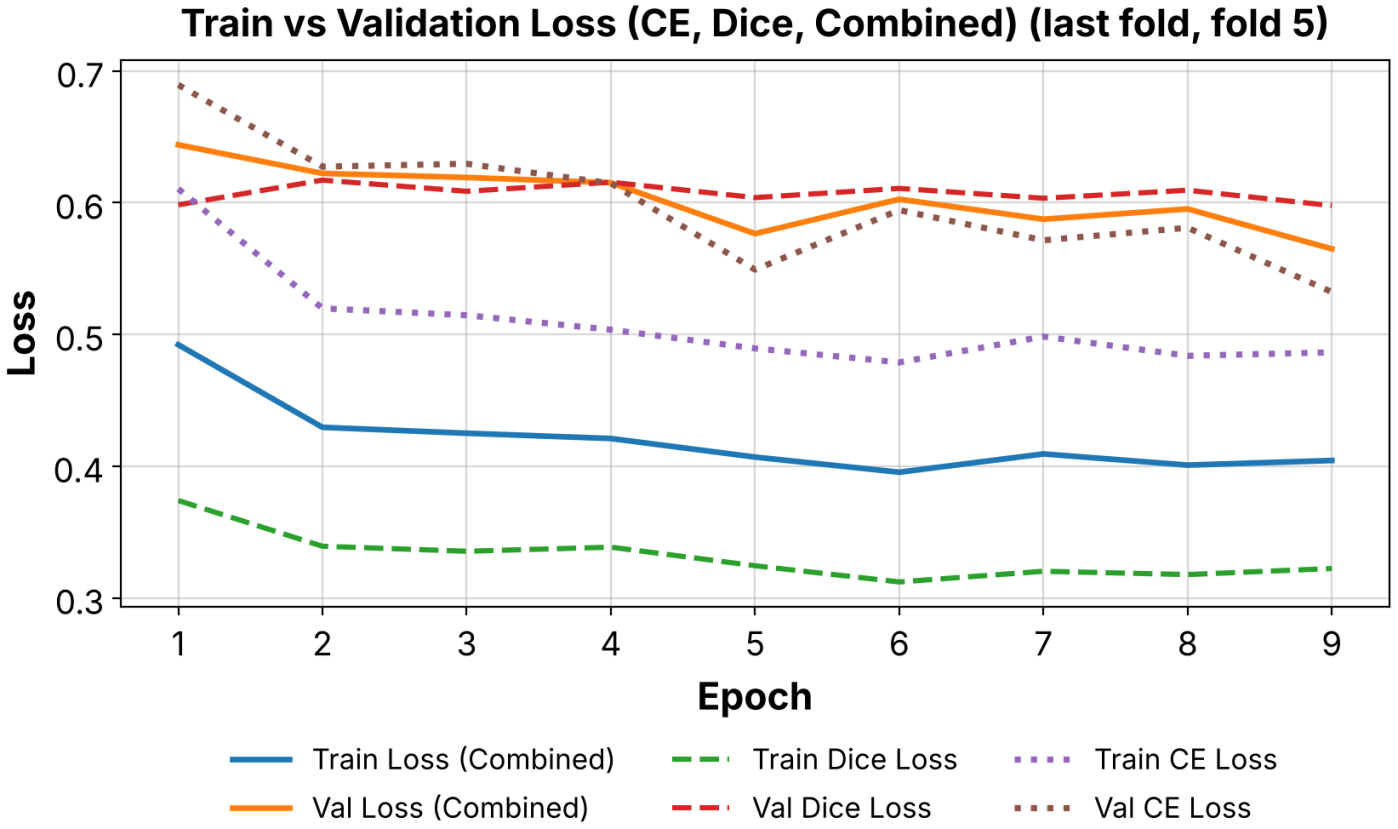
F1-score: 86.26%

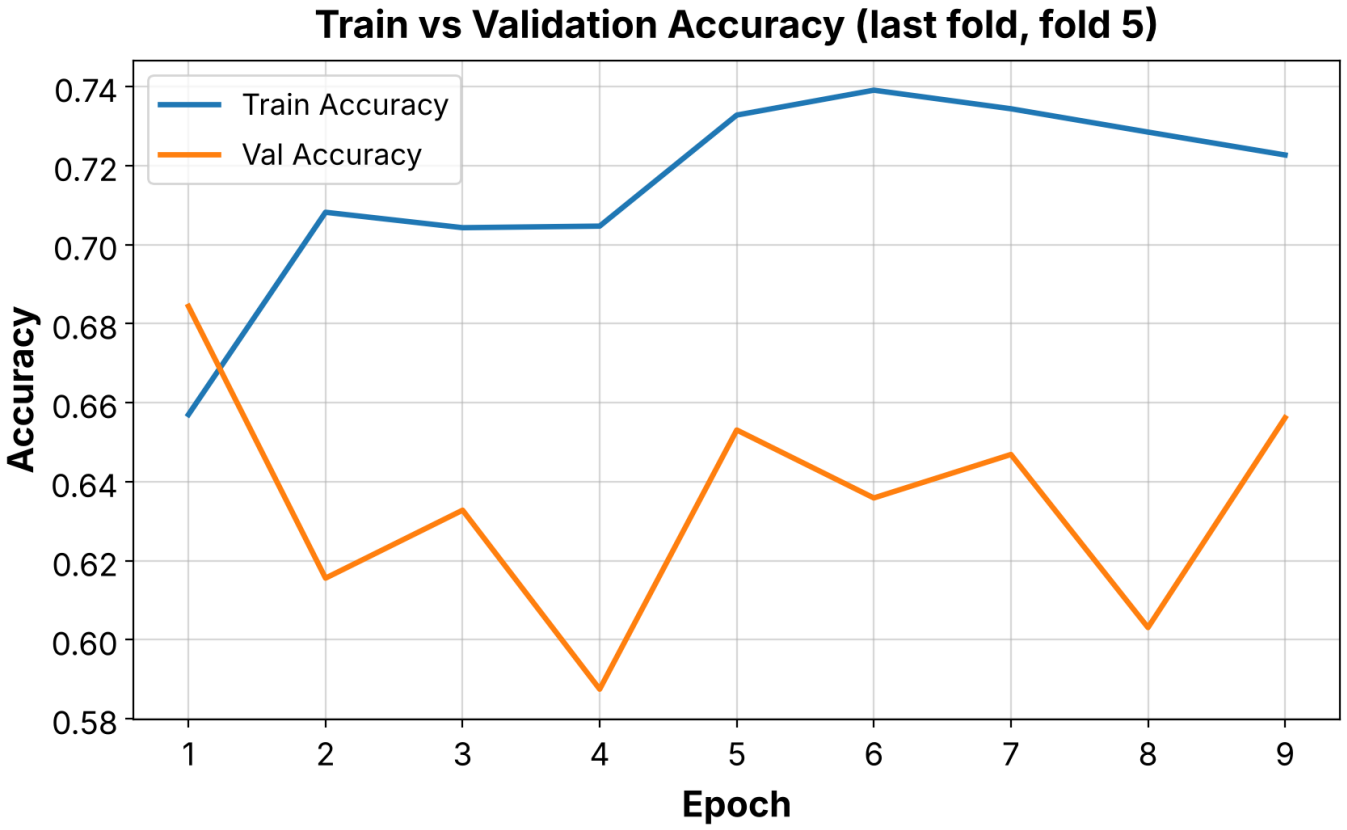
Confusion Matrix



33. MobileNetV2 — Direct — 5-fold CV — graphs (1/2)

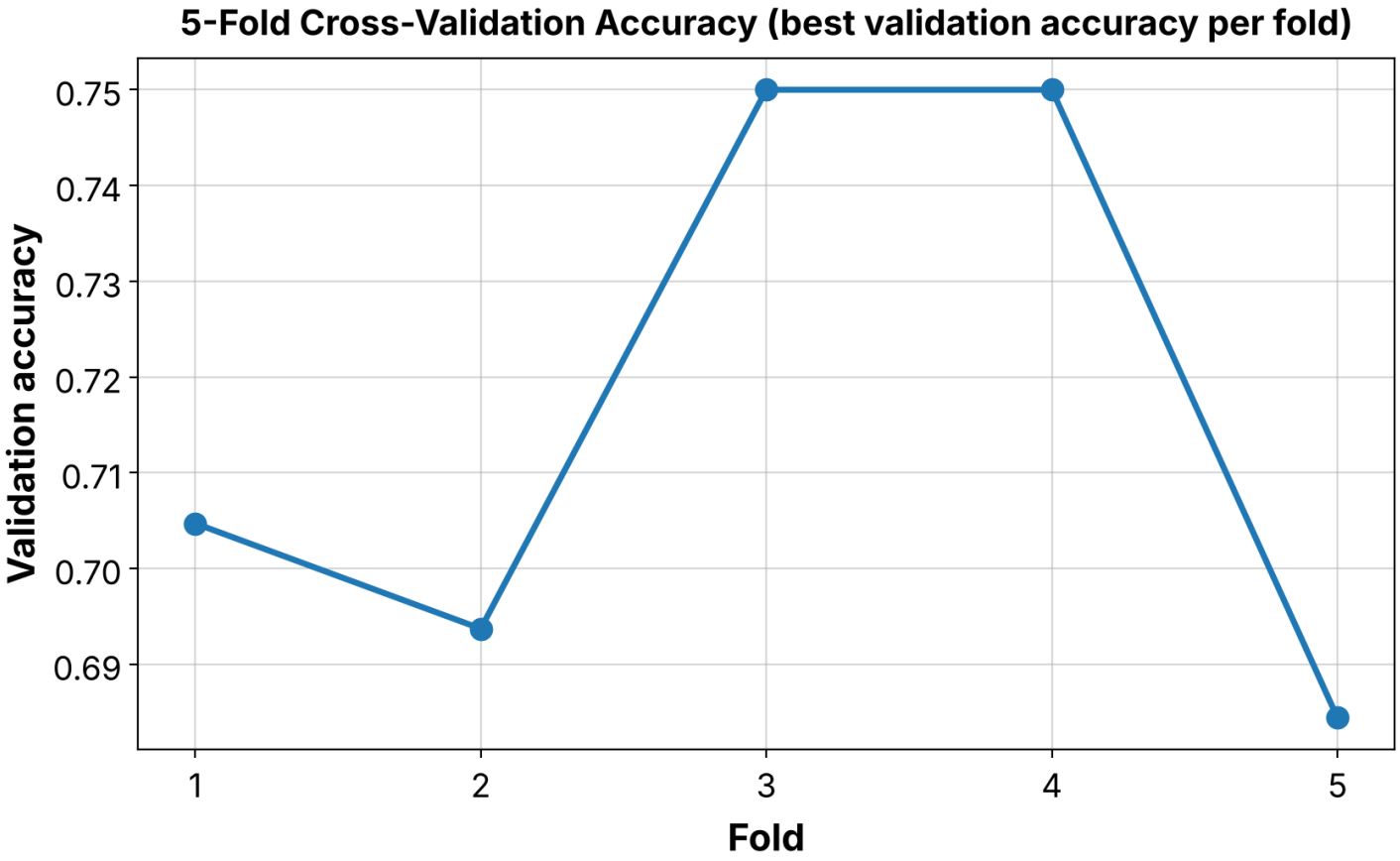
Training / Validation Graphs





33. MobileNetV2 — Direct — 5-fold CV — graphs (2/2)

Training / Validation Graphs



34. InceptionV3 + AdaBoost

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

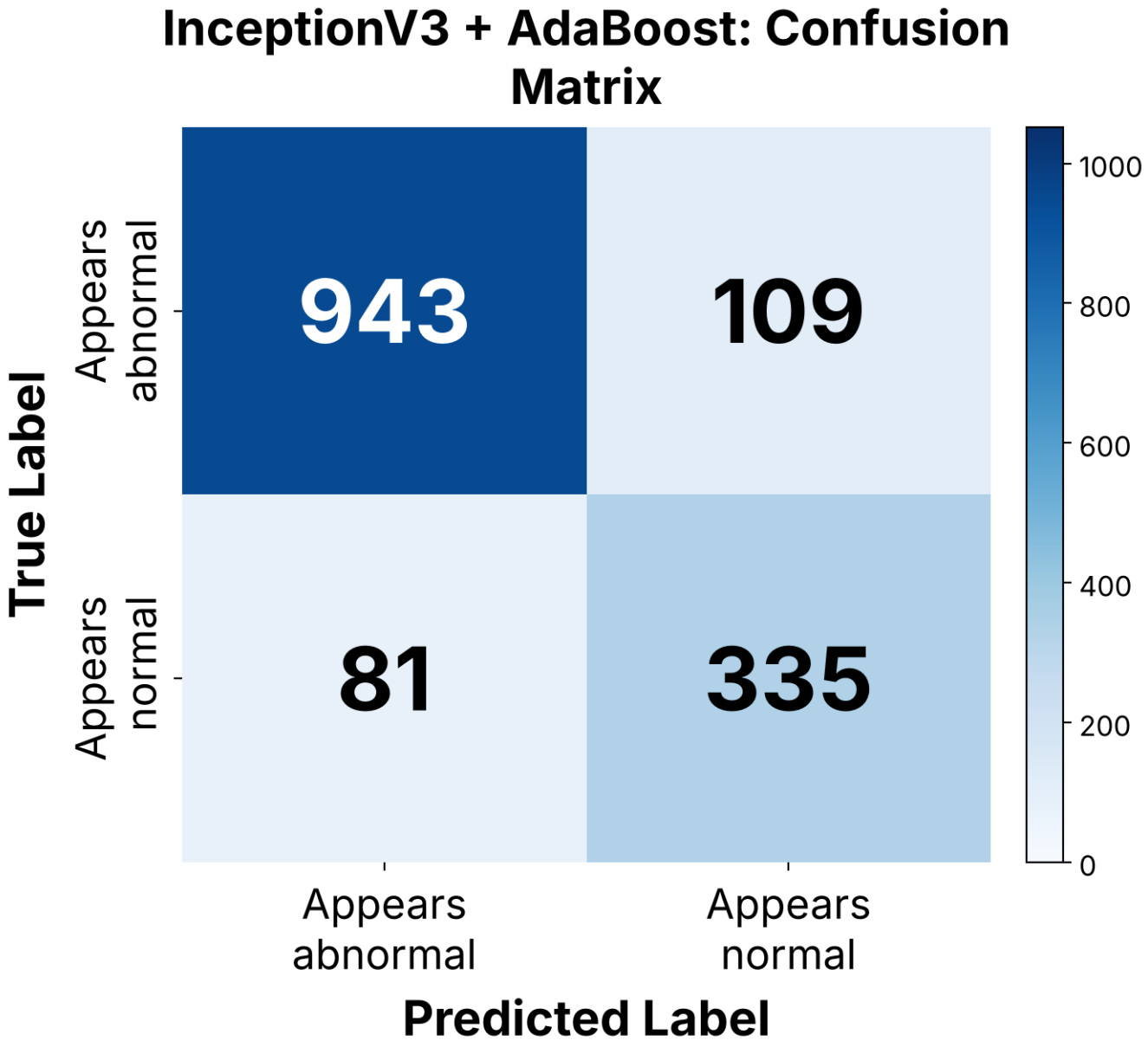
Accuracy: 87.06%

Precision: 87.37%

Recall: 87.06%

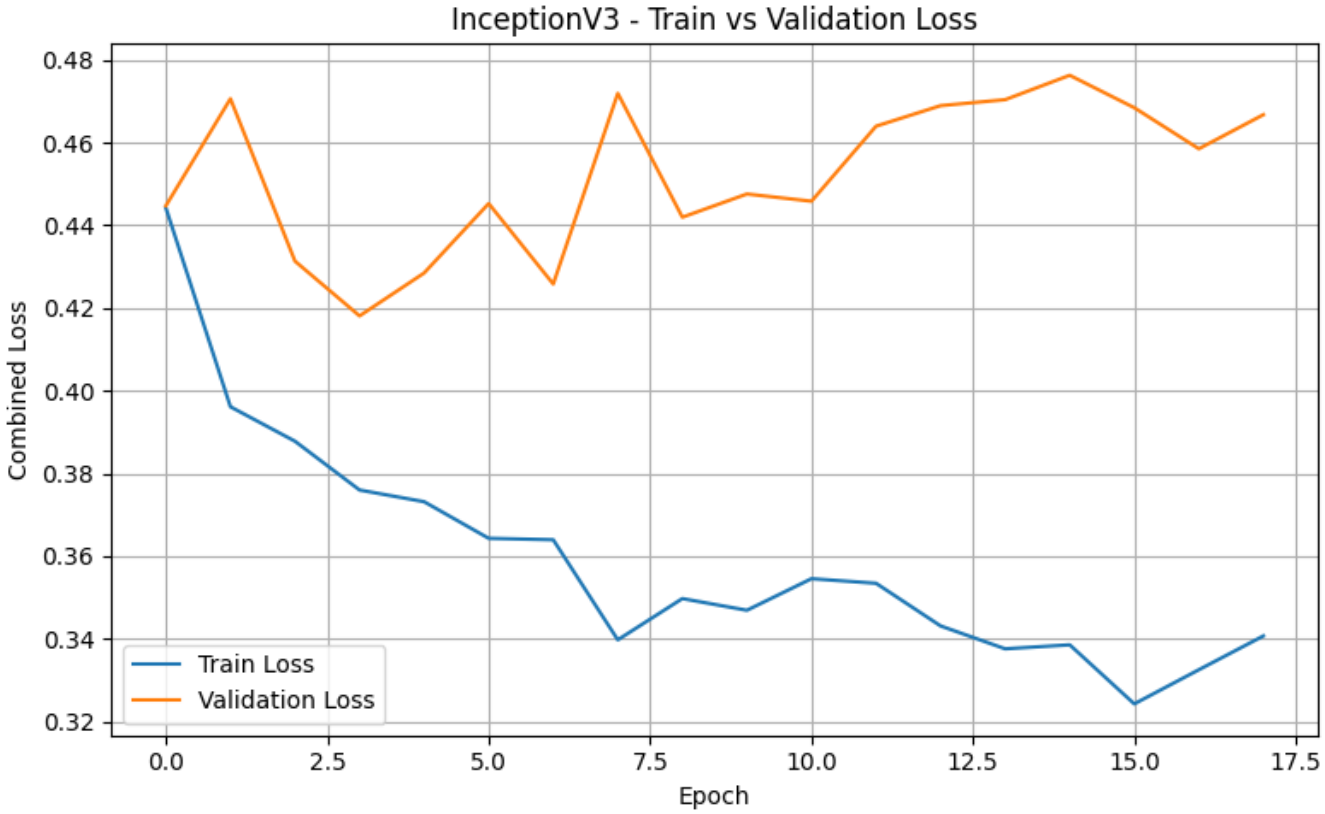
F1-score: 87.18%

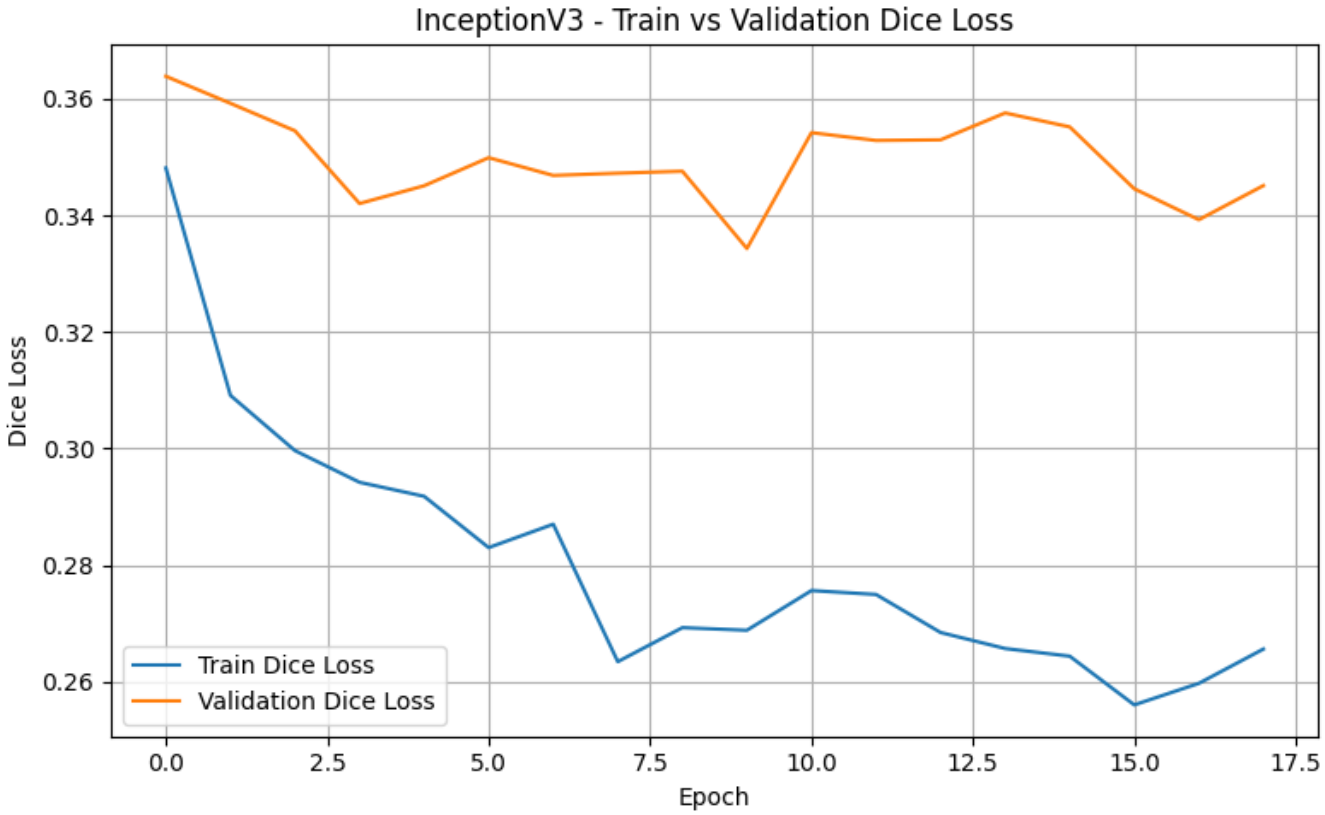
Confusion Matrix



34. InceptionV3 + AdaBoost — graphs (1/2)

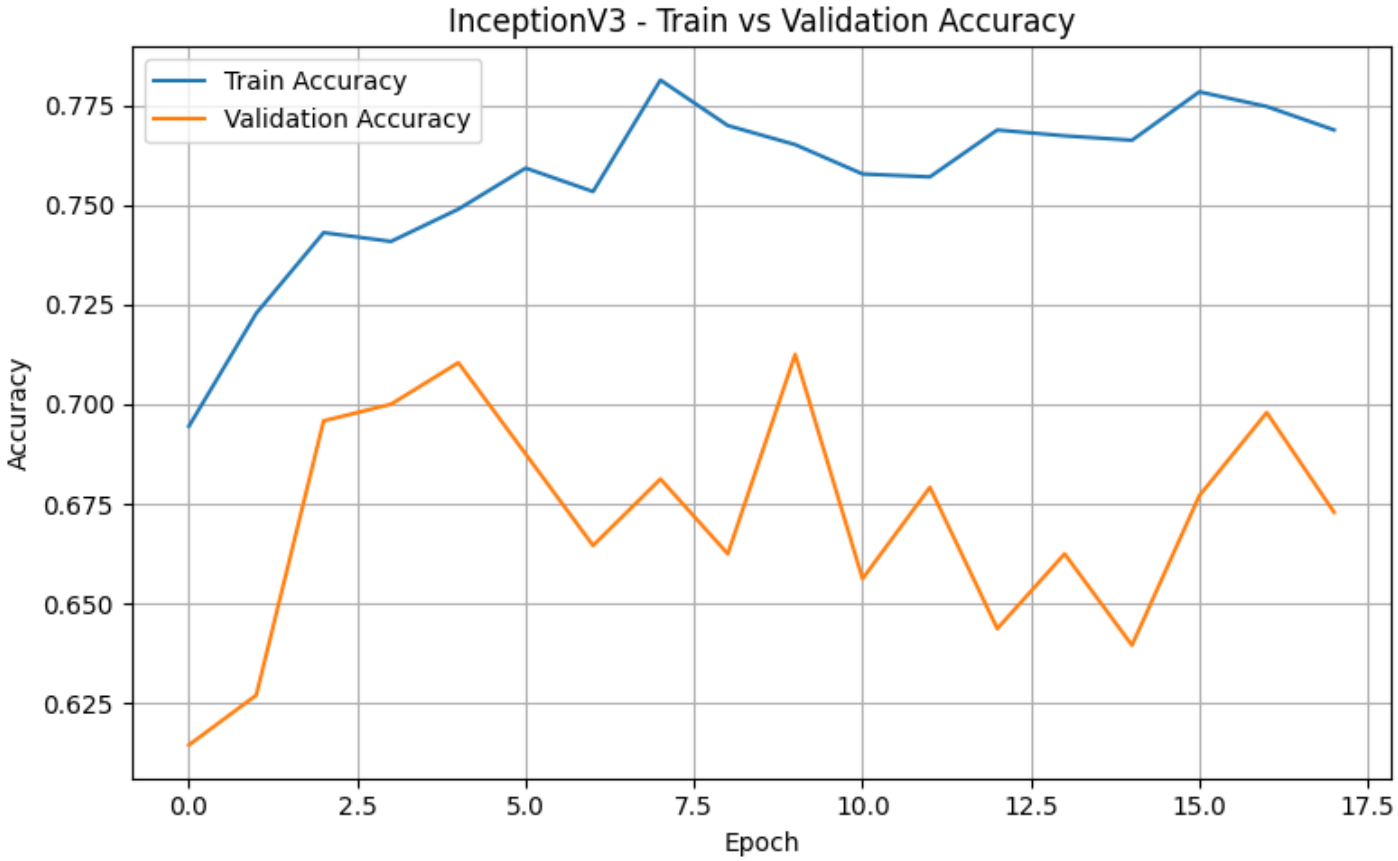
Training / Validation Graphs





34. InceptionV3 + AdaBoost — graphs (2/2)

Training / Validation Graphs



35. MiT-B2 + MobileNetV2 + Random Forest

Source notebook: MIT+ MobileNet+ ML.ipynb (inside MIT + MobileNet.zip)

Accuracy: 87.06%

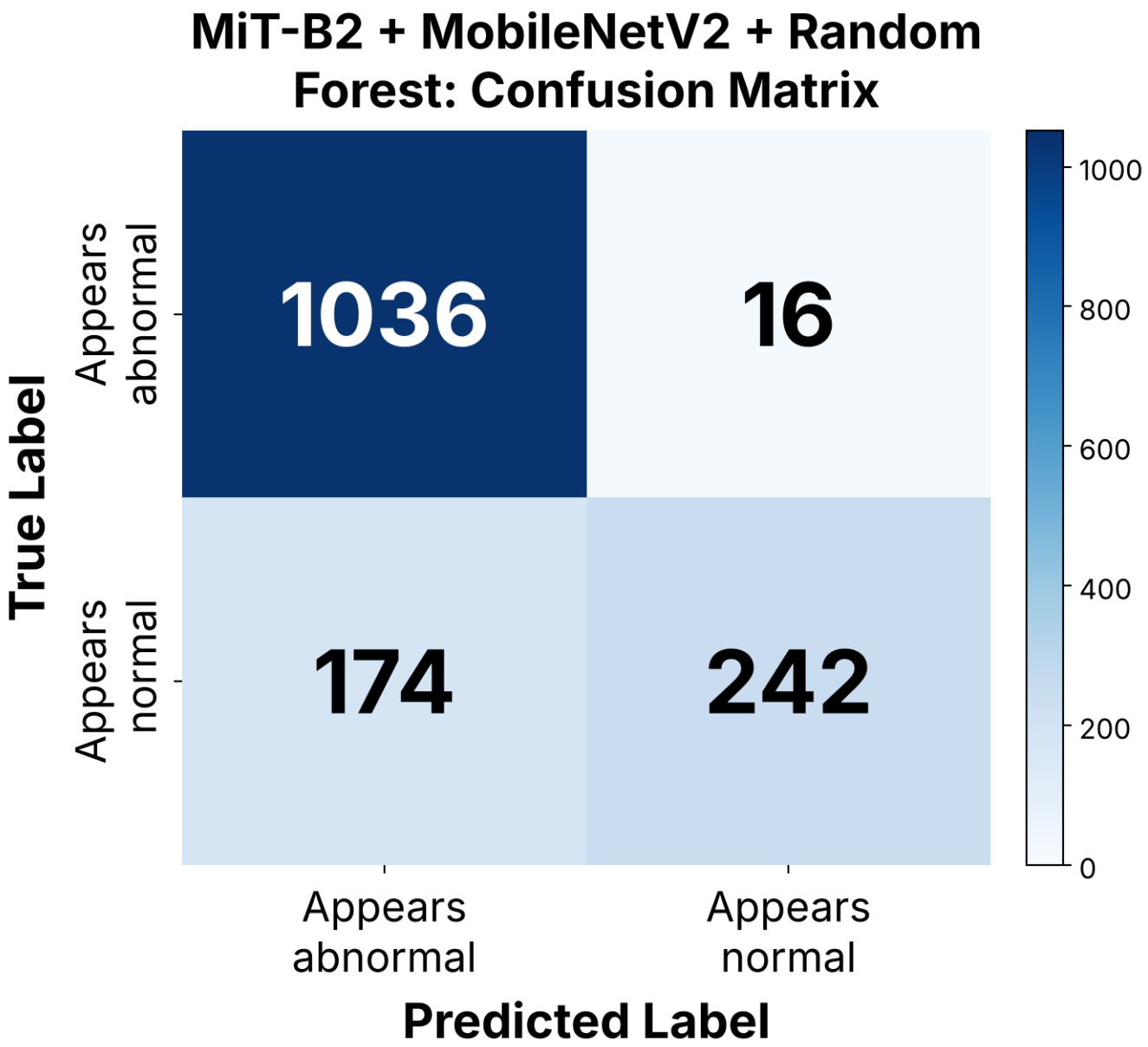
Precision: 87.94%

Recall: 87.06%

F1-score: 85.99%

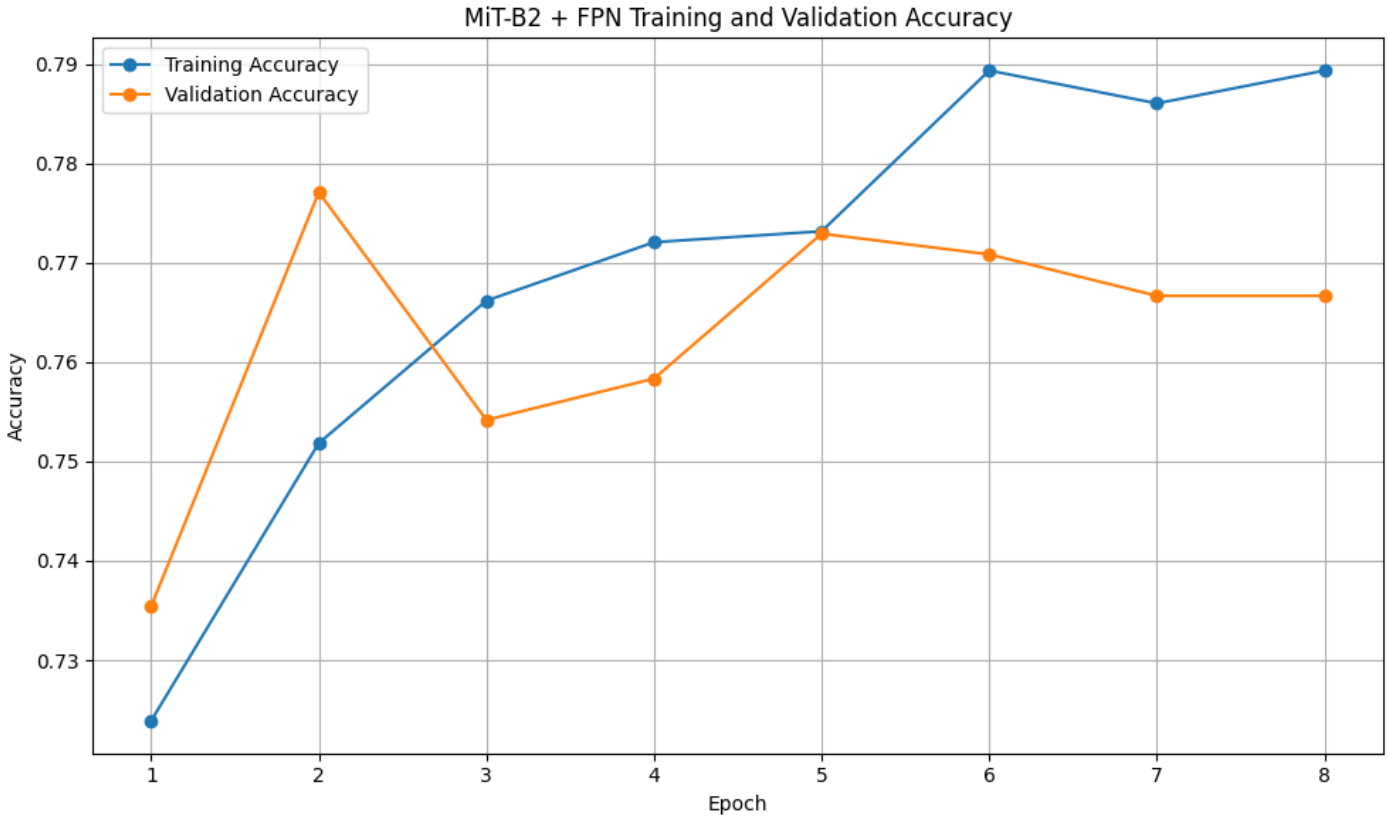
Same pair also run elsewhere with lower accuracy: 86.85% (not shown).

Confusion Matrix



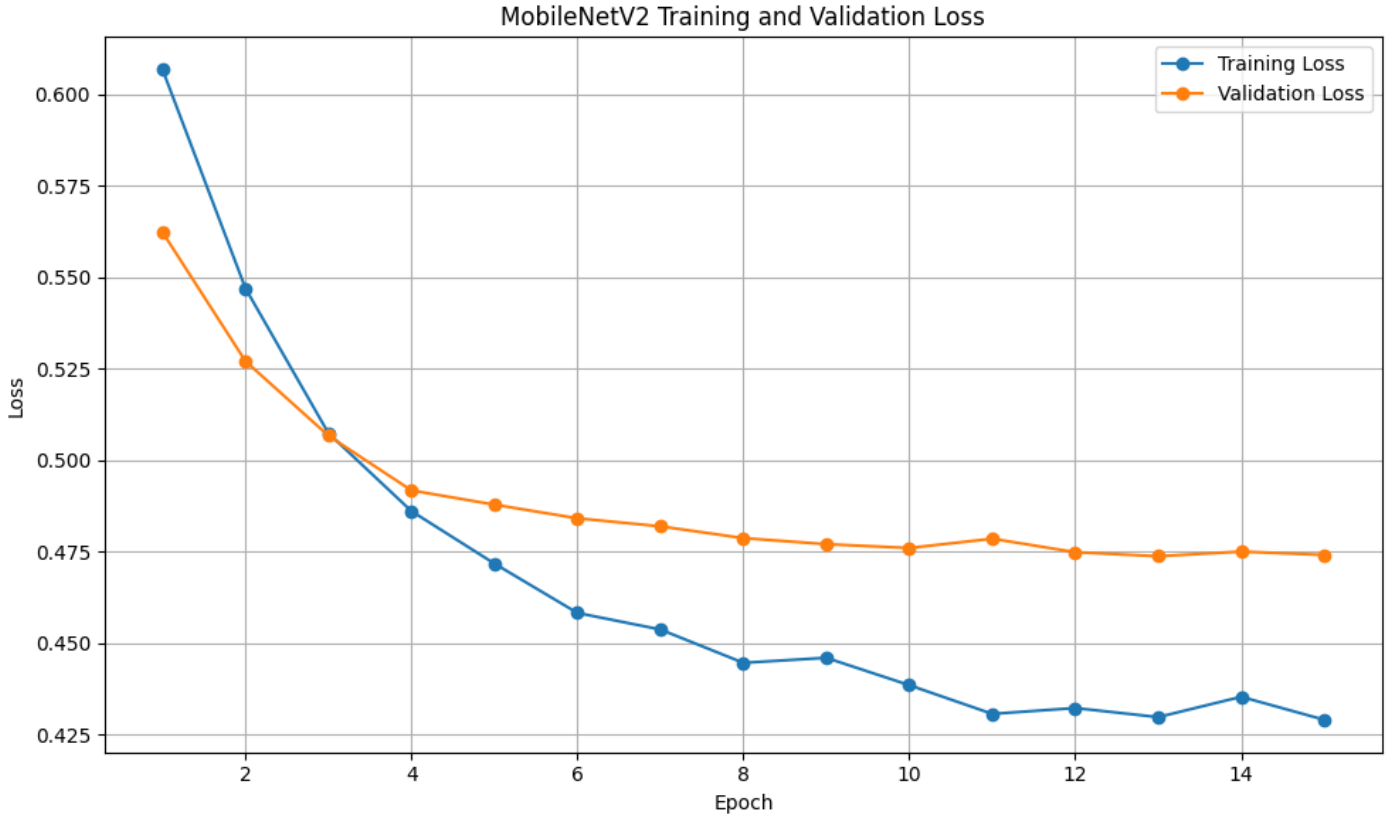
35. MiT-B2 + MobileNetV2 + Random Forest — graphs (1/2)

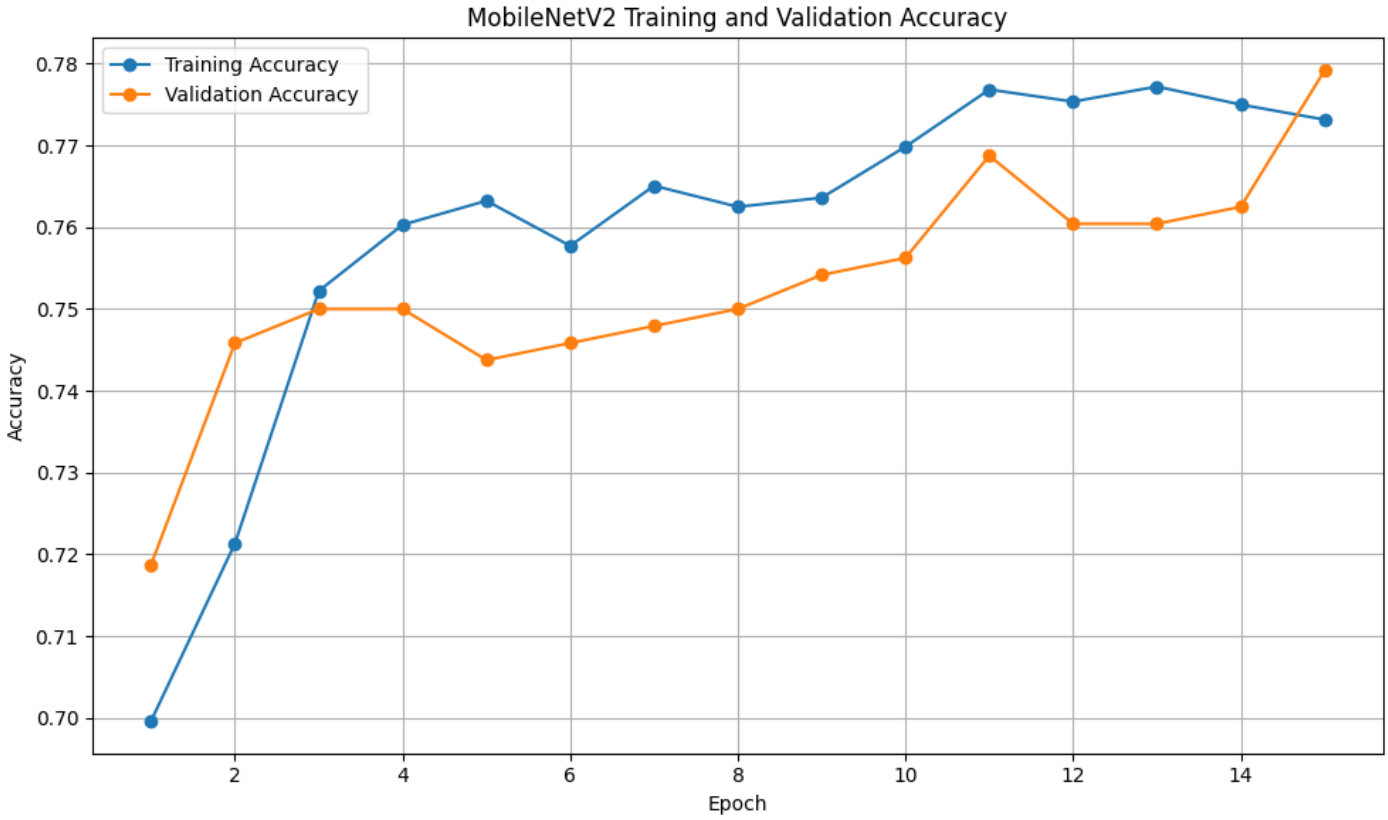
Training / Validation Graphs



35. MiT-B2 + MobileNetV2 + Random Forest — graphs (2/2)

Training / Validation Graphs





36. VGG16 + XGBoost

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

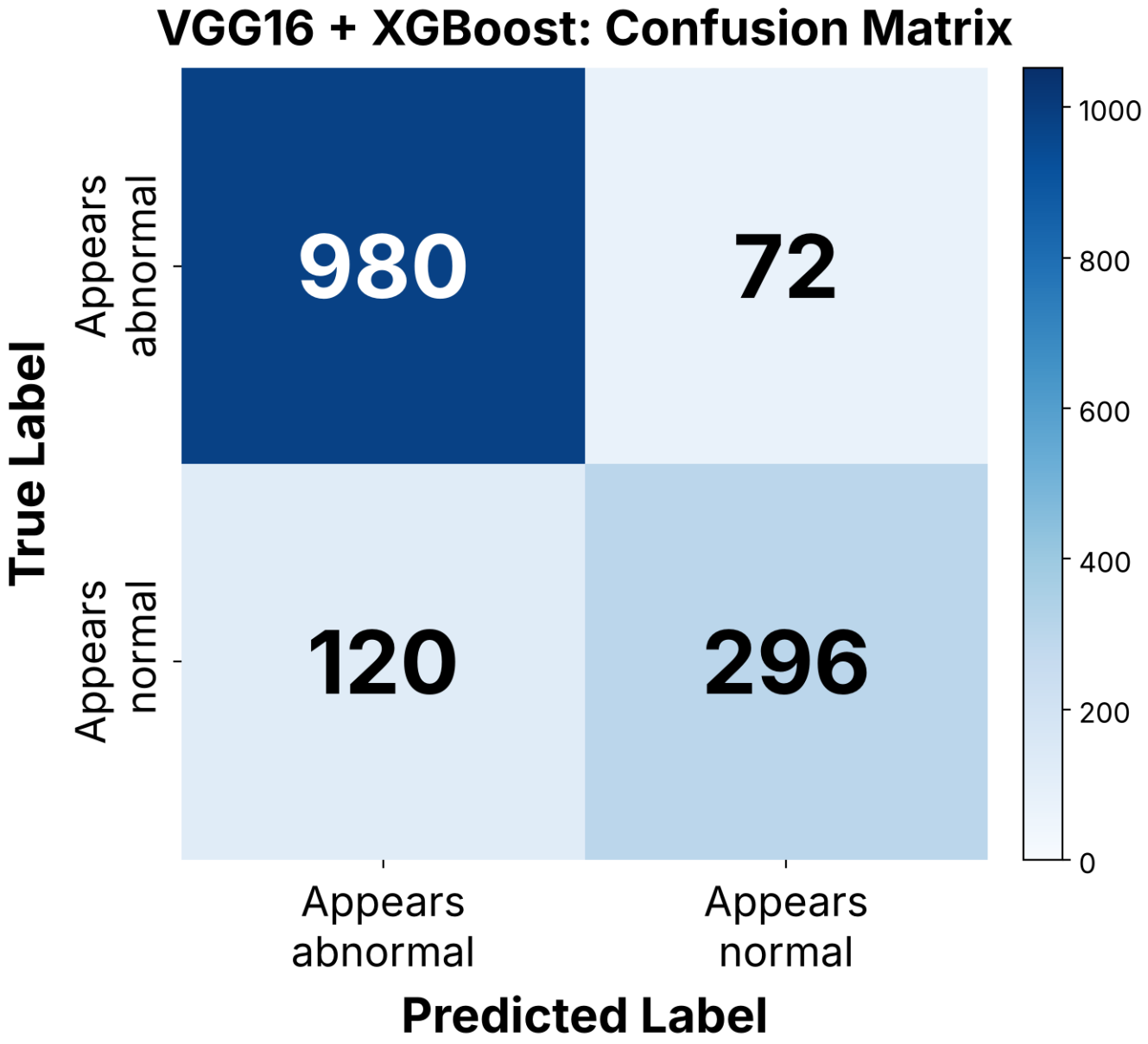
Accuracy: 86.92%

Precision: 86.64%

Recall: 86.92%

F1-score: 86.67%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

37. InceptionV3 + XGBoost

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

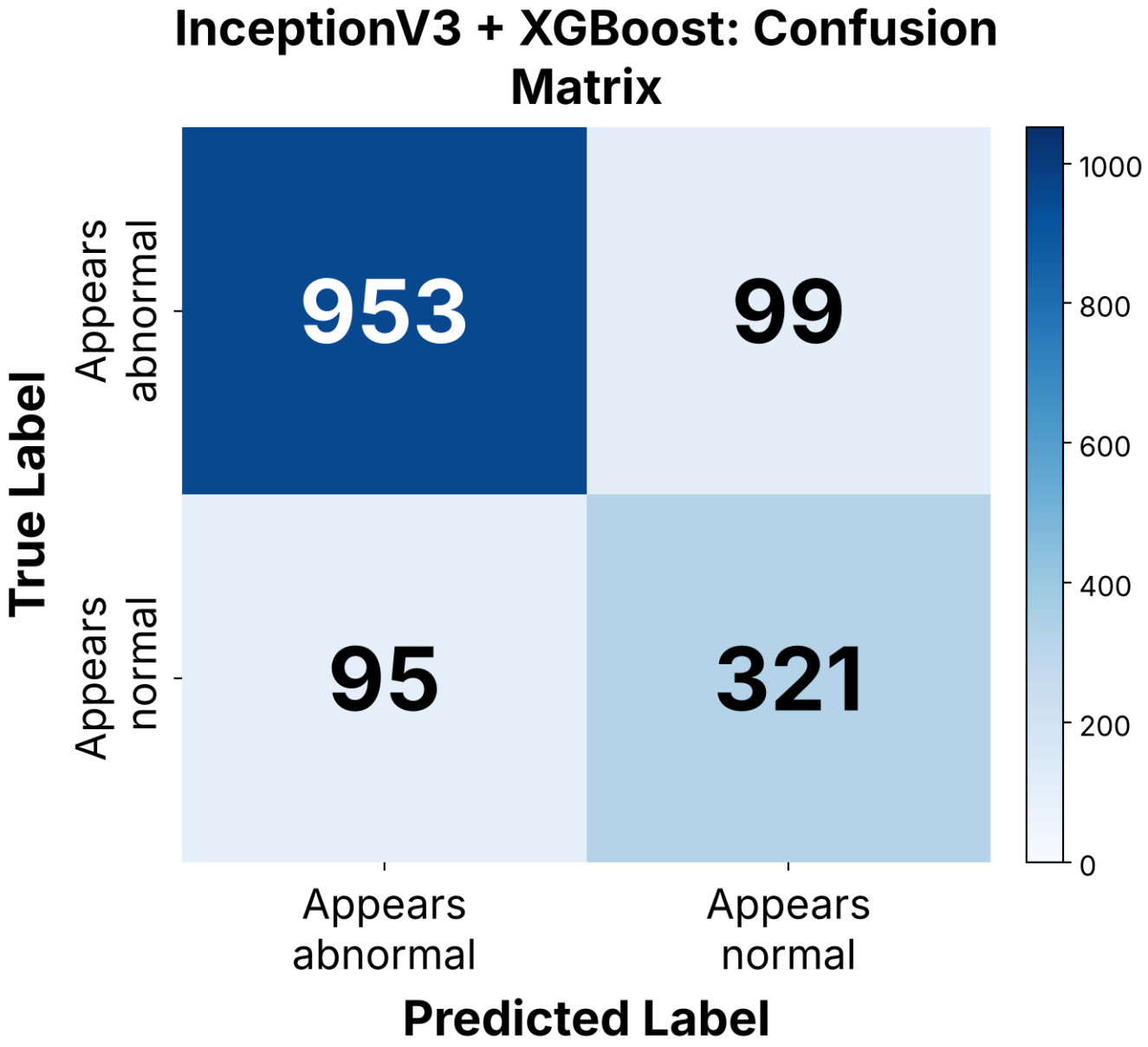
Accuracy: 86.78%

Precision: 86.82%

Recall: 86.78%

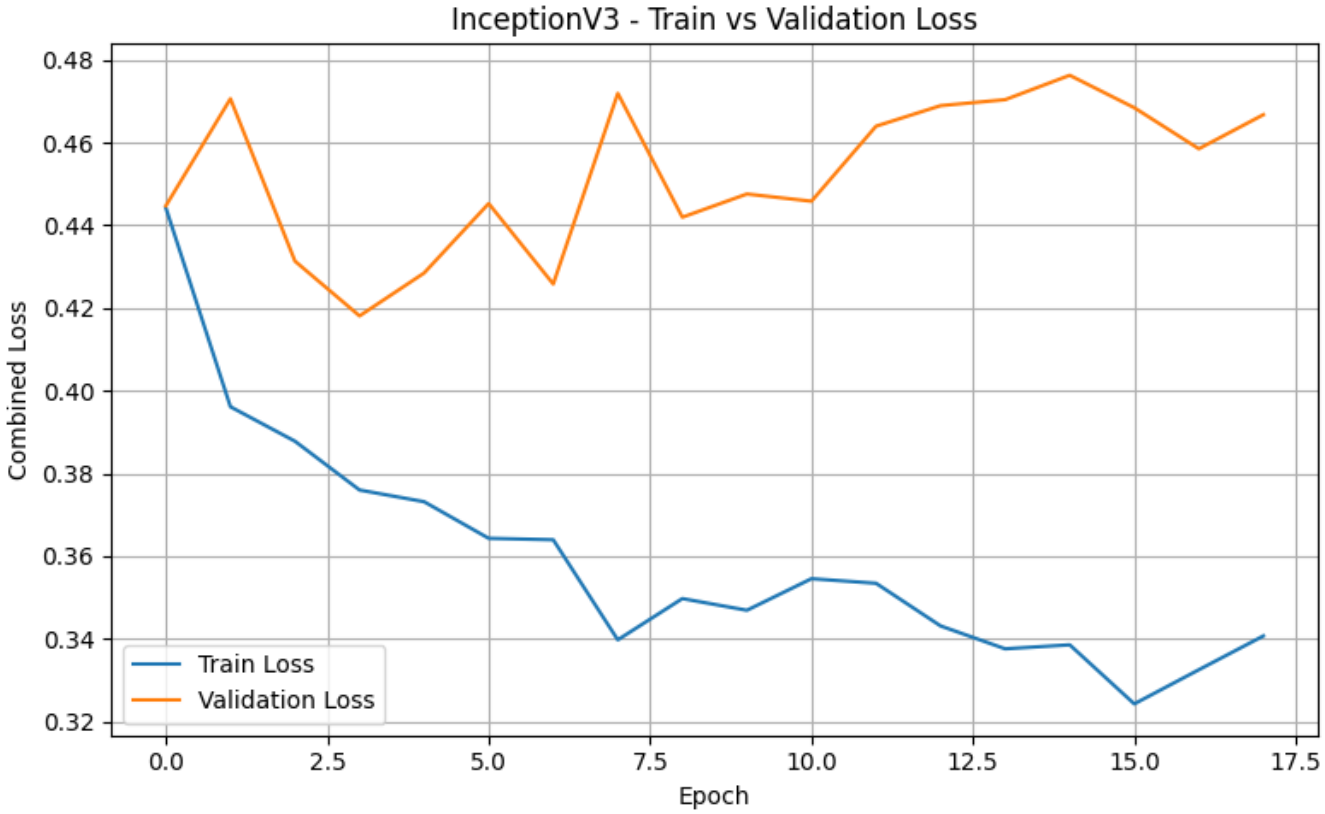
F1-score: 86.80%

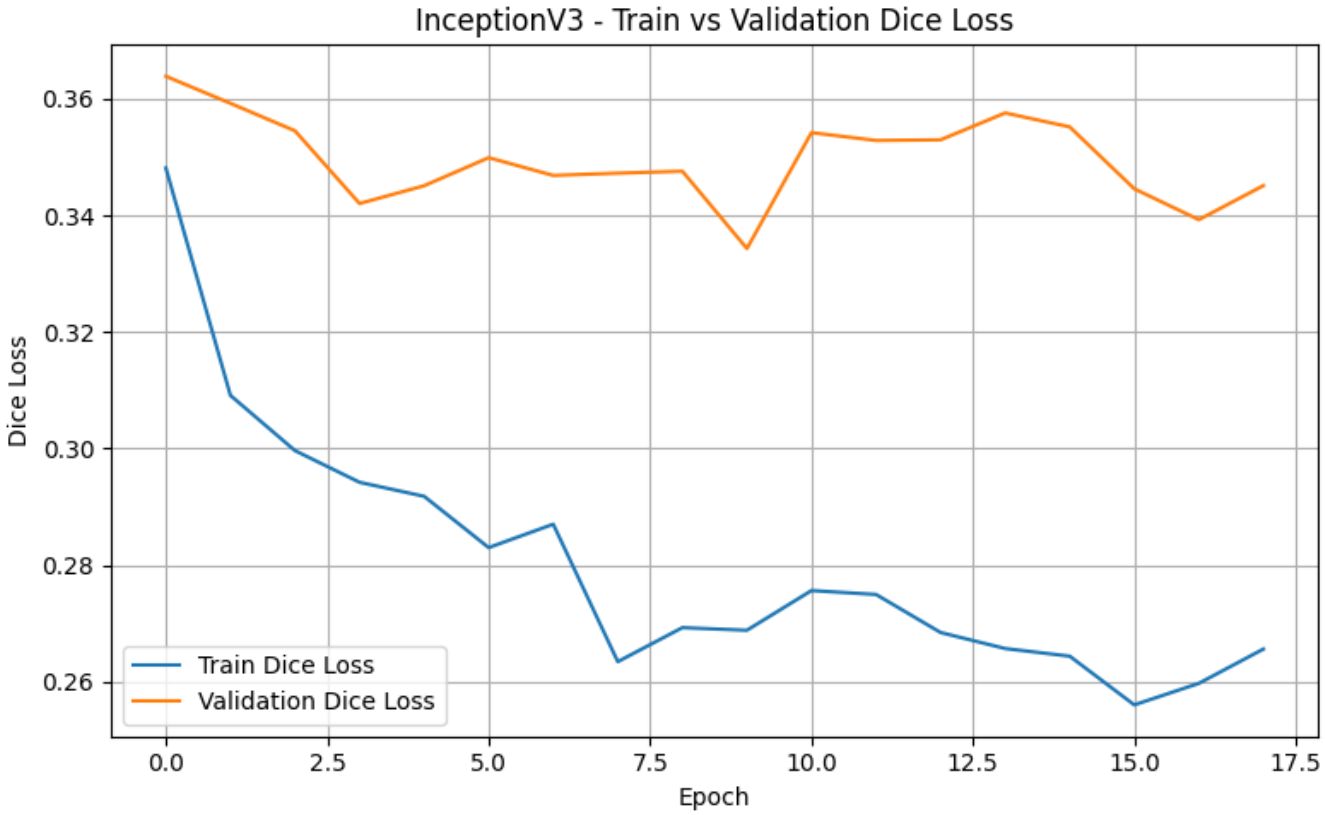
Confusion Matrix



37. InceptionV3 + XGBoost — graphs (1/2)

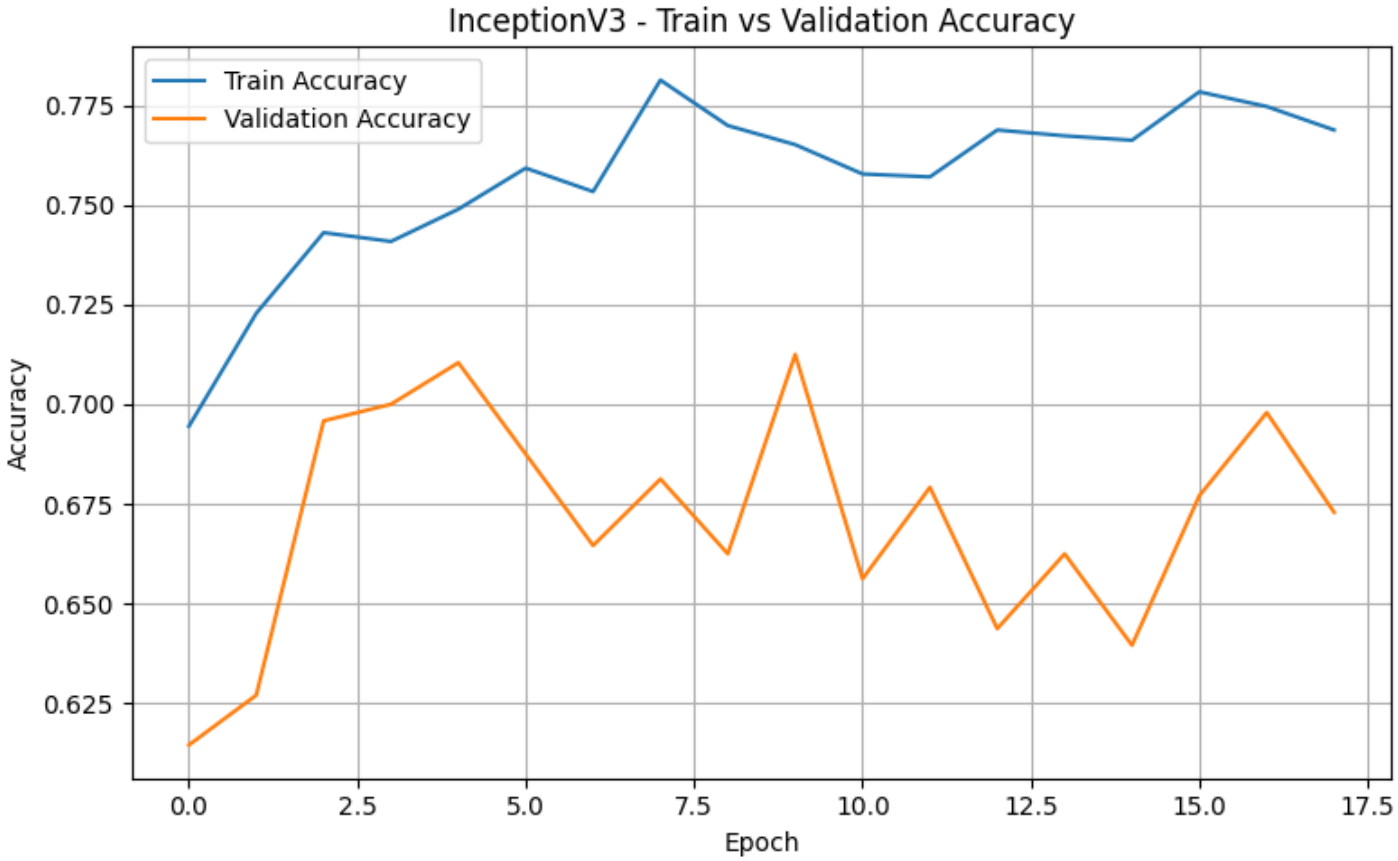
Training / Validation Graphs





37. InceptionV3 + XGBoost — graphs (2/2)

Training / Validation Graphs



38. ResNet18 + SVM

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

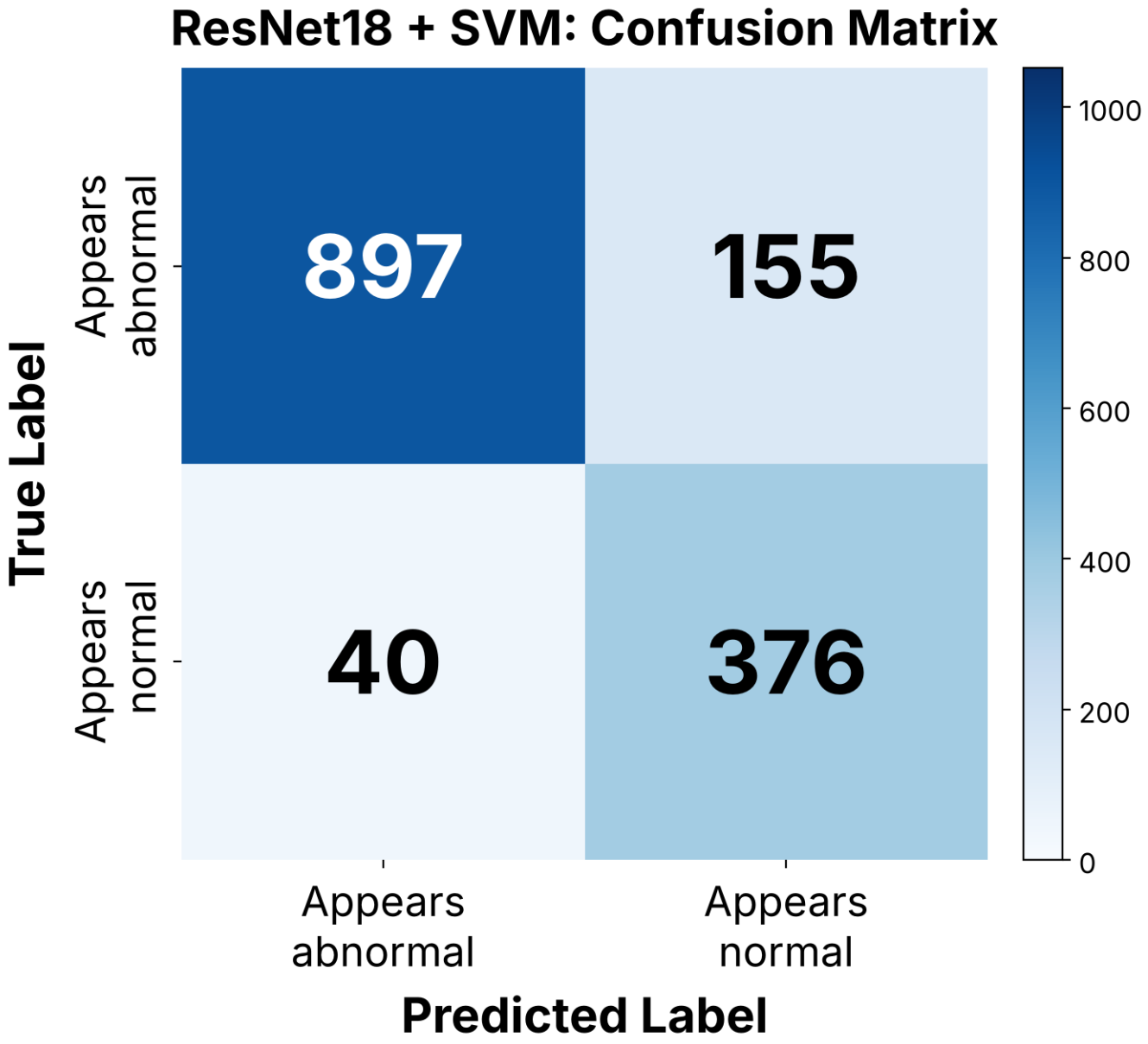
Accuracy: 86.72%

Precision: 88.67%

Recall: 86.72%

F1-score: 87.14%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

39. MobileNetV2 + XGBoost

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

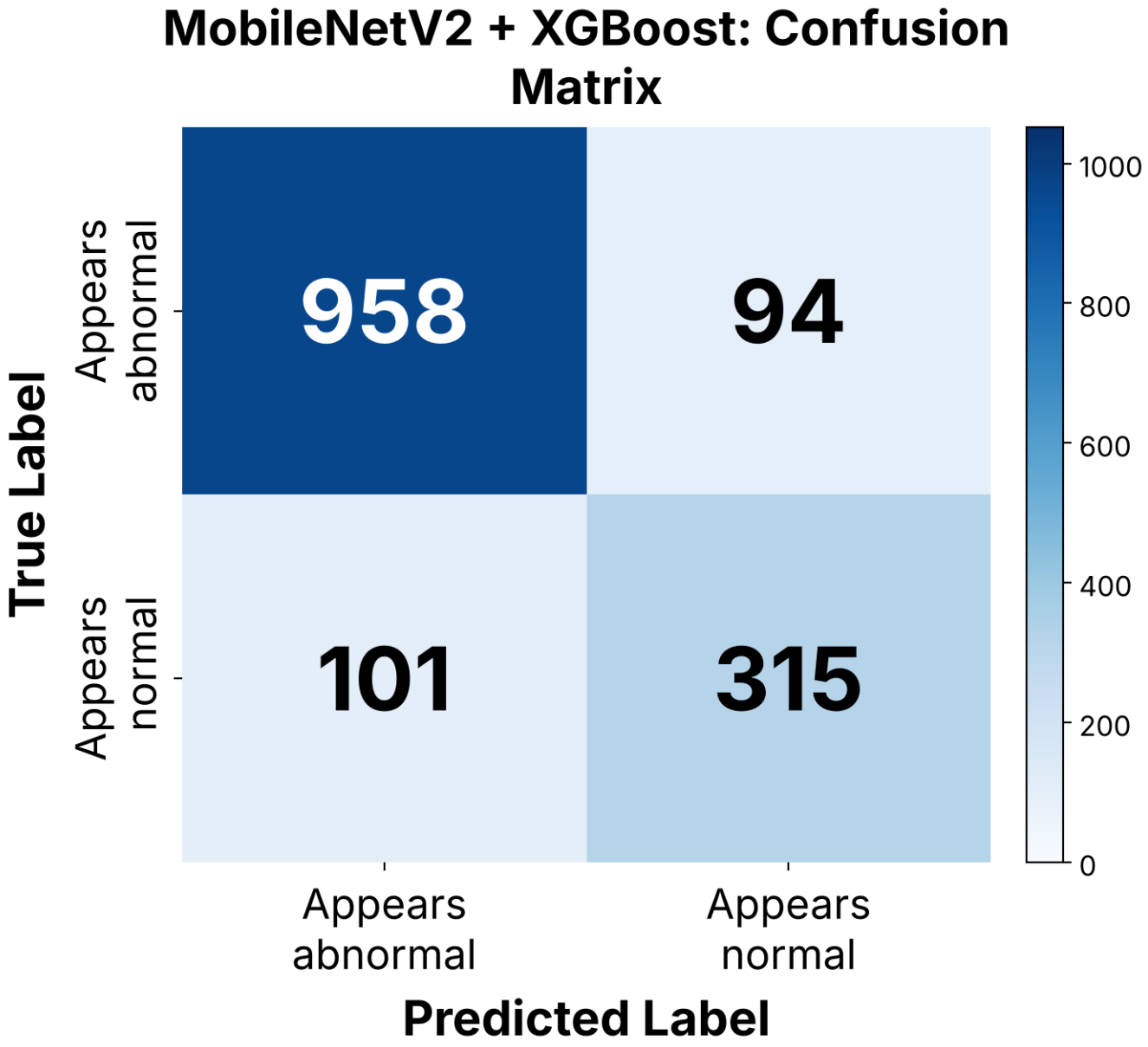
Accuracy: 86.72%

Precision: 86.65%

Recall: 86.72%

F1-score: 86.68%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

40. AlexNet — Direct — 5-fold CV

Source notebook: AlexNet with kfold.ipynb (inside Models_DL.zip)

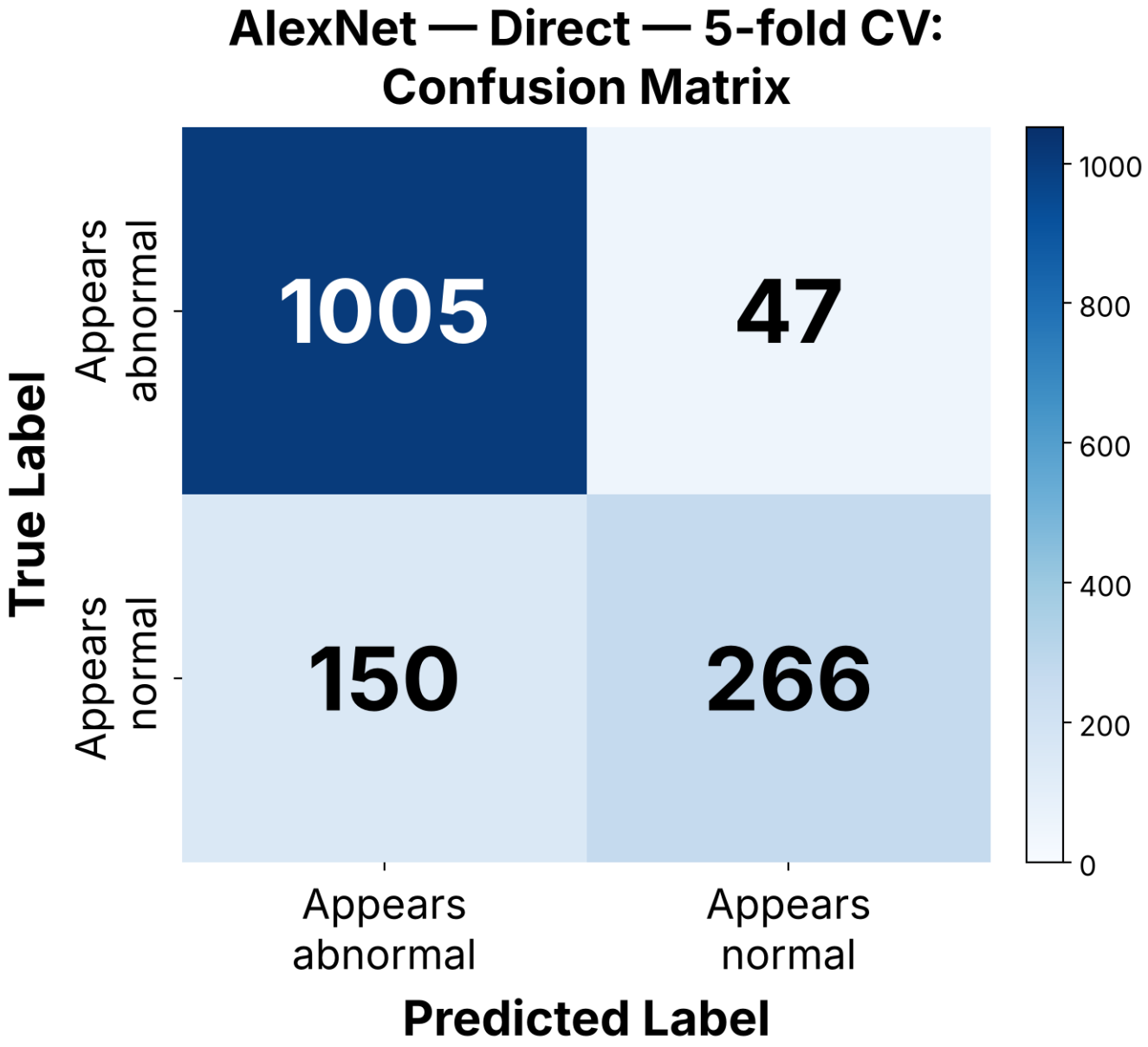
Accuracy: 86.58%

Precision: 86.44%

Recall: 86.58%

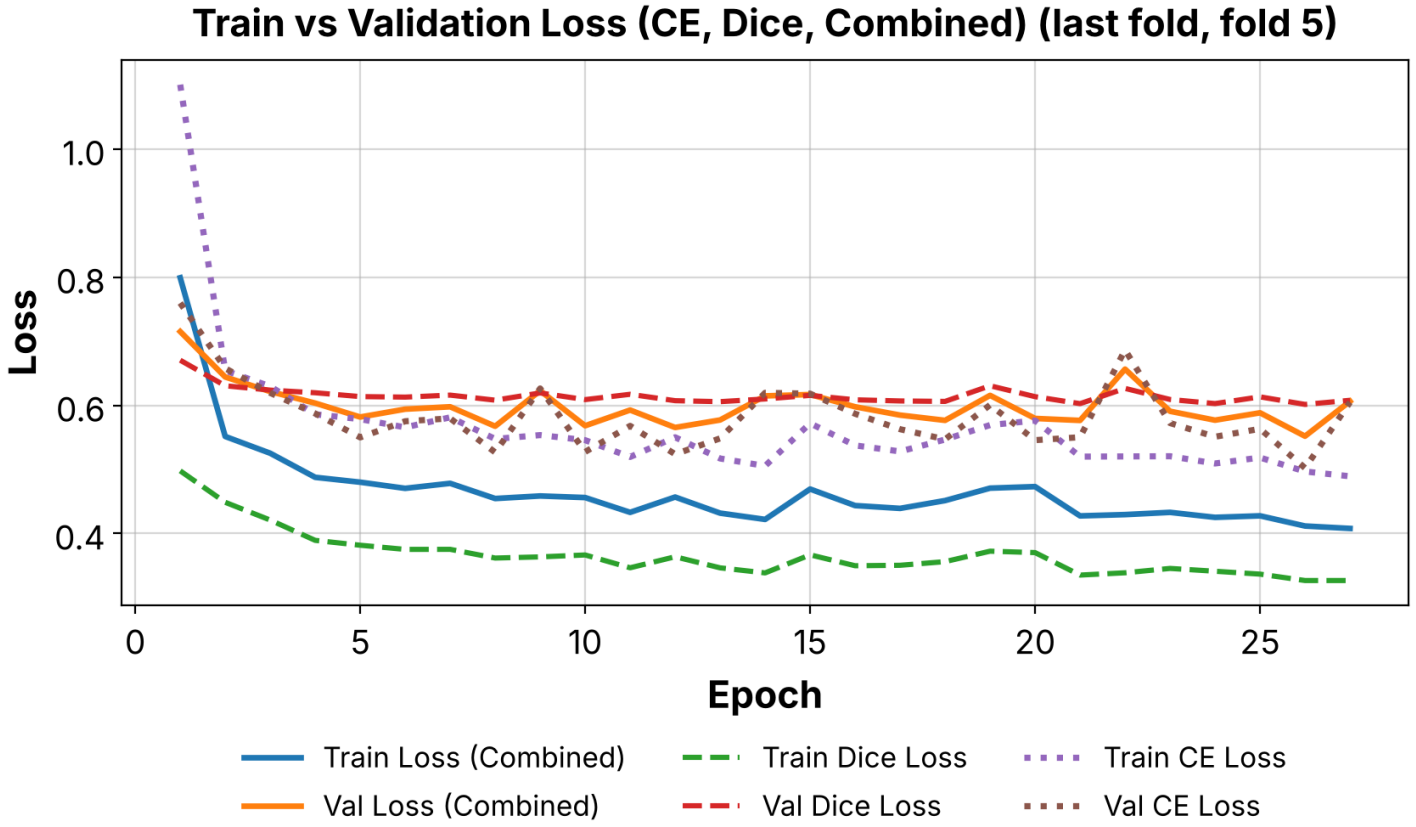
F1-score: 85.95%

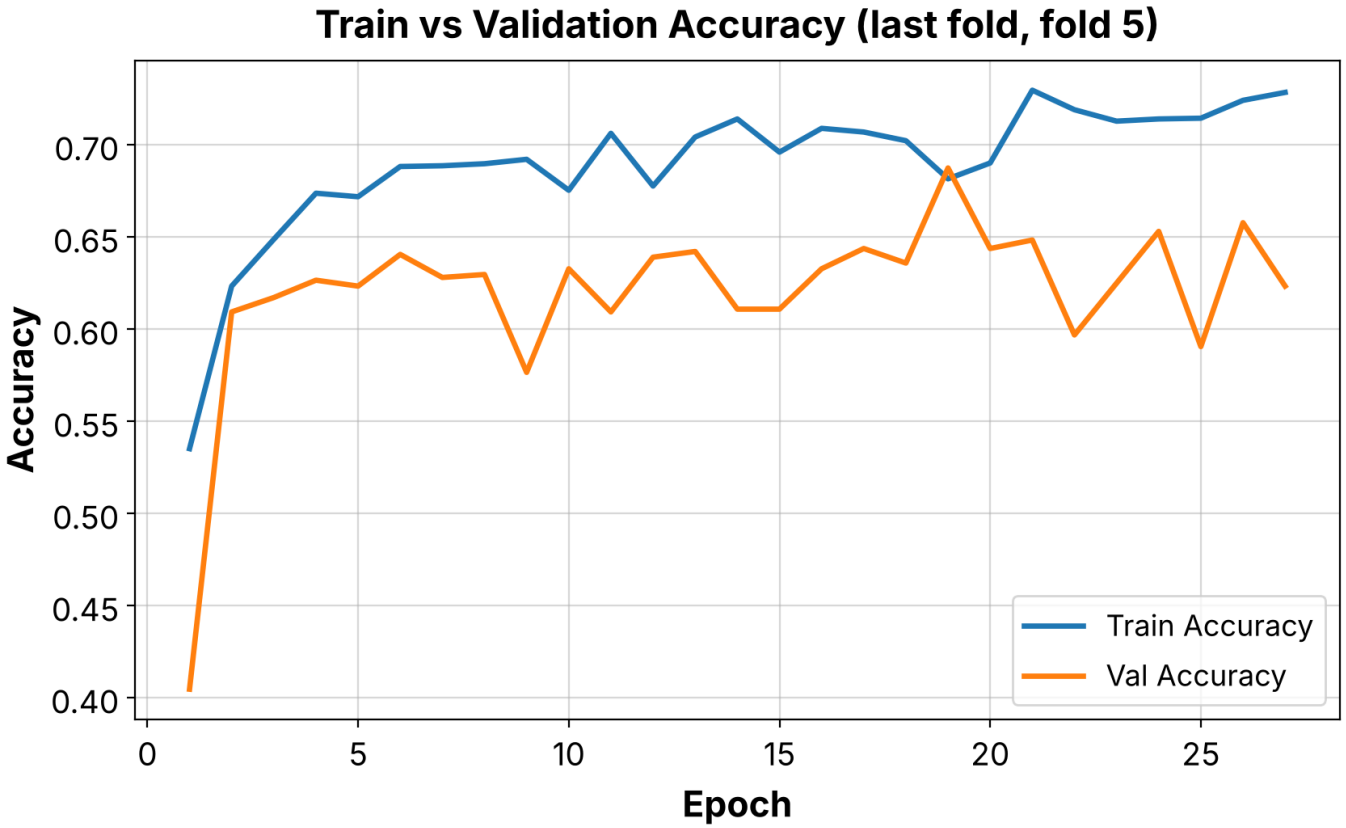
Confusion Matrix



40. AlexNet — Direct — 5-fold CV — graphs (1/2)

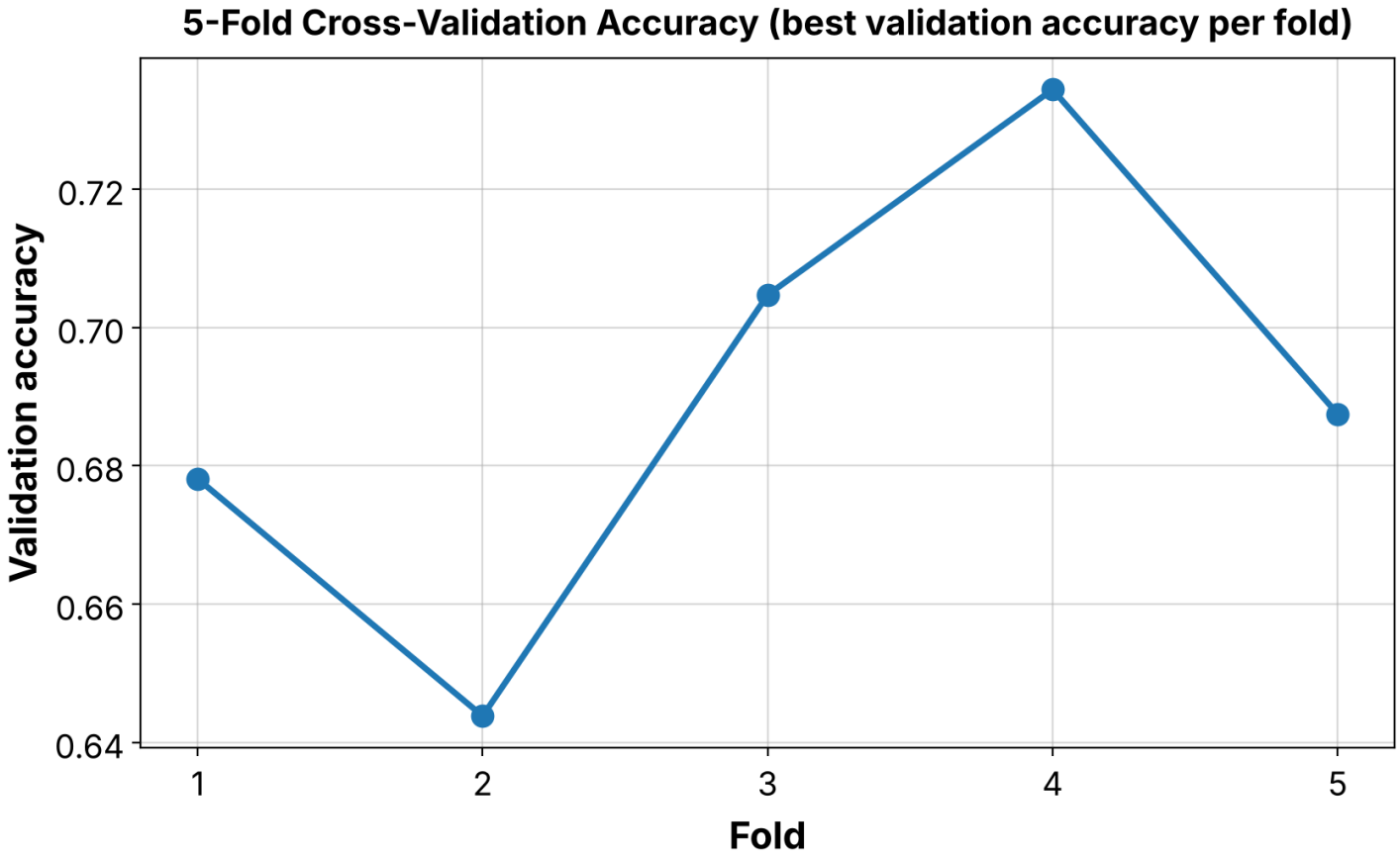
Training / Validation Graphs





40. AlexNet — Direct — 5-fold CV — graphs (2/2)

Training / Validation Graphs



41. MiT-B2 + MobileNetV2 + AdaBoost

Source notebook: MIT+ MobileNet+ ML.ipynb (inside MIT + MobileNet.zip)

Accuracy: 86.51%

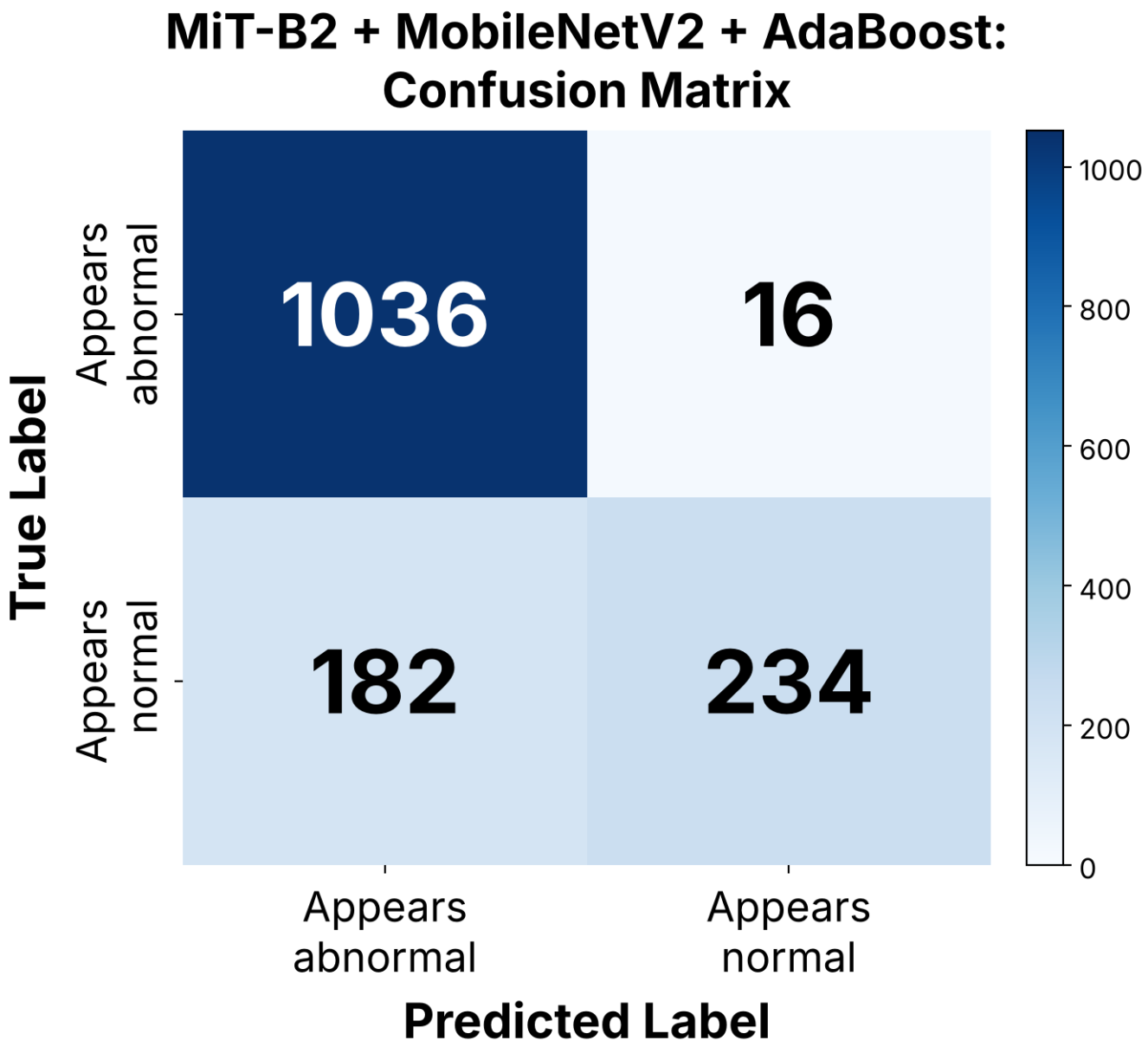
Precision: 87.48%

Recall: 86.51%

F1-score: 85.32%

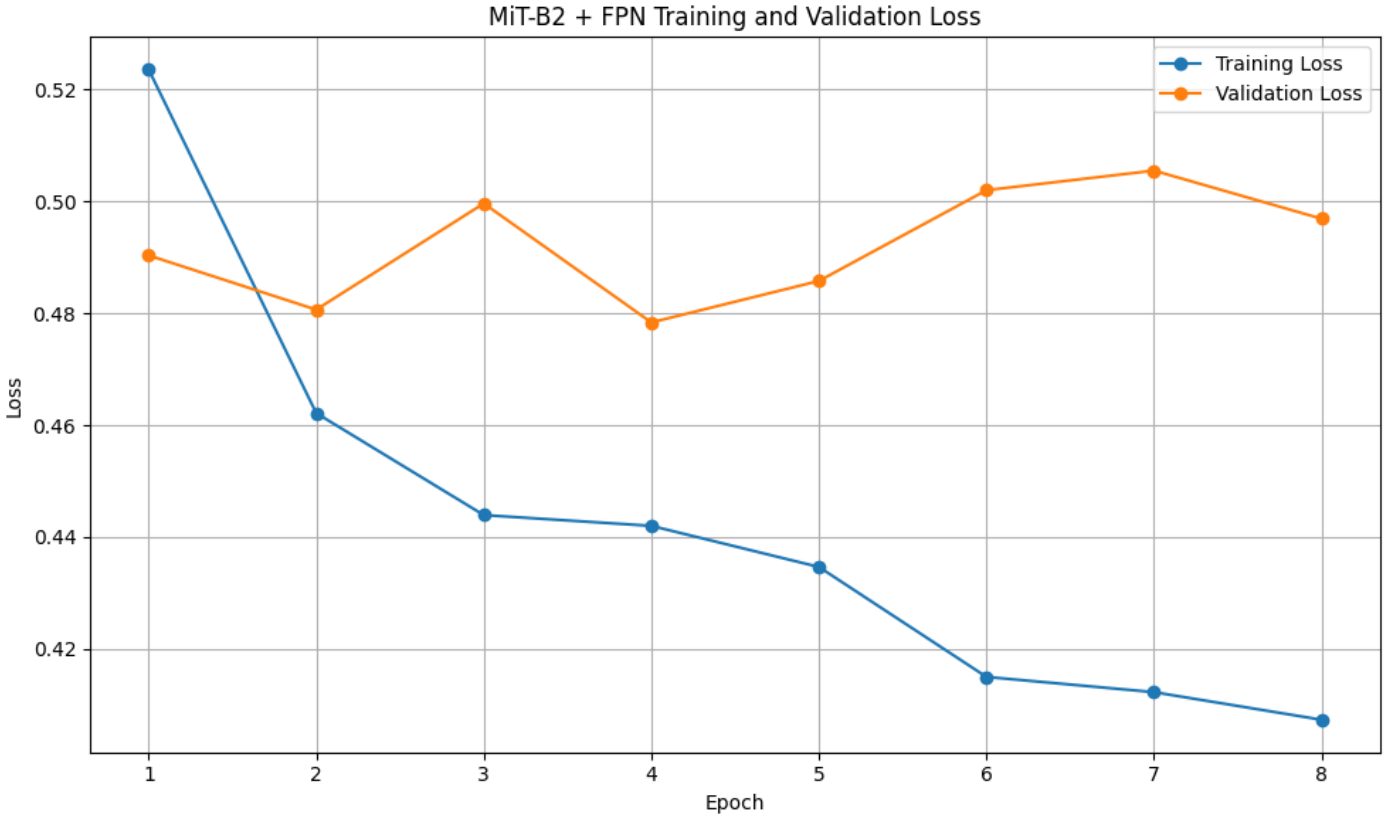
Same pair also run elsewhere with lower accuracy: 86.31% (not shown).

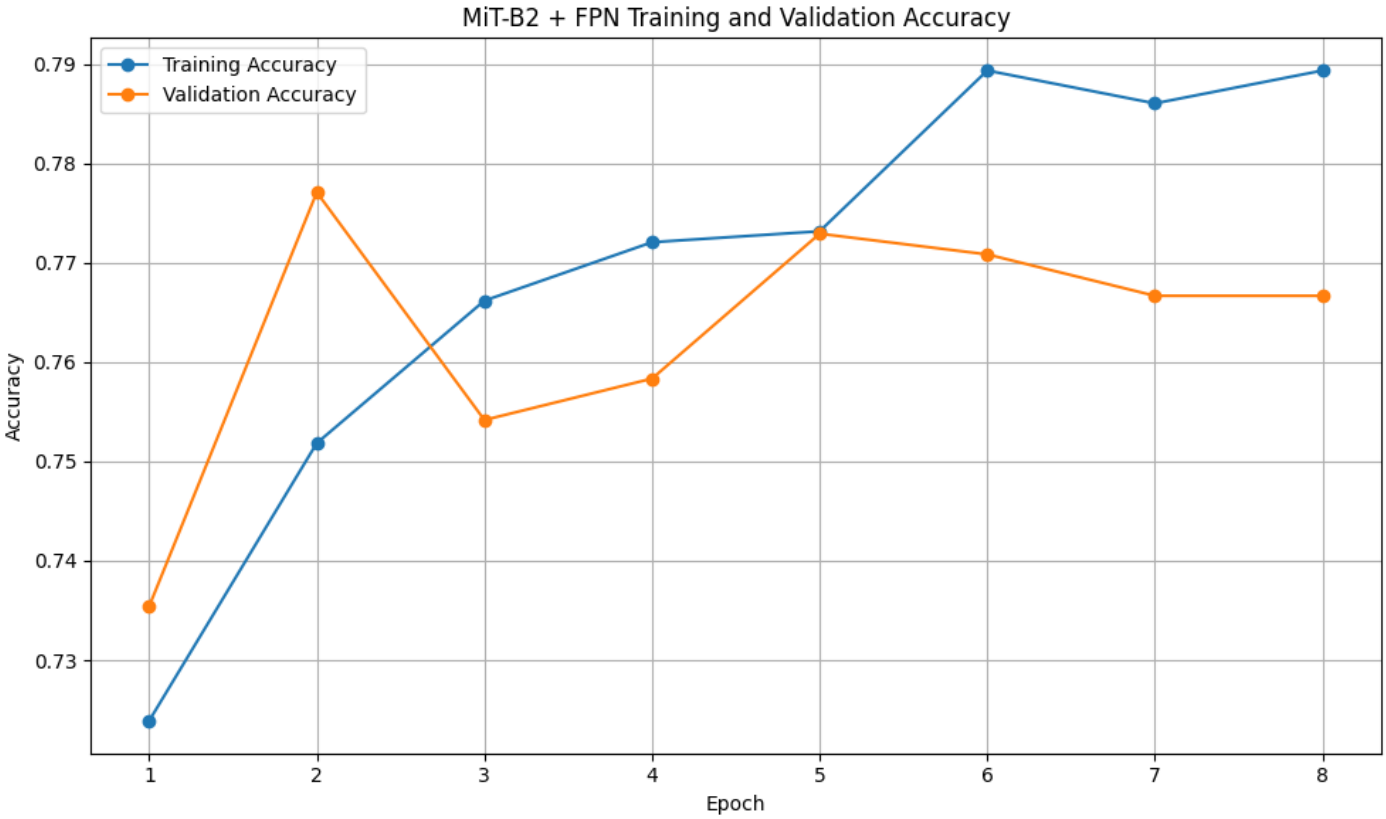
Confusion Matrix



41. MiT-B2 + MobileNetV2 + AdaBoost — graphs (1/2)

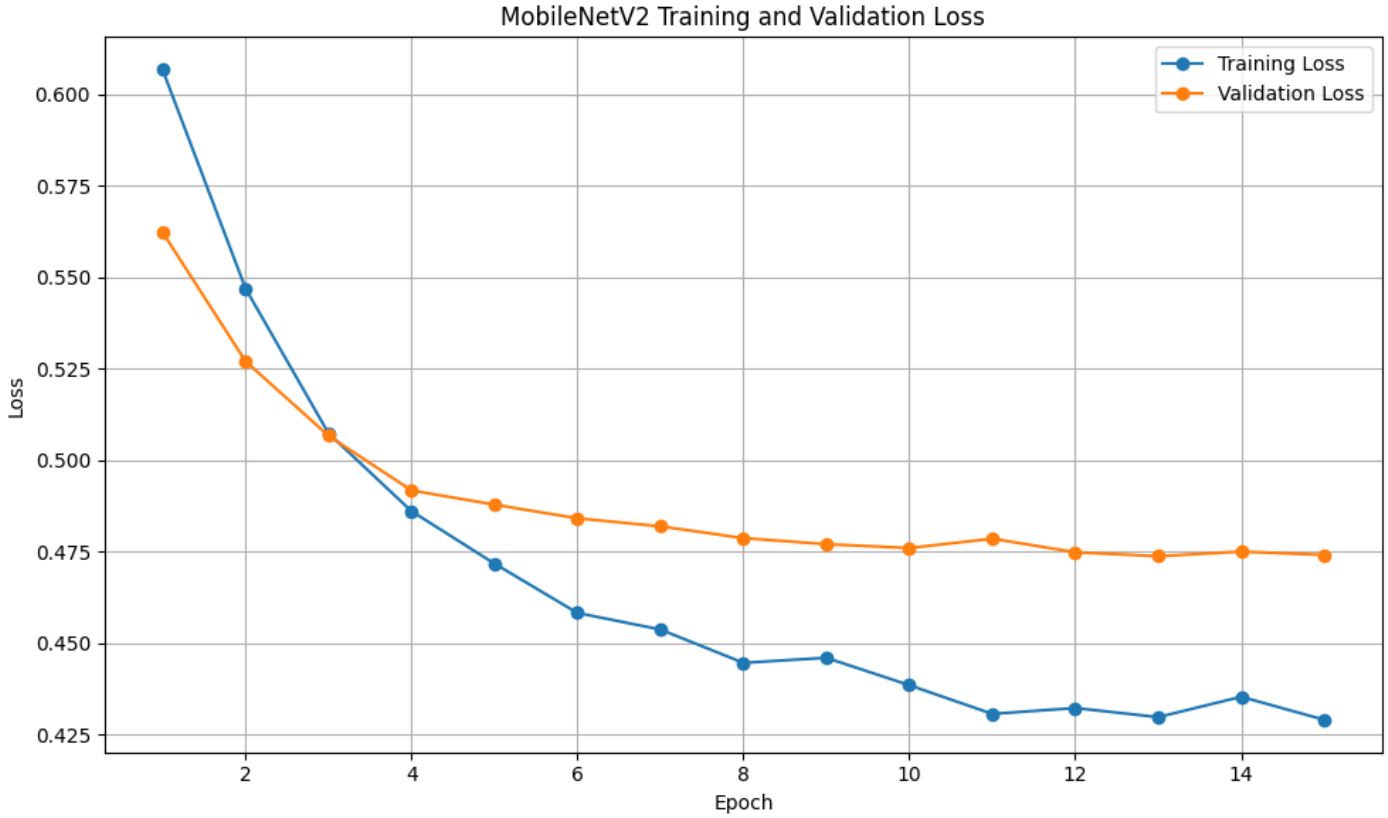
Training / Validation Graphs

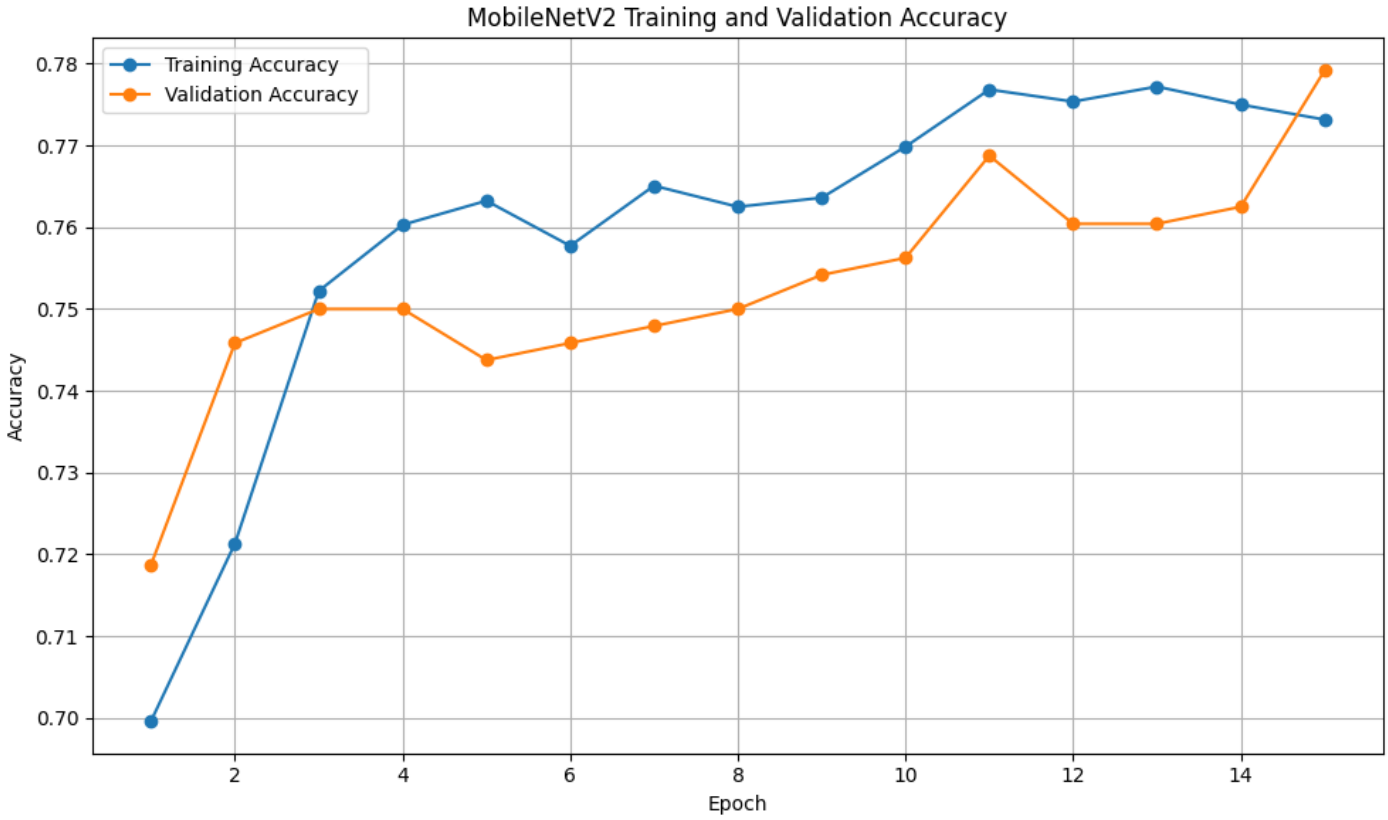




41. MiT-B2 + MobileNetV2 + AdaBoost — graphs (2/2)

Training / Validation Graphs





42. EfficientNet-B3 + Gaussian Naive Bayes

Source notebook: efficinetnet b3 baseline and ml.ipynb (inside Models_DL.zip)

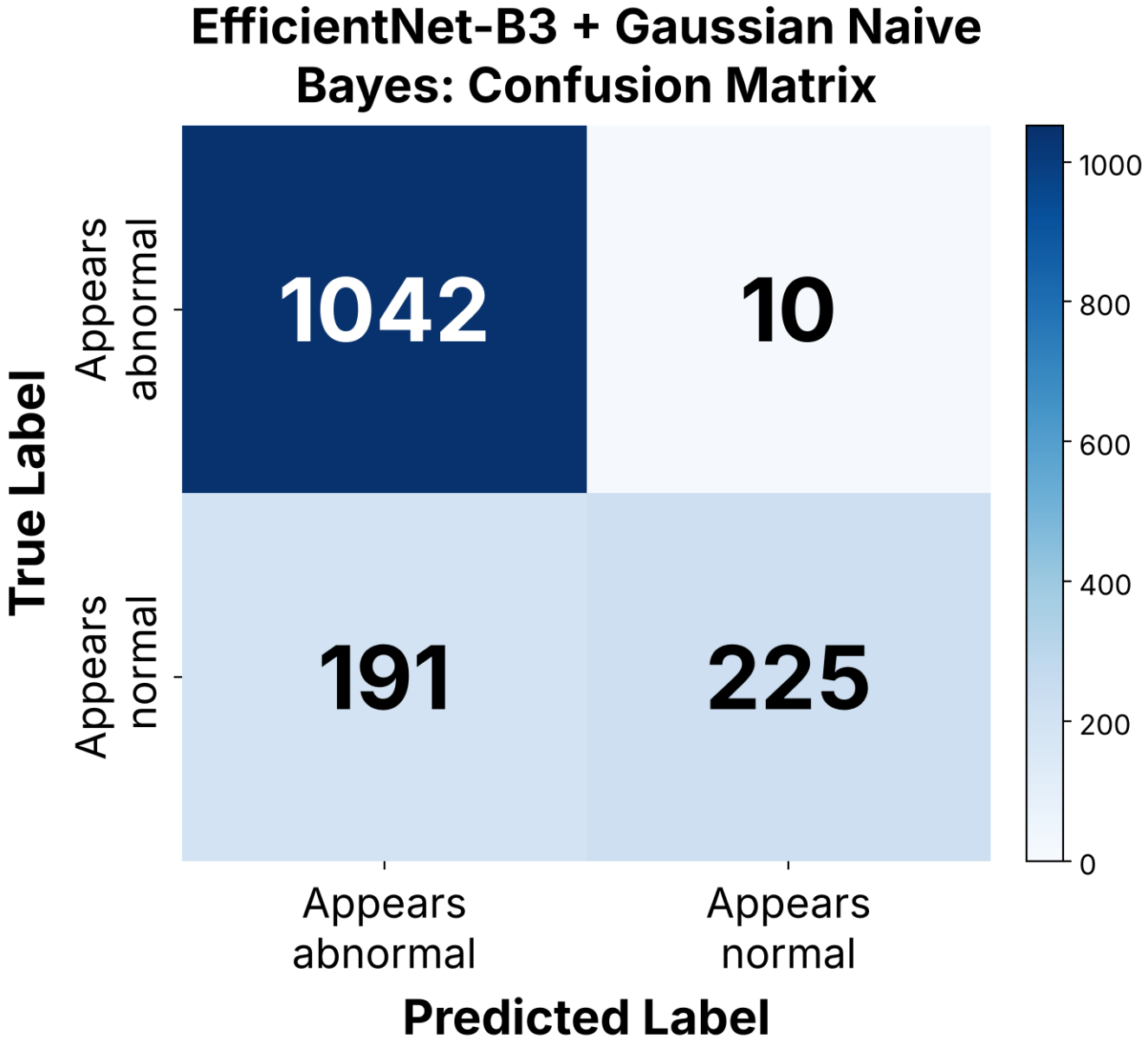
Accuracy: 86.31%

Precision: 87.69%

Recall: 86.31%

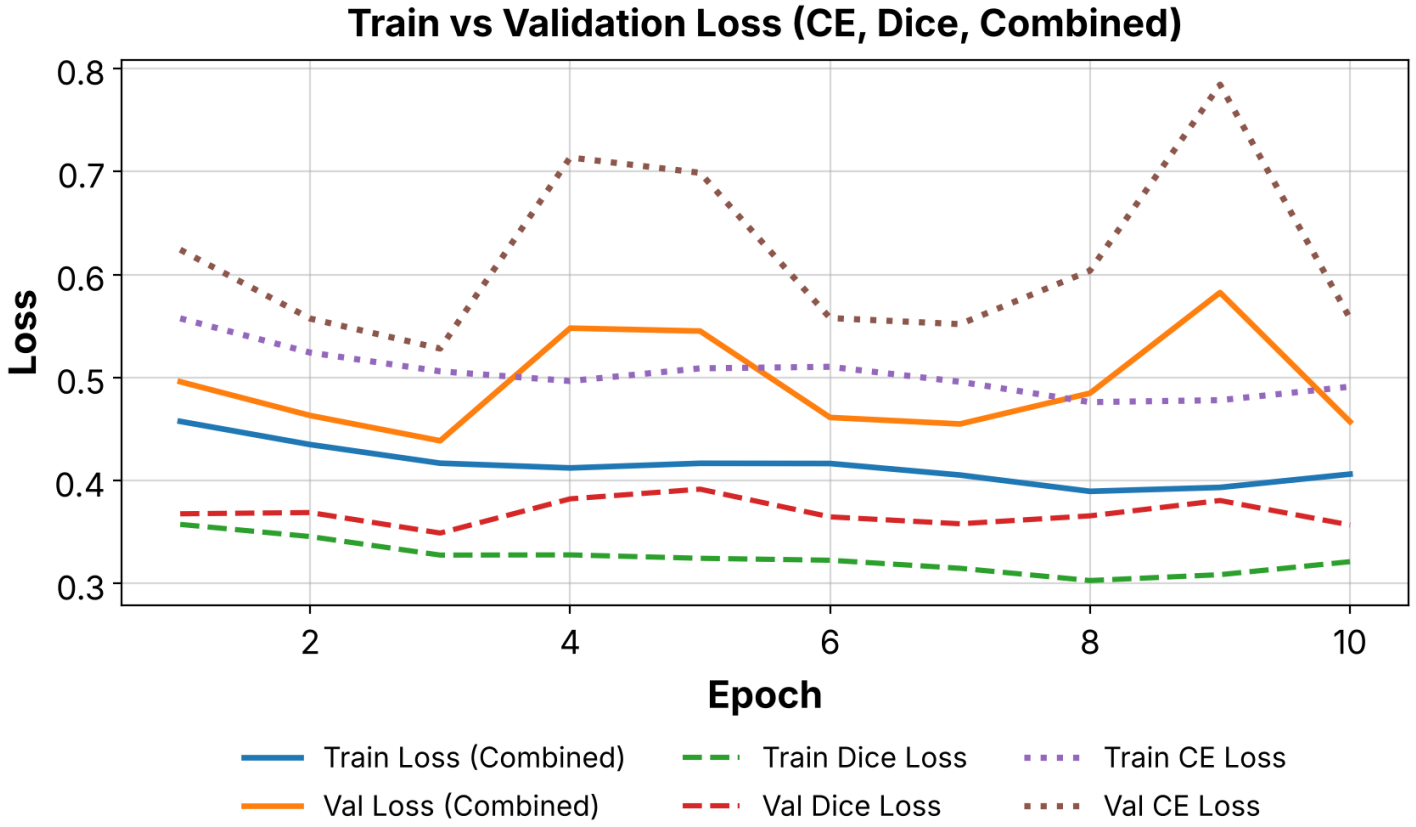
F1-score: 84.95%

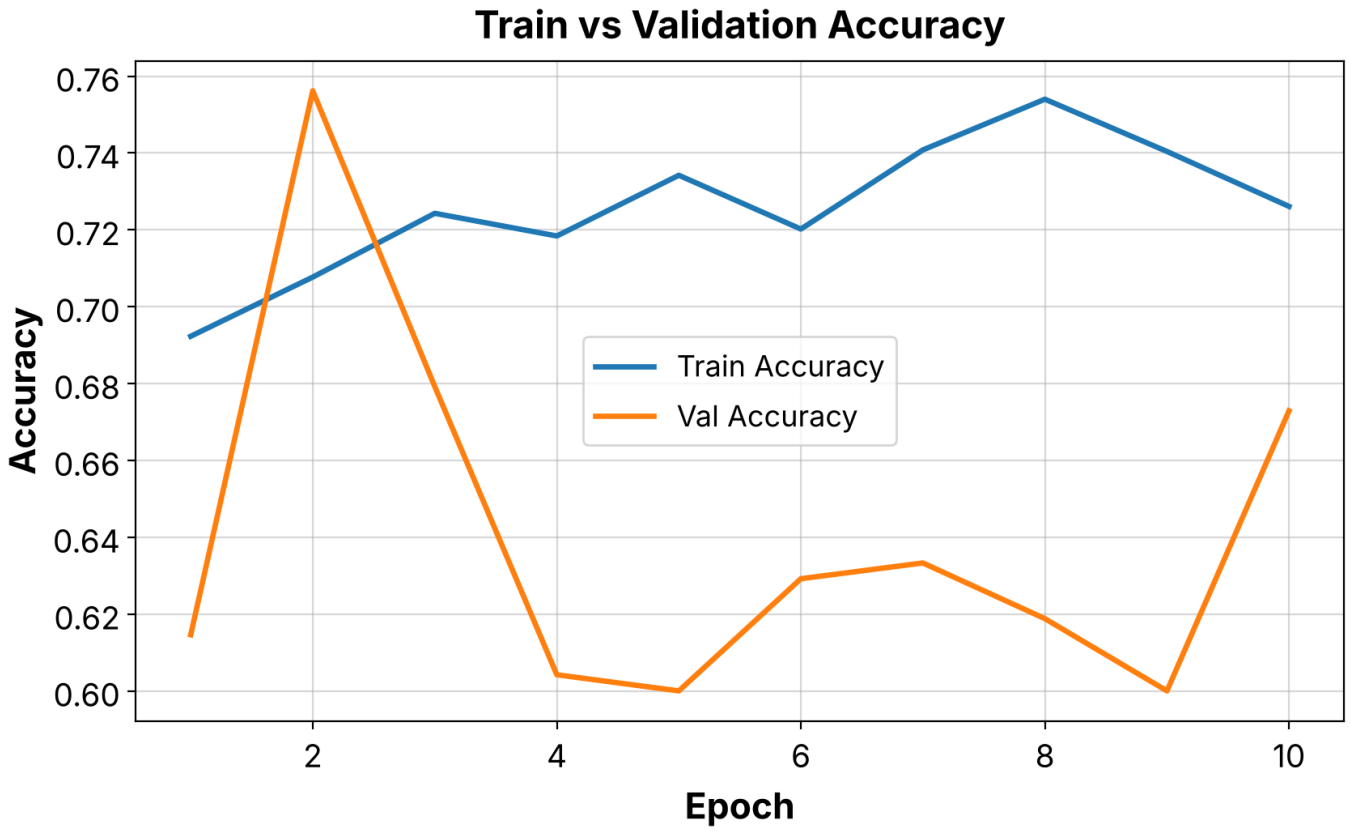
Confusion Matrix



42. EfficientNet-B3 + Gaussian Naive Bayes — graphs

Training / Validation Graphs





43. MiT-B2 + FPN — Direct (fine-tuned CNN)

Source notebook: MIT + RF and MIT only.ipynb (inside MIT + MobileNet.zip)

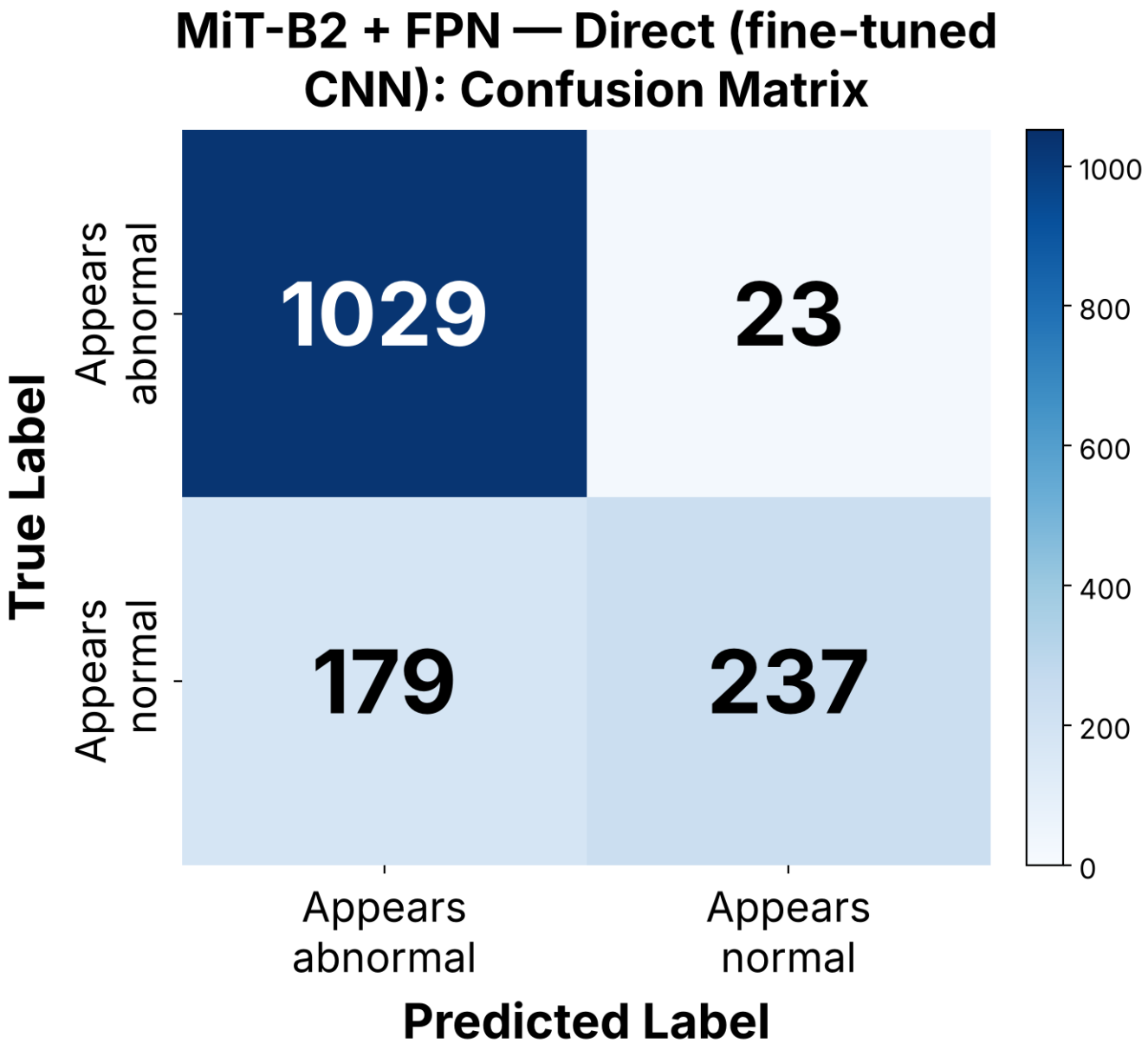
Accuracy: 86.24%

Precision: 86.87%

Recall: 86.24%

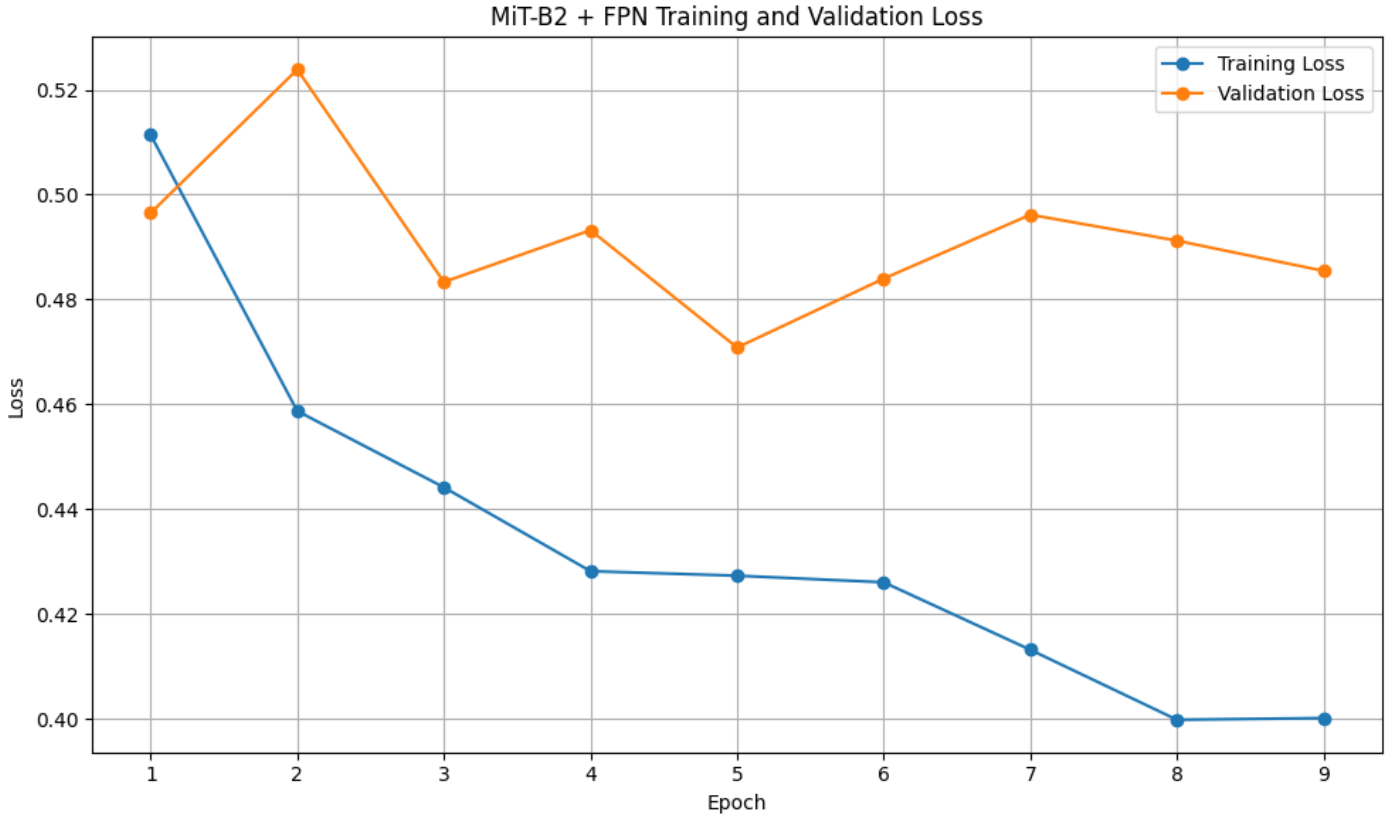
F1-score: 85.13%

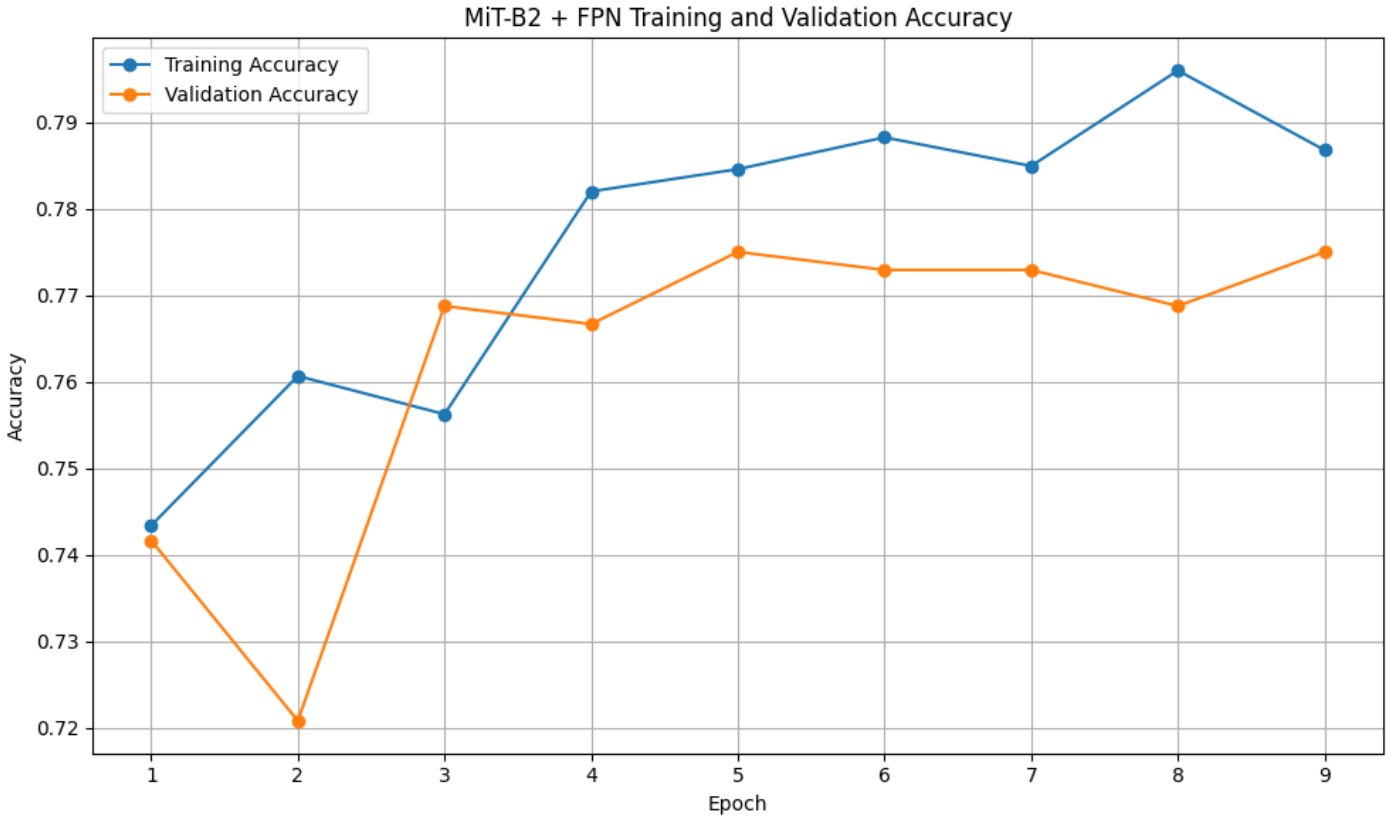
Confusion Matrix



43. MiT-B2 + FPN — Direct (fine-tuned CNN) — graphs

Training / Validation Graphs





44. AlexNet + XGBoost

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

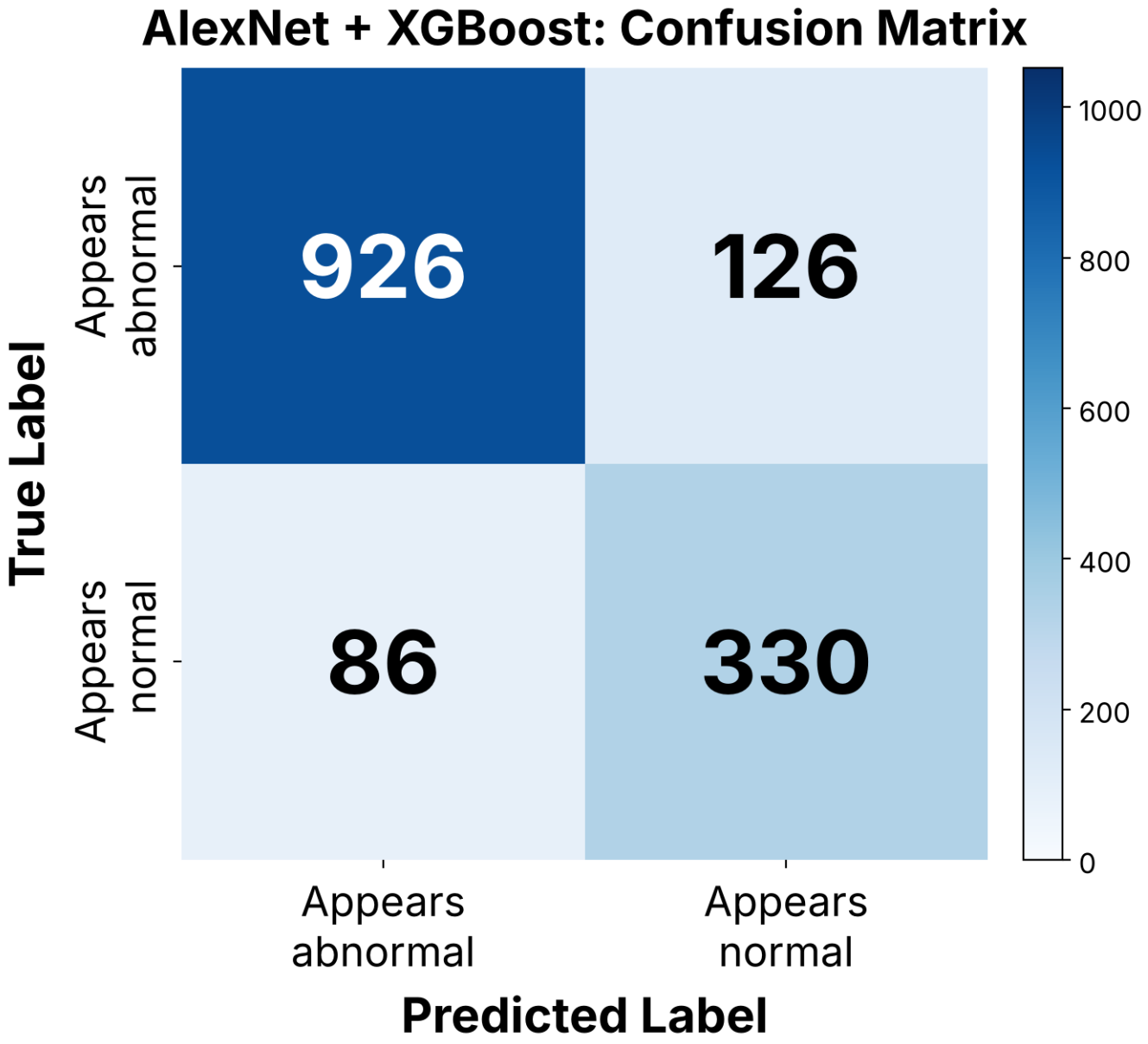
Accuracy: 85.56%

Precision: 86.08%

Recall: 85.56%

F1-score: 85.75%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

45. AlexNet — Direct (fine-tuned CNN)

Source notebook: AlexNet with Dice Loss (Baseline).ipynb (inside Models_DL.zip)

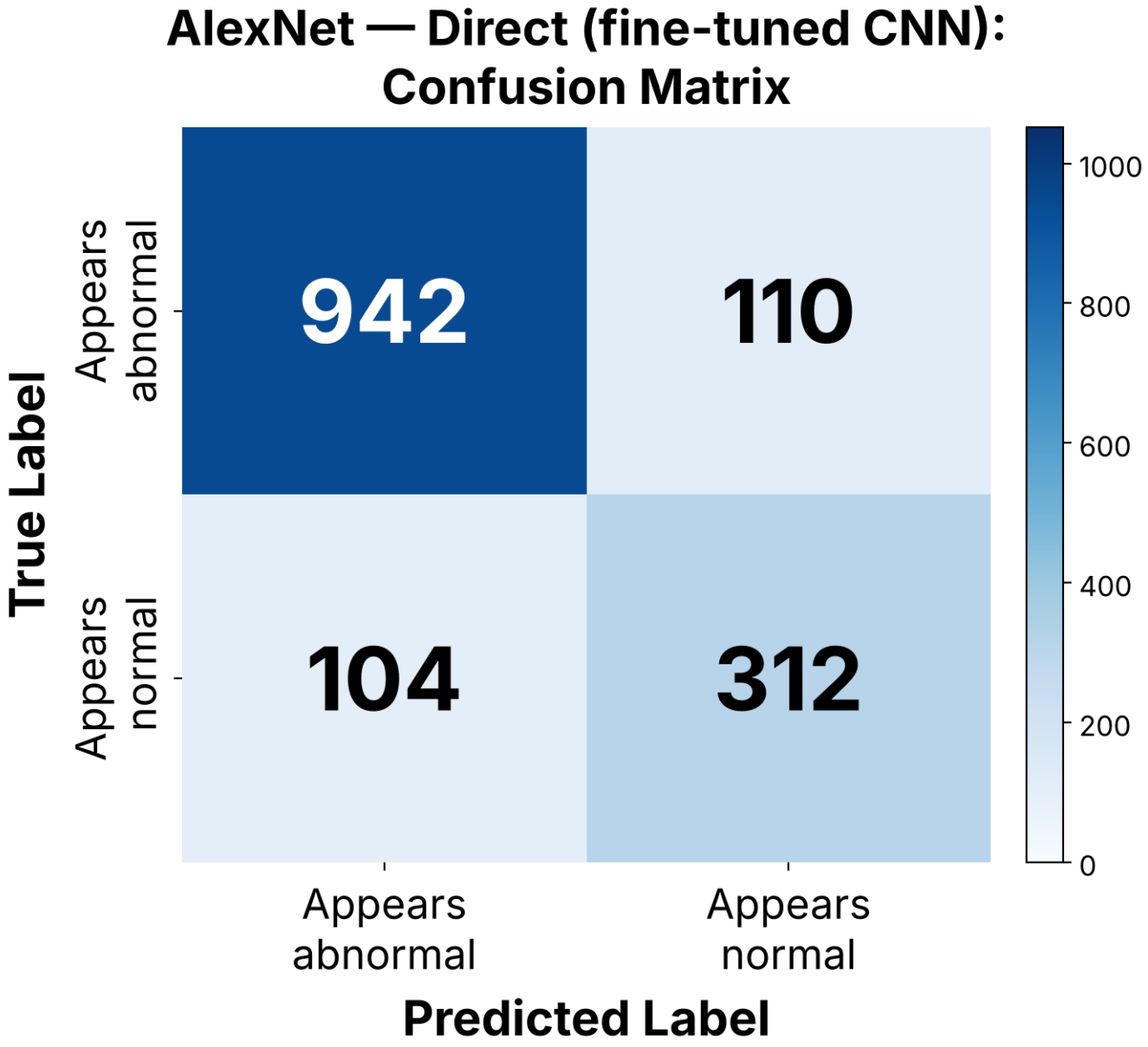
Accuracy: 85.42%

Precision: 85.49%

Recall: 85.42%

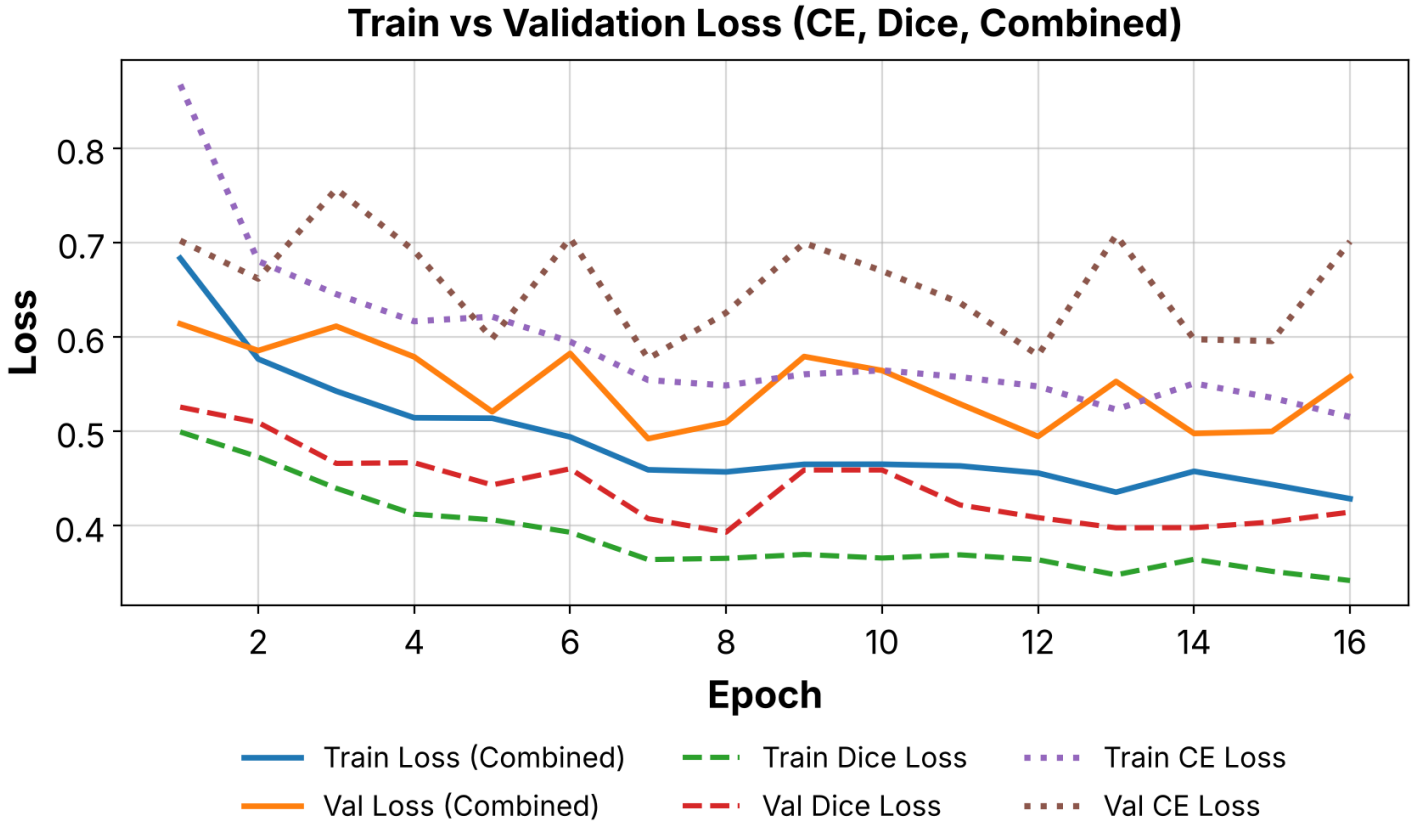
F1-score: 85.45%

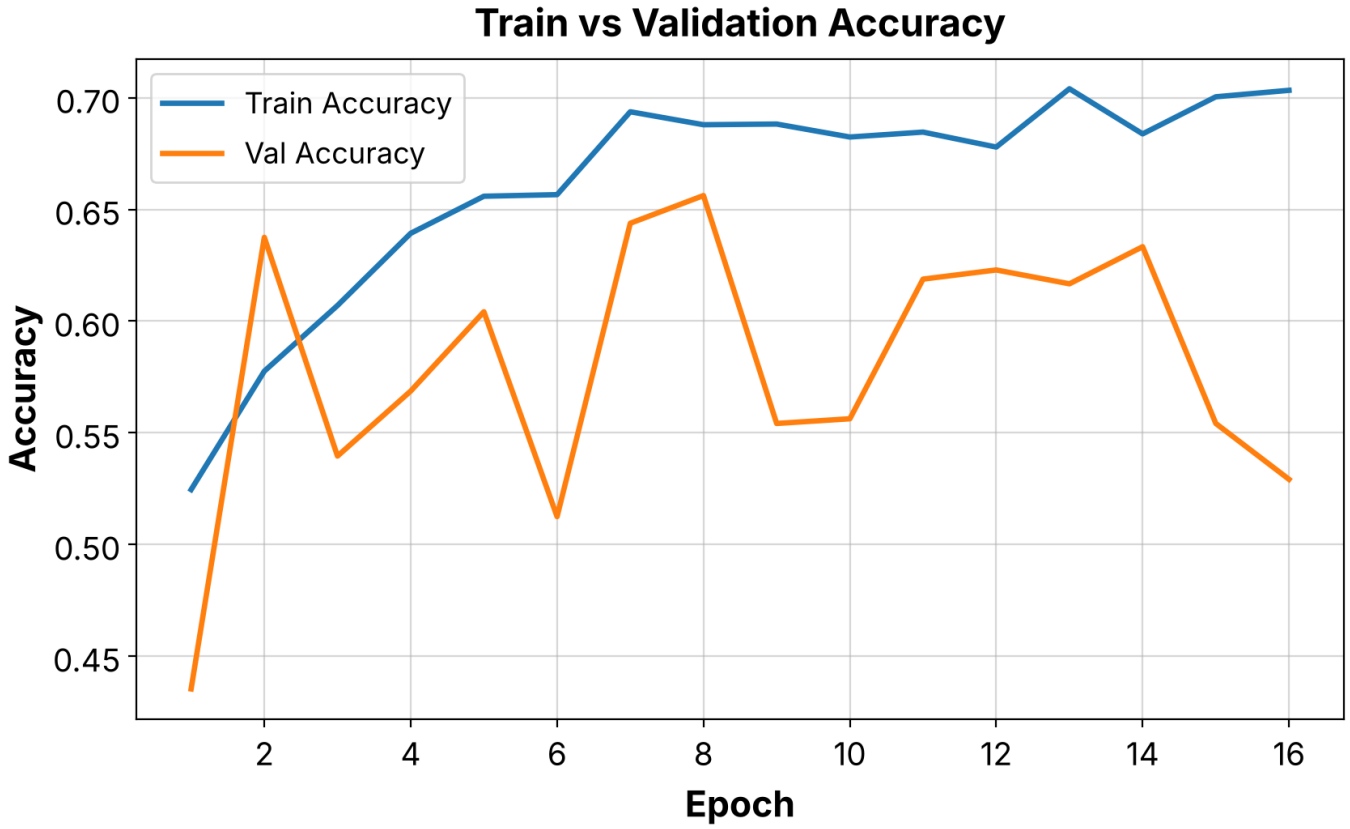
Confusion Matrix



45. AlexNet — Direct (fine-tuned CNN) — graphs

Training / Validation Graphs





46. InceptionV3 + KNN

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

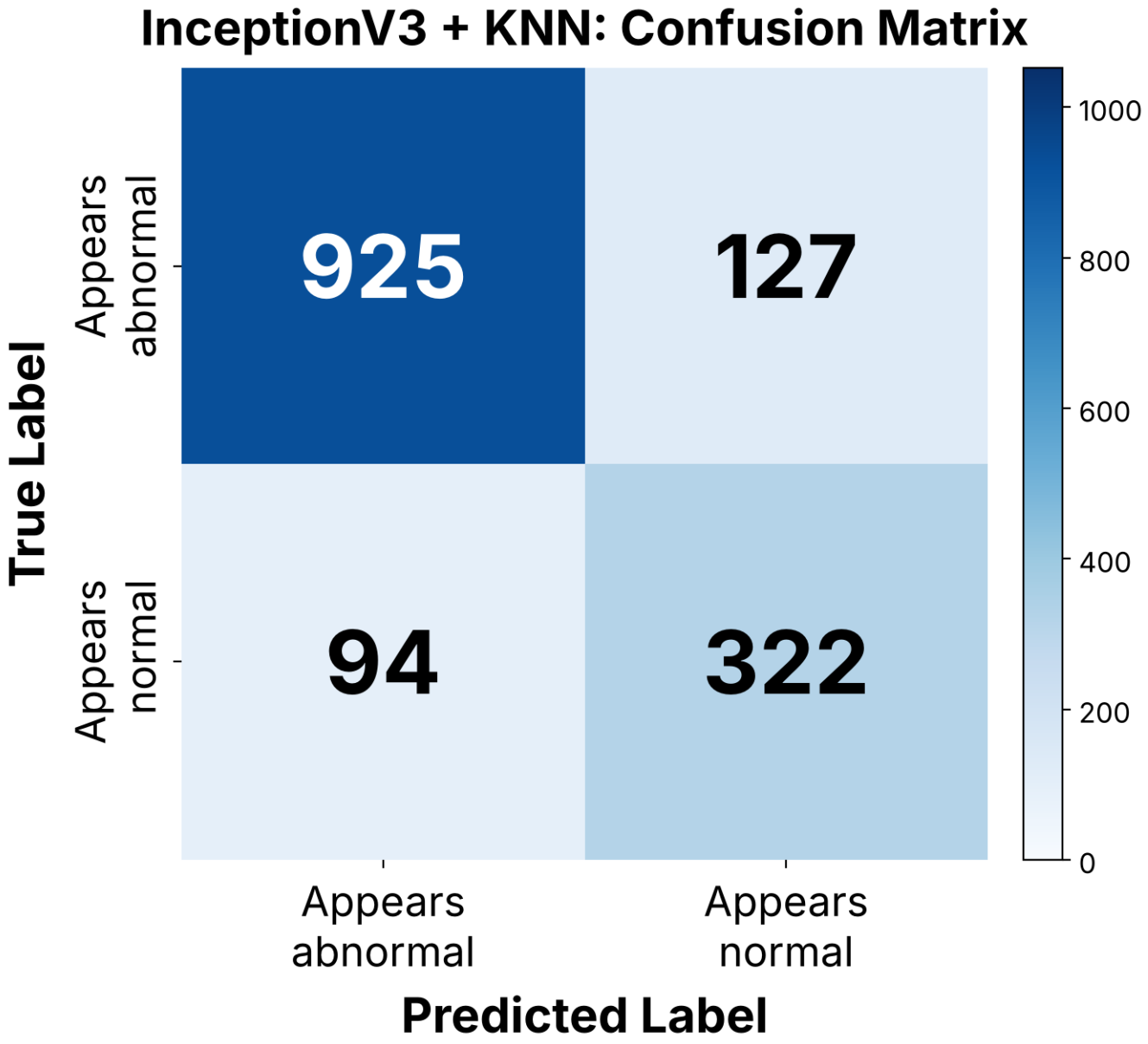
Accuracy: 84.95%

Precision: 85.37%

Recall: 84.95%

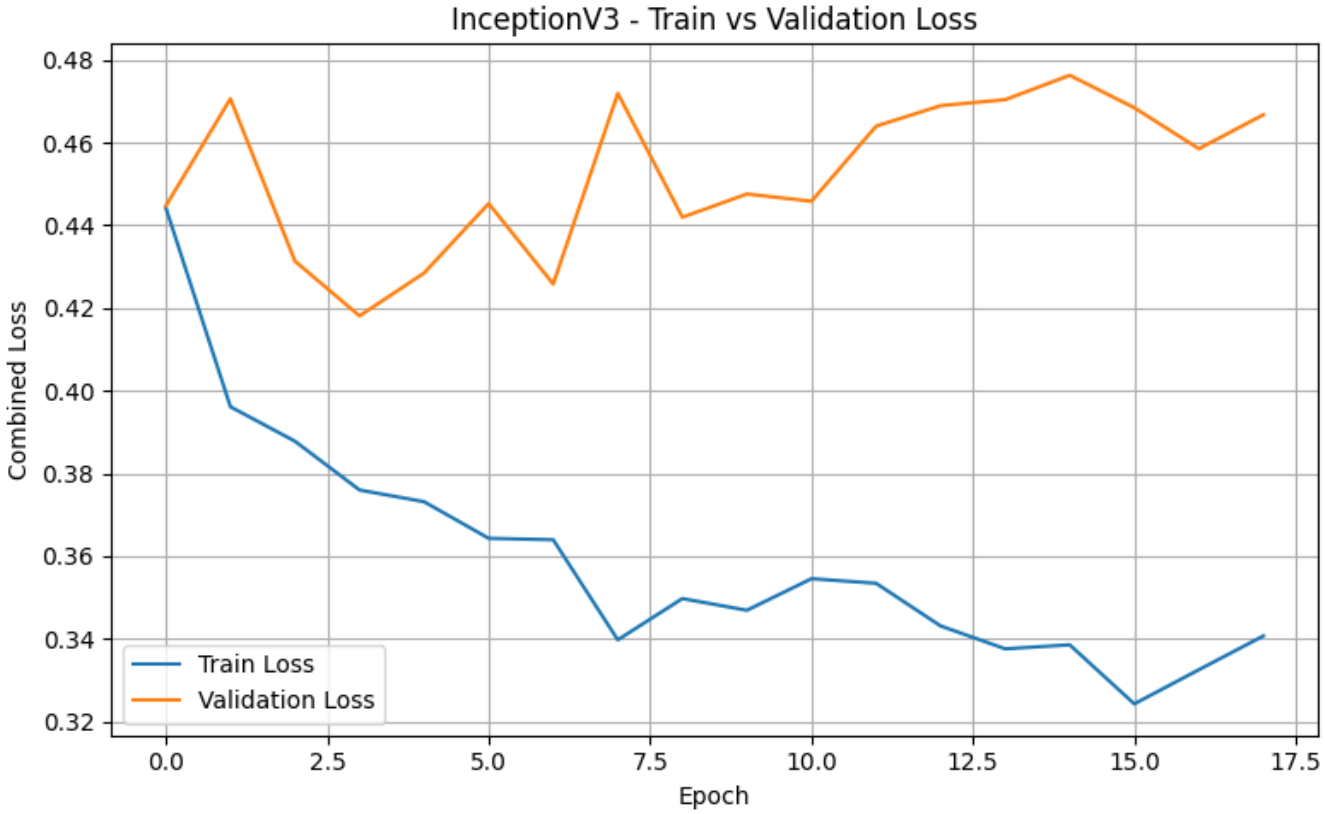
F1-score: 85.11%

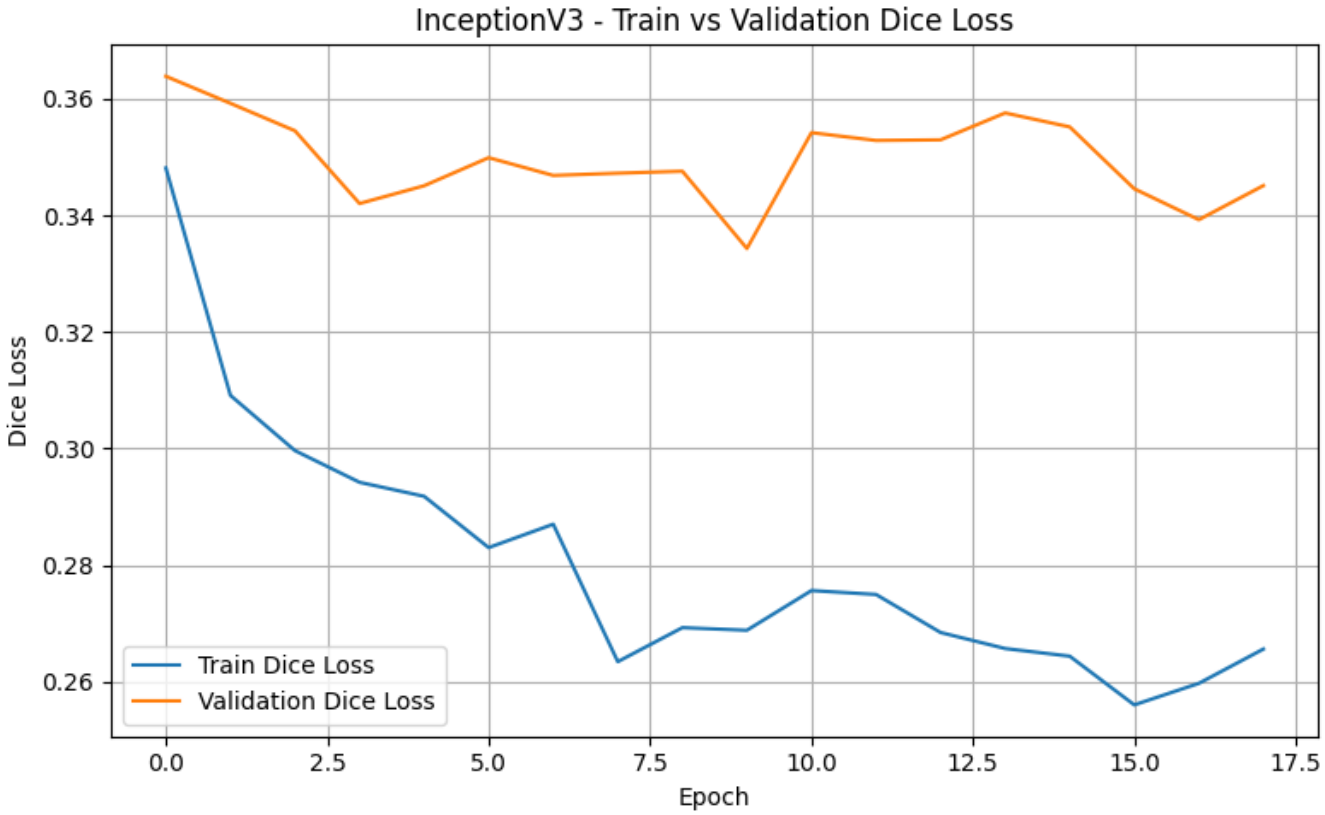
Confusion Matrix



46. InceptionV3 + KNN — graphs (1/2)

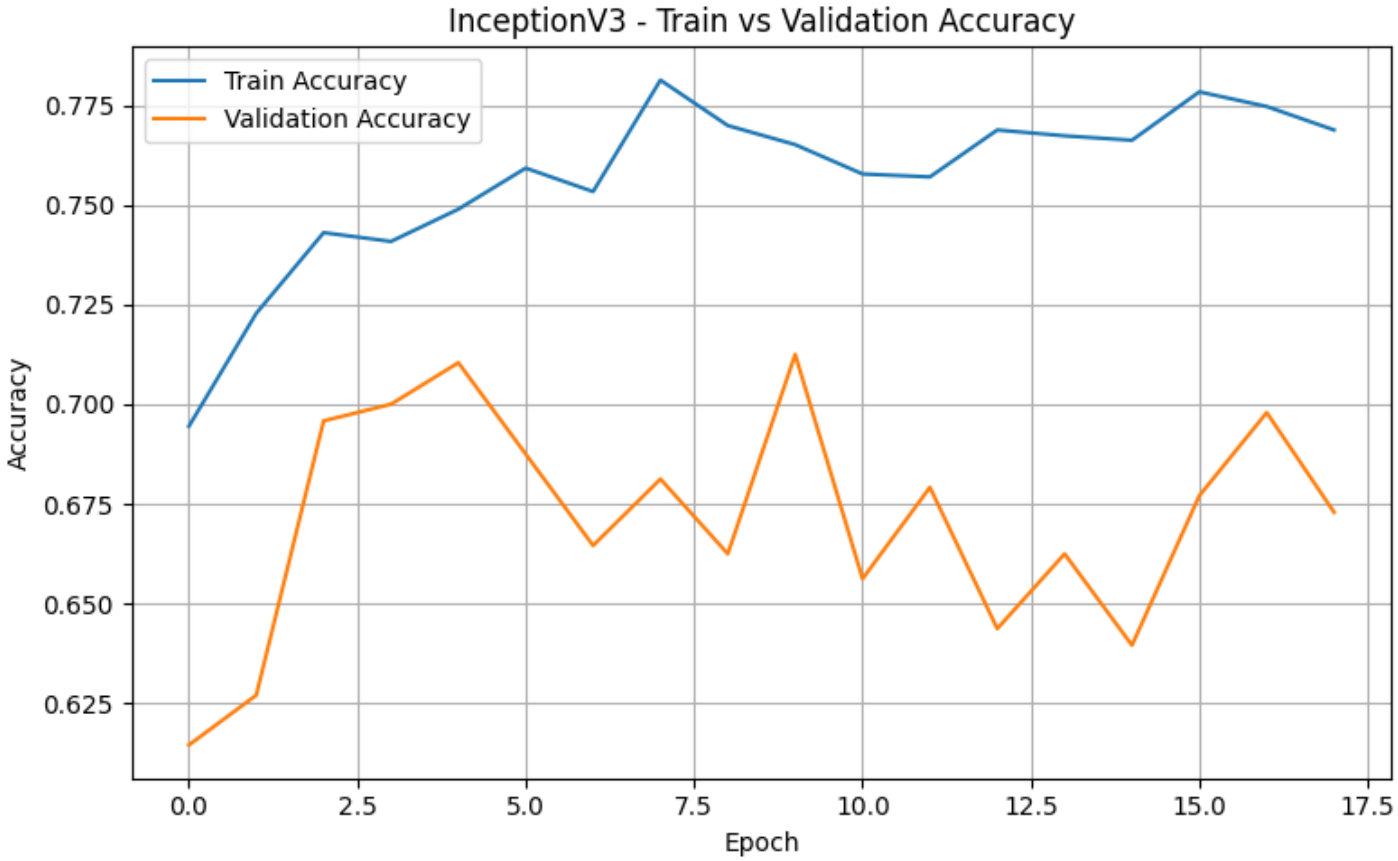
Training / Validation Graphs





46. InceptionV3 + KNN — graphs (2/2)

Training / Validation Graphs



47. AlexNet + SVM

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

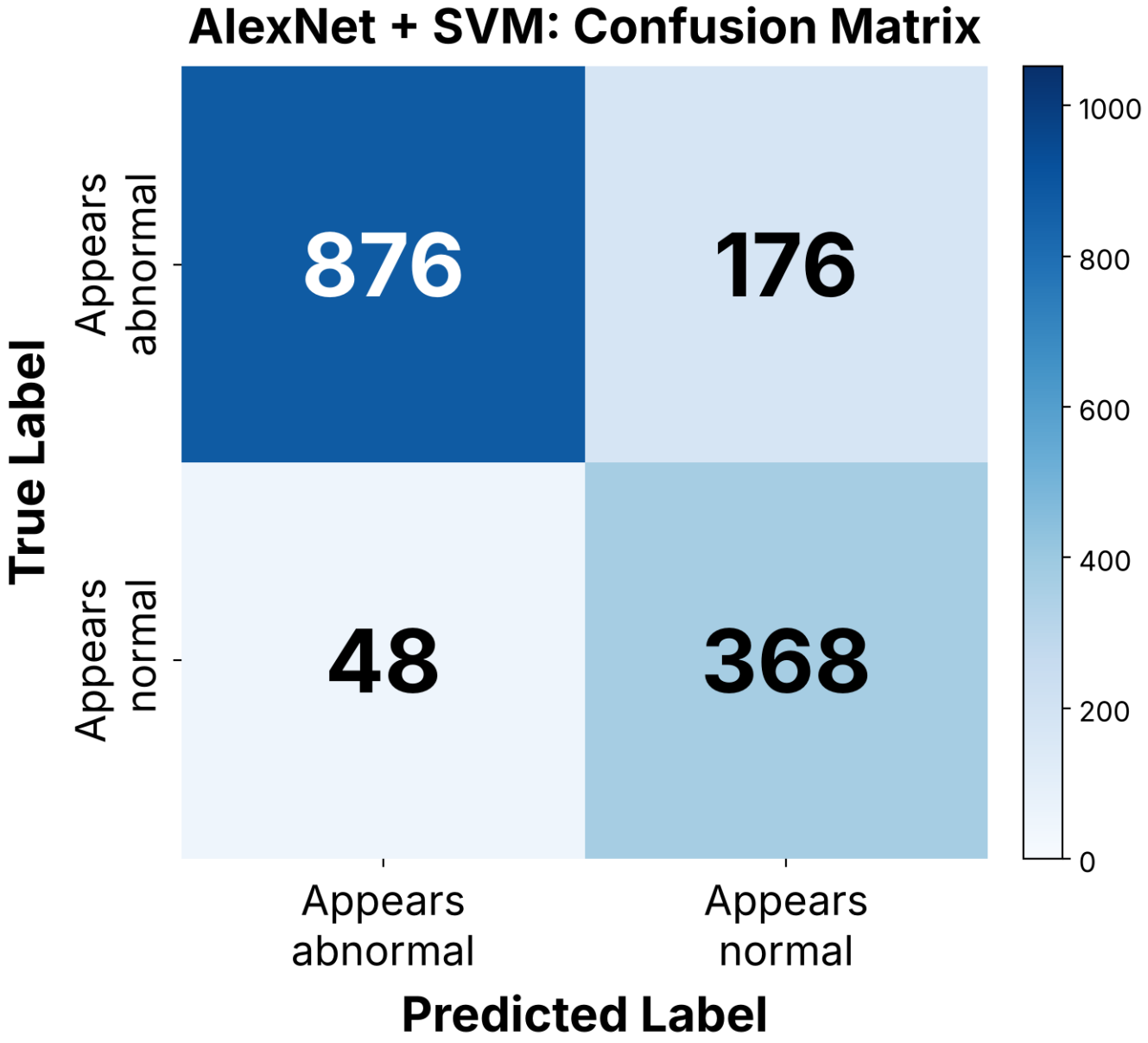
Accuracy: 84.74%

Precision: 87.11%

Recall: 84.74%

F1-score: 85.26%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

48. VGG16 — Direct — 5-fold CV

Source notebook: Vgg kfold.ipynb (inside Models_DL.zip)

Accuracy: 84.20%

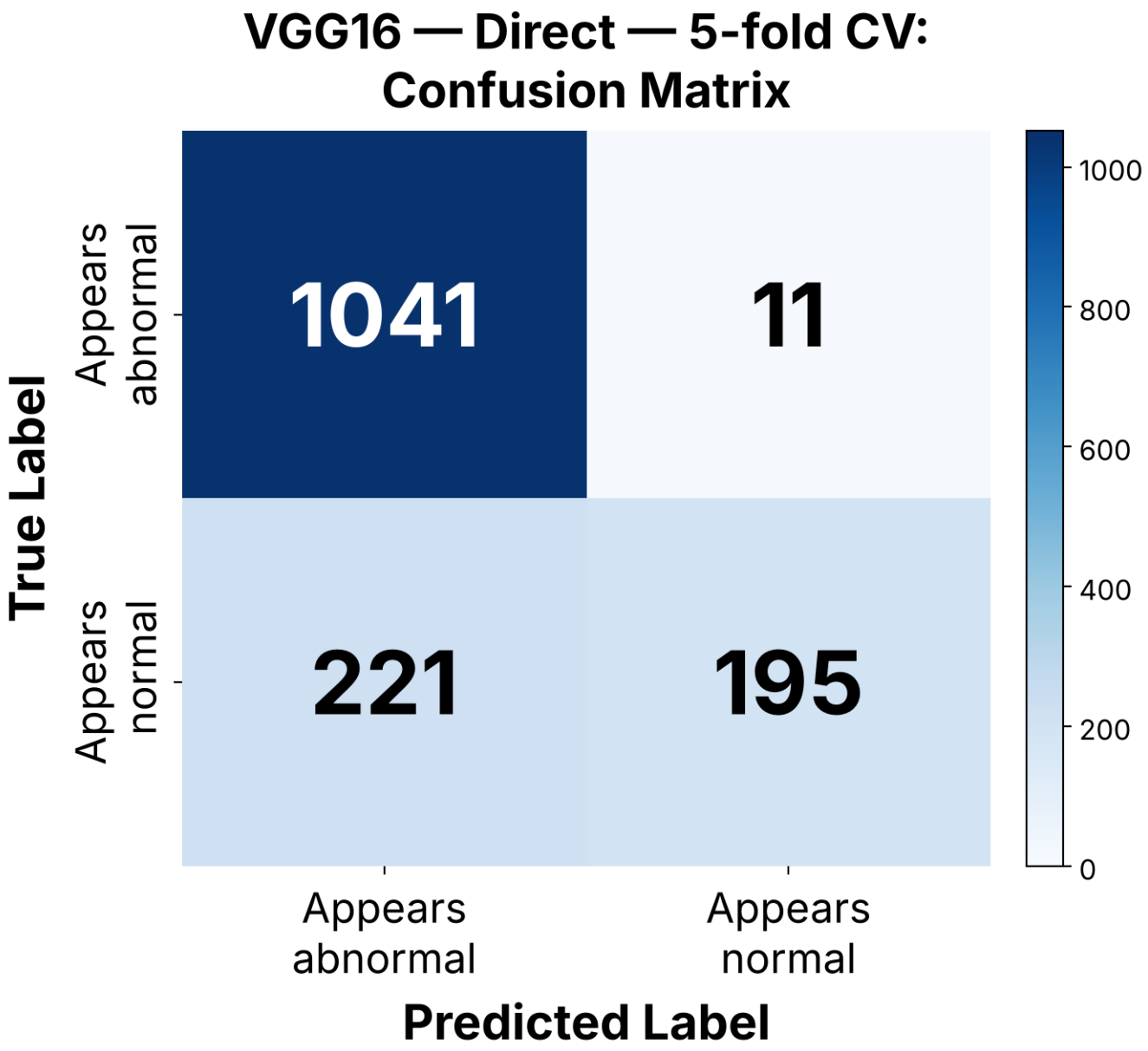
Precision: 85.94%

Recall: 84.20%

F1-score: 82.25%

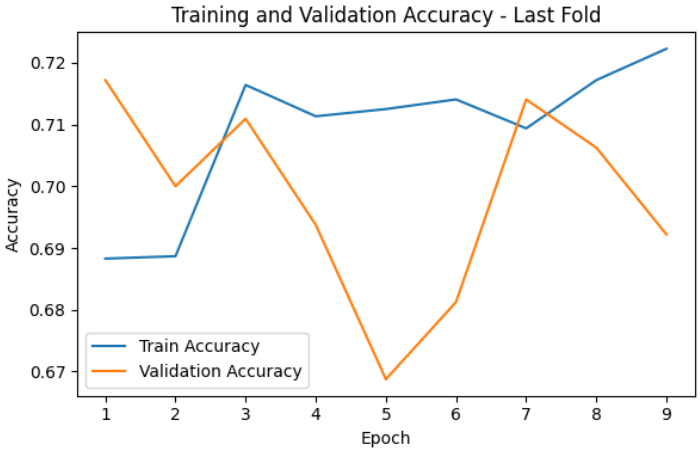
Note: the notebook prints 81.06% as its test / “Final Test Accuracy”, which does not agree with the notebook’s own printed confusion matrix and weighted recall (in the Dice-loss notebook this number is simply the best *validation* accuracy). The test accuracy shown here is the one implied by the confusion matrix and equals the weighted recall.

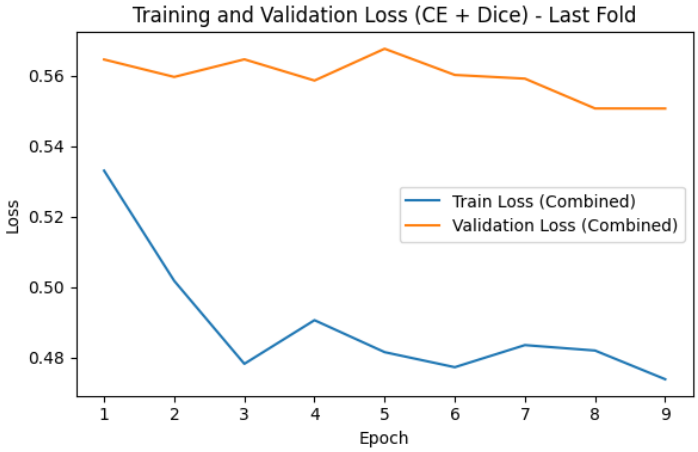
Confusion Matrix

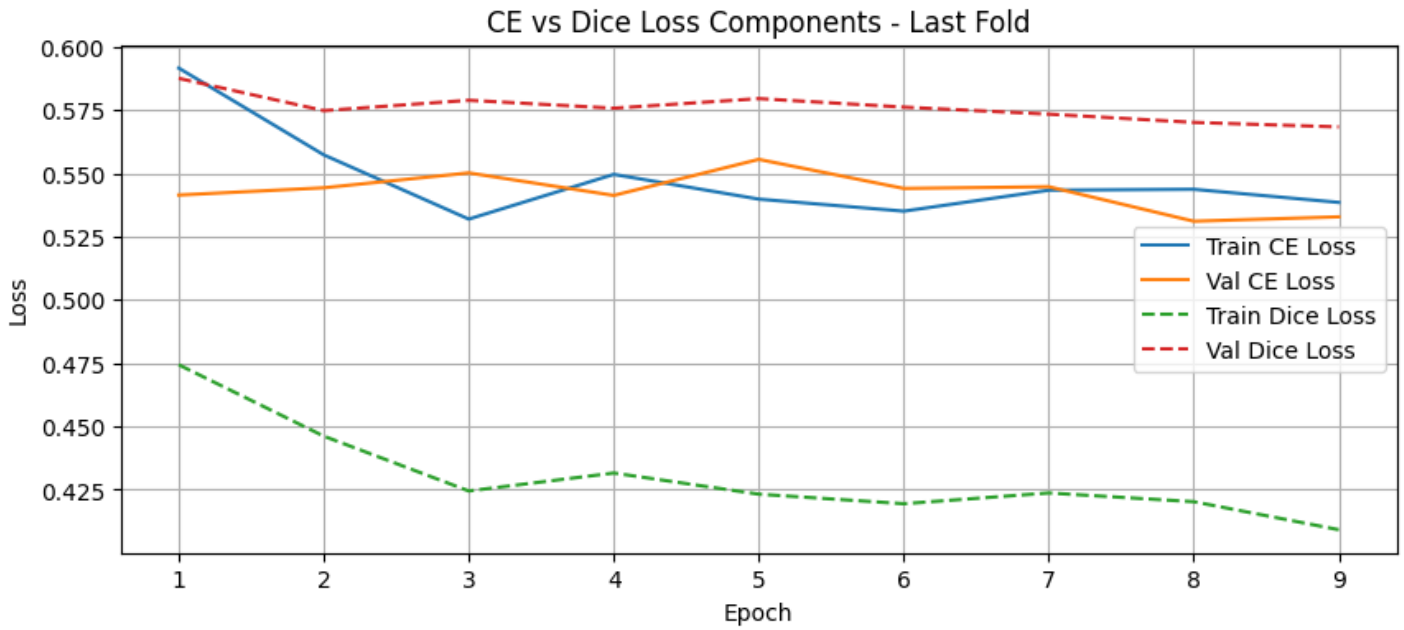


48. VGG16 — Direct — 5-fold CV — graphs (1/2)

Training / Validation Graphs

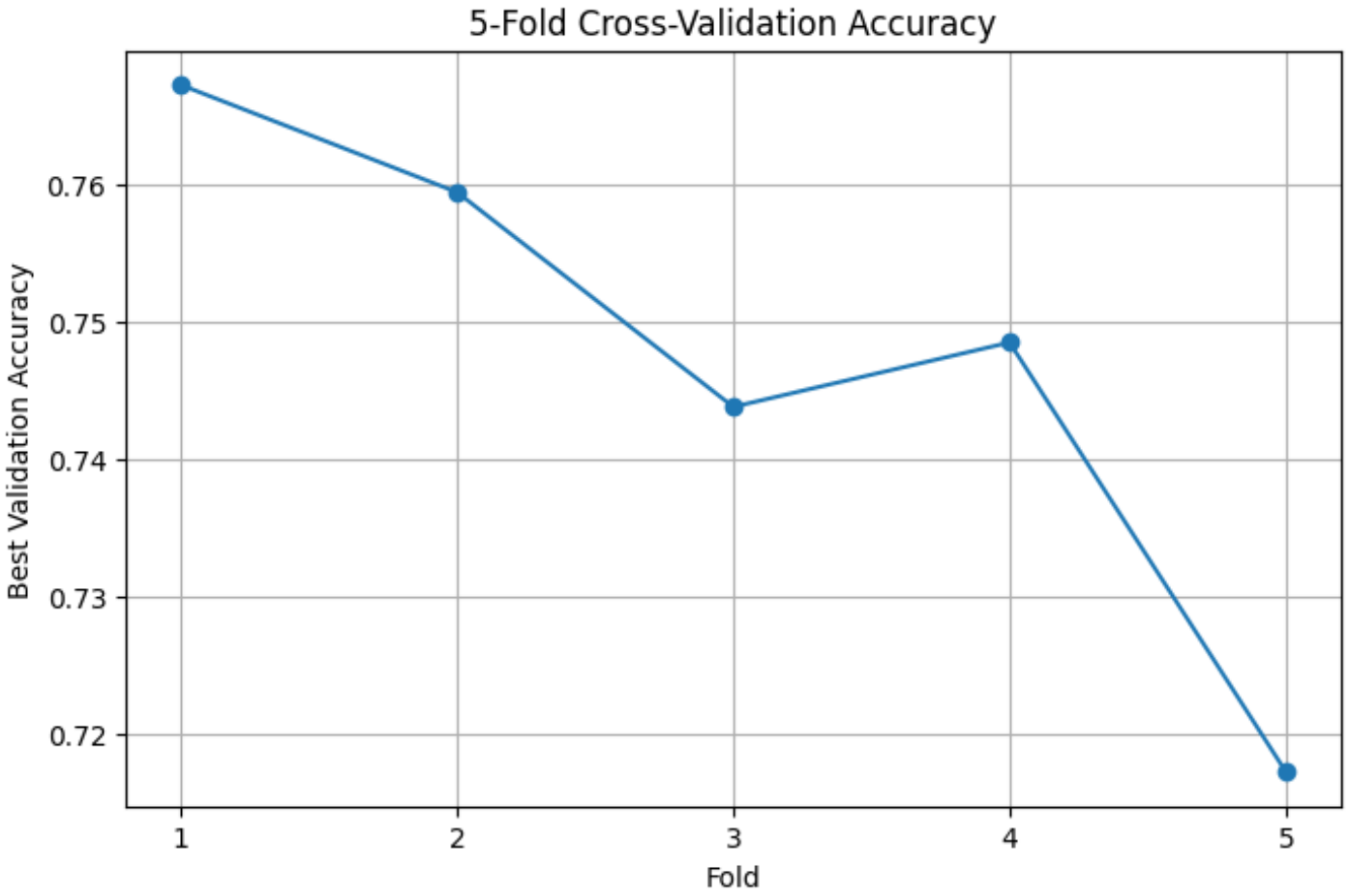






48. VGG16 — Direct — 5-fold CV — graphs (2/2)

Training / Validation Graphs



49. InceptionV3 + Gaussian Naive Bayes

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

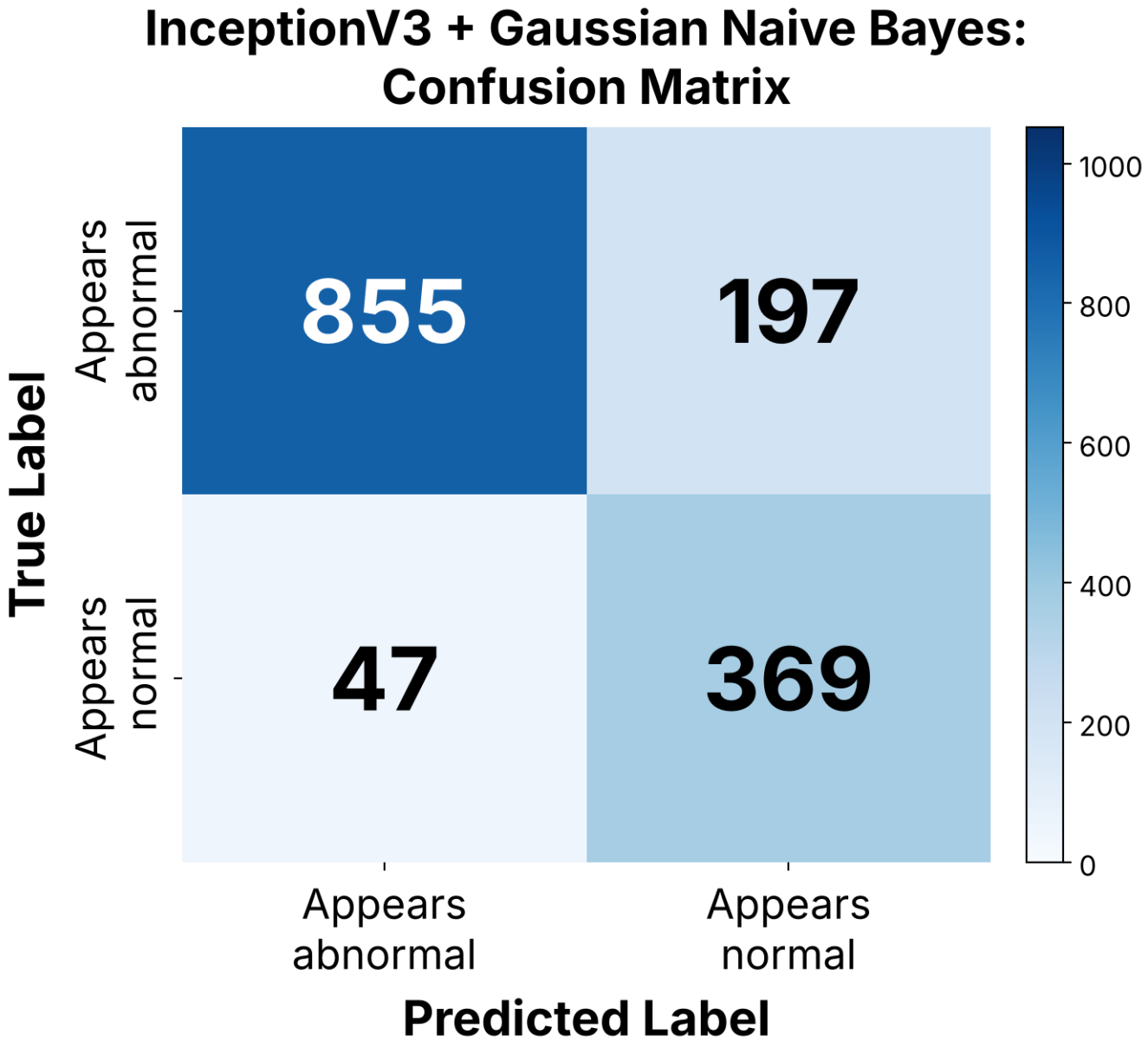
Accuracy: 83.38%

Precision: 86.40%

Recall: 83.38%

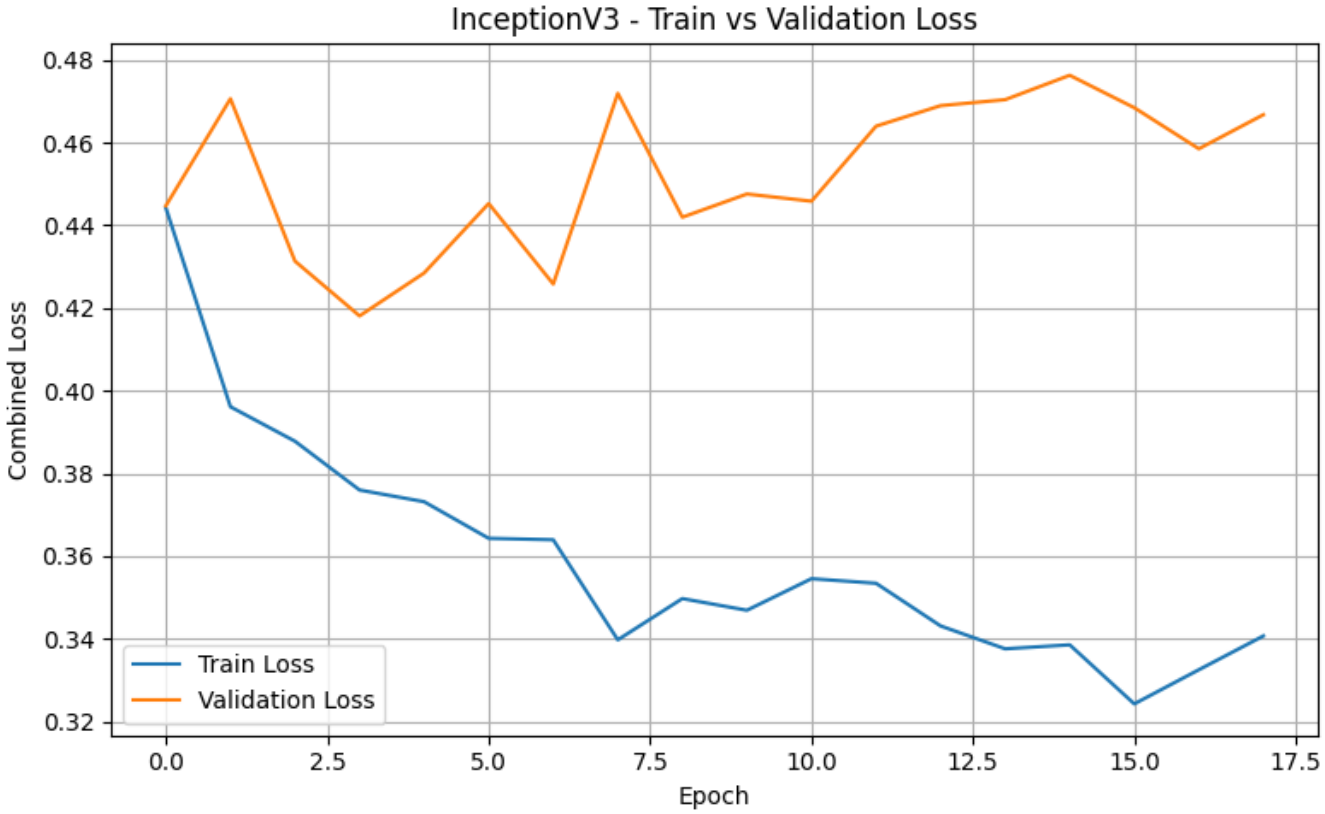
F1-score: 84.01%

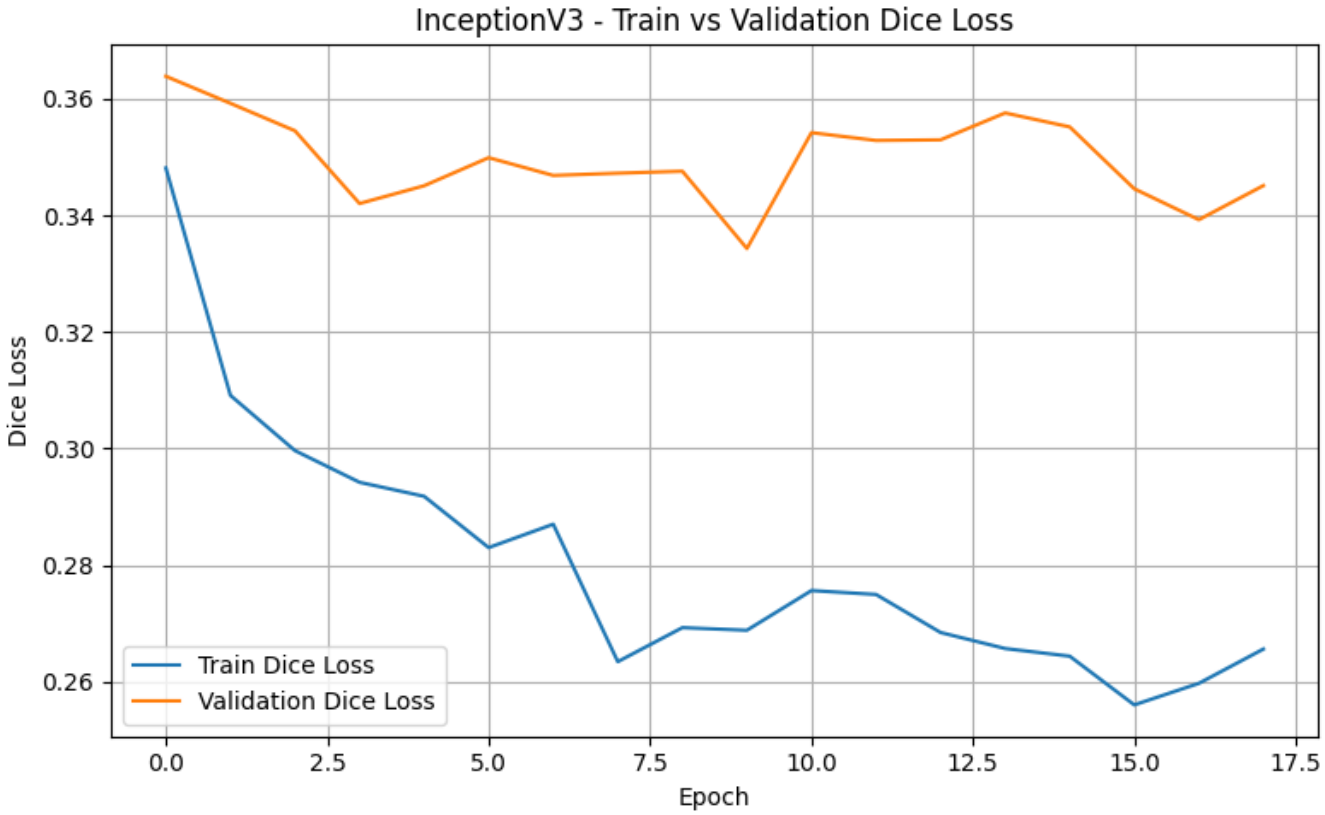
Confusion Matrix



49. InceptionV3 + Gaussian Naive Bayes — graphs (1/2)

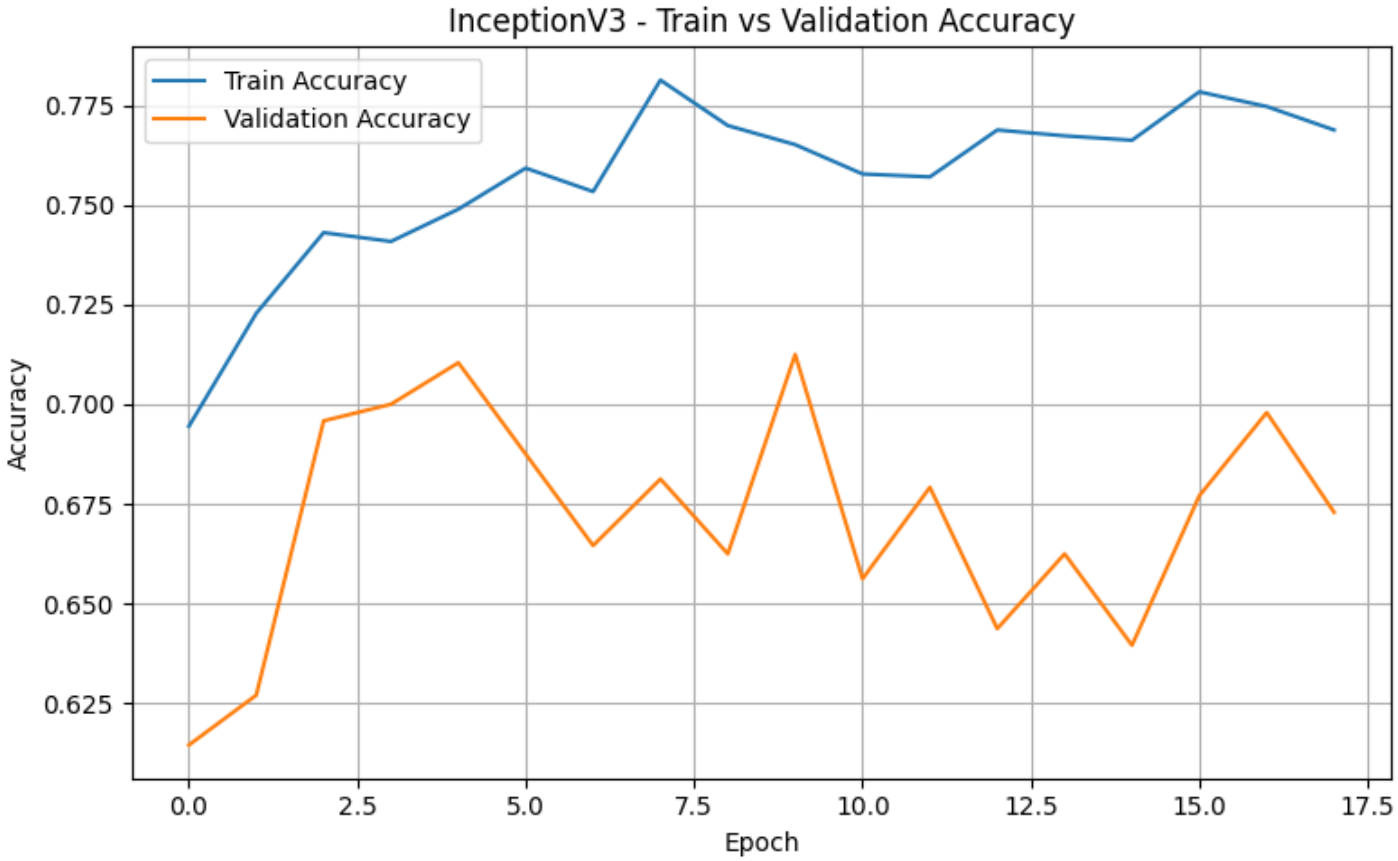
Training / Validation Graphs





49. InceptionV3 + Gaussian Naive Bayes — graphs (2/2)

Training / Validation Graphs



50. VGG16 + Gaussian Naive Bayes

Source notebook: Vgg with ML.ipynb (inside Models_DL.zip)

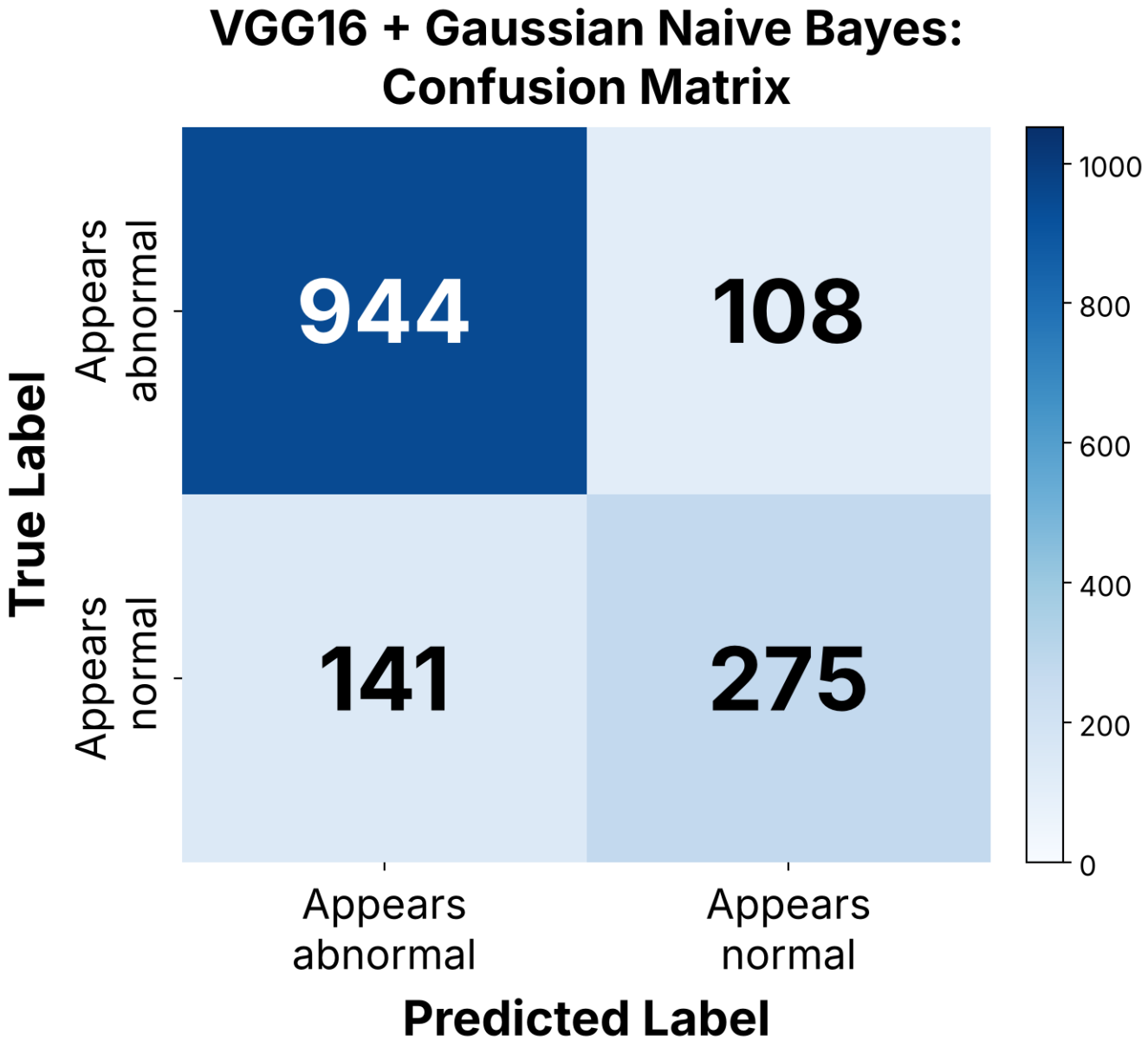
Accuracy: 83.04%

Precision: 82.70%

Recall: 83.04%

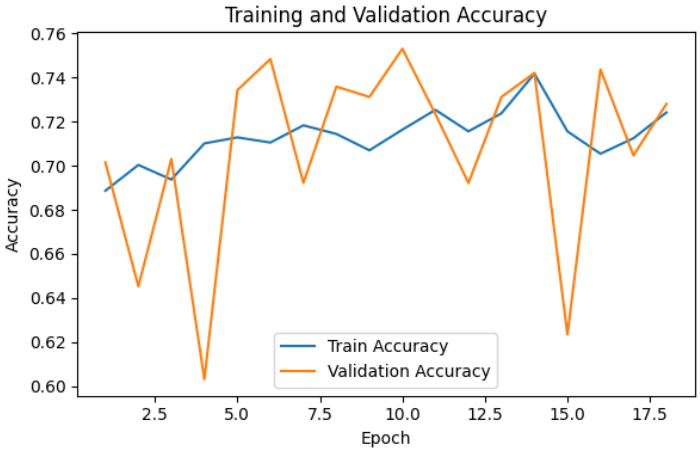
F1-score: 82.82%

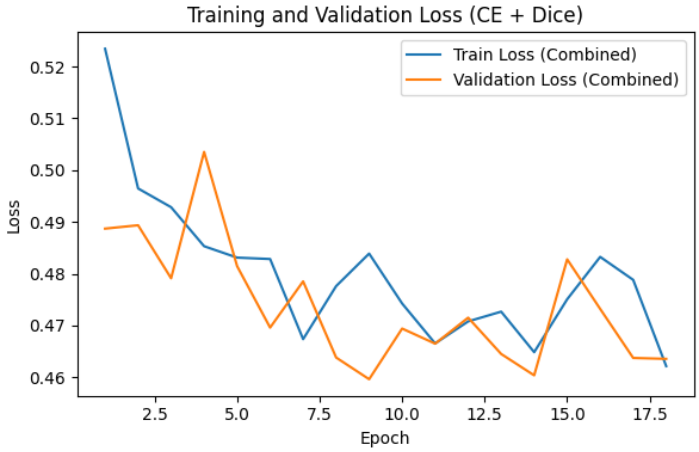
Confusion Matrix

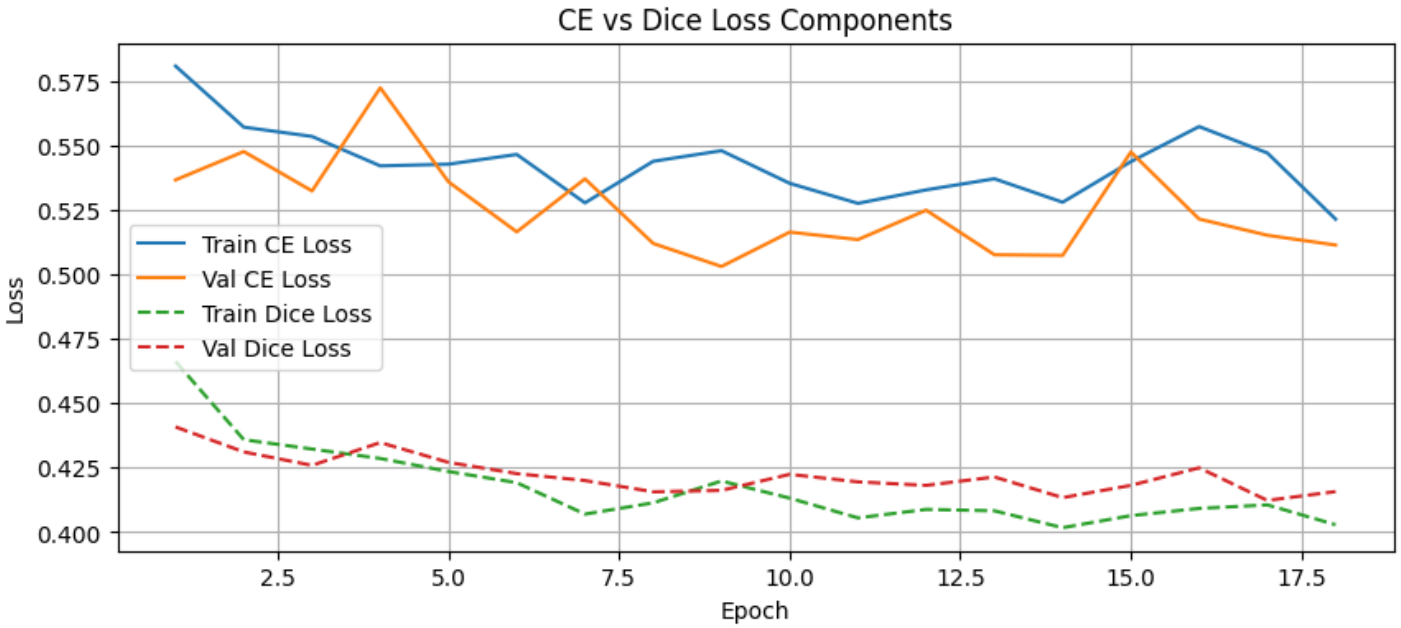


50. VGG16 + Gaussian Naive Bayes — graphs

Training / Validation Graphs







51. ResNet18 — Direct — 5-fold CV

Source notebook: ResNet wthi Kfold.ipynb (inside Models_DL.zip)

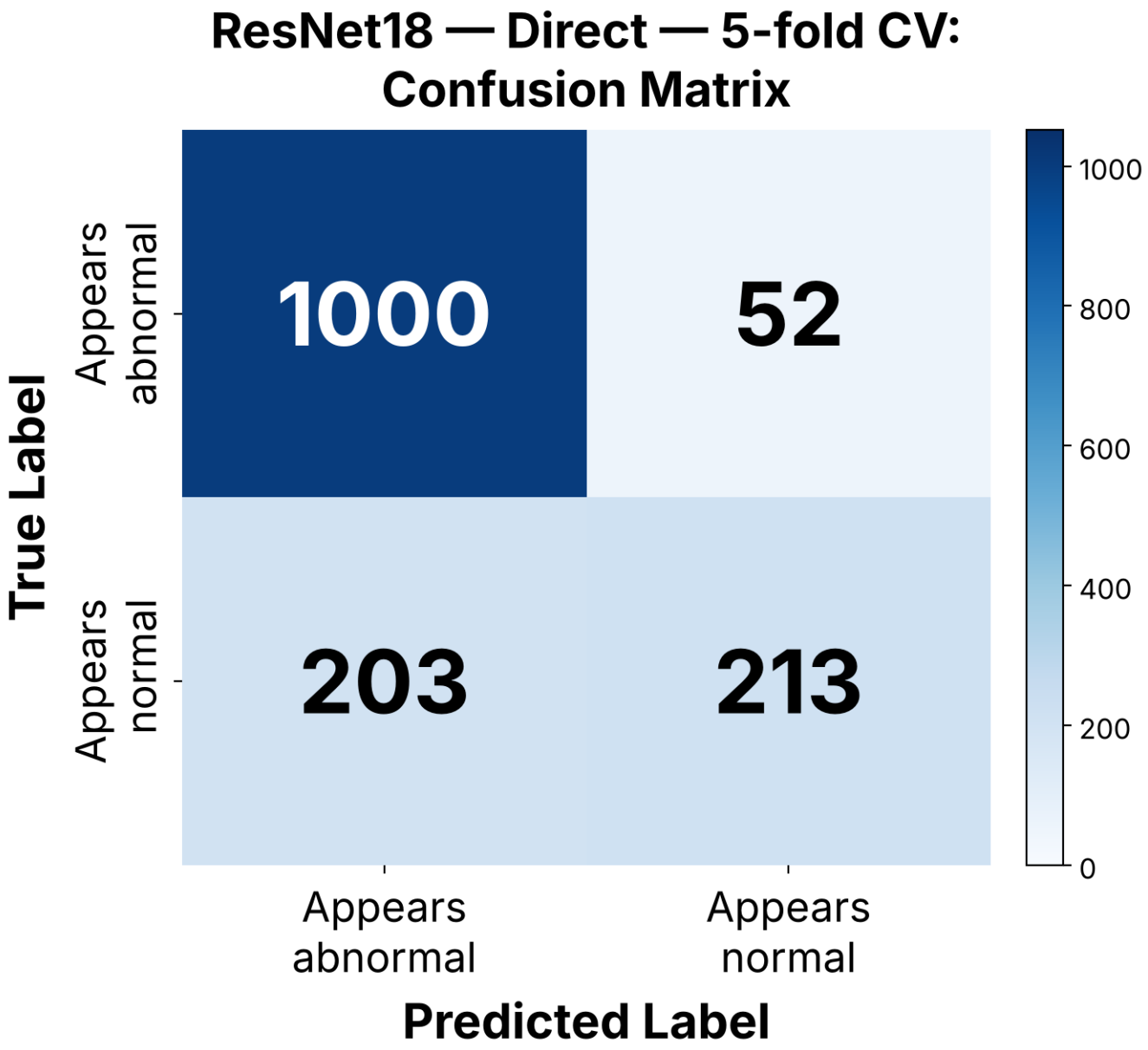
Accuracy: 82.63%

Precision: 82.35%

Recall: 82.63%

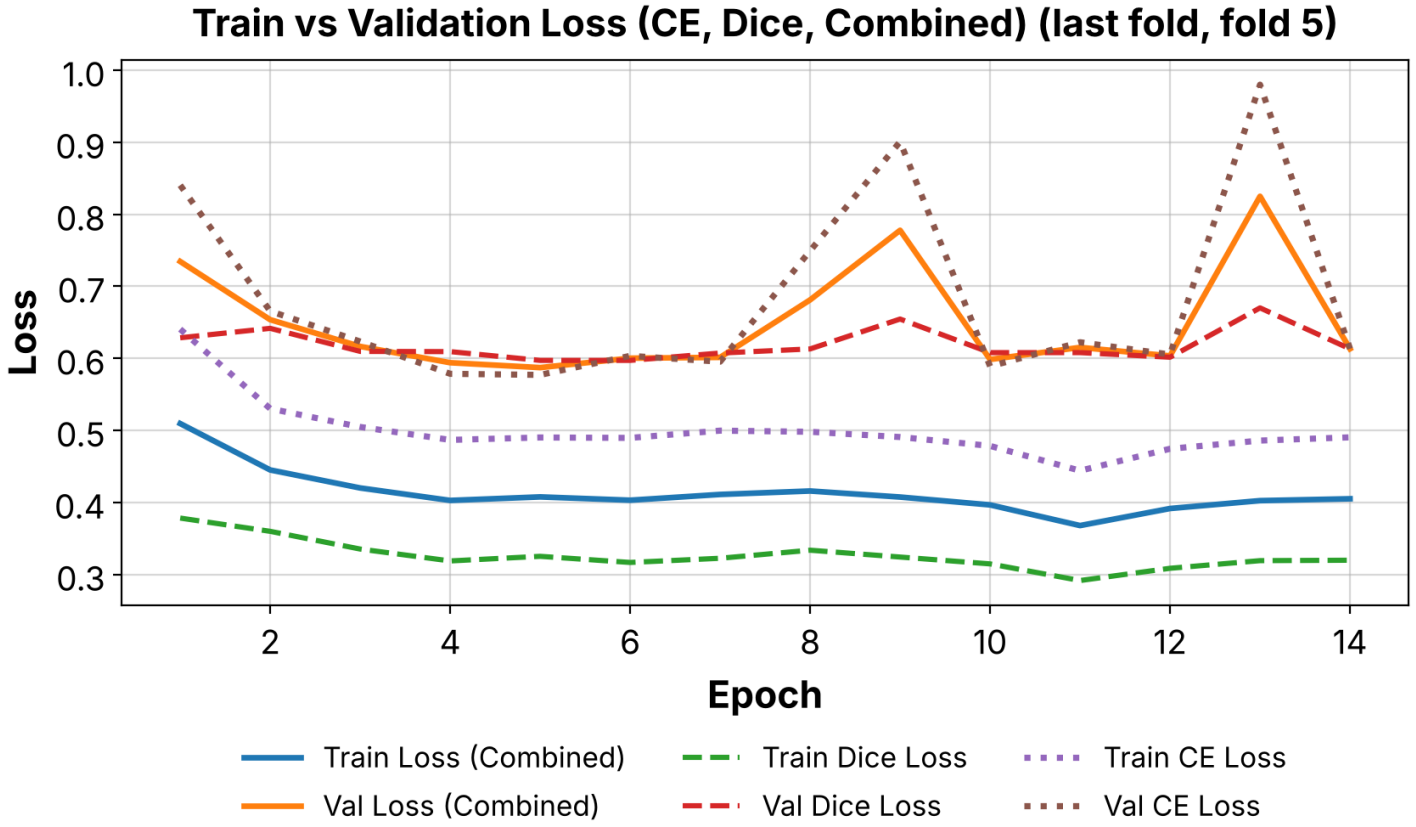
F1-score: 81.29%

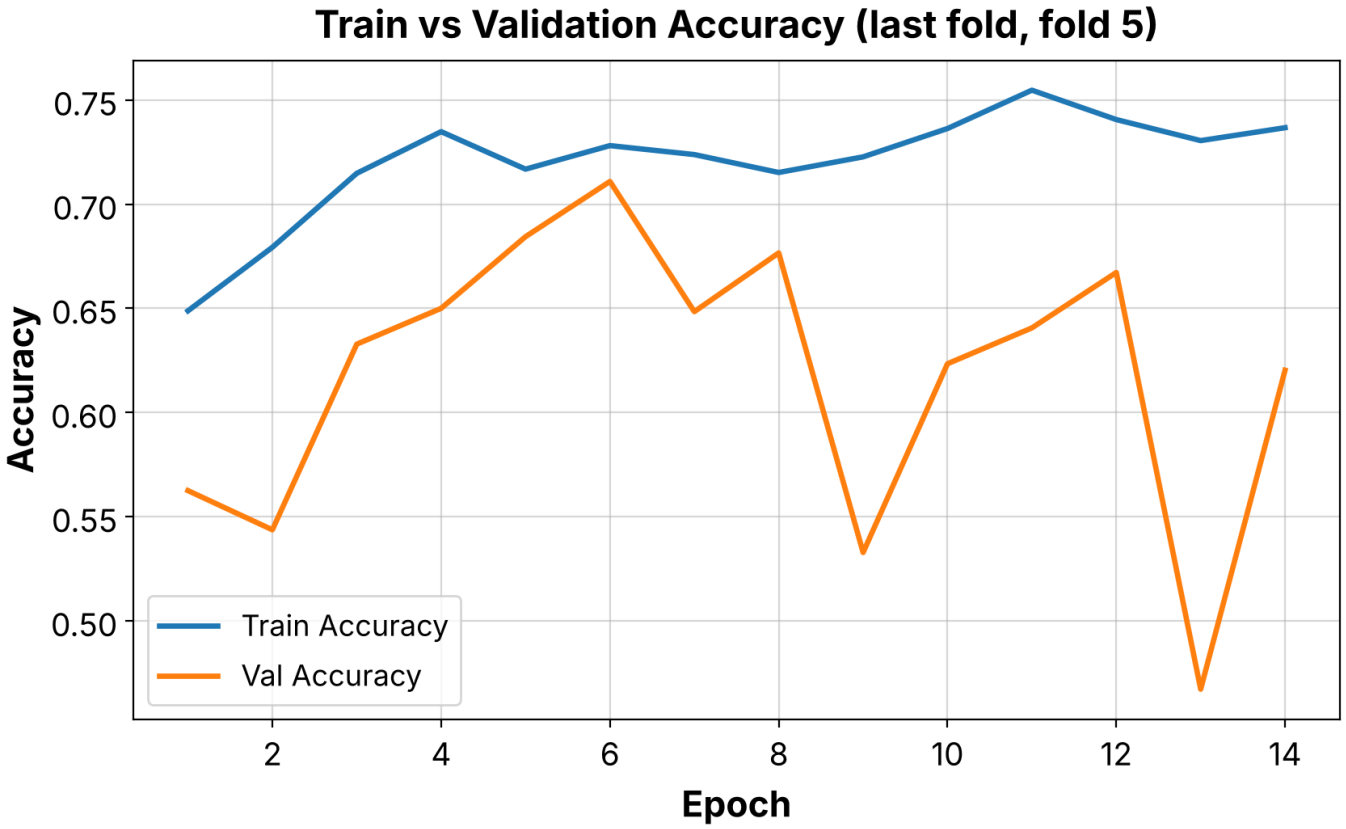
Confusion Matrix



51. ResNet18 — Direct — 5-fold CV — graphs (1/2)

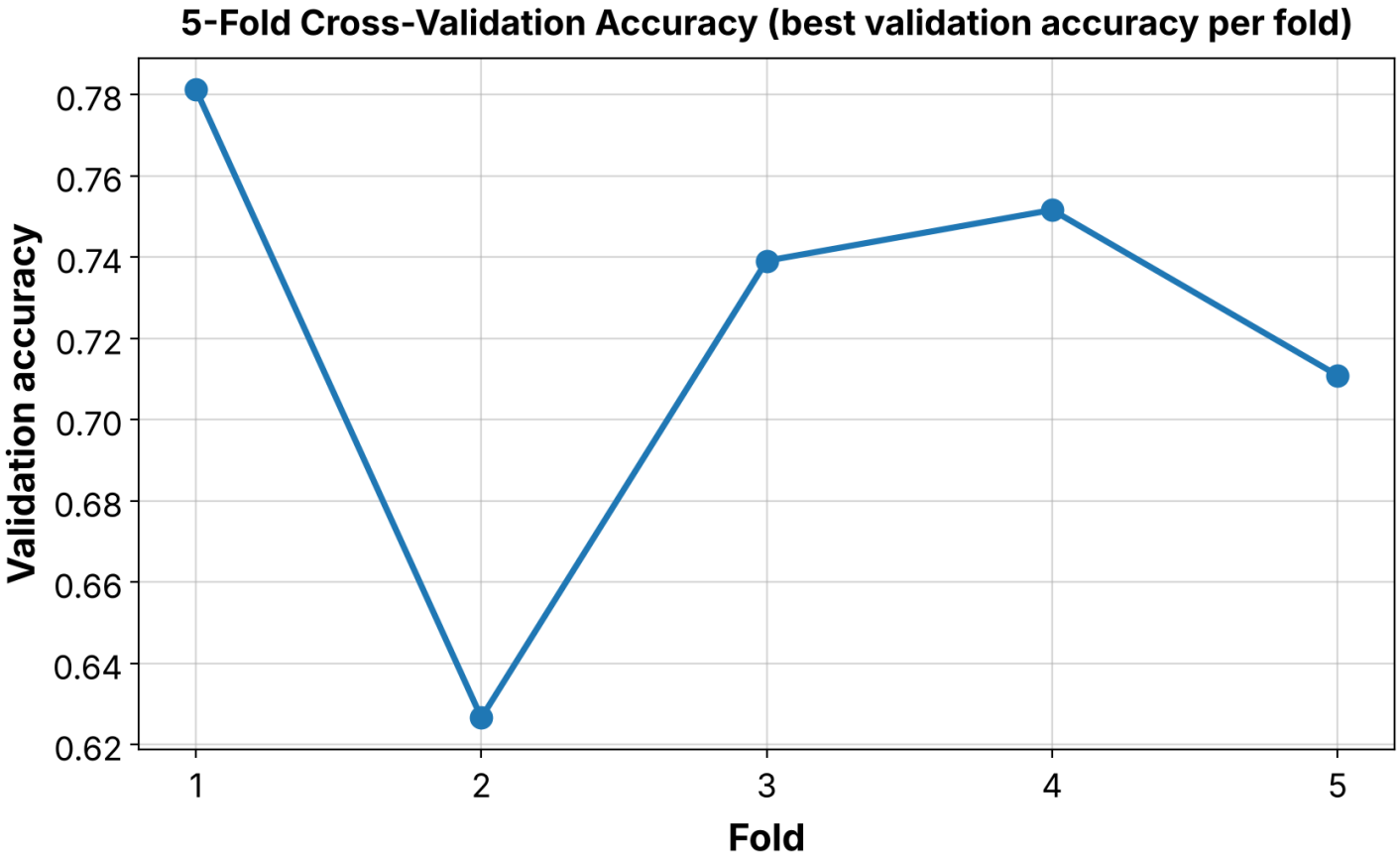
Training / Validation Graphs





51. ResNet18 — Direct — 5-fold CV — graphs (2/2)

Training / Validation Graphs



52. AlexNet + KNN

Source notebook: AlexNet with ML.ipynb (inside Models_DL.zip)

Accuracy: 82.36%

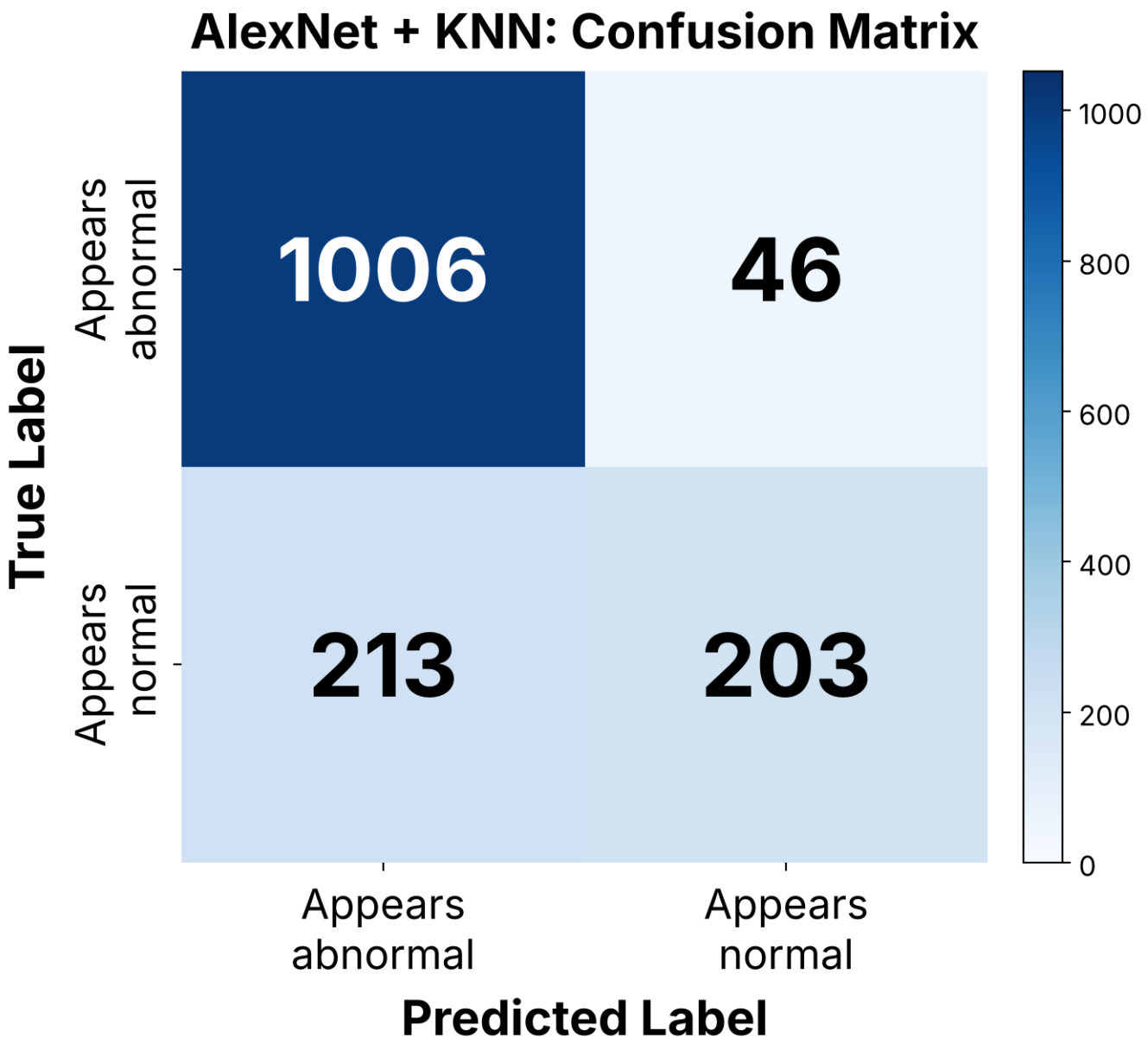
Precision: 82.24%

Recall: 82.36%

F1-score: 80.79%

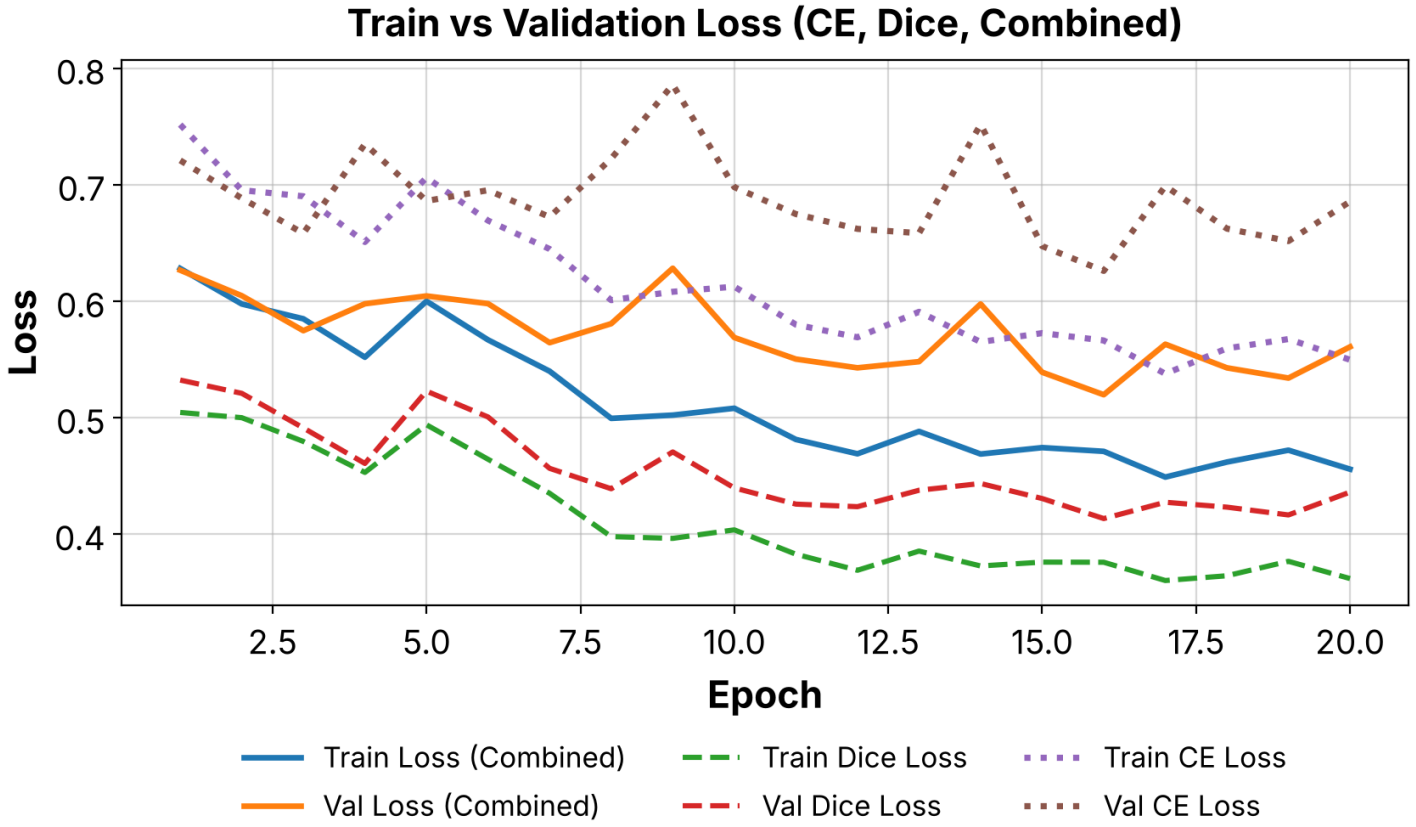
Same pair also run elsewhere with lower accuracy: 80.25% (not shown).

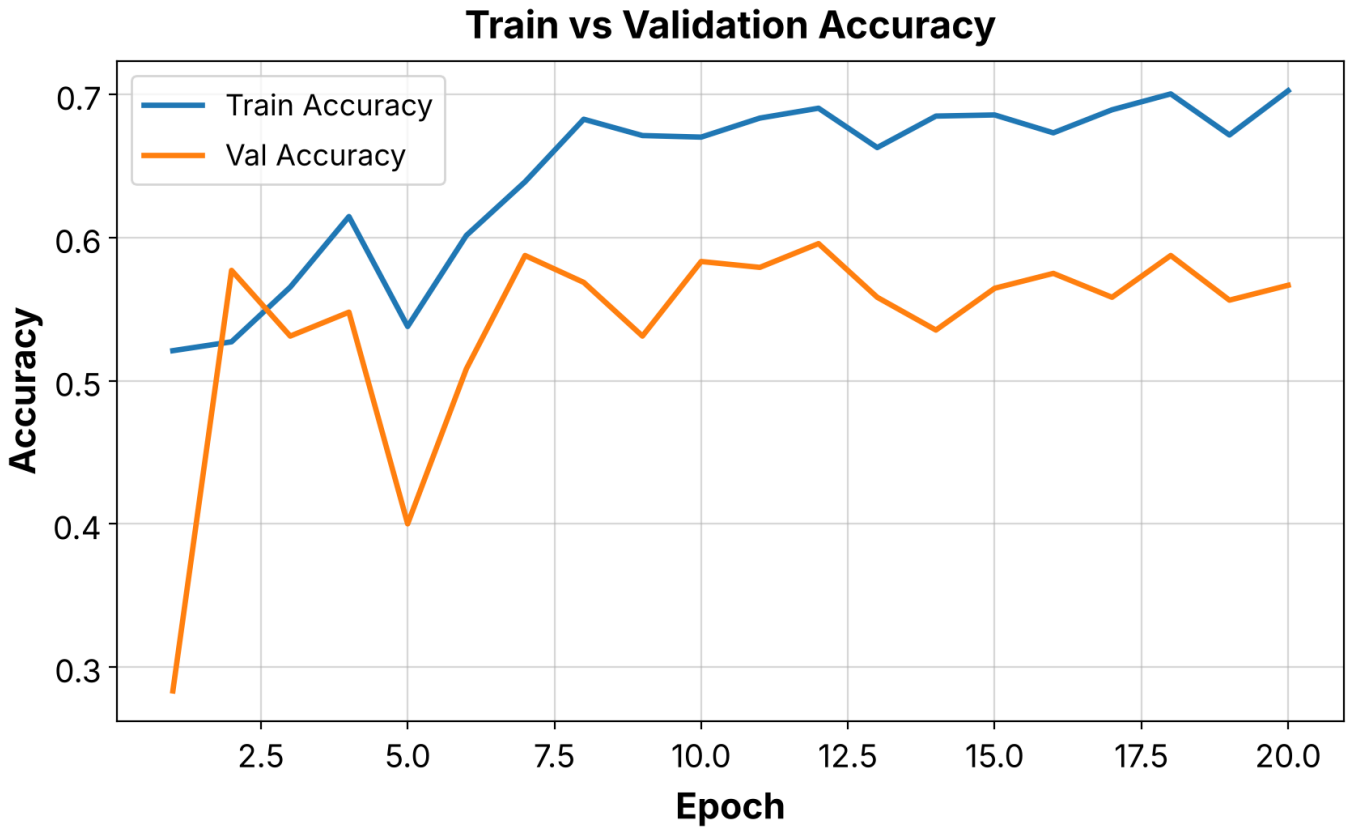
Confusion Matrix



52. AlexNet + KNN — graphs

Training / Validation Graphs





53. AlexNet + Random Forest

Source notebook: AlexNet with ML.ipynb (inside Models_DL.zip)

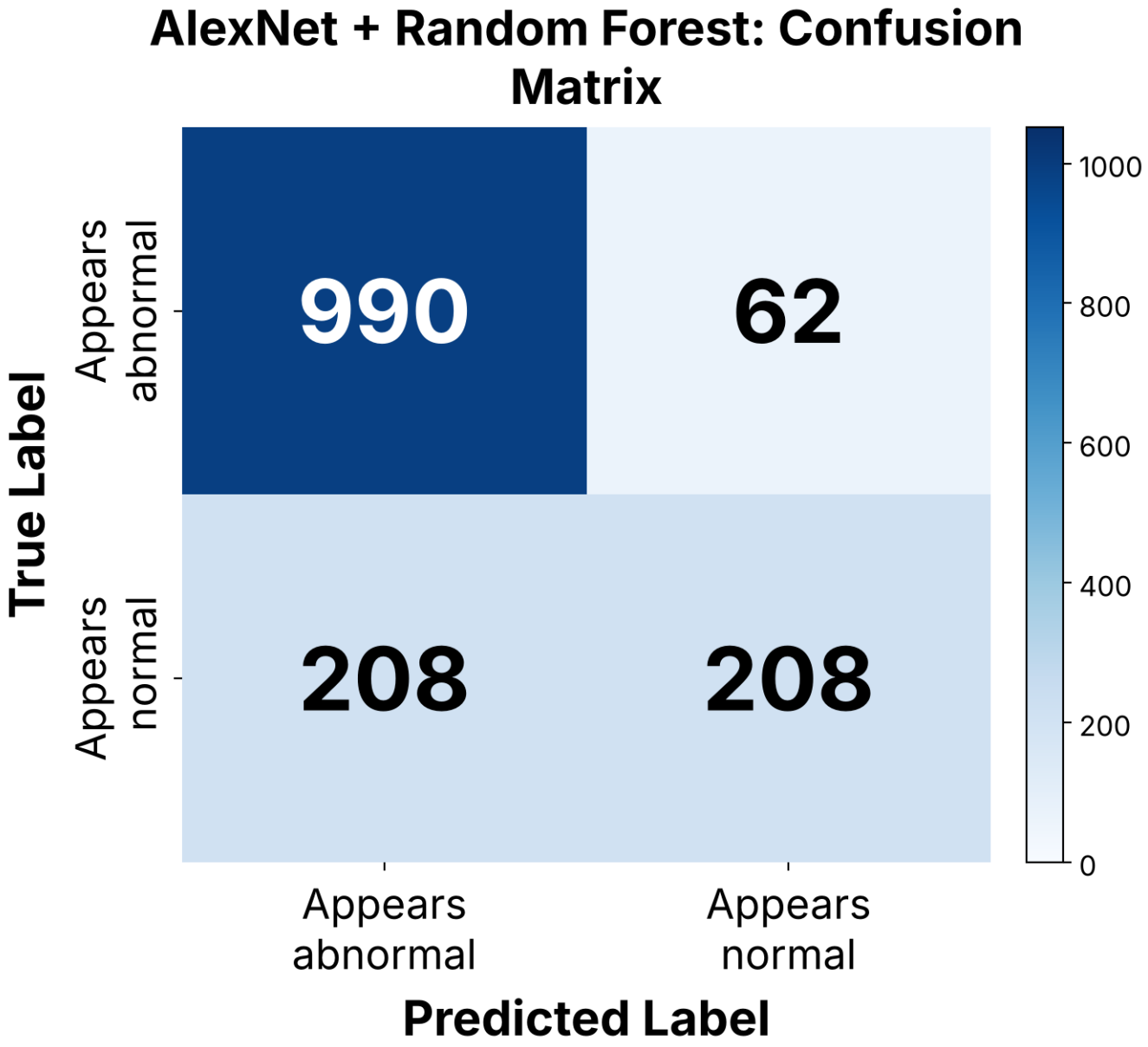
Accuracy: 81.61%

Precision: 81.05%

Recall: 81.61%

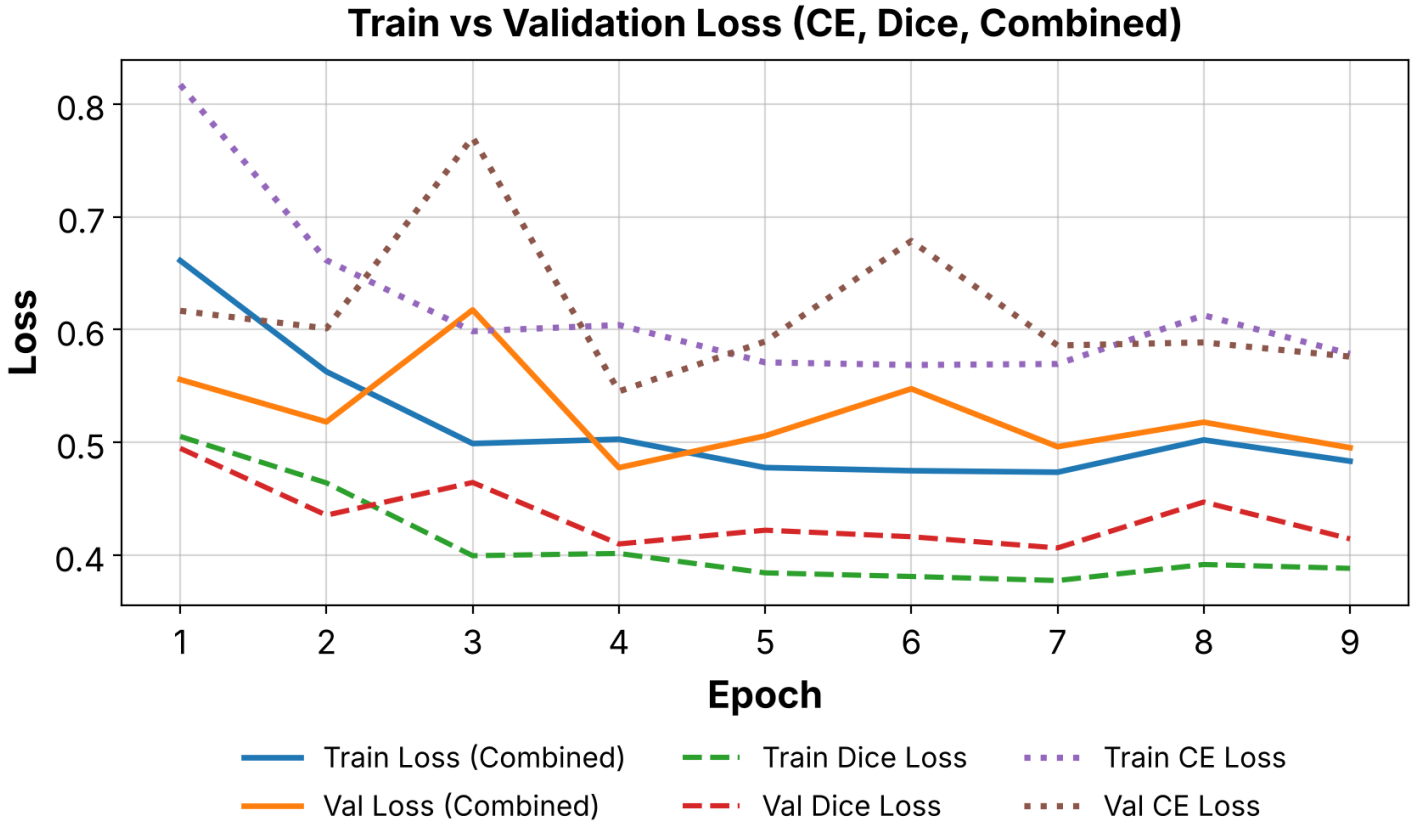
F1-score: 80.25%

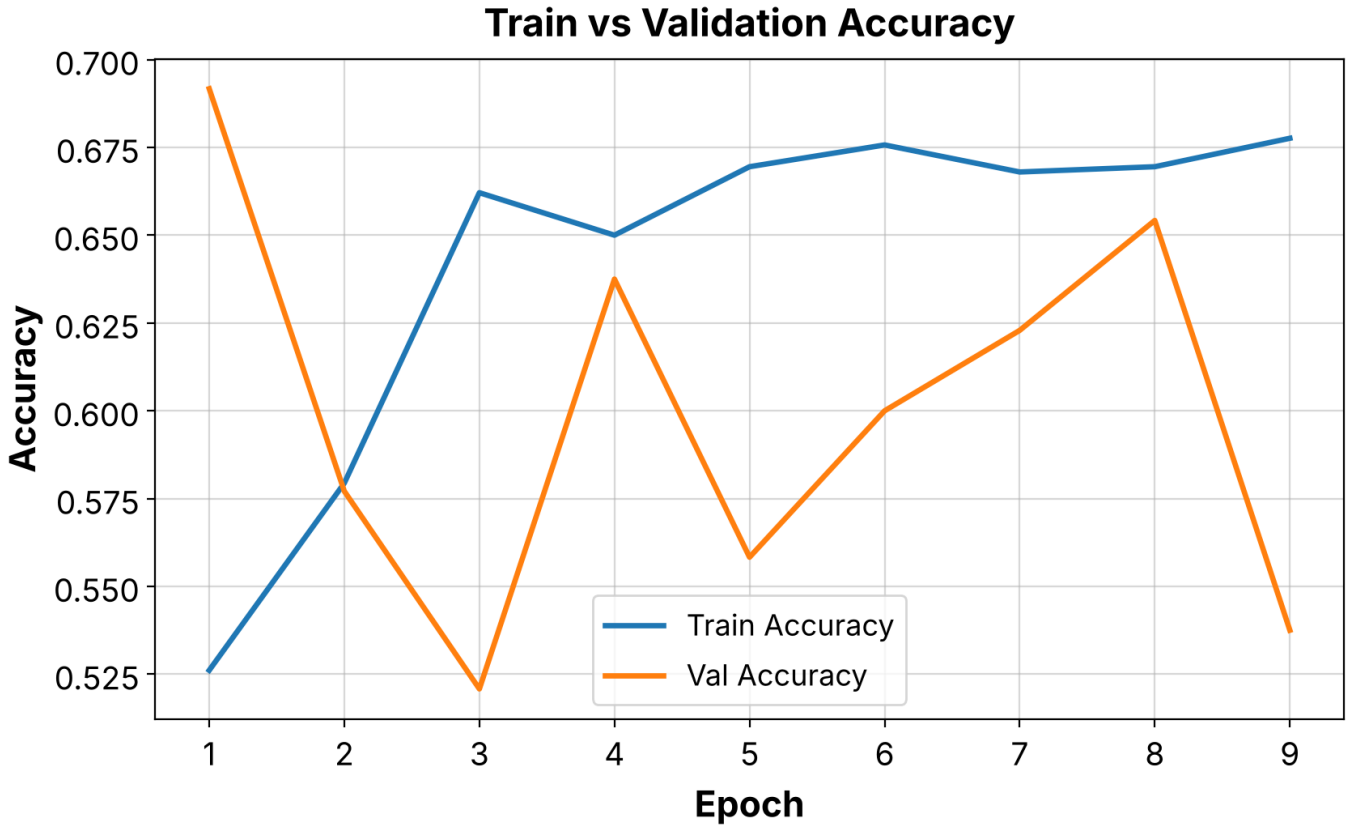
Confusion Matrix



53. AlexNet + Random Forest — graphs

Training / Validation Graphs





54. VGG16 + AdaBoost

Source notebook: Vgg with ML.ipynb (inside Models_DL.zip)

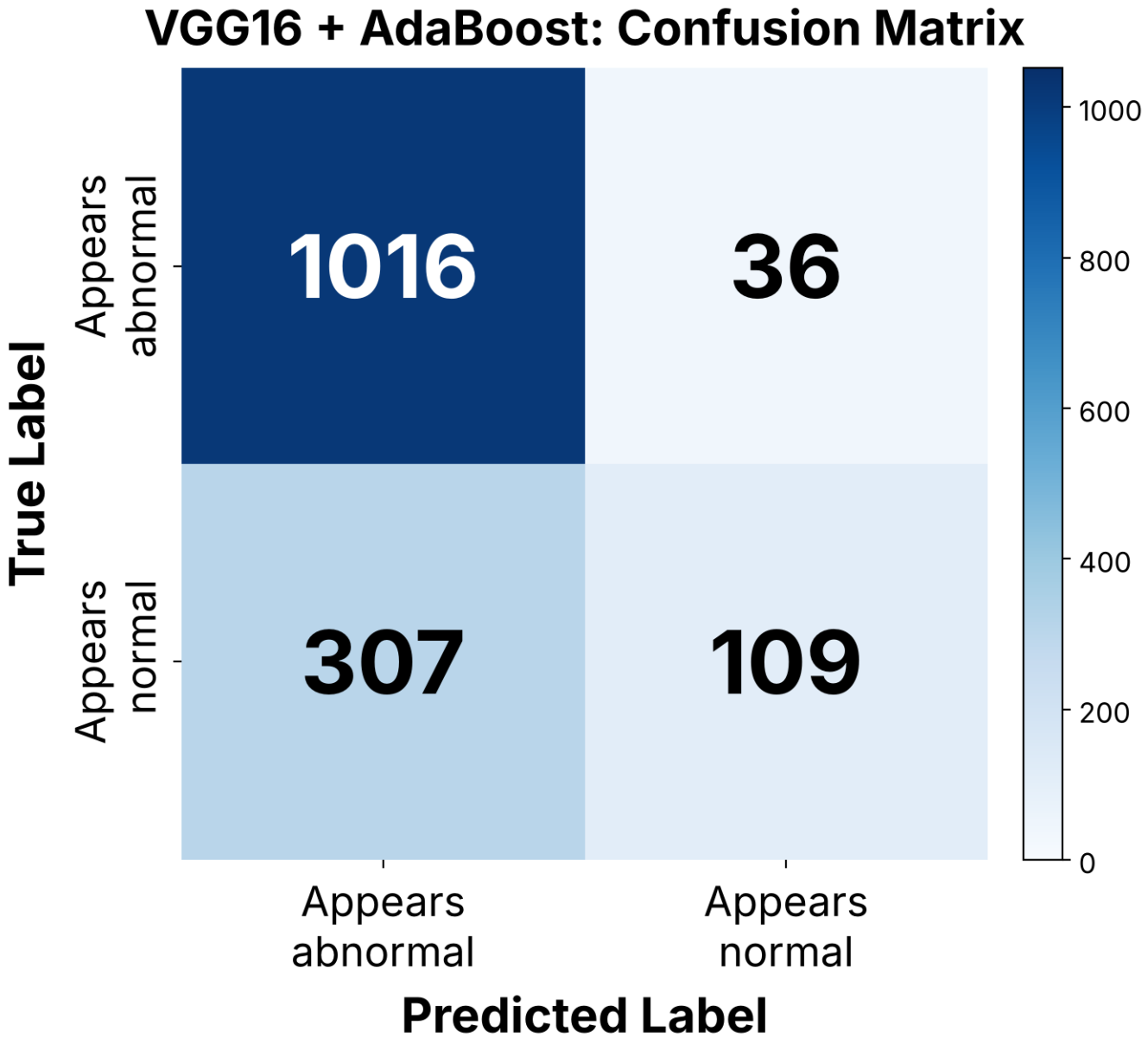
Accuracy: 76.63%

Precision: 76.34%

Recall: 76.63%

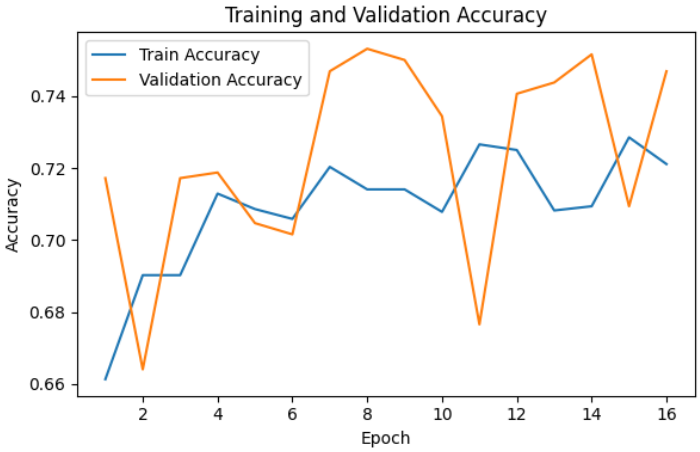
F1-score: 72.32%

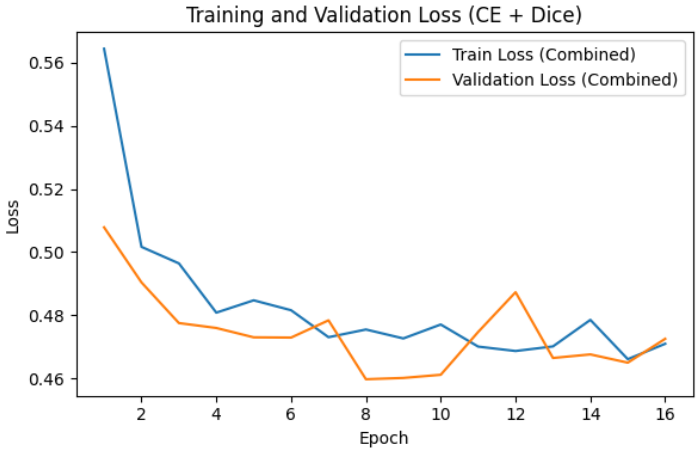
Confusion Matrix

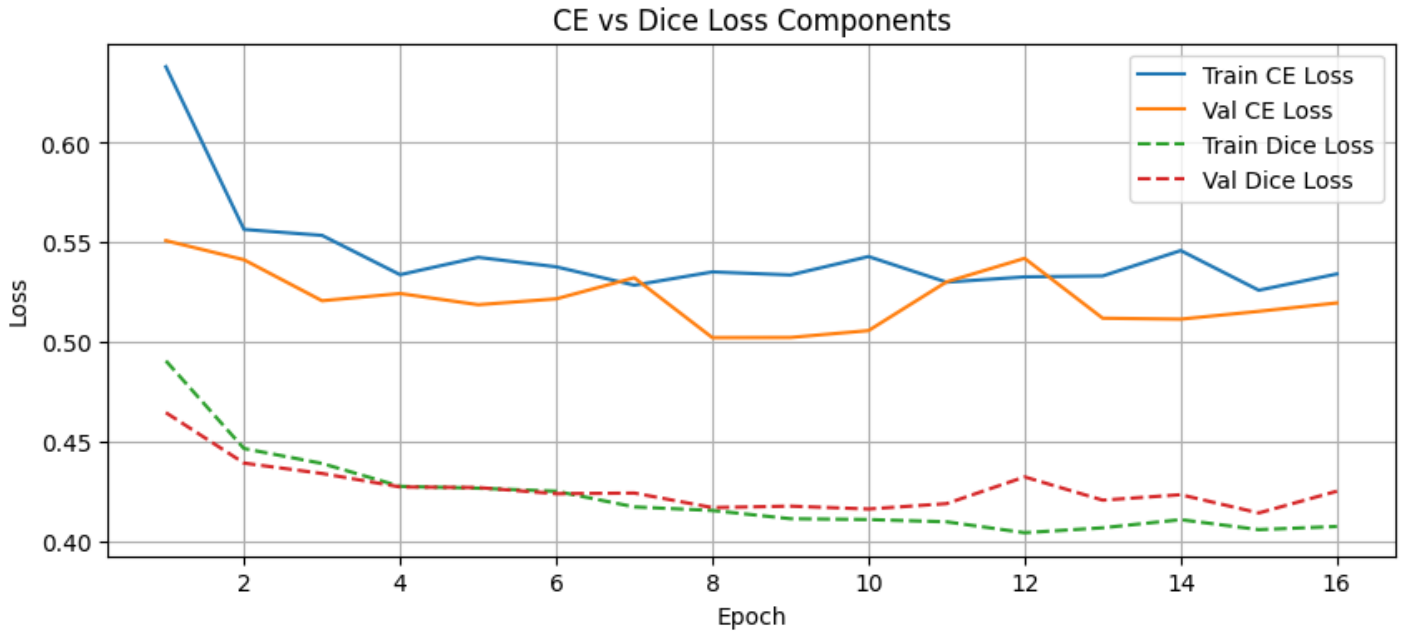


54. VGG16 + AdaBoost — graphs

Training / Validation Graphs







55. AlexNet + Gaussian Naive Bayes

Source notebook: AlexNet with ML.ipynb (inside Models_DL.zip)

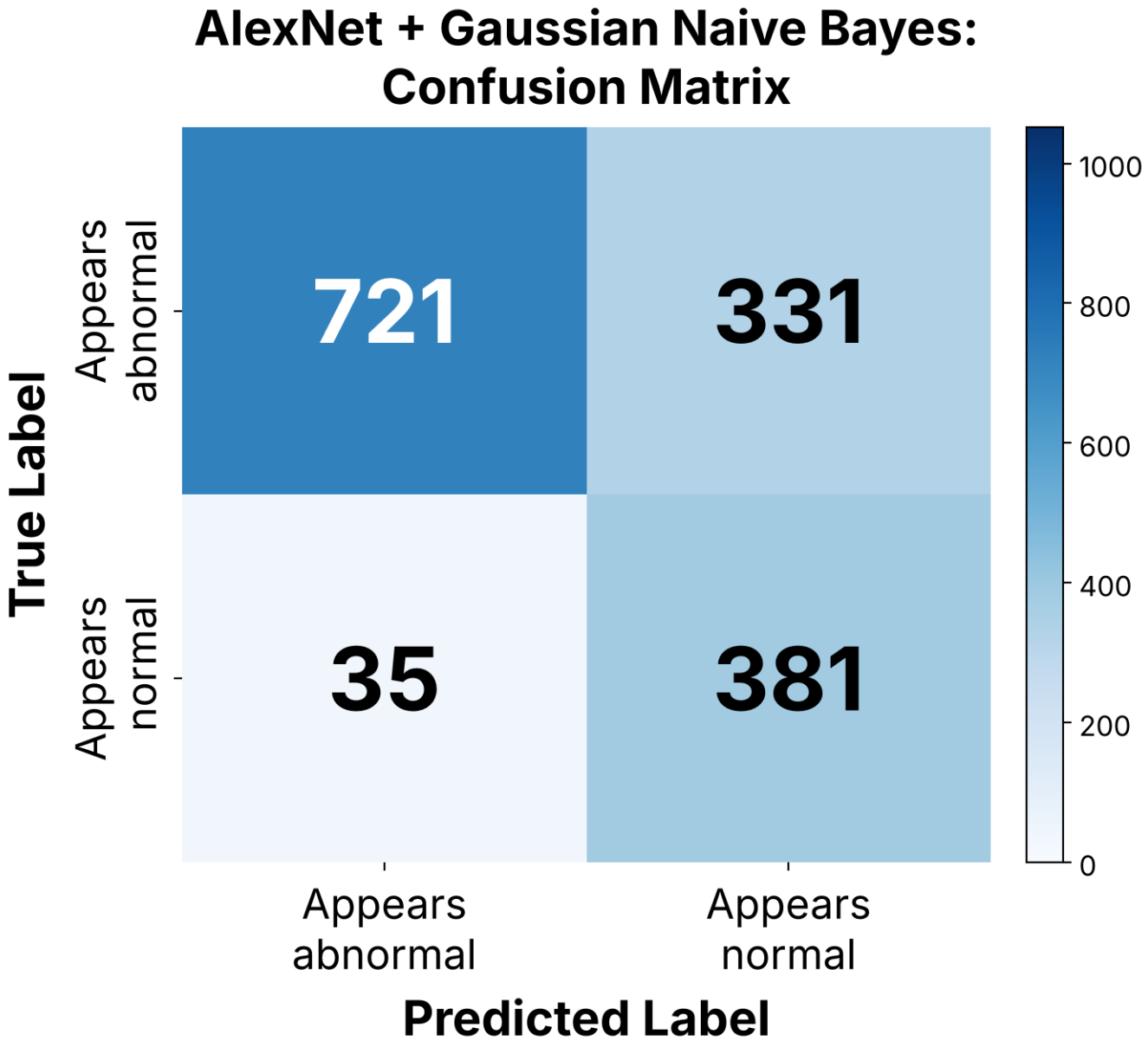
Accuracy: 75.07%

Precision: 83.51%

Recall: 75.07%

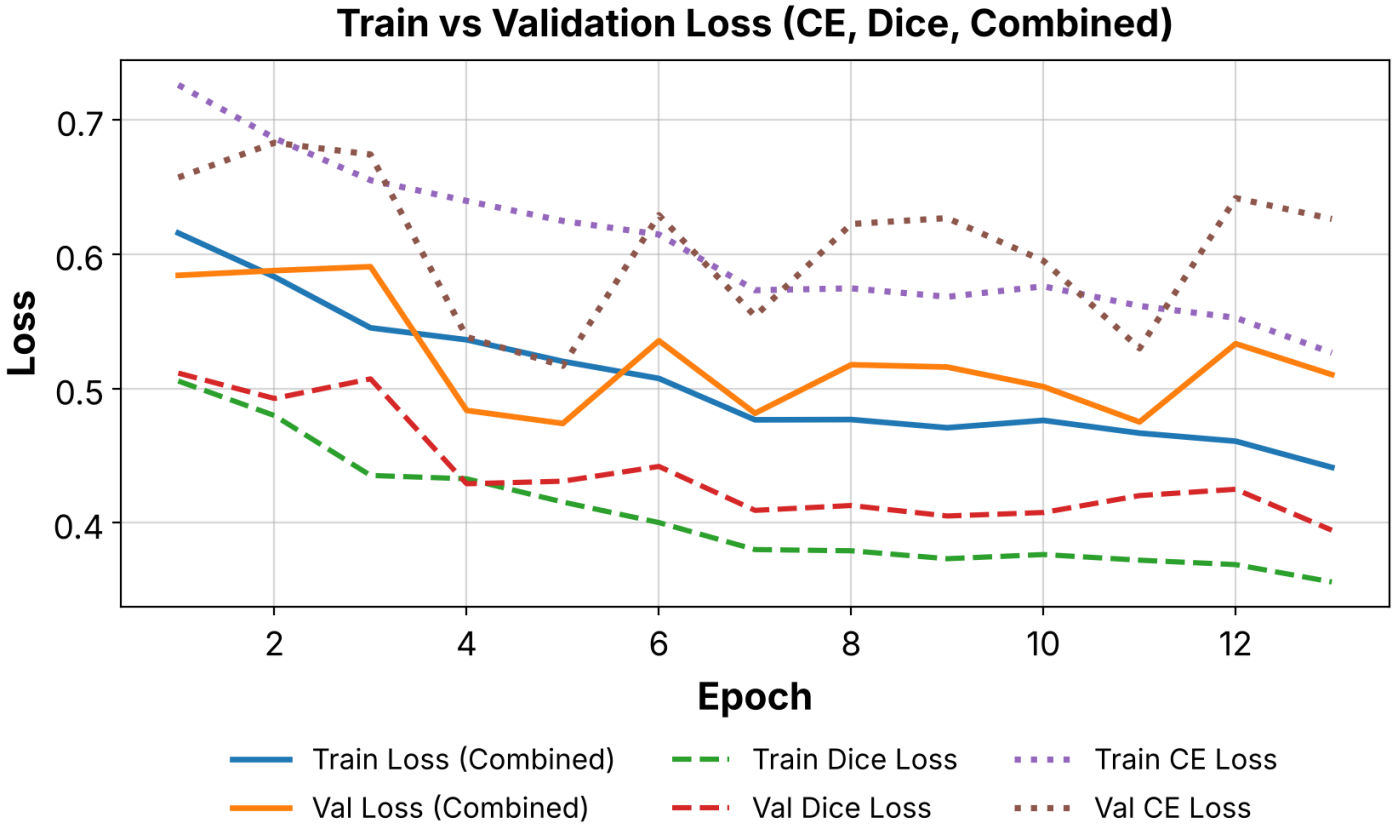
F1-score: 76.30%

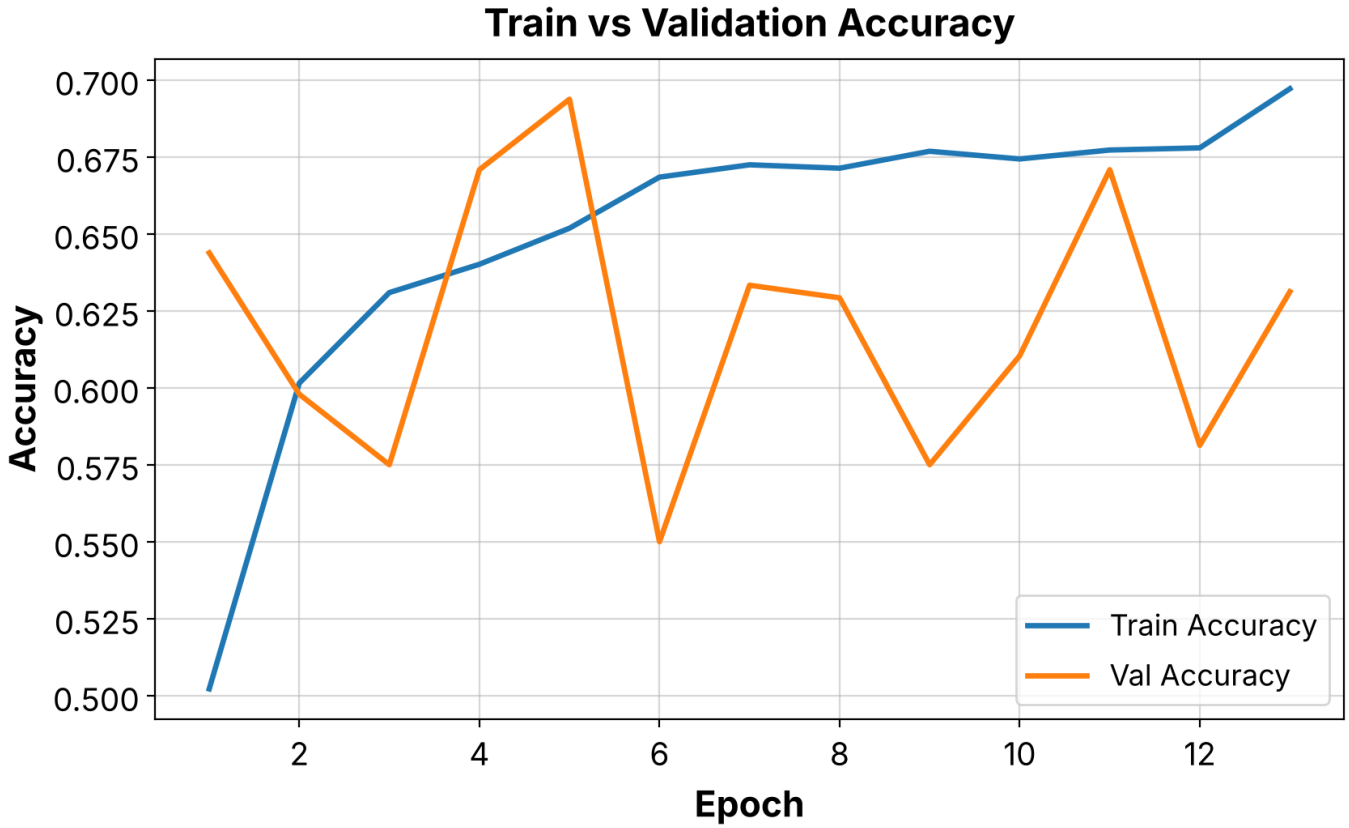
Confusion Matrix



55. AlexNet + Gaussian Naive Bayes — graphs

Training / Validation Graphs





56. VGG16 + Random Forest

Source notebook: Vgg with ML.ipynb (inside Models_DL.zip)

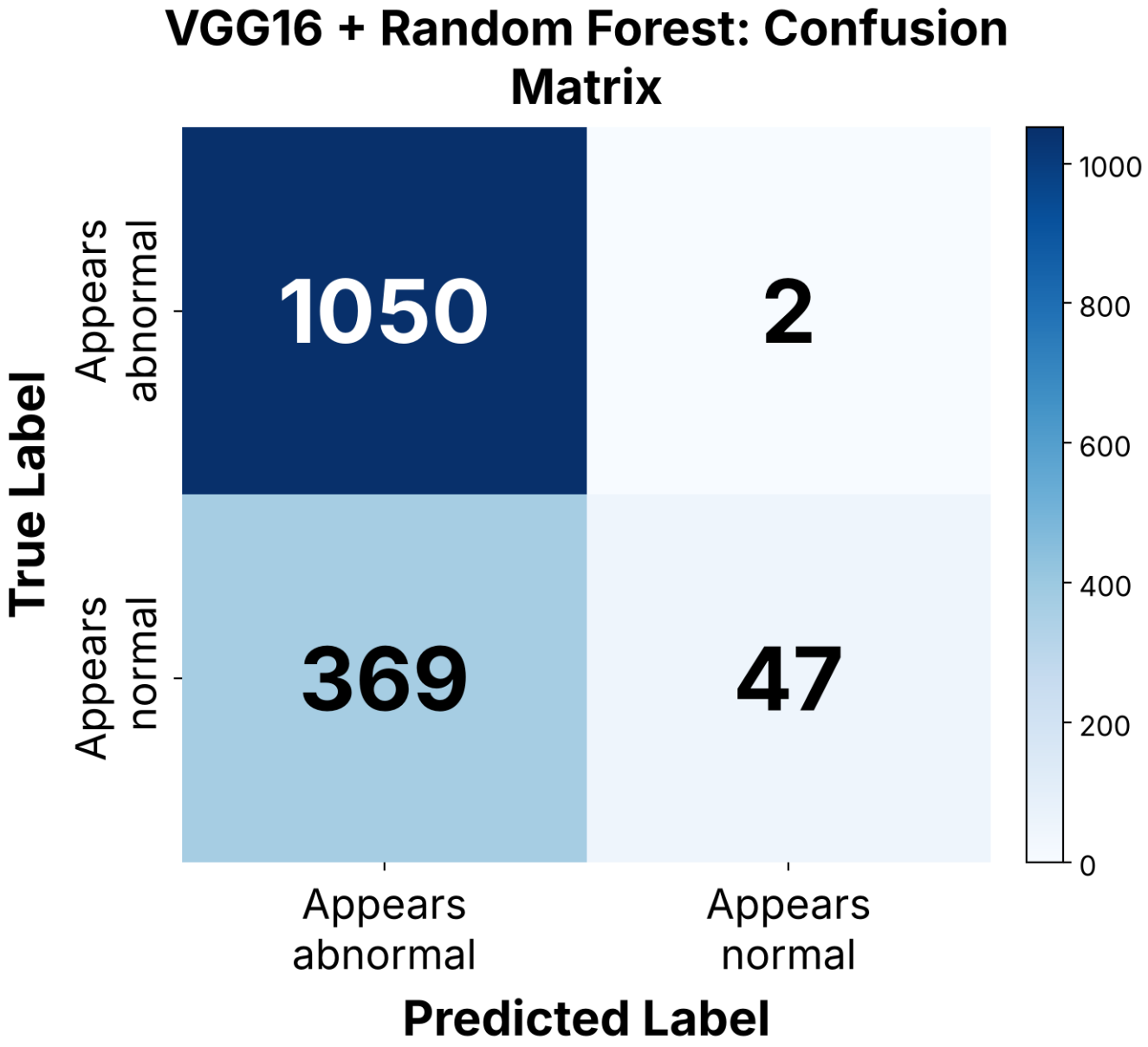
Accuracy: 74.73%

Precision: 80.21%

Recall: 74.73%

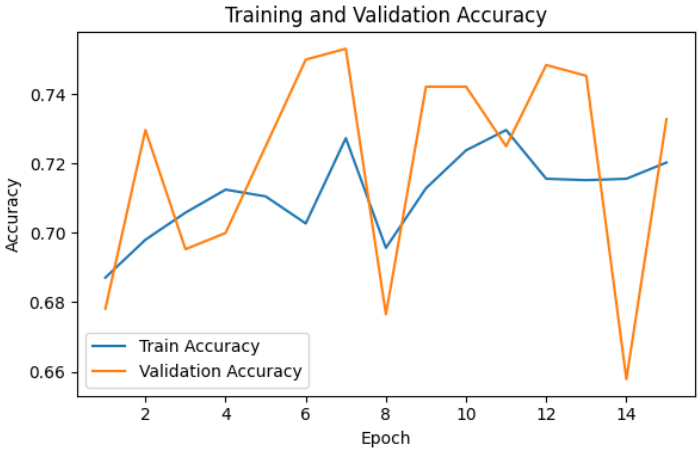
F1-score: 66.63%

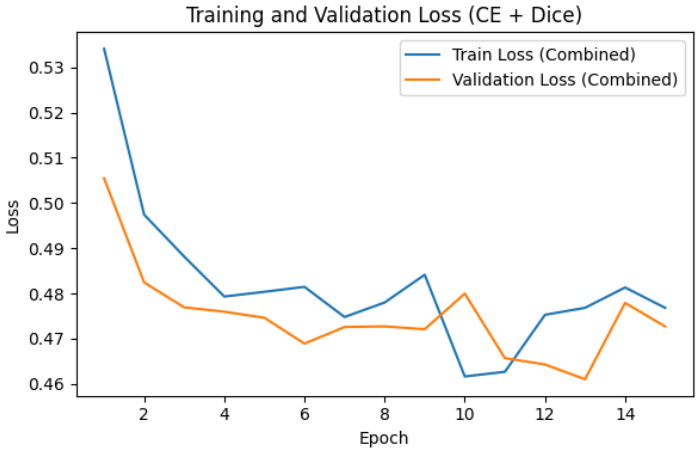
Confusion Matrix

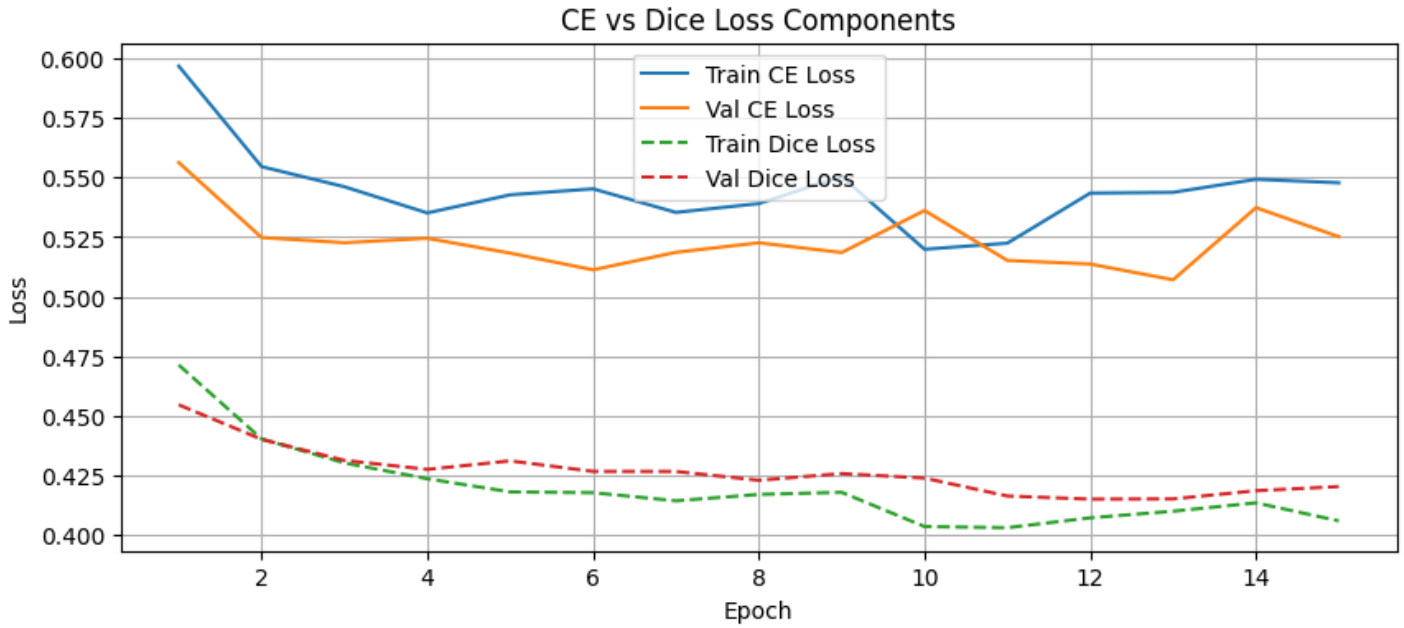


56. VGG16 + Random Forest — graphs

Training / Validation Graphs







57. ResNet18 + Gaussian Naive Bayes

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

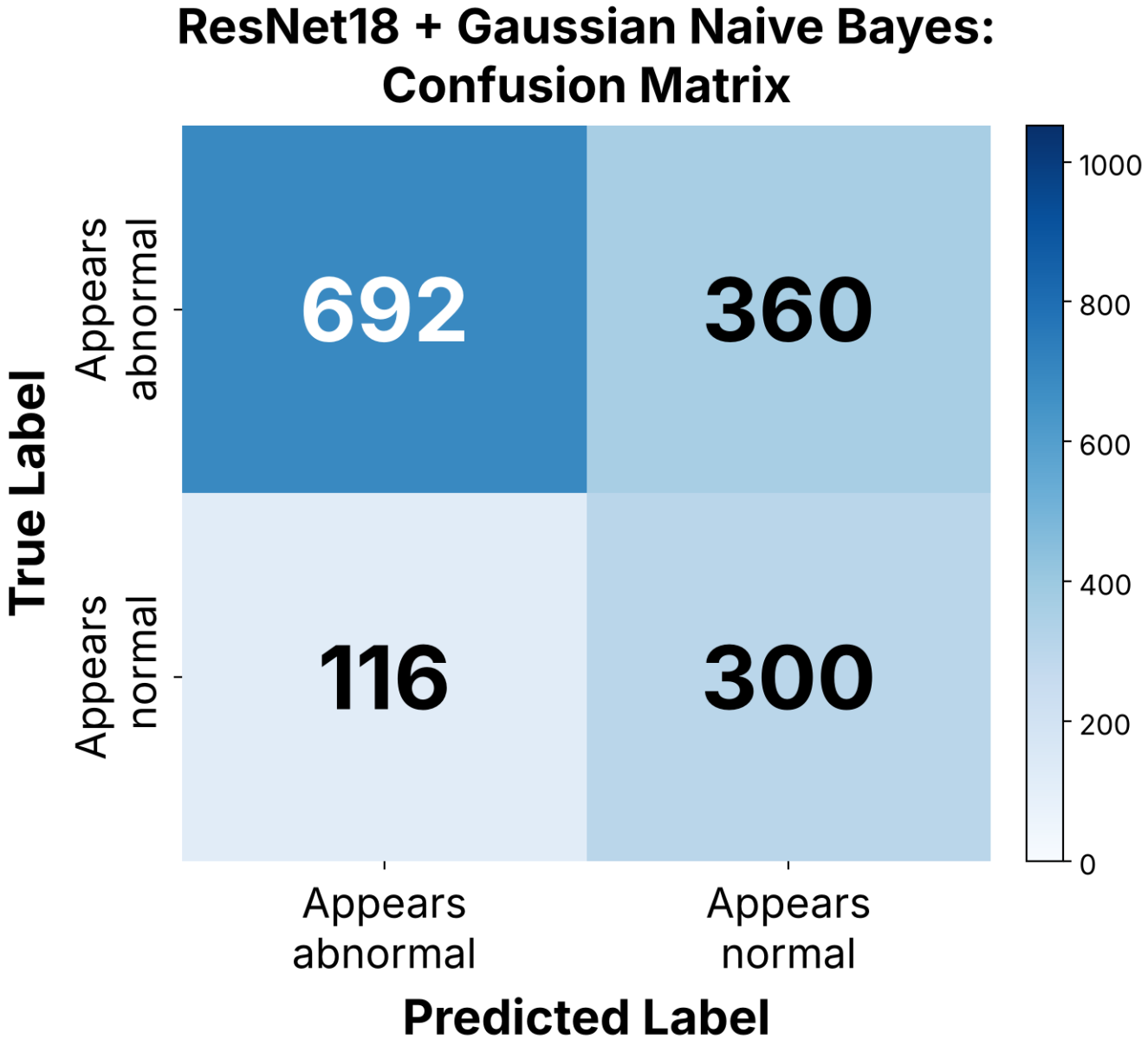
Accuracy: 67.57%

Precision: 74.25%

Recall: 67.57%

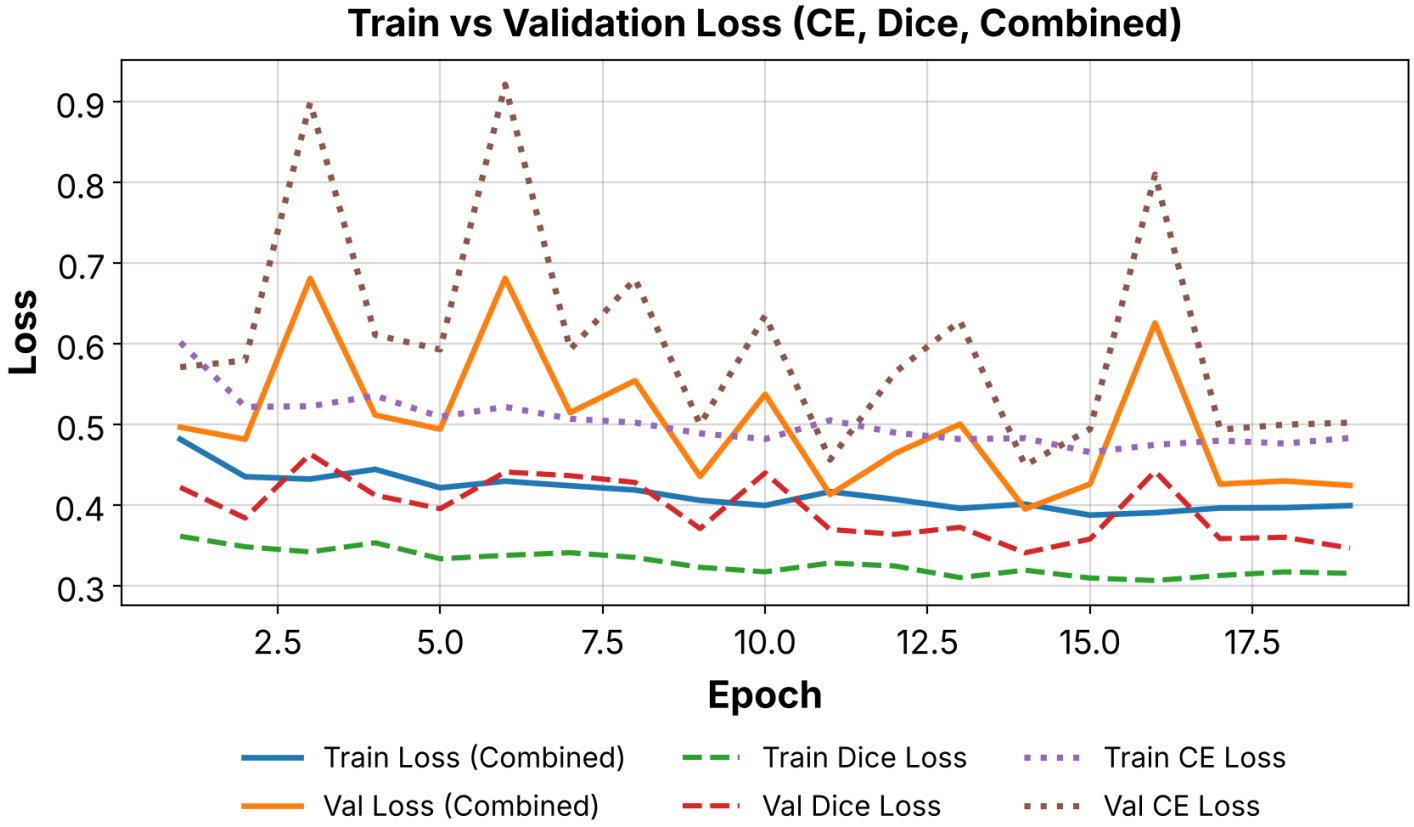
F1-score: 69.12%

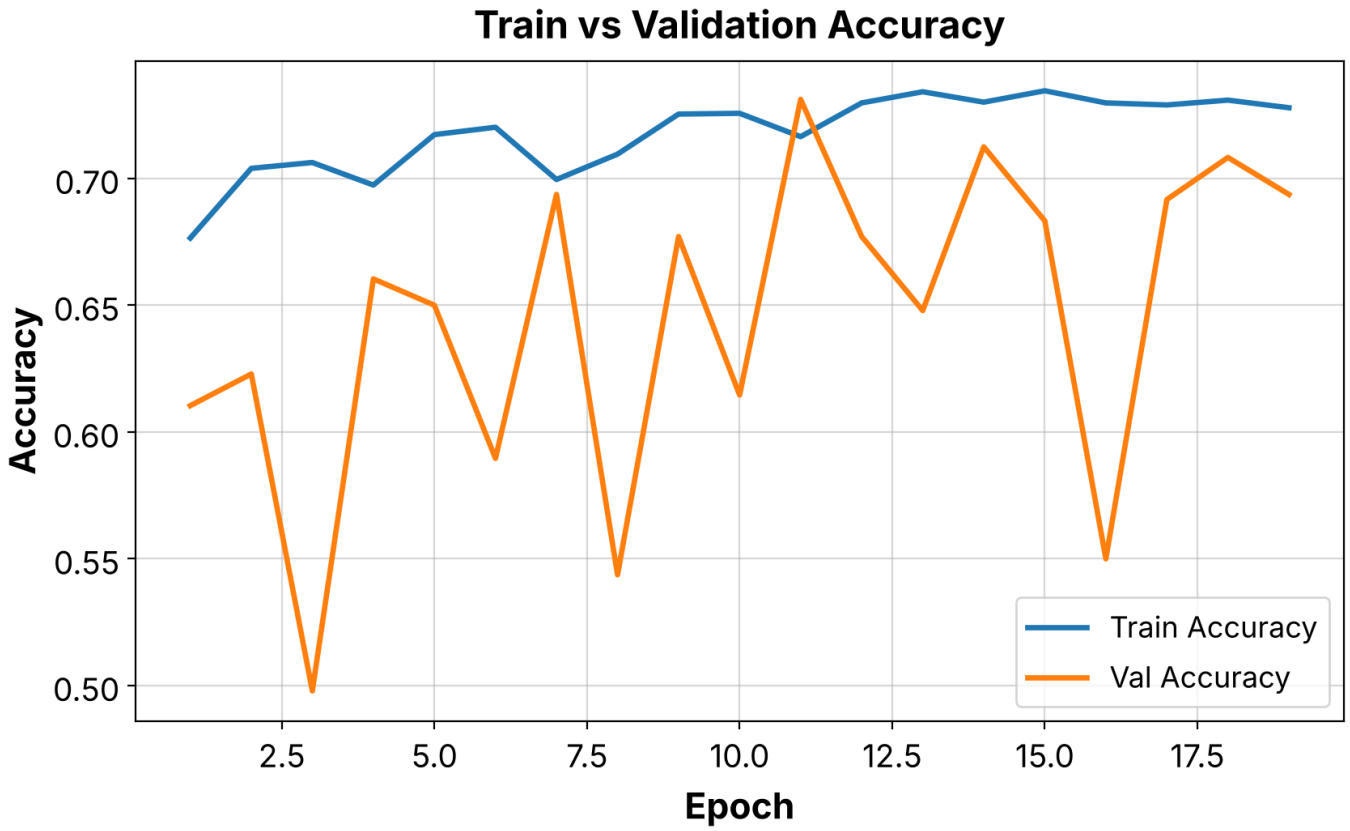
Confusion Matrix



57. ResNet18 + Gaussian Naive Bayes — graphs

Training / Validation Graphs





3. Notes

- Sources: the 17 notebooks of Models_DL.zip, the 4 notebooks of MIT__MobileNet.zip the 3 seeded notebooks (3_Seed_exp, seeded EfficientNet-B3 + AdaBoost, seeded MobileNetV2 / EfficientNet-B0 + AdaBoost) and the executed (seeded) Ensemble notebook. Every value was read from the notebooks' saved outputs; nothing was re-run.
- De-duplication: 68 results from the ZIP notebooks plus 6 seeded results and 4 ensemble results were reduced to 57 unique rows. Each network has one "Direct (fine-tuned CNN)" row (best of its baseline / Dice-loss run, the direct CNN runs inside the classifier sections, and the preprocessing run) and, where the notebooks contain a 5-fold CV run, one "Direct — 5-fold CV" row (AlexNet, MobileNetV2, ResNet18, VGG16). Each model + classifier pair appears once with its highest test accuracy.
- 3-seed rows (green): each seed has its own page. The three seeds are 2622715460, 512 and 2, all with the same architecture and protocol. The headline number is the seed with the highest test accuracy (it is also the highest on precision, recall and F1); brackets give the mean ± sample SD over the three seeds, taken from the notebook's own mean ± SD block and re-computed to check.
- Ensembles (green rows named "Ensemble"): for each seed the AdaBoost test probabilities of the member models (MobileNetV2, EfficientNet-B0, EfficientNet-B3) are averaged image by image (soft voting) and the class with the highest averaged probability is predicted. All members use the same seed, the same train / validation split and the same test set. Before the ensembles were computed, the notebook re-checked all 9 single-model AdaBoost test accuracies (3 models × 3 seeds) against the earlier 3-seed run, and all 9 were identical to four decimals. As for the other 3-seed rows, the headline value is the best seed and the brackets give mean ± sample SD over the three seeds.
- Reproducible = results that are seed-fixed and were re-obtained identically in a second, independent notebook (MobileNetV2 and EfficientNet-B0 seed 2622715460; EfficientNet-B3 seed 512 — same accuracy, precision, recall and F1 to four decimals), with the split and weights saved for packaging. All other notebooks either set no torch seed (only sklearn random_state = 42) or, for the MiT-B2 notebooks, set seed 42 without any repeat run or saved artefacts, so they are reported as not reproducible.
- VGG16 5-fold CV uses Vgg kfold.ipynb (84.20%). The 5-fold section inside MobileNet with Preprocessing.ipynb (85.35%) actually trains VGG16, not MobileNetV2; by request it is left out of the table. That notebook's normal-split section does train MobileNetV2 with preprocessing and is merged into the MobileNetV2 Direct row.
- AlexNet with ML.ipynb: the last section is titled "Adaboost" but the code trains KNN; it is therefore counted as AlexNet + KNN.
- Two VGG16 notebooks print a test accuracy that disagrees with their own confusion matrix: Vgg with Dice Loss prints 88.84% (this is the best validation accuracy), and Vgg kfold prints 81.06% (it does not match its own recall of 84.20%). The table uses the true test accuracy, which equals the weighted recall and the accuracy of the notebook's own printed confusion matrix (90.33% and 84.20%).
- Source column: "cell N" is the position of the cell inside the .ipynb file counted from 0 over all cells (the first cell is cell 0). Some notebooks also contain their own "# CELL n" comments (for example the MiT notebooks), which use a different numbering and are not used here. Eight rows (VGG16 SVM and XGBoost, MobileNetV2 SVM and XGBoost, ResNet18 SVM and XGBoost, AlexNet SVM and XGBoost) come from "Xgboost and SVM and all 4 models.ipynb", which prints only the confusion matrix and metrics and no training log, so those rows have no training graphs.
- Confusion matrices: copied from the notebook output where it prints a matrix; where only an image or per-class report exists, the counts were recovered exactly from per-class recall × support (1,052 / 416) and cross-checked with precision and accuracy. All matrices are redrawn with enlarged fonts.
- Graphs: only values printed in the notebooks' training logs are plotted — nothing is estimated, smoothed or interpolated. Where a notebook printed a complete epoch log (checked: one early-stopping line per run, epoch count and best validation accuracy matching the log), the graph is redrawn from those values with larger fonts, distinct colours and the Train/Val CE curves added; for 5-fold runs the last fold is shown, as in the notebooks. This covers the 3-seed rows and the AlexNet, ResNet18, MobileNetV2 and EfficientNet-B3 rows. For the InceptionV3, VGG16 and MiT-B2 rows the notebooks' original figures are kept unchanged (MiT-B2 / MobileNetV2 inside the MiT notebooks used plain CrossEntropyLoss only; for InceptionV3 the notebook printed only the combined loss and accuracy). "CE" is torch.nn.CrossEntropyLoss (not binary cross-entropy); the combined loss is 0.5 × CE + 0.5 × Dice loss, which holds on every logged epoch. For MobileNetV2 seed 2622715460, the 3-seed notebook's saved text log skips epochs 8–16; the complete 19-epoch log of the same seed comes from the separate seeded MobileNet / EfficientNet-B0 notebook, and it agrees exactly with the 3-seed log on every epoch that both contain. In all scripts the validation images carry the training augmentations, so validation accuracy is lower and noisier than test accuracy.
- Confusion matrices: all are drawn with the same colour map and the same fixed colour scale (0–1,052 images), so equal colours mean equal counts across the report.